

The Two-Time Moving-Defect Problem for Dispersing Billiards

Pair-safe selection lifts, occurrencewise propagated currents, product-time face control, global incidence Cauchy, two-cutoff convergence, and a strongly conditional pathwise CM2 criterion

Working note

14 July 2026 (version 46)

Abstract

We isolate the two-time estimate needed to differentiate an infinite correlation susceptibility under a genuine motion of dispersing scatterers. The moving first defect contains regular $M \times M$ terms and singular one-dimensional physical faces. Neither a projective-cone trace estimate nor a nuclear graph-current representation is available. We avoid both. In the central two-time band we first remove contracted coordinates. The physical face is then pushed to its two scalar dominant coordinates, smoothed only on that quotient, and lifted on two same-branch homogeneous sections. The branchwise contracted-coordinate estimate intertwines the original graph pairing with the section pairing without averaging over other symbolic pasts or singularity cuts. Thus a two-dimensional Fourier polynomial is never declared to be a four-dimensional density by a fictitious chart inverse. At finite cutoff the lifted low block is exactly a sum of products of measured standard-curve currents. Oscillatory section densities are decomposed into positive regular standard families. We retain their actual recovery times, split labels into jointly recovered and one-sided bad tails, and use separate conditional recovery moments rather than multiplying the two initial boundary complexities. Current centering and its test-side adjoint remain distinct typed operations. A measure-trivializing transport puts every moving collision map on one fixed probability space, so both operations, endpoint unfolding, and standard-family coupling use the same invariant reference. At finite difference-quotient level we keep the fixed-gauge operator target: coordinate commutators are not cancelled against transported observables, which would change the differentiated object. Instead the complete operator quotient is decomposed exactly into regular, physical-face, product-current, and one-derivative response blocks. Regular and response blocks are double-projected individually, whereas every graph face is kept together with the exact product-current correction generated by its two marginals. Three nested distribution spaces separate the two jobs that a single weak space cannot perform: an intermediate response space gives the uniform CM2 majorant, while a weaker difference-quotient space gives operator convergence and hence each fixed two-time limit. Dominated convergence then exchanges the limit and the CM2 double sum. The regular and physical-face blocks use the same two-level logic: a strong ledger gives the uniform CM2 majorant, while respectively a weaker Sobolev topology and a weak oriented-current topology give fixed-term convergence. Uniform boundary mass control upgrades the latter convergence to the branchwise tests generated by a fixed finite singularity refinement. A separate two-cutoff theorem then passes from this fixed core to the growing depth $L(s)$: an s -uniform positive coarea/return bound removes the intermediate triangular band, with no $|s|^{-1}$ loss, before the existing deep remainder is used. The complementary dominant modes are first unfolded to the genuine past/future physical endpoints. Thus the Fourier coefficients belong to the original smooth observables, while all time

dependence stays in the physical phase; no regularity of an evolved observable is assumed. Off the central band a signed proper-family theorem supplies forward and reverse coupling. This removes graph Banach spaces, tensor renewal, and nuclear decompositions of the regular defect.

The high-frequency face term is controlled by a physical slope energy at the same prescribed resolving depth used by the phase. We use Leclerc's native unstable coordinate and his cutting theorem modulo every fixed-degree polynomial as a nonvacuity strategy. The theorem itself assumes the exact actual-SRB physical polynomial escape estimate that the phase argument needs. It does not infer that estimate from a fixed negative-relative-pressure clean subsystem: conditional normalization controls bad fractions among survivors but cannot turn exponentially small survival mass into a logarithmic-time physical cover. An adaptive full-mass finite alphabet with fixed rough witnesses is recorded only as a separately falsifiable route to this primitive input. No affine jet is absorbed into the polynomial defining the event, and no infinite fractal conditional is compared with an absolutely continuous one. A single-source conditional identity freezes one oriented endpoint and places the other endpoint in the context polynomial. An actual-SRB first-stopping antichain selects native intervals whose physical diameters are comparable with the energy scale; its complete positive tree law carries all context and mark moments. A mass-preserving resolving buffer and Jensen then give the reduced-slope small-ball estimate. Before any context, stopping, buffer, gate, or continuation label is discarded, all branches are placed in one complete positive energy tree, with a first-failure label for each loss. Each loss is then represented by an exact stopped positive source. Its source density and remaining propagation norm are dominated by a predictable cemetery envelope; future factors are set to one only when that remaining propagation is a positive total-variation contraction. The survivor weight and the five cemetery envelopes form one piecewise full event weight, so a tilted failure probability controls the actual discarded current rather than only an auxiliary label law. A full-product L^{pw} lemma on the complete tree, including a genuine joint cemetery-envelope moment, gives a nonvacuity route from unweighted tails, with every global aggregate exponent explicitly degraded by $1/p'_W$; mere L^1 integrability is never used to preserve a tail exponent. Uniform parent-context tails instead use a normalized kernel reverse-Hölder interface on every frozen actual, split, cemetery, and source-incidence parent. A global complete-tree moment is explicitly not promoted to this conditional statement. The cemetery factor is never conditioned on a sigma-algebra on which it is already measurable. Probability clocks and additive currents are not identified. First-hit disjointization controls only the bad-event union and survivor mass. Every actual additive source occurrence instead lives on one fixed standard-Borel marked source space. Two finite positive kernels over each complete-tree label have distinct duties: one records the exact positive-component coefficient mass and the other the source-norm, recovery, opposite-amplitude, and operator envelope. Every signed source is first measurably decomposed into positive regular components with occurrence-constant polarity; no arbitrary carrier-dependent sign is multiplied into a previously costed positive section. Their Radon-Nikodym density assembles the exact current against the envelope kernel and a normalized weakly measurable carrier section. Global envelope integrability of this one propagated kernel makes the scalar weak pairing a bounded functional on one fixed test space. Thus continuous tree labels are integrated rather than nakedly summed, while two distinct currents with identical support are still counted twice by their occurrence marks. An exhaustive measurable owner partition and a separate resolution-dependent main clock-coverage identity prevent unowned or unlocked marked mass. Tonelli controls the positive envelope before signed recombination. Source-incidence complexity, rather than support-event complexity, is subtracted from every raw source gap. The central clock system is a clock-passage matrix: polynomial-in-energy, prescribed PPE depth, and genuine two-time depth each have separate Jensen and TV columns, while word depth is handled jointly by the independent face-time, face-depth, and deep ledgers. The theorem assumes aggregate summability of both the probability union

and the additive source mass, or derives it from strict complexity gaps before applying Hölder. A Jensen cell now carries two separate quantities: its total base occurrence mass and its fibre energy. One global Cauchy inequality on the entire finite incidence measure converts these into a value-source tail. No square root is taken occurrence by occurrence. The main resonant source is removed from the primitive source clock. Its energy lift starts from the unselected collision–SRB pair law, adds a pointwise normalized proposal mark kernel before the main selector, and places the selector, propagated occurrence weight, reciprocal proposal, and carrier ratios in one unnormalized source likelihood. Different carriers require an actual pushforward reference law, a Radon–Nikodym bound, and target-scale slope compatibility. Thus probability PPE is never conditioned on a main cell for free. All likelihood and base-mass losses enter explicit source-energy and value exponents before this one global Cauchy step. Only genuine two-time TV tails enter the central exponent directly; all Jensen rates are the already-derived base–energy gaps. The choice $\sigma = r^{1/(1+\alpha_L)}$ separates Leclerc’s cutting exponent from the billiard $C^{1+\alpha_B}$ remainder. On the nonresonant set a direct $C^{1+\alpha}$ fractional nonstationary-phase estimate yields a polynomial Fourier saving. The dominant lattice is divided into three ranges: a finite low block below $K = e^{\eta_{lo} \max\{m,n\}}$, an intermediate phase block ending at $H = e^{\eta_{hi} \max\{m,n\}}$, and an ultra-high block estimated only by the Fourier tail of the original observables. Thus no prescribed escape estimate is called beyond the finite spacetime resolving depth. The raw lattice cutoff is supplemented by an upper-scale grazing ledger: endpoint derivative growth is charged jointly to the encountered homogeneity indices and to an initial face/section/chart/coarea scale mark. The excess tail is stated as a survivor subprobability estimate under the complete event-weight tilt, rather than inferred from an unweighted probability tail and mere integrability of the mark. No depth- N endpoint observable is read on an earlier cemetery branch. Labels with excessive weighted growth are removed by that tilted survivor tail. All buffer, gate, stopping, and continuation losses enter the final phase exponent explicitly rather than being hidden in an unnamed exponential remainder.

For radial faces we index every connected component and normalize its mass before forming the boundary functional. A full tangent one-dimensional area formula replaces the vertical two-flux formula, which is retained only as a special case. It is combined with an oriented component ledger; no component is silently discarded. The resulting standard families recover after $O(\log Z)$ iterates. Qualitative residual transversality is insufficient for the infinite-depth limit, so the theorem is stated for analytic or circular table deformations satisfying an explicit pathwise jet-tail condition with a strict return/grazing margin. A word depth $L(s) \asymp \log(1/|s|)$ and two independent time thresholds are used. Shallow short times use endpoint Fourier analysis only while all intermediate-frequency resolving depths lie below $L(s)$; shallow long times are removed directly as finite- s positive-envelope collars. Their positive source mass is normalized before the two one-sided recovery estimates and multiplied back afterwards, so an unrecorded recovery prefactor cannot enter the off-diagonal cones. Deep short times are controlled by positive source mass, and long-time central collars by a boundary-crossing estimate for one positive envelope that simultaneously dominates the actual multi-time law and every functional-correlation split law, followed by one common N -dependent cylinder approximation; off-diagonal collars use moving standard-family coupling after the deep remainder is normalized and shown to have a uniform recovery exponential moment. The small total mass is multiplied back only after this normalized estimate. This gives an $o(1)$ tail in the absolute CM2 double sum, not merely pointwise convergence. The analytic realization requires a fiberwise Remez/valency condition in normalized Fourier coordinates; finite-branch jet surjectivity alone is not used as a sublevel theorem. Circular families retain the quantitative jet condition. Under the primitive hyperbolicity, single-source, actual endpoint–slope, native-stopping, physical-polynomial-escape, exact unselected product-pair energy, complete-weighted-energy, dual-mass constant-polarity source-kernel, post-propagation carrier domination, resolution-exhaustive main-clock coverage, selection-safe joint occurrence–carrier

likelihood, parent-kernel reverse-Hölder, branch-section, geometric-flux, endpoint-upper-scale, split-standard-family, connected-pullback, shallow spacetime refinement, two-cutoff face convergence, either a product-time positive face-envelope or a recovered signed core with an unrecovered positive tail, bounded physical-test and physical/tree matching, normalized shallow/deep-recovery, and pathwise inputs listed below, including the stopped-source domination and clock-passage interfaces, the complete first defect satisfies CM2. For a nontrivial fixed-radius relative translation of one scatterer we then reduce the six hardest nonvacuity interfaces to three concrete geometric gates: a direct finite-holonomy prescribed-depth jet estimate or a shorter terminal rough-gate alternative, an exact coarea physical/source likelihood, and oriented face recovery with weighted tails. The matching gate has a diagonal realization on one common unselected physical tree, for which the only likelihood loss is the total source envelope. Once a moving coarea face has entered a measured unstable family, one propagated power moment of its combined density/boundary-complexity mark implies the face recovery tail and both normalized shallow/deep recovery moments with an explicit common rate window. The cited literature supplies the uniform hyperbolic and coupling package, but not these three gates. No unconditional generic circular-table or relative-translation claim is made.

1 The exact target

Throughout, $\{\widehat{T}_s : |s| < s_0\}$ is a C^{r+1} compact family of finite-horizon planar dispersing billiard collision maps generated by a C^{r+2} normal deformation of strictly convex scatterer boundaries, with collision spaces M_s and invariant collision probabilities $\widehat{\mu}_s$. Write $M_s = \bigsqcup_{i=1}^{d_0} M_{s,i}$, where $M_{s,i} = \partial O_{s,i} \times (-\pi/2, \pi/2)$. A near-identity diffeomorphism cannot transfer mass between these connected components, so the following compatibility is a genuine restriction, not a coordinate construction:

$$\widehat{\mu}_s(M_{s,i}) = \frac{|\partial O_{s,i}|}{\sum_j |\partial O_{s,j}|} = \widehat{\mu}_0(M_{0,i}), \quad 1 \leq i \leq d_0. \quad (1)$$

We work only in deformation charts satisfying (1). On such a chart we impose the following measure-trivializing transport interface. There are C^{r+1} diffeomorphisms $A_s : M = M_0 \rightarrow M_s$, $A_0 = \text{id}$, such that

$$A_s^* \widehat{\mu}_s = \mu_0, \quad T_s := A_s^{-1} \widehat{T}_s A_s. \quad (2)$$

Consequently every T_s preserves the *same* probability μ_0 . For the collision measure this is a componentwise smooth mass-coordinate transport after the component masses have been matched by (1); a transport inside one component cannot repair a changing component mass. Existence with the regularity and chart compatibility used below is part of the interface. Thus independent radius changes of several circular scatterers and unconstrained independent boundary modes are not covered. Admissible examples include center motions with fixed shapes and constrained shape/radius paths whose normalized perimeter vector is constant. The chosen analytic-boundary or finite-dimensional circular deformation is subject to QT_{path} , specified in Section 7.2. The table at $s = 0$ contains the unobstructed clean cycle used in Proposition 6.8. We consider only the complete *first* moving-table defect: regular Jacobian/density derivatives and all same-label physical one-sided faces are assembled before centering. The tests f, g belong to $C^r(M)$, with r above the Fourier summability threshold, and $\mu_0(g) = 0$. Second shape defects and unassembled point atoms are outside the statement.

On this fixed probability space we use the Perron convention relative to μ_0 :

$$P_s h = h \circ T_s^{-1}, \quad P_s^* f = f \circ T_s, \quad P_s \mathbf{1} = \mathbf{1}. \quad (3)$$

Thus P_s is the transfer operator on $L^1(\mu_0)$ and P_s^* is its $L^\infty(\mu_0)$ adjoint for every s . Let

$$\Pi h = \mu_0(h)\mathbf{1}, \quad Q = P_0 - \Pi,$$

and, for every finite s , use the *same* projection

$$\Pi_0 h = \mu_0(h)\mathbf{1}, \quad Q_s = P_s - \Pi_0.$$

Since $P_s \Pi_0 = \Pi_0 P_s = \Pi_0$, for every $m \geq 0$ and $n \geq 1$,

$$Q_s^m g = g \circ T_s^{-m} \quad \text{if } \mu_0(g) = 0, \quad (Q_s^*)^n f = f \circ T_s^n - \mu_0(f). \quad (4)$$

All finite-parameter estimates below are required uniformly on the common spacetime refinement of the compact table chart. Let K_V denote the complete first moving-table defect after all regular terms, same-label physical faces, and grazing blocks have been assembled. The two-time estimate needed for a prescribed centered source g is

$$|\langle Q^n K_V Q^m g, f \rangle| \leq C \tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r}, \quad m, n \geq 0, \quad (5)$$

for some $\tau, \vartheta < 1$. For the difference quotient it is enough to have the same estimate on words of depth at most $L(s)$ together with an $o(1)$ CM2 tail beyond $L(s)$; this is the pathwise diagonal criterion below.

The operator must be centered as a whole:

$$K_V = \Pi^\perp \partial_s P_s|_{s=0} \Pi^\perp. \quad (6)$$

Centering separate faces is neither invariant nor sufficient. Without (6), rank-one terms generally give an additive bound $O(\tau^m + \vartheta^n)$, whose double sum diverges.

Let $\mathfrak{R}_R := \mathcal{M}(M)$ be the strong Banach space of finite signed Borel measures on one collision coordinate, with total-variation norm. Let $\mathfrak{R}_{BL}$ be its completion for the genuinely weaker bounded-Lipschitz dual norm $\|r\|_{BL^*} := \sup_{\|h\|_{BL} \leq 1} |r(h)|$. This norm is fixed as follows. Choose once and for all a bounded compatible complete metric $d_M \leq 1$ on the collision space in the measure-trivializing gauge, with distinct collision components at distance one, and put

$$\|h\|_{BL} := \|h\|_\infty + \sup_{x \neq y} \frac{|h(x) - h(y)|}{d_M(x, y)}. \quad (7)$$

All one-coordinate BL^* statements use this same metric; no resolution- or parameter-dependent chart metric is substituted for it. The inclusion $\mathfrak{R}_R \hookrightarrow \mathfrak{R}_{BL}$ is continuous, $\mu_0 \in \mathfrak{R}_R$, and

$$|r(h)| \leq \|r\|_{\mathfrak{R}_R} \|h\|_\infty, \quad r \in \mathfrak{R}_R, \quad h \in C^0(M). \quad (8)$$

If J is a finite signed graph measure on $M \times M$, its two marginal maps are the actual pushforwards and obey

$$\|(\text{pr}_\pm)_* J\|_{\mathfrak{R}_R} \leq \|J\|_{TV}, \quad |J(1, 1)| \leq \|J\|_{TV}. \quad (9)$$

Thus no independently stronger response norm is hidden in the fourth type. For later bookkeeping let $\mathfrak{P}_R$ be the space of finite product currents

$$P = r^- \otimes \mu_0 + \mu_0 \otimes r^+ + c \mu_0 \otimes \mu_0, \quad \|P\|_{\mathfrak{P}_R} := \|r^-\|_{\mathfrak{R}_R} + \|r^+\|_{\mathfrak{R}_R} + |c|, \quad (10)$$

The representation is taken modulo the obvious constant overlap, with the infimum norm. These three-dimensional mixed products are neither four-dimensional $W^{R,1}$ densities nor one-dimensional graph faces. They therefore form a fourth type even though their pairing with a product of two individually centered endpoint tests is zero.

Definition 1.1 (Fixed-gauge operator finite-difference interface MT_{DQ}). *Let U_s denote the measure-trivializing pullback in (2). The differentiated object is the fixed-gauge operator*

$$K_s^{\text{DQ}} := \Pi_0^\perp \frac{P_s - P_0}{s} \Pi_0^\perp. \quad (11)$$

The tests f, g are fixed in this gauge. In particular coordinate commutators are not cancelled against derivatives of transported observables: such a cancellation would compute a different physical bilinear response. Put $C_0^r(M) = \{h \in C^r(M) : \mu_0(h) = 0\}$ and let $\mathcal{B}_2 \hookrightarrow \mathcal{B}_1 \hookrightarrow \mathcal{B}_0$ be three fixed distribution spaces, called respectively the strong, response, and difference-quotient spaces. The maps Π_0 and $\Pi_0^\perp$ extend continuously to all three; a subscript 0 denotes the corresponding zero-mass subspace. Condition MT_{DQ} requires the following finite- s statements before any absolute value is taken.

(MT1) *Before projection the quotient has one exact typed decomposition. A raw regular or response block is projected individually:*

$$\widehat{K}_{s,j}^{\text{DQ}} := \Pi_0^\perp \widetilde{K}_{s,j}^{\text{DQ}} \Pi_0^\perp, \quad j \in \{\text{reg}, \text{sw}\}. \quad (12)$$

In contrast, let $J_{s,\text{face}}^{\text{DQ}}$ be the assembled raw graph current and let $m_{s,\text{face}}^\pm$ be its two marginals. Its product-current correction and centered cluster are, exactly,

$$P_{s,\text{prod}}^{\text{DQ}} := -m_{s,\text{face}}^- \otimes \mu_0 - \mu_0 \otimes m_{s,\text{face}}^+ + J_{s,\text{face}}^{\text{DQ}}(1, 1) \mu_0 \otimes \mu_0, \quad (13)$$

$$K_{s,\text{face-cl}}^{\text{DQ}} := J_{s,\text{face}}^{\text{DQ}} + P_{s,\text{prod}}^{\text{DQ}} = D_{\text{cur}} J_{s,\text{face}}^{\text{DQ}}. \quad (14)$$

Thus the graph and product summands are distinct distributional types but one jointly centered cluster. Neither summand is asserted to have zero marginals, and $P_{s,\text{prod}}^{\text{DQ}}$ is not projected again (doing so would kill it). Marginal and rank-one terms are never relabelled as regular densities or graph faces. In the notation of (10), $r_s^- = -m_{s,\text{face}}^-$, $r_s^+ = -m_{s,\text{face}}^+$, and $c_s = J_{s,\text{face}}^{\text{DQ}}(1, 1)$. With this convention the complete quotient is

$$K_s^{\text{DQ}} = \widehat{K}_{s,\text{reg}}^{\text{DQ}} + J_{s,\text{face}}^{\text{DQ}} + P_{s,\text{prod}}^{\text{DQ}} + \widehat{K}_{s,\text{sw}}^{\text{DQ}} + E_s, \quad E_s = \Pi_0^\perp E_s \Pi_0^\perp.$$

Here the regular kernel obeys

$$\sup_{0 < |s| < s_0} \sum_{\chi} \|\zeta_\chi k_{s,\text{reg}}^{\text{DQ}}\|_{W^{R,1}} < \infty, \quad (15)$$

while the graph-supported part belongs to the existing physical face atlas and obeys the uniform strong face ledger

$$\sup_{0 < |s| < s_0} \sum_{\lambda} M_{s,\lambda} \|w_{s,\lambda}\|_{A^\alpha} \mathfrak{d}_{s,\lambda}^{A_{\text{dom}}} \text{Mark}(s, \lambda)^2 \{1 + Z(\mathcal{G}_{s,\lambda}^-) + Z(\mathcal{G}_{s,\lambda}^+)\} < \infty. \quad (16)$$

This is the face summand of (245), not a convergence assertion. Finally,

$$\sup_{0 < |s| < s_0} \left(\|\widehat{K}_{s,\text{sw}}^{\text{DQ}}\|_{\mathcal{B}_2 \rightarrow \mathcal{B}_1} + \|E_s\|_{\mathcal{B}_2 \rightarrow \mathcal{B}_1} \right) < \infty. \quad (17)$$

The product block has a typed marginal representation and satisfies

$$\sup_{0 < |s| < s_0} \|P_{s,\text{prod}}^{\text{DQ}}\|_{\mathfrak{P}_R} < \infty. \quad (18)$$

Its two one-coordinate factors and its scalar coefficient are the actual marginals and total mass in (13); no unrelated finite measure may be substituted. Their bounds are precisely the contractive marginal estimates (9) applied before the positive face ledger is summed. A graph-supported boundary contribution of the commutator is included in $J_{s,\text{face}}^{\text{DQ}}$; it is never called an $M \times M$ density.

(MT2) Uniformly for $|s| < s_0$, the centered evolution has the two genuine loss-of-memory bounds

$$\|Q_s^m h\|_{\mathcal{B}_2} \leq C\theta_-^m \|h\|_{C^r}, \quad h \in C^r(M), \quad \mu_0(h) = 0, \quad (19)$$

$$\|Q_s^n u\|_{(C_0^r)^*} \leq C\theta_+^n \|u\|_{\mathcal{B}_1}, \quad u \in \mathcal{B}_{1,0}, \quad (20)$$

with $0 < \theta_{\pm} < 1$. These are Banach-space estimates, not an assumed CM2 estimate for $\widehat{K}_{s,\text{sw}}^{\text{DQ}}$.

(MT3) The same typed organization, including the exact face-product cluster (14), holds for the limiting fixed-gauge operator K_V , with $\|\widehat{K}_{V,\text{sw}}\|_{\mathcal{B}_2 \rightarrow \mathcal{B}_1} \leq C$, and

$$\|\widehat{K}_{s,\text{sw}}^{\text{DQ}} - \widehat{K}_{V,\text{sw}}\|_{\mathcal{B}_2 \rightarrow \mathcal{B}_0} + \|E_s\|_{\mathcal{B}_2 \rightarrow \mathcal{B}_0} \longrightarrow 0. \quad (21)$$

The product-current factors converge in their one-coordinate weak topology:

$$r_s^{\pm} \longrightarrow r_V^{\pm} \quad \text{in } \mathfrak{R}_{\text{BL}}, \quad c_s \longrightarrow c_V. \quad (22)$$

The other two types have their own explicitly weaker fixed-term topologies. The regular kernels satisfy

$$\sum_{\chi} \|\zeta_{\chi}(k_{s,\text{reg}}^{\text{DQ}} - k_{V,\text{reg}})\|_{W^{R-1,1}} \longrightarrow 0. \quad (23)$$

For the face type, first fix (m, n) and a finite depth L , and let $\mathcal{J}_{s,L}^{\text{face}}$ and $\mathcal{J}_{V,L}^{\text{face}}$ be the assembled, oriented graph currents on the common UCR_{sh} atlas. More precisely, let $\mathfrak{A}_{L,m,n}^{\text{face}}$ be the finite disjoint union of the component-indexed parameter arcs in that atlas, equipped with its fixed chart metric and Borel record map into $M \times M$; currents and tests below are pulled back to this common metric space before their bounded-Lipschitz norms are taken. Throughout, $\mathcal{J}_{s,L}^{\text{face}} := \mathcal{J}_{s,\leq L}^{\text{face}}$ and $\mathcal{J}_{V,L}^{\text{face}} := \mathcal{J}_{V,\leq L}^{\text{face}}$; the subscript is a cutoff depth, never an exact-depth label. With $\|\cdot\|_{\text{BL}^*}$ denoting the bounded-Lipschitz dual norm on that finite atlas, require

$$\|\mathcal{J}_{s,L}^{\text{face}} - \mathcal{J}_{V,L}^{\text{face}}\|_{\text{BL}^*} \longrightarrow 0, \quad (24)$$

$$\sup_{0 < |s| < s_0} |\mathcal{J}_{s,L}^{\text{face}}|([\mathcal{S}_{L,m,n}]_{\rho}) + |\mathcal{J}_{V,L}^{\text{face}}|([\mathcal{S}_{L,m,n}]_{\rho}) \leq C_{L,m,n} \rho^{\theta_{L,m,n}}, \quad 0 < \rho < 1. \quad (25)$$

Here $\mathcal{S}_{L,m,n} \subset \mathfrak{A}_{L,m,n}^{\text{face}}$ is the finite union of genuine singularity and common-atlas boundaries encountered by the depth- L face words and by the branchwise tests $Q_s^m g, Q_s^{*n} f$. In particular, bounded-Lipschitz convergence together with (25) implies convergence against every bounded test that is branchwise continuous off $\mathcal{S}_{L,m,n}$. The chart maps, orientations, amplitudes, and connected-component indices used in these currents are those of the common atlas; symbolic cylinders without component indices do not define this topology. For each fixed n , uniformly near $s = 0$,

$$\|Q_s^n u\|_{(C_0^r)^*} \leq C_n \|u\|_{\mathcal{B}_0}, \quad u \in \mathcal{B}_{0,0}. \quad (26)$$

In addition, for every fixed m, n ,

$$\|Q_s^m - Q_0^m\|_{C_0^r \rightarrow \mathcal{B}_2} \longrightarrow 0, \quad \|Q_s^n - Q_0^n\|_{\mathcal{B}_{0,0} \rightarrow (C_0^r)^*} \longrightarrow 0. \quad (27)$$

For the regular and face pairings, the corresponding fixed-time tests converge branchwise on the same finite atlas away from $\mathcal{S}_{L,m,n}$, with uniform sup norm and the

boundary tightness in (25). Hence (23) and (24) give the regular limit and the fixed-core face limit. Passage from this core to $L(s)$ is not part of MT_{DQ} ; it is exactly $\text{FACE}_{2\text{CUT}}$ below. No spectral gap is required on $\mathcal{B}_0$: it is used only for each fixed two-time limit. All constants are uniform on the shallow spacetime cells used below. For every fixed L, m, n and $\rho > 0$, after identifying the common finite atlas, put $\mathfrak{A}_{L,m,n,\rho}^{\text{off}} := \mathfrak{A}_{L,m,n}^{\text{face}} \setminus [\mathcal{S}_{L,m,n}]_\rho$. The moving face tests satisfy the genuinely uniform clause

$$\begin{aligned} \sup_{|s| < s_0} \|\Phi_{s,m,n}\|_{\text{BL}(\mathfrak{A}_{L,m,n,\rho}^{\text{off}})} &< \infty, \\ \|\Phi_{s,m,n} - \Phi_{0,m,n}\|_{\text{BL}(\mathfrak{A}_{L,m,n,\rho}^{\text{off}})} &\longrightarrow 0. \end{aligned} \quad (28)$$

In particular the family has a common modulus of continuity away from the boundary. Pointwise branchwise convergence with only a uniform supremum bound is not accepted.

The interface depends on the chosen measure-trivializing gauge, exactly as the operator K_V does. A material or Eulerian physical susceptibility may be recovered only after adding the corresponding observable-transport terms; that is a separate theorem and is not called operator CM2 here.

Definition 1.2 (Two-cutoff face difference-quotient interface $\text{FACE}_{2\text{CUT}}$). For every fixed (m, n) let $\Phi_{s,m,n}$ be the double-centered branchwise test induced by $Q_s^m g$ and $Q_s^{*n} f$ on the common oriented face atlas. Decompose the coarea current by genuine word depth,

$$\mathcal{J}_{s,\leq L}^{\text{face}} = \sum_{\ell \leq L} \mathcal{J}_{s,\ell}^{\text{face}}, \quad \mathcal{J}_{V,\leq L}^{\text{face}} = \sum_{\ell \leq L} \mathcal{J}_{V,\ell}^{\text{face}}.$$

Besides the fixed-core convergence and boundary tightness (24)–(25), require constants $c_{\text{ret}} > A_{\text{face}} \geq 0$ such that the positive, pre-cancellation coarea ledger obeys, for every $\ell \geq 0$,

$$\begin{aligned} \sup_{0 < |s| < s_0} \sum_{\text{depth}(\lambda) = \ell} \int_0^1 \int |\partial_s G_\lambda(\theta s, x)| \delta(G_\lambda(\theta s, x)) \mathcal{M}_{\lambda,m,n}(\theta, s, x) dx d\theta \\ \leq C_{m,n} e^{-(c_{\text{ret}} - A_{\text{face}})\ell}. \end{aligned} \quad (29)$$

The identical bound is required for the limiting positive current at $s = 0$. Here $\mathcal{M}_{\lambda,m,n}$ is the positive mark dominating the oriented amplitude, the two fixed-time tests, component multiplicity, and every actual/split constituent. It is evaluated in the exact coarea identity before signed recombination. It is not a parallel physical-mass ledger: the positive coarea measure with density $\mathcal{M}_{\lambda,m,n}$ is required to be dominated, up to the displayed fixed-time constant $C_{m,n}$, by the coarea pushforward of the restriction of the propagated envelope $d\overline{M}_{s,\mathfrak{d}}^q = d\Pi_{s,\mathfrak{d}} q_\lambda$ to $I_{\text{face/word}}(\varrho) \cup I_{\text{main}}^{\text{face/word}}(\varrho)$. Hence every operator, recovery, and opposite-amplitude exponent is already contained in A_{face} and c_{face} . The domination is uniform in every suppressed entry of $\mathfrak{d}$, including the depth- ℓ and (m, n) coordinates. In particular (29) contains no $|s|^{-1}$ factor. Put $c_{\text{face}} := c_{\text{ret}} - A_{\text{face}} > 0$. Summing the positive per-depth bound gives the uniform-integrability consequence

$$\begin{aligned} \sup_{0 < |s| < s_0} \left| \left\langle \mathcal{J}_{s,(L_0,L(s))}^{\text{face}}, \Phi_{s,m,n} \right\rangle \right| + \left| \left\langle \mathcal{J}_{V,>L_0}^{\text{face}}, \Phi_{0,m,n} \right\rangle \right| \\ \leq C_{m,n} e^{-c_{\text{face}} L_0}, \quad L_0 \geq 0. \end{aligned} \quad (30)$$

The terminal part deeper than $L(s)$ is not included in (30); it is the positive-envelope remainder of $\text{TAIL}_{\text{env}}^+/\text{NREC}_{\text{deep}}$ and is required to be $o(1)$ in the CM2 norm by Theorem 7.18. Thus the fixed core, middle band, and deep remainder are three different estimates. The no- $|s|^{-1}$ estimate (29) is an independent positive-current primitive; neither a common refinement nor UCR_{sh} implies it by itself.

Definition 1.3 (Positive-envelope product-time face interface $\text{FACE_TIME}_{\text{CM2}}$). For the resolution datum belonging to (s, m, n) put

$$I_{\text{fw}}(s, m, n) := I_{\text{face/word}}(s, m, n) \sqcup I_{\text{main}}^{\text{face/word}}(s, m, n).$$

This is the union of all active main and nonmain face/word restrictions at the linked query. The nonmain term is the shallow part of the single static word owner in (149), while the main term is the word restriction in (188); no depth-zero or fixed-core part is removed. Both terms are record-compatible by (148) and (189); the face-time handler cannot be populated by changing a clock record after q is fixed. Condition $\text{FACE_TIME}_{\text{CM2}}$ requires constants $c_{\text{fw}} > 0$ and $C < \infty$, uniform in s, m, n and every suppressed master coordinate, such that the restriction of the same propagated master envelope satisfies

$$\overline{M}_{s, \mathfrak{d}}^q(I_{\text{fw}}(s, m, n)) \leq C e^{-c_{\text{fw}} \max\{m, n\}}. \quad (31)$$

Here $\mathcal{E}_{\mathfrak{d}}(f, g)$ denotes the single bounded bilinear physical-test lift into the master test product; its definition and norm bound are part of this interface if they have not already been supplied globally. The lift and occurrencewise current domination must hold on this restriction with no time-dependent constant:

$$|J_{\text{fw}, s, m, n}(\mathcal{E}_{\mathfrak{d}}(f, g))| \leq C \overline{M}_{s, \mathfrak{d}}^q(I_{\text{fw}}(s, m, n)) \|f\|_{C^r} \|g\|_{C^r}. \quad (32)$$

Equivalently, one may replace the right side of (31) by a common array $M_{m, n}^{\text{fw}}$ with $\sum_{m, n} M_{m, n}^{\text{fw}} < \infty$; the exponential form is required when a numerical CM2 exponent is claimed. Put

$$c_{\text{face, time}} := c_{\text{fw}}/2 > 0. \quad (33)$$

This interface is orthogonal to $\text{FACE}_{2\text{CUT}}$: $\text{FACE}_{2\text{CUT}}$ controls the word-depth limit for each fixed (m, n) , whereas $\text{FACE_TIME}_{\text{CM2}}$ supplies the missing product-time majorant. Neither condition implies the other.

Definition 1.4 (Recovered-core/positive-tail face-time interface $\text{FACE_TIME}_{\text{REC}}$). The positive-envelope condition (31) is sufficient but is generally too strong for a moving face of order-one positive mass: mixing makes a centered signed pairing small, not the mass of the face itself. The following alternative keeps those two mechanisms separate. Put $N = \max\{m, n\}$ and require a jointly Borel disjoint decomposition

$$I_{\text{fw}}(s, m, n) = I_{\text{fw}}^{\text{tail}}(s, m, n) \sqcup \bigsqcup_{\substack{\mathfrak{o} \in \{-, +\} \\ 0 \leq R \leq \delta N}} I_{\text{fw}}^{\mathfrak{o}, R}(s, m, n), \quad 0 < \delta < 1. \quad (34)$$

Here R is the recovery time in the selected pre-recorded factorization of the positive standard-family component. The immutable field $h_{\text{rec}}(a)$ declares which recovery sides were proved before the linked query was chosen. Put

$$\text{Acc}(a) := \begin{cases} \{-\}, & h_{\text{rec}}(a) = -, \\ \{+\}, & h_{\text{rec}}(a) = +, \\ \{-, +\}, & h_{\text{rec}}(a) = \text{both}, \\ \emptyset, & h_{\text{rec}}(a) = \text{none}. \end{cases} \quad (35)$$

For every side in $\text{Acc}(a)$ the base record contains the corresponding typed alternative recovery factorization $\mathcal{P}_{\text{rec}}^{\mathfrak{o}}(a)$ from (131) below. In particular $h_{\text{rec}} = \text{both}$ means two pre-certified representations of the same coefficient occurrence and physical current, not two additive occurrences whose unused half would enter the tail. The query selects one recorded representation but may not manufacture one after (m, n) is known. Every recovered cell satisfies the typed inclusion

$$I_{\text{fw}}^{\mathfrak{o}, R}(s, m, n) \subset \{a : R_{\text{rec}}^{\mathfrak{o}}(a) = R, \mathfrak{o} \in \text{Acc}(a), d_{\mathfrak{o}}(m, n) = N\}. \quad (36)$$

Thus the active orientation is assigned only to an accessible long side: with $d_{-}(m, n) = m$ and $d_{+}(m, n) = n$,

$$I_{\text{fw}}^{\mathfrak{o}, R}(s, m, n) \neq \emptyset \implies d_{\mathfrak{o}}(m, n) = N. \quad (37)$$

This is a filter of the datum-independent occurrence kernel, not a new decomposition chosen after (m, n) . All components whose access set misses every active long side, or whose recovery time exceeds δN , belong to the positive tail. For constants $c_{\text{tail}, \text{fw}}, c_{\text{mix}} > 0$ and $a_{\text{fw}} \geq 0$, require, on the same propagated q -measure,

$$\overline{M}_{s, \mathfrak{o}}^q(I_{\text{fw}}^{\text{tail}}(s, m, n)) \leq C e^{-c_{\text{tail}, \text{fw}} N}, \quad \overline{M}_{s, \mathfrak{o}}^q(I_{\text{fw}} \setminus I_{\text{fw}}^{\text{tail}}) \leq C e^{a_{\text{fw}} N}. \quad (38)$$

For every recovered cell, the occurrencewise grouped current, after the carrier has been integrated and with the physical endpoint test centered, obeys the signed proper-family estimate

$$\begin{aligned} \left| J_{s, m, n; I_{\text{fw}}^{\mathfrak{o}, R}}(\mathcal{E}_{\mathfrak{o}}(f, g)) \right| \\ \leq C e^{-c_{\text{mix}}(d_{\mathfrak{o}}(m, n) - R)} \overline{M}_{s, \mathfrak{o}}^q(I_{\text{fw}}^{\mathfrak{o}, R}) \|f\|_{C^r} \|g\|_{C^r}. \end{aligned} \quad (39)$$

The tail pairing is bounded by its positive q -mass through POST_CARRIER_DOM and PHYS_TEST_LIFT. Thus cancellation is used only in (39), whereas every discarded or unrecovered piece is estimated positively. Put

$$c_{\text{fw}}^{\text{rec}} := \min\{c_{\text{tail}, \text{fw}}, (1 - \delta)c_{\text{mix}} - a_{\text{fw}}\} > 0, \quad c_{\text{face}, \text{time}} := c_{\text{fw}}^{\text{rec}}/2. \quad (40)$$

This interface is designed for verification by a growth lemma followed by forward or reverse standard-family coupling. It does not assert that an order-one positive moving-face mass decays.

Lemma 1.5 (Face/word product-time majorant). *Under either FACE_TIME_CM2 or FACE_TIME_REC, uniformly in s and for the limiting current,*

$$|B_{s, m, n}^{\text{fw}}| \leq C e^{-c_{\text{face}, \text{time}}(m+n)} \|f\|_{C^r} \|g\|_{C^r}. \quad (41)$$

Hence the face/word channel is absolutely summable in product time.

Proof. For FACE_TIME_CM2, combine (31) with (32). For FACE_TIME_REC, sum (39) over $R \leq \delta N$ and both orientations, use (37) and the second bound in (38), and estimate the tail by the first bound there. In both cases the result is $C e^{-2c_{\text{face}, \text{time}} N}$. Finally $N = \max\{m, n\} \geq (m+n)/2$. $\square$

Theorem 1.6 (Growing-depth face limit). *For every fixed (m, n) , assume MT_DQ and FACE_2CUT. The complete face difference-quotient pairing converges to the limiting face pairing after fixing the core depth and then letting $s \rightarrow 0$. More precisely,*

$$\left\langle \mathcal{J}_{s, \leq L(s)}^{\text{face}}, \Phi_{s, m, n} \right\rangle \longrightarrow \left\langle \mathcal{J}_V^{\text{face}}, \Phi_{0, m, n} \right\rangle, \quad (42)$$

If, in addition, all hypotheses of Theorem 7.18 hold, then adding the part deeper than $L(s)$ changes the left side by $o(1)$ in the complete CM2 sum. This last assertion is not a consequence of MT_{DQ} and $\text{FACE}_{2\text{CUT}}$ alone.

Proof. Given $\varepsilon > 0$, choose L_0 so that the right side of (30) is below ε . With this L_0 fixed, bounded-Lipschitz convergence on the common atlas and (25) give convergence of the core pairing at depth at most L_0 . Equation (30) removes both the finite- s triangular band $L_0 < \ell \leq L(s)$ and the limiting tail $\ell > L_0$, uniformly in s . Finally under its additional hypotheses, Theorem 7.18 removes the terminal part $\ell > L(s)$. The order is therefore L_0 first, $s \rightarrow 0$ second, and, when invoked, the deep positive-envelope estimate last. A fixed-depth topology is never used as a substitute for growing-depth uniform integrability. $\square$

Lemma 1.7 (Fixed-gauge centering and typed finite-DQ convergence). *Under MT_{DQ} , the same projection Π_0 centers the finite- s operators, endpoint observables, branch-section modes, and both standard-family couplings. Every term has precisely one of the regular, face, product-current, or response types in Definition 1.1; the face and product types are the two inseparable summands of the centered cluster (14). Moreover*

$$|\langle Q_s^n \widehat{K}_{s,\text{sw}}^{\text{DQ}} Q_s^m g, f \rangle| \leq C \theta_-^m \theta_+^n \|g\|_{C^r} \|f\|_{C^r}, \quad (43)$$

and the same uniform majorant holds for E_s . For every fixed (m, n) the product-current block converges in (22) and pairs to zero with the two individually centered endpoint tests. Hence the centered face-product cluster has exactly the same pairing as its graph-face summand on those tests; this does not make either summand separately centered. The response block converges to $\widehat{K}_{V,\text{sw}}$ and the E_s pairing tends to zero. The regular pairing converges in (23); the face pairing converges on every fixed core in (24), and its growing-depth limit is supplied by Theorem 1.6. Consequently, once the regular and face estimates below provide their summable uniform majorants, one common dominated-convergence argument exchanges the limit with the absolute CM2 double sum for all four types. Explicitly, if $B_{s,m,n}^j$ denotes the pairing of type $j \in \{\text{reg}, \text{face}, \text{prod}, \text{sw}\}$ and the corresponding ledger gives

$$|B_{s,m,n}^j| \leq M_{m,n}^j, \quad \sum_j \sum_{m,n \geq 0} M_{m,n}^j < \infty, \quad (44)$$

then

$$\lim_{s \rightarrow 0} \sum_j \sum_{m,n \geq 0} B_{s,m,n}^j = \sum_j \sum_{m,n \geq 0} \lim_{s \rightarrow 0} B_{s,m,n}^j.$$

Proof. Equation (2) gives $\mu_0(P_s h) = \mu_0(h)$, hence $P_s \Pi_0 = \Pi_0 P_s = \Pi_0$ and $\partial_s \Pi_0 = 0$. Apply (19), then (17), and finally (20); duality gives (43). Replacing $\widehat{K}_{s,\text{sw}}^{\text{DQ}}$ by E_s gives the same uniform majorant. For fixed (m, n) add and subtract the expressions with Q_0^m and Q_0^n . Equations (21)–(27) show that the response difference and the E_s pairing tend to zero. The individual double projection (12) makes the regular and response outputs eligible for the zero-mass estimates. The face and product summands are instead used only through their exact jointly centered cluster (14); no zero-mass estimate is applied to either one alone. Dominated convergence with (43) completes the claim. The regular claim follows from (23) and fixed-time branchwise continuity. The product-current pairing vanishes because each tensor summand contains μ_0 in one slot while both endpoint tests are centered; its separate topology prevents this fact from being used to misclassify it as a density or a graph face, and the equality with the clustered pairing follows before absolute values are taken. For the face core, use (28) outside $[\mathcal{S}_{L,m,n}]_\rho$; then use (25) for

the two discarded boundary neighbourhoods and let $\rho \downarrow 0$. Then apply Theorem 1.6 to pass from the fixed core to the complete face pairing. This is the standard weak-current plus boundary-tightness upgrade; it is not supplied by the uniform face norm (16). The regular and face CM2 majorants are proved later from their separate ledgers, after which the same dominated-convergence theorem applies. $\square$

The divergence formula for smooth Anosov transfer derivatives in [3] maps a strong anisotropic space to an intermediate response space, whereas the finite difference quotients converge only in a still weaker operator topology. The spectral gap is used on the response space and is not imposed on the weakest space. This is the three-space type model behind MT_{DQ} ; the derivative is not an $M \times M$ Sobolev density. The fixed zero-mass organization is consistent with the measure-valued-section response framework of [23] and the extended-space loss-of-memory framework of [26]. None of these references supplies the singular moving-billiard decomposition or its face atlas; those remain explicit interfaces here. A pure auxiliary conjugacy provides a decisive type test: the full physical pairing with transported observables may have zero derivative while the fixed-gauge commutator is nonzero. This is why the two targets are not identified. The moving-singularity obstruction in [4] is a further reason not to infer this interface from smooth-map response theory. For the face topology, [22, Lemma 5.4] gives the closest available model: weak-* limits of standard families with uniform boundary Z may still be tested by bounded functions whose discontinuities lie on a fixed finite singularity set. Equations (24)–(25) require the corresponding statement for the oriented moving-face currents themselves; the cited lemma does not prove that moving-billiard DQ input.

2 Why the direct projective-cone bridge fails

The projective cone for dispersing billiards controls integrals of a density against positive dynamically Hölder tests on stable curves, short-leaf mass, and Hölder comparison of such integrals between nearby stable curves. A moving inverse-branch boundary instead produces, before physical assembly, a coefficient of the form

$$h \longmapsto w(h|_{I^+} - h|_{I^-}) \delta_{\Gamma}, \quad (45)$$

where $I^\pm$ are one-sided terminal preimage arcs. In the radial collision atlas these arcs land on the grazing seam and are transversal to the stable test foliation. Thus (45) asks for a transverse trace, not a stable-leaf average.

The following elementary local model rules out the missing implication.

Proposition 2.1 (Stable-leaf control does not dominate transverse trace). *Let $R = (-2, 2)^2$, let the local stable leaves be the horizontal segments $W_y = (-2, 2) \times \{y\}$, and let the transversal face be $I = \{0\} \times (-1, 1)$. Fix $0 < q < 1$ and $0 < \gamma < 1$. There are smooth functions u_ε supported in an ε -tube about I such that*

$$u_\varepsilon|_I = b \neq 0,$$

while, uniformly over bounded Hölder test functions ψ on stable subsegments $W \subset W_y$,

$$\left| \int_W u_\varepsilon \psi \, dm_W \right| \leq C \min\{|W|, \varepsilon\} \|\psi\|_\infty, \quad (46)$$

$$\sup_{0 < |W| \leq 1} |W|^{-q} \left| \int_W u_\varepsilon \psi \, dm_W \right| \leq C \varepsilon^{1-q} \|\psi\|_\infty, \quad (47)$$

and the Hölder comparison of these integrals between nearby parallel stable leaves is $O(\varepsilon)$. Consequently no norm or order estimate using only the stable-leaf quantities in (46)–(47) and their transverse Hölder comparisons can bound $\|u\|_I$.

Proof. Choose $\phi \in C_c^\infty((-1, 1))$ with $\phi(0) = 1$ and $b \in C_c^\infty((-1, 1))$, $b \neq 0$, and set

$$u_\varepsilon(x, y) = \phi(x/\varepsilon)b(y).$$

The trace assertion is immediate. A horizontal subsegment meets the support in length at most $C \min\{|W|, \varepsilon\}$, proving (46). If $|W| \leq \varepsilon$, multiplication by $|W|^{-q}$ gives at most $C\varepsilon^{1-q}$; if $|W| \geq \varepsilon$, it gives at most $C\varepsilon|W|^{-q} \leq C\varepsilon^{1-q}$, proving (47). Replacing y by y' changes the integral by at most $C\varepsilon|y - y'| \|b\|_{C^1} \|\psi\|_{C^0}$, with the usual additional Hölder test-comparison term. The trace is independent of ε , so the asserted domination is impossible. $\square$

Corollary 2.2 (Failure of the proposed positive cone operator). *The local face map (45) is not known to be positive for the Demers–Liverani cone order, and its boundedness cannot be inferred from projective contraction. In particular, the implication*

$$\Theta(P_0^m f_+, P_0^m f_-) \leq C\chi^m \implies \|K_V P_0^m(f_+ - f_-)\|_Y \leq C'\chi^m$$

is invalid without an additional transverse-trace theorem.

Proof. The cone order is tested through stable-leaf integrals. The face coefficient contains precisely the trace excluded by Proposition 2.1. Projective contraction therefore supplies no order comparison for the two output face measures. $\square$

This is a go/no-go conclusion, not merely a missing estimate: the direct “past cone $\rightarrow$ face $\rightarrow$ future family” proof must be abandoned unless physical assembly first converts the terminal trace into an integral functional controlled by the cone. The existing adjoint dyadic bound does not perform that conversion.

3 A sufficient two-sided bridge

The complete defect is a kernel located at the event time. It must be controlled from the past and the future simultaneously. The following criterion is an exact sufficient condition, but not a characterization of CM2.

Definition 3.1 (Centered nuclear recovery bridge). *Let X_- be a Banach space of centered past observables and let Y_+ be a Banach space of centered signed measured unstable families. A centered operator $K : X_- \rightarrow Y_+$ has a nuclear recovery presentation if*

$$K = \sum_{j \geq 1} \ell_j \otimes \eta_j, \quad \sum_{j \geq 1} \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+} < \infty, \quad (48)$$

where $\ell_j(\mathbf{1}) = 0$ and $\eta_j(\mathbf{1}) = 0$ after the complete regular and physical-face assembly.

Theorem 3.2 (Two-sided bridge implies CM2). *Assume*

$$\|Q^m g\|_{X_-} \leq C_- \tau^m \|g\|_{X_-}, \quad (49)$$

$$|\eta(\mathcal{U}_0^n f)| \leq C_+ \vartheta^n \|\eta\|_{Y_+} \|f\|_{C^1}, \quad (50)$$

for centered g and centered η . If K has the presentation (48), then

$$|(KQ^m g)(\mathcal{U}_0^n f)| \leq C_- C_+ \mathfrak{N}(K) \tau^m \vartheta^n \|g\|_{X_-} \|f\|_{C^1},$$

where $\mathfrak{N}(K) = \sum_j \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+}$.

Proof. Using (48), (49), and (50),

$$\begin{aligned} |(KQ^m g)(\mathcal{U}_0^n f)| &\leq \sum_j |\ell_j(Q^m g)| |\eta_j(\mathcal{U}_0^n f)| \\ &\leq C_- C_+ \tau^m \vartheta^n \|g\|_{X_-} \|f\|_{C^1} \sum_j \|\ell_j\|_{X_-^*} \|\eta_j\|_{Y_+}. \end{aligned}$$

□

Remark 3.3. *Used globally, a one-gap functional-correlation estimate gives only $O(\tau^m + \vartheta^n)$ and does not by itself prove CM2. It is nevertheless exactly sufficient in the central band, where $m \asymp n$: after smoothing the event kernel, one decoupling gives $e^{-c \max\{m, n\}}$, hence an $e^{-c'(m+n)}$ bound there. Forward and reverse standard-family coupling handle the two off-diagonal regions; Lemma 6.2 assembles the product estimate. Thus no operator-ideal estimate is used in the principal proof.*

Remark 3.4 (The nuclear criterion is generally too strong). *A physical face current is supported on a graph in a stable–unstable product chart. Its kernel is therefore analogous to a pullback or a Fourier integral operator, not to a smoothing operator. Even after double centering, a graph delta need not be nuclear between natural past and future spaces. Thus failure of (48) or of the projective-tensor condition below does not imply failure of CM2. Product decay may instead arise dynamically, from oscillation and transversality after the graph is acted on by both time directions.*

4 Recovery-time weighted future space

For a measured unstable component (W, ρ) let $R(W)$ be the first time at which its canonical forward subdivision is a proper standard family. For $\omega > 0$, define schematically

$$\|\eta\|_{Y_+} := \inf \sum_j |a_j| e^{\omega R(W_j)} (1 + \text{Reg}(\rho_j)), \quad \eta = \sum_j a_j (W_j, \rho_j). \quad (51)$$

The infimum also allows the positive and negative Jordan parts to be split by recovery rank. Standard-family coupling and recovery give (50) once the constants in (51) are chosen within the recovery tail.

For a radial grazing block b , the physical flux factorization gives a mass bound $M_b \leq C2^{-b}$. If its geometric deterioration is polynomial in 2^b , the recovery lemmas give

$$R_b \leq C_0 + C_1 b.$$

Hence

$$\sum_{b \geq 0} M_b e^{\omega R_b} < \infty \quad \text{whenever} \quad \omega C_1 < \log 2. \quad (52)$$

Thus initial finite Z is unnecessary. What remains is not future recovery, but the past functional ℓ_j in (48).

4.1 Recovery windows and the actual summability target

For differentiability of the susceptibility, a uniform pointwise CM2 bound for every grazing block is stronger than necessary. The required conclusion is absolute summability of the complete block series. This distinction removes an artificial exponential loss in the grazing rank.

Proposition 4.1 (Recovery-window summability). *Let $B_b(m, n)$ be the scalar pairing produced by block b at past and future times m, n . Suppose that there are masses M_b , recovery times R_b^-, R_b^+ , constants $C < \infty$, and $0 < \tau, \vartheta < 1$ such that*

$$|B_b(m, n)| \leq CM_b \tau^{(m-R_b^-)_+} \vartheta^{(n-R_b^+)_+}, \quad m, n \geq 0. \quad (53)$$

Then

$$\sum_{b, m, n} |B_b(m, n)| \leq C' \sum_b M_b (1 + R_b^-)(1 + R_b^+). \quad (54)$$

In particular, if $M_b \leq C2^{-\alpha b}$ for some $\alpha > 0$ and $R_b^\pm \leq C_0 + C_1 b$, then the full threefold series converges. The same conclusion holds uniformly for parameter difference quotients if the constants and block masses are uniform.

The deterministic recovery times may be replaced by a countable refinement $B_b = \sum_j B_{b,j}$ with masses $M_{b,j}$ and times $R_{b,j}^\pm$, provided

$$\sum_j M_{b,j} (1 + R_{b,j}^-)(1 + R_{b,j}^+) \leq CM_b (1 + b)^2. \quad (55)$$

It is enough for the two-sided recovery tail, after its deterministic $O(b)$ entrance time, to be exponential.

Proof. For every $R \geq 0$ and $0 < q < 1$,

$$\sum_{k \geq 0} q^{(k-R)_+} \leq R + 1 + \frac{1}{1-q}.$$

Sum (53) first in m, n and then in b . Exponential block mass dominates the quadratic recovery window. Applying the same calculation to every refined piece gives (55). An exponential tail for $R_{b,j}^- + R_{b,j}^+ - O(b)$ has uniformly bounded polynomial moments, proving the last assertion. $\square$

Lemma 4.2 (Two marginal growth tails imply bi-recovery). *Let a block of total variation M_b be cut by the common refinement of its past and future recovery trees, with refined masses $M_{b,j}$ and recovery times $R_{b,j}^\pm$. Assume cutting preserves total variation and, for some $r_b \leq C_0 + C_1 b$, $0 < \rho < 1$,*

$$\sum_{j: R_{b,j}^\pm > r_b + t} M_{b,j} \leq CM_b \rho^t, \quad t \geq 0,$$

for each sign. Then (55) holds.

Proof. If $R_{b,j}^- + R_{b,j}^+ > 2r_b + t$, at least one marginal time is larger than $r_b + t/2$. The union bound therefore gives an exponential tail for the sum of the two recovery times under the normalized weights $M_{b,j}/M_b$. Its second moment is $O((1+r_b)^2)$. Since $(1+R^-)(1+R^+) \leq (1+R^-+R^+)^2$, the claim follows. $\square$

For a general graph current the relevant geometric input is therefore a *proper bi-recovery* statement, not an estimate of a Fourier constant on the original grazing curve. For the clean first radial current this input is Corollary 6.82. A central two-time theorem then gives (53). The fully unrecovered rectangle is handled by total variation, the two mixed rectangles by the corresponding recovered one-sided estimate, and the fully recovered

rectangle by the product bound. The growth-lemma tail sums these windows through (55).

Consequently even the generic $M_b = O(2^{-b})$ assembly margin is sufficient once proper bi-recovery is proved; the stronger first-radial $O(2^{-(3-\epsilon_*)b})$ margin gives additional room but is no longer needed to absorb an exponential Fourier constant in b . This decouples the grazing rank from the return-rank truncation in Proposition 6.4.

5 Billiard-specific block problem

For each genuine physical block b , the complete difference quotient must be written, after artificial-face cancellation, as

$$K_{V,b} = \sum_{k \geq 1} \ell_{b,k} \otimes \eta_{b,k}, \quad (56)$$

with

$$\sum_{b,k} \|\ell_{b,k}\|_{X_-^*} \|\eta_{b,k}\|_{Y_+} < \infty. \quad (57)$$

The desired estimate follows from Theorem 3.2. Three sufficient mechanisms for proving (57) are:

- (i) a reverse-time conditional-mixing theorem for the terminal slice, furnishing X_-^* directly;
- (ii) a two-sided Young-tower lift of the complete centered defect, with an event kernel that is nuclear between the past and future tower norms;
- (iii) a smoothing/nuclear decomposition of the terminal restriction, with a quantitative bunching inequality strong enough to sum the smoothing error over the homogeneity blocks.

The projective-cone contraction alone supplies none of these three inputs. These mechanisms are not exhaustive because a non-nuclear graph current can still equidistribute under the two-sided dynamics.

5.1 A two-sided tower formulation

The natural extension gives a sharper version of alternative (ii). Let Δ be a two-sided Young tower for the fixed billiard, with past and future cylinder filtrations $\mathcal{F}_i^-$ and $\mathcal{F}_j^+$. Denote the associated martingale differences by

$$D_i^- = \mathbb{E}(\cdot \mid \mathcal{F}_i^-) - \mathbb{E}(\cdot \mid \mathcal{F}_{i-1}^-), \quad D_j^+ = \mathbb{E}(\cdot \mid \mathcal{F}_j^+) - \mathbb{E}(\cdot \mid \mathcal{F}_{j-1}^+).$$

A lifted event kernel $\widehat{K}$ has summable mixed cylinder variation if

$$\sum_{i,j \geq 1} \|D_i^- D_j^+ \widehat{K}\|_{\mathcal{B}_-^* \widehat{\otimes}_\pi \mathcal{B}_+} < \infty. \quad (58)$$

The indices start at one: complete double centering removes the two axis terms $D_i^- D_0^+ \widehat{K}$ and $D_0^- D_j^+ \widehat{K}$.

Proposition 5.1 (Mixed-cylinder criterion). *Suppose the two-sided tower has exponential tails and the past and future transfer actions contract the centered martingale blocks at rates τ^m and ϑ^n . If the complete lifted moving defect satisfies (58), then it has a nuclear recovery presentation (48) and hence satisfies CM2.*

Proof. Expand

$$\widehat{K} = \sum_{i,j \geq 1} D_i^- D_j^+ \widehat{K}$$

in the projective tensor product. Each summand is a norm-convergent sum of rank-one past/future tensors by the definition of $\widehat{\otimes}_\pi$. Summability in (58) therefore gives (48). Shifting m steps into the past and n steps into the future contracts the two centered tensor factors separately. Apply Theorem 3.2. $\square$

This criterion identifies one strong geometric issue. A smooth two-dimensional density on a tower rectangle has summable mixed cylinder variation. A face current is instead supported on a one-dimensional graph in the local stable–unstable product chart. Its graph delta is not automatically in the projective tensor product. Proving (58) is therefore a genuine conditional-mixing theorem, not routine tower bookkeeping. The finite-strip central face supplies uniform transversality and one-sided recovery, but these facts alone do not prove the mixed tensor estimate.

6 The intrinsic dynamic graph-current formulation

Let $\gamma_b : I_b \rightarrow M$ parametrize one side of a complete physical face in block b , let $F_b \gamma_b$ denote its assembled terminal image, and let w_b be the signed coarea weight after artificial-face cancellation. The face contribution to CM2 has the model form

$$B_{m,n}^{(b)}(g, f) = \int_{I_b} g(T^{-m} \gamma_b(t)) f(T^n F_b \gamma_b(t)) w_b(t) dt. \quad (59)$$

Equivalently, if $J_{V,b}$ is the graph current on $M \times M$ induced by $t \mapsto (\gamma_b(t), F_b \gamma_b(t))$, then

$$B_{m,n}^{(b)}(g, f) = ((T^{-m} \times T^n)_* J_{V,b})(g \otimes f). \quad (60)$$

Complete double centering removes the two marginals of $J_{V,b}$. The intrinsic CM2 statement is therefore two-parameter equidistribution of a centered graph current under $T^{-1} \times T$ at a product rate.

This formulation explains both the difficulty and the opportunity. A graph curve is one-dimensional, whereas the unstable bundle of the product map is two-dimensional, so ordinary standard-family coupling does not apply directly. On the other hand, the current need not be statically nuclear: hyperbolic stretching sends its Fourier frequencies to exponentially large scales, and polynomial Fourier decay of the induced slice is already enough to produce exponential decay in collision time.

Proposition 6.1 (Fourier reduction on one induced rectangle). *Assume that an induced return map has a stable–unstable product chart in which the centered face slice induces a finite signed measure ν_b on the contracting coordinate. Suppose that, uniformly over admissible inverse words and over the relevant C^2 phase maps F ,*

$$|\widehat{F_* \nu_b}(\xi)| \leq C_b (1 + |\xi|)^{-\eta} \quad (61)$$

for some $\eta > 0$. Suppose also that, after excluding phase cancellation, the expanding frequencies generated by m past returns and n future returns are bounded below by

$$c \max\{\Lambda_-^m, \Lambda_+^n\},$$

apart from cylinder pieces whose total centered mass is bounded by $C\tau_0^m\vartheta_0^n$. Then, for sufficiently regular centered cylinder observables,

$$|B_{m,n}^{(b)}(g, f)| \leq CC_b \tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r} \quad (62)$$

for some $\tau, \vartheta < 1$.

Proof. Decompose g and f into Fourier modes in the product chart and split the sum into low and high frequencies. Centering and cylinder mixing control the low-frequency part. In the high-frequency part, hyperbolicity converts the two return depths into the lower frequency scale $c \max\{\Lambda_-^m, \Lambda_+^n\}$. Since

$$\max\{\Lambda_-^m, \Lambda_+^n\} \geq (\Lambda_-^m \Lambda_+^n)^{1/2},$$

(61) gives the product factor $\Lambda_-^{-\eta m/2} \Lambda_+^{-\eta n/2}$. Summability of the Fourier coefficients of C^r tests and the exceptional-cylinder bound complete the estimate. The loss of one half in each exponent is harmless. $\square$

This preliminary proposition is not used as an assumption in the final proof. Uniform (61) is supplied below by the cohomological local slope-energy theorem, the dominant-coordinate reduction, and the physical fractional nonstationary-phase estimate. One-sided mixing alone would not supply it.

6.1 A three-region reduction

Fourier analysis is not needed uniformly over the whole (m, n) quadrant. One-sided standard-family recovery gives a forward estimate with a past regularity cost, and reversibility gives its backward analogue. The following elementary lemma isolates the remaining diagonal band.

Lemma 6.2 (Off-diagonal coupling plus diagonal oscillation). *Suppose*

$$|B_{m,n}| \leq C e^{\beta_- m - \alpha_+ n}, \quad (63)$$

$$|B_{m,n}| \leq C e^{-\alpha_- m + \beta_+ n}, \quad (64)$$

with all four exponents positive. Choose $A > \beta_-/\alpha_+$ and $D > \beta_+/\alpha_-$. Suppose in addition that

$$|B_{m,n}| \leq C e^{-\gamma \max\{m, n\}}$$

whenever $D^{-1}m < n < Am$. Then there are $\tau, \vartheta \in (0, 1)$ such that $|B_{m,n}| \leq C' \tau^m \vartheta^n$ for all m, n .

Proof. If $n \geq Am$, (63) is bounded by $e^{-c_+(m+n)}$ for some $c_+ > 0$. If $m \geq Dn$, use (64). The remaining indices lie in the stated diagonal band, where $\max\{m, n\} \geq (m+n)/2$. Taking the minimum of the three positive exponents proves the claim. $\square$

For billiards, (63) follows from signed standard-family coupling after cutting the past-weighted density into homogeneous pieces: the growth lemma gives the coupling factor $e^{-\alpha_+ n}$, while differentiating the m -step past weight costs at most $e^{\beta_- m}$. The marked-amplitude moment sums the homogeneous pieces. Applying the same argument to the reversible collision map gives (64). All constants are uniform on the fixed compact atlas; recovered grazing pieces enter only after their recorded recovery times; these are the signed

standard-family forms of the usual growth and coupling estimates [32, 2]. Consequently the genuinely new slope/Fourier theorem is needed only for $m \asymp n$.

This also corrects the naive ambient phase test. The graph $t \mapsto (T^{-m}\gamma(t), T^n F\gamma(t))$ is one-dimensional in a four-dimensional product. Its first two derivatives cannot be uniformly transverse to every Fourier covector. In the diagonal band one must first discard the contracted coordinates and pass to the dominant past-stable and future-unstable coordinates. The resulting two-dimensional plane curve is governed by the stable–unstable temporal-distance/holonomy function. A finite-difference slope-value QNL is appropriate for this reduced phase. If one insists on working in the full four-dimensional chart, a finite-type condition involving derivatives through order at least four is required.

6.2 A verified radial nonlinearity seed

The first radial face is not an abstract graph. For two circular scatterers with radii R_j, R_i and centre distance $d > R_j + R_i$, put the source centre at the origin, the target centre at $(d, 0)$, and parametrize the source boundary by

$$q_j(\alpha) = R_j(\cos \alpha, \sin \alpha).$$

Let $r(\alpha) = |(d, 0) - q_j(\alpha)|$ and $\beta(\alpha) = \arg((d, 0) - q_j(\alpha))$. The two directions tangent to the target circle are

$$\theta_\sigma(\alpha) = \beta(\alpha) + \sigma \arcsin \frac{R_i}{r(\alpha)}, \quad \sigma \in \{-1, 1\}.$$

Thus the radial physical face in Birkhoff coordinates is locally $\varphi_\sigma(\alpha) = \theta_\sigma(\alpha) - \alpha$.

Lemma 6.3 (The circular tangency face is genuinely curved). *At the collinear source point $\alpha = 0$, write $r_0 = d - R_j$. Then*

$$\varphi''_\sigma(0) = -\sigma \frac{R_i d R_j}{r_0^2 \sqrt{r_0^2 - R_i^2}} \neq 0. \quad (65)$$

Moreover the angular coarea derivative of

$$G = |((d, 0) - q_j) \times v_\theta|^2 - R_i^2$$

is nonzero there. Hence each tangency choice gives a smooth, non-affine central face germ with a smooth coarea density.

Proof. Direct differentiation gives

$$r'(0) = 0, \quad r''(0) = \frac{dR_j}{r_0}, \quad \beta''(0) = 0.$$

Since

$$\frac{d}{dr} \arcsin \frac{R_i}{r} = -\frac{R_i}{r \sqrt{r^2 - R_i^2}},$$

equation (65) follows. On the tangency face, $|((d, 0) - q_j) \times v_\theta| = R_i$ while the longitudinal component has magnitude $\sqrt{r_0^2 - R_i^2}$; therefore $|\partial_\theta G| = 2R_i \sqrt{r_0^2 - R_i^2} > 0$. $\square$

This proves a useful but limited fact: the radial face supplies a concrete one-step nonlinearity seed and no Baker-type affine resonance is forced by the circular geometry. It does not prove uniform quantitative nonlinearity after arbitrary past and future words. Near the diagonal where the two terms in the reduced phase

$$\Phi_{m,n,\xi,\zeta}(t) = \xi \cdot T^{-m}\gamma(t) + \zeta \cdot T^n F\gamma(t) \quad (66)$$

have comparable size, their derivatives can cancel; this is the only time regime in which a genuine Dolgopyat/sum-product argument is needed. Off that diagonal one time direction dominates and ordinary nonstationary phase applies.

6.3 A countable-alphabet truncation lemma

The latest smooth result closest to the required mechanism is Leclerc's Fourier-decay theorem for equilibrium states of nonlinear, area-preserving smooth Axiom-A surface diffeomorphisms. Its quantitative nonlinearity condition is expressed through a temporal-distance function and is generic, with an explicit periodic-point cocycle test. A billiard return scheme is piecewise smooth with a countable alphabet, so that theorem is not directly applicable. The countable alphabet itself, however, is not the fundamental obstruction.

Proposition 6.4 (Exponential-tail reduction to uniform finite truncations). *Let an induced symbolic system have a return/singularity rank R with*

$$\mu\{R > L\} \leq Ce^{-aL}.$$

Suppose a Fourier proof at frequency $|\xi|$ uses at most $N(\xi) \leq c_0 \log(2 + |\xi|)$ successive symbols. Assume that the subsystem with all ranks at most L satisfies, uniformly in the truncation,

$$\left| \int e^{i\xi\psi} \chi d\mu_L \right| \leq Ce^{cL} |\xi|^{-\eta}.$$

Then the full countable system has polynomial Fourier decay, provided $a, \eta > 0$ and the finite-truncation constants have the displayed exponential bound.

Proof. The mass of orbit words which see a rank larger than L is at most $CN(\xi)e^{-aL}$. On the complement apply the finite-truncation estimate. Choose $L = \delta \log |\xi|$ with $0 < \delta < \eta/c$ (with no upper restriction if $c = 0$). The good contribution is $O(|\xi|^{-\eta+c\delta})$ and the bad contribution is $O(\log |\xi| |\xi|^{-a\delta})$. Both have a positive power saving. $\square$

Thus this fallback would require a *uniform* singular version of Leclerc's quantitative-nonlinearity estimate on finite homogeneity/return truncations. Exponential tower tails could then remove the truncation, but the truncation-uniform constants are not supplied by that route. The direct gate-resolved slope argument below bypasses this fallback entirely.

6.4 Uniform QNL requires anchored big images

Inspection of the QNL proof shows that a single nonzero periodic cocycle is not, by itself, a uniform truncation theorem. The power exponent also depends on a cutting constant: at every scale one removes a child interval carrying at least an η_{cut} fraction of its parent's conditional mass. For an arbitrary exhaustion by finite subshifts, η_{cut} may tend to zero; then the QNL exponent tends to zero and Proposition 6.4 is useless.

The billiard inducing scheme has a possible cure. A Gibbs–Markov return map has a finite big-images/preimages core, and on a compact central magnet the SRB conditional densities on homogeneous unstable leaves are bounded above and below. The truncations must therefore be *anchored*: retain a fixed finite family of full-image connector branches, including the nonlinear periodic witness, and add all branches up to rank L . The desired uniform cutting statement is

$$\nu_{x,L}(I_{\text{child}}) \geq \eta_0 \nu_{x,L}(I_{\text{parent}}), \quad \eta_0 > 0 \quad (67)$$

with η_0 independent of x, L , for at least one QNL-separated child at each scale. If (67) and a uniform UNI constant $\kappa_0 > 0$ hold, Leclerc’s tree argument gives a QNL exponent $\Gamma > 0$ independent of L ; only the multiplicative constant is allowed to grow like e^{cL} .

Rühr’s pressure inequalities quantify weak convergence of Gibbs measures on countable Markov shifts to equilibrium measures on finite subsystems by the square root of the pressure gap. They support the truncation step, but do not prove (67) or uniform UNI. Strong positive recurrence/BIP controls the symbolic escape; the billiard geometry must still provide the nonlinear separated child.

For the collision SRB measure on a compact homogeneous magnet, the unstable conditional is better than a general fractal Gibbs conditional: in arclength coordinates its density is bounded above and below. In that case the measure-theoretic part of the QNL argument is elementary.

Lemma 6.5 (Uniform interval UNI implies power non-concentration). *Let ν_L be measures on an interval whose densities satisfy $0 < c_- \leq d\nu_L/dt \leq c_+ < \infty$, uniformly in L . Let X_L be a family of functions with the following scale-invariant UNI property: for every interval J of length $\sigma < \sigma_0$ and every polynomial P in a fixed finite-dimensional class, there is a subinterval $J' \subset J$ with $|J'| \geq \kappa|J|$ such that*

$$|X_L(t) - P(t)| \geq \kappa|J| \quad (t \in J'),$$

where $\kappa > 0$ is independent of J, P, L . Then for every fixed $\alpha > 0$ there are $C, \Gamma > 0$, independent of L , such that

$$\nu_L\{t \in J : |X_L(t) - P(t)| \leq |J|^{1+\alpha}\} \leq C|J|^\Gamma \nu_L(J).$$

Proof. Partition every surviving interval into a bounded number of comparable children and discard a child contained in the UNI interval J' . The density bounds imply that at least a fixed fraction $\eta_0 > 0$ of the conditional mass is discarded at each generation. Iterate until the child scale is comparable to $|J|^{1+\alpha}$. The number of generations is $c_\alpha |\log |J|| + O(1)$, so the surviving mass is at most $(1 - \eta_0)^{c_\alpha |\log |J||} \nu_L(J) \leq C|J|^\Gamma \nu_L(J)$. $\square$

Thus for the central homogeneous face the principal task can be reduced further: establish a uniform geometric UNI statement for the stable–unstable temporal-distance function. Finite-subsystem approximation is then needed for long singular itineraries and difference quotients, but not to manufacture the QNL exponent on the central magnet.

6.5 A fixed two-branch UNI certificate

Bertozi’s 2026 countable-Markov exponential-mixing construction provides a useful template: after sufficient inducing, two fixed inverse branches are repeated and the derivative of their Birkhoff-roof difference is bounded away from zero uniformly in the iteration depth. The same device can be formulated for the billiard temporal-distance increment.

Let h_0, h_1 be two regular inverse return branches on the central magnet, with a common contraction bound $\lambda < 1$, and let τ denote the reduced stable–unstable phase increment (for the geometric application, $\tau = \log J^u F$ after the stable coboundary normalization). Define

$$\Psi_N(x) = \sum_{k=1}^N [\tau(h_0^k x) - \tau(h_1^k x)], \quad \Psi_\infty = \lim_{N \rightarrow \infty} \Psi_N. \quad (68)$$

Proposition 6.6 (Anchored two-branch certificate). *Suppose the series in (68) converges in C^1 and*

$$\inf_{x \in I} |\Psi'_\infty(x)| \geq c_0 > 0$$

on a central unstable interval I . Then, for all sufficiently large N ,

$$\inf_{x \in I} |\Psi'_N(x)| \geq c_0/2.$$

If the two branches and their fixed connector branches are retained in every anchored truncation, this UNI constant is independent of the truncation rank.

Proof. C^1 convergence makes $\|\Psi'_N - \Psi'_\infty\|_\infty < c_0/2$ for all large N . The second assertion follows because the calculation uses only the anchored branches, not the additional symbols admitted by a larger truncation. $\square$

Corollary 6.7 (First-term domination after acceleration). *Assume $\|Dh_i\| \leq \lambda < 1$, $\|D\tau\|_\infty \leq M$, and*

$$\inf_I |D(\tau \circ h_0 - \tau \circ h_1)| \geq d_0 \lambda.$$

If

$$\frac{2M\lambda^2}{1-\lambda} \leq \frac{1}{2}d_0\lambda,$$

then $\inf_I |\Psi'_\infty| \geq d_0\lambda/2$.

Proof. The derivative of the sum of all terms with $k \geq 2$ is bounded by $2M \sum_{k \geq 2} \lambda^k = 2M\lambda^2/(1-\lambda)$. Subtract this from the first-term lower bound. $\square$

This changes the next billiard calculation from an infinite statistical problem into a finite geometric one. Choose two central return words whose one-step radial tangency germs have different curvature coefficients in (65); attach fixed regular connectors to the magnet; and compute Ψ'_∞ (or a rigorously bounded finite approximation plus its geometric tail). A nonzero lower bound is open under small table perturbations and gives the uniform UNI input for Lemma 6.5. The non-equality of the two one-step curvatures is a promising seed, but it is not by itself the required lower bound because the stable coboundary and all subsequent return terms enter Ψ_∞ . After accelerating both branches, Corollary 6.7 shows that it is enough to retain a lower bound of order λ for the first-term curvature separation; the remaining infinite temporal-distance tail is only $O(\lambda^2)$. This remains a useful branchwise fallback. The next calculation gives a more direct fixed periodic witness for the smooth finite truncations.

6.6 An exact Anosov-cocycle witness from a normal two-cycle

There is a more direct way to certify the geometric nonlinearity needed by the smooth finite truncations. Leclerc's criterion for a smooth area-preserving Axiom-A surface map says that QNL follows if the stable or unstable distribution is not C^2 . At a hyperbolic fixed point, this can be checked by the Anosov cocycle. In determinant-one adapted coordinates, write the map as (F, G) with derivative $\text{diag}(\lambda, \mu)$, where $\lambda > 1$, $\mu = \lambda^{-1}$. The relevant scalar is

$$\mathfrak{A}(F, G) = \frac{G_{xyy}}{\mu} - \frac{F_{xy}G_{xy}}{\lambda - 1} - \frac{G_{xx}F_{yy}}{1 - \mu^3} - \frac{G_{xy}G_{yy}}{\mu^2}, \quad (69)$$

with all derivatives evaluated at the fixed point.

Proposition 6.8 (Exact circular normal-cycle witness). *Consider two unit circular scatterers whose facing boundary points are separated by a unit free-flight gap. Let $\mathcal{R} = T^2$ be the billiard return map to the first component near the unobstructed normal period-two orbit, in Birkhoff coordinates (s, p) with $p = \sin \varphi$. Then*

$$D\mathcal{R}(0) = \begin{pmatrix} 7 & 4 \\ 12 & 7 \end{pmatrix}.$$

For the determinant-one eigenbasis

$$S = \begin{pmatrix} 1 & -\sqrt{3}/6 \\ \sqrt{3} & 1/2 \end{pmatrix}, \quad \det S = 1,$$

write $S^{-1}\mathcal{R}S = (F, G)$. Then

$$\begin{aligned} \lambda &= 7 + 4\sqrt{3}, & \mu &= 7 - 4\sqrt{3}, \\ F_{xy} &= F_{yy} = G_{xx} = G_{xy} = G_{yy} = 0, & G_{xyy} &= -\mu. \end{aligned}$$

Consequently

$$\mathfrak{A}(F, G) = -1.$$

In particular, the local return dynamics has a nonzero QNL witness which is stable under sufficiently small clean perturbations of the table.

Proof. Place the circle centres at $(0, 0)$ and $(3, 0)$ and parametrize the first boundary near the facing point by $q(s) = (\cos s, \sin s)$. For outgoing Birkhoff data (s, p) , set

$$n = (\cos s, \sin s), \quad t = (-\sin s, \cos s), \quad v = \sqrt{1 - p^2} n + pt.$$

For a ray based at q and a target circle with centre c and radius one, put $w = q - c$, $b = w \cdot v$, and $c_0 = |w|^2 - 1$. The first collision time and reflected velocity are

$$t_* = -b - \sqrt{b^2 - c_0}, \quad v^+ = v - 2(v \cdot n_*)n_*, \quad n_* = q + t_*v - c.$$

Apply this formula first to the second circle and then to the first. Taylor expansion through degree three gives the displayed linear matrix. Reflection in the line of centres conjugates (s, p) to $(-s, -p)$ and commutes with the two-collision return, so the return map is odd and all its quadratic jets vanish. Conjugating the cubic jet by S gives $G_{xyy} = -(7 - 4\sqrt{3})$; substitution in (69) gives -1 . $\square$

The calculation is scale free: if both radii and the free-flight gap are multiplied by the same positive number, use dimensionless arclength s/R and obtain the same return jet. Thus the local configuration can be inserted at an arbitrarily small geometric scale into a finite-horizon table, provided the normal two-cycle remains unobstructed. More generally, on any fixed clean finite itinerary the return jet and $\mathfrak{A}$ depend real-analytically on the circular radii, centres, and collision points. Proposition 6.8 therefore implies that the nonzero-cocycle tables form an open dense subset of every connected analytic parameter chart containing this sample.

For an anchored finite horseshoe which contains this normal two-cycle and a regular connector branch, the same fixed point occurs in every larger anchored truncation. Hence the geometric nonlinearity witness itself no longer deteriorates with truncation rank. What still requires proof is a uniform smooth extension/Markov realization of those finite billiard cores, uniform cutting mass for the SRB conditionals, and transfer of Leclerc's QNL bound from the equilibrium state to the weighted double-centered face slice. The exact cocycle closes the local geometry, not these singular-exhaustion steps.

6.7 Transfer of QNL to a central weighted face slice

The equilibrium-state/face-current mismatch is harmless on one compact homogeneous return rectangle. Let W^u be a reference unstable leaf and let $H : \Gamma \rightarrow W^u$ be stable holonomy from a central smooth face arc. Write m_Γ for arclength and let w include the physical face velocity, coarea factor, and the smooth branch Jacobians.

Lemma 6.9 (Weighted face-slice transfer). *Assume, uniformly on a compact clean family, that*

- (i) H is a $C^{1+\alpha}$ diffeomorphism onto its image and $0 < c_H \leq JH \leq C_H$;
- (ii) the SRB conditional μ^u on W^u has arclength density between $c_u > 0$ and $C_u < \infty$;
- (iii) $w \in C^\alpha(\Gamma)$ and $\|w\|_\infty \leq C_w$.

Define the signed face measure

$$\nu = H_*(w dm_\Gamma).$$

If, on every interval $J \subset W^u$, the temporal-distance phase X satisfies

$$\mu^u\{t \in J : |X(t) - P(t)| \leq \epsilon\} \leq C_0 \epsilon^\Gamma \mu^u(J),$$

then

$$|\nu|\{t \in J : |X(t) - P(t)| \leq \epsilon\} \leq C_1 \epsilon^\Gamma |J|, \quad C_1 = C_0 \frac{C_w C_u}{c_H c_u},$$

after increasing C_1 by a uniform chart constant. The same conclusion holds for a finite signed sum of physical face pieces, with C_1 multiplied by the sum of their total-variation bounds.

Proof. The change-of-variables formula gives

$$\frac{d|\nu|}{dm_{W^u}}(Hz) = \frac{|w(z)|}{JH(z)} \leq \frac{C_w}{c_H}.$$

The density bounds for μ^u compare arclength and μ^u in both directions. Apply the assumed small-value estimate and make those two comparisons. Total variation and the triangle inequality handle signed pieces. $\square$

Lemma 6.3 supplies the central coarea transversality, while standard clean-rectangle holonomy estimates supply (i)–(ii). Thus, once the exact cocycle is placed in an anchored smooth horseshoe, the central weighted face slice inherits the same QNL exponent. For pieces whose return word or homogeneity rank tends to infinity, this local transfer is combined below with the central marked moment and the two-sided recovery sum; Proposition 7.25 keeps those constants on one common refinement.

6.8 A countable-IFS theorem removes the return truncation

There is a more direct route than passing through smooth finite horseshoes. Leclerc–Paukkonen–Sahlsten [29] prove power Fourier decay for Gibbs measures of finite or countable $C^{1+\alpha}$ conformal IFSs, allowing overlaps. In the countable case their assumptions are uniform contraction, bounded distortion, summable variations, and the light-tail condition

$$\sum_a \sup_{x \in I} e^{\phi(h_a x)} |h'_a(x)|^{-\gamma_0} < \infty \quad \text{for some } \gamma_0 > 0. \quad (70)$$

Their Theorem 1.1 says that QNL gives power Fourier decay, including for Hölder amplitudes and nonstationary phases with a small polynomial loss in their norms. Sections 6.1–6.2 prove QNL by finite-difference cutting arguments for finite MNL and for countable UNI systems with a light tail. No finite-alphabet exhaustion or smooth extension of the branch maps is used.

This theorem is unusually well matched to the present induced problem. Let $F : I \rightarrow I$ be the unstable quotient of a full-branch Gibbs–Markov return to a central billiard magnet and let $\{h_a : I \rightarrow I\}_{a \in \mathcal{A}}$ be its inverse branches. After accelerating once if necessary, the desired dictionary is

countable-IFS input	billiard input	
h_a	homogeneous inverse return branch	
ϕ	normalized induced unstable-Jacobian potential	
uniform contraction	uniform hyperbolicity after inducing	(71)
bounded distortion	homogeneous return distortion	
light tail	return tail plus a small grazing moment	
QNL	a fixed clean finite-difference gate.	

There is a small but important point in the first line of this dictionary. The one-sided moving-face labels must not be inserted by cutting every return branch into labelled subbranches: those pieces need not have full image and would destroy the hypothesis to which the countable-IFS theorem is applied. The correct object is a full-branch dynamical quotient with a marked amplitude fiber.

For an interval amplitude use the affine scale-normalized norm

$$\|w\|_{\mathcal{A}^\alpha(J)} := \|w\|_{\infty, J} + |J|^\alpha [w]_{C^\alpha(J)}. \quad (72)$$

It is unchanged up to a fixed factor by affine normalization and cannot increase under restriction. Every connected component is charged separately; hence endpoint and component complexity is visible in the sum rather than hidden in a piecewise-Hölder notation.

Proposition 6.10 (Marked full-branch quotient). *Let Λ be a Young return rectangle with product structure and let $\{\Lambda_a\}_{a \in \mathcal{A}}$ be a return partition such that $F|_{\Lambda_a} = T^{R_a}$ maps the stable subset Λ_a across Λ with the usual stable/unstable Markov property. Fix a reference unstable leaf I and let $\pi^s : \Lambda \rightarrow I$ be stable holonomy. Then*

$$\bar{F} \circ \pi^s = \pi^s \circ F$$

defines a one-state full-branch expanding map: for $I_a = \pi^s(\Lambda_a)$, the map $\bar{F} : I_a \rightarrow I$ is onto and has an inverse branch $h_a : I \rightarrow I_a$.

Suppose now that the lift of a complete physical face through Λ_a is refined into one-sided marked pieces $E_{a,\ell}$, after artificial-face cancellation. On the stable quotient its action can be written

$$\nu(\varphi) = \sum_{a,\ell} \int_{J_{a,\ell}} \varphi(h_a x) w_{a,\ell}(x) d\mu^u(x), \quad (73)$$

where the $J_{a,\ell} \subset I$ are amplitude pieces and the sign is included in $w_{a,\ell}$. Thus the quotient dynamical alphabet remains $\{h_a\}_{a \in \mathcal{A}}$; the face label ℓ is a fiber mark and does not have to define a full-image branch. If

$$\sum_{a,\ell} \|w_{a,\ell}\|_{\mathcal{A}^\alpha(J_{a,\ell})} \mu^u(J_{a,\ell}) < \infty,$$

then the marked current is a summable piecewise-Hölder amplitude over the same full-branch Gibbs system.

Proof. The Markov property says that every unstable leaf in Λ_a is mapped across every stable leaf of Λ . Quotienting stable leaves therefore makes $\bar{F}|_{I_a}$ onto I ; contraction of inverse branches and bounded distortion are the usual quotient estimates. Disintegrate the coarea face measure first over return atoms and then over its one-sided physical marks. Stable holonomy transports each piece to the reference leaf and gives (73). No restriction $h_a|_{J_{a,\ell}}$ is declared to be a dynamical branch. The last assertion is the triangle inequality for the Hölder-amplitude norm. Matched artificial cuts contribute opposite weights and disappear before this disintegration. $\square$

Consequently the standard Young quotient and the moving-face bookkeeping are compatible provided the marked-amplitude sum is finite. The construction of the quotient is standard. Proposition 6.11 supplies this summability on the central return alphabet, while the bi-recovery refinement supplies the grazing tail.

The SRB potential part of this audit is now supported by a direct billiard result. Lemmas 5.2–5.3 of Lima–Obata–Poletti [31] prove that the collision Lebesgue/SRB measure is an equilibrium state of the geometric potential

$$\phi = -\log \|DT|_{E^u}\|, \quad P(\phi) = 0,$$

and that ϕ becomes Hölder on their countable Markov coding for finite-horizon dispersing billiards. Hence the induced potential has summable variations. This does not by itself give the one-state return tail, but it removes the former ambiguity between an MME coding and the required collision-SRB Gibbs weights.

Proposition 6.11 (Central amplitude moment). *Assume the Young return scheme has*

$$\sum_a \sup_I p_a e^{\epsilon R_a} < \infty$$

and restrict first to a compact regular homogeneity core. For a complete first moving-face defect, suppose every collision insertion has uniformly bounded C^α coarea and stable-holonomy factors. Then the marked amplitude disintegration (73) satisfies

$$\sum_{a,\ell} e^{\epsilon' R_a} \|w_{a,\ell}\|_{\mathcal{A}^\alpha(J_{a,\ell})} \mu^u(J_{a,\ell}) < \infty \quad (0 < \epsilon' < \epsilon).$$

After high-homogeneity insertions are assigned to the bi-recovery blocks of Corollary 6.82, the same conclusion holds for the entire assembled first defect.

Proof. A return word of length R_a has at most R_a possible physical insertion times. On the compact regular core each insertion contributes at most a uniform constant times the Gibbs branch weight. Hence the left-hand side is bounded by

$$C \sum_a R_a e^{\epsilon' R_a} \sup_I p_a,$$

which is finite because $Re^{\epsilon' R} \leq C_{\epsilon, \epsilon'} e^{\epsilon R}$. Artificial insertions cancel before taking total variation. The removed high-homogeneity pieces satisfy the common two-sided recovery moment and are summed by Proposition 4.1. $\square$

The following consequence records the exact countable-IFS interface rather than hiding it in a finite-horseshoe approximation.

Proposition 6.12 (Finite-difference countable-IFS bridge). *Let the central unstable quotient be a one-state full-branch $C^{1+\alpha}$ IFS with uniform inverse contraction, bounded distortion, a normalized Gibbs potential of summable variations, and (70). For words $\mathbf{a}, \mathbf{b}$ of length N and $x, y \in I$, put*

$$\Delta_N((\mathbf{a}, x), (\mathbf{b}, y)) = \{\log |h'_{\mathbf{a}}(x)| - \log |h'_{\mathbf{b}}(x)|\} - \{\log |h'_{\mathbf{a}}(y)| - \log |h'_{\mathbf{b}}(y)|\}.$$

Assume the LPS quantitative nonlinearity estimate: for some $C > 0$, $\Theta > 0$, and $\rho \in (0, 1)$,

$$\sum_{\mathbf{a}, \mathbf{b}, \mathbf{c}, \mathbf{d} \in \mathcal{A}^N} p_{\mathbf{a}} p_{\mathbf{b}} p_{\mathbf{c}} p_{\mathbf{d}} \mathbf{1}_J(\Delta_N((\mathbf{a}, x_{\mathbf{c}}), (\mathbf{b}, x_{\mathbf{d}}))) \leq C\{|J|^{\Theta} + \rho^N\} \quad (74)$$

for every interval $J \subset \mathbb{R}$; Gibbs weights may be evaluated at arbitrary reference points in their cylinders. Then the induced SRB Gibbs measure has the weighted nonstationary-phase Fourier estimate of Leclerc–Paukkonen–Sahlsten for every phase with the stated C^{α} log-derivative bounds. The conclusion remains valid for the marked physical face amplitudes of Lemma 6.9.

Proof. Equation (74) is precisely the QNL hypothesis in Theorem 1.1 of [29]. That theorem is formulated for finite or countable $C^{1+\alpha}$ systems and Hölder amplitudes; it neither differentiates $\log h'_a$ nor assumes a C^2 holonomy. The light-tail moment controls the countable alphabet. The weighted-face conclusion follows from Lemma 6.9 and the summable scale-normalized amplitude norm (72). $\square$

The former C^2 multiplier/UNI route is deliberately not used. For a billiard quotient the stable holonomy is generally only $C^{1+\alpha}$, so the derivative of $\log h'_a$ need not exist. The correct gate input is the finite-difference estimate (74).

Proposition 6.13 (Clean homoclinic horseshoe supplies a finite QNL gate). *Let p be the regular fixed point of the squared billiard map associated with the normal period-two orbit in Proposition 6.8. If its Anosov cocycle satisfies $\mathfrak{A}(p) \neq 0$, then there is a compact finite clean horseshoe Ω_p , contained in regular branches and containing p , such that in at least one time orientation its temporal distance satisfies QNL for every Hölder equilibrium state. Thus Ω_p supplies a finite topological QNL seed; Proposition 6.19 transfers its polynomial escape to the physical collision-SRB bracket.*

Proof. Lemma 5.1 of Lima–Obata–Poletti [31] says that finite-horizon dispersing billiards have a single homoclinic class at regular hyperbolic points. Choose a second regular hyperbolic periodic point q . The theorem gives clean transverse intersections $W^u(p) \pitchfork$

$W^s(q)$ and $W^u(q) \cap W^s(p)$. The inclination lemma then gives a nontrivial clean transverse homoclinic point of p ; this avoids the vacuous substitution $x = y = p$. The Smale–Birkhoff construction, using a finite segment of this clean homoclinic orbit, gives a compact finite horseshoe Ω_p on which the billiard map is smooth. In symplectic Birkhoff coordinates the map is area preserving; the local dynamics near Ω_p may be viewed as a smooth Axiom–A basic set.

By the contrapositive of the Anosov cocycle obstruction, together with Theorem 4.3 of Leclerc [28], in one of the two time orientations the temporal distance on Ω_p satisfies QNL for every Hölder equilibrium state, in particular for the equilibrium state of the geometric potential restricted to this finite subsystem. No comparison of that horseshoe measure with collision SRB is made here. $\square$

There is a regularity caveat. The stable quotient of a smooth horseshoe is in general only $C^{1+\alpha}$, precisely because the stable holonomy need not be C^2 . Therefore the finite QNL conclusion above cannot be fed directly into the C^2 non-linearizability criterion of Algom–Rodriguez Hertz–Wang. The route below instead propagates the physical gate’s finite-difference gap to the countable $C^{1+\alpha}$ inducing scheme. The LPS theorem is recorded as a background alternative; it is not invoked in the principal fractional-phase proof.

The light-tail hypothesis has a concrete billiard reduction. If a branch a has return time R_a and grazing rank k_a , bounded distortion gives, schematically,

$$|h'_a|^{-\gamma_0} \leq C e^{c\gamma_0 R_a} (1 + k_a)^{q\gamma_0}.$$

Thus an exponential return moment and any strictly positive polynomial SRB tail in k_a imply (70) after choosing $\gamma_0 > 0$ sufficiently small. This is strictly weaker than uniform control of every rank- L smooth extension.

Lemma 6.14 (Return and grazing moments imply the LPS light tail). *Assume the normalized branch weights $p_a(x) = e^{\phi(h_a x)}$ satisfy*

$$\sum_a \sup_x p_a(x) e^{\epsilon R_a} (1 + k_a)^\delta < \infty$$

for some $\epsilon, \delta > 0$, and suppose $|h'_a|^{-1} \leq C e^{cR_a} (1 + k_a)^q$. Then (70) holds for every

$$0 < \gamma_0 < \min\{\epsilon/c, \delta/q\}.$$

Proof. Raise the inverse-derivative estimate to γ_0 and absorb the two factors into the assumed exponential and polynomial moment. $\square$

There is also a pressure-based way to obtain the required moment. The geometric-potential theory for finite-horizon dispersing billiards has a spectral gap for $-t \log J^u T$ on every compact subinterval of $(0, t_*)$, where $t_* > 1$, and its modified one-step expansion is uniform for $t \geq t_0 > 0$ [2]. Since the SRB value is $t = 1$, the LPS factor $|h'_a|^{-\gamma_0}$ merely changes the geometric exponent from 1 to $1 - \gamma_0$.

Proposition 6.15 (Geometric-pressure route to the light tail). *Let a homogeneous full-branch inducing scheme have an exponential return tail at the SRB parameter. Assume its induced distortion is uniform and that the geometric transfer estimates hold for $t \in [1 - \epsilon, 1]$. Then, after decreasing $\epsilon > 0$ if necessary, its normalized induced SRB potential satisfies (70) for every sufficiently small $\gamma_0 > 0$.*

Proof. Group branches by return time $R_a = n$. Up to the uniform distortion coboundary, the summand in (70) is the inverse unstable Jacobian to the power $1 - \gamma_0$. The geometric transfer estimate at $t = 1 - p\gamma_0$ bounds the corresponding p -moment by $Ce^{c(p\gamma_0)n}$, with $c(p\gamma_0) \rightarrow 0$ as $\gamma_0 \rightarrow 0$. Hölder's inequality combines this with the SRB return tail Ce^{-an} to give

$$\sum_{R_a=n} \sup_x e^{\phi(h_a x)} |h'_a(x)|^{-\gamma_0} \leq Ce^{-an/q + c(p\gamma_0)n/p}.$$

Choose $p, q > 1$ conjugate and then γ_0 so that the exponent is negative; summing over n proves the claim. Homogeneity-strip subdivision is already included in the geometric one-step expansion. $\square$

Lemma 6.16 (Fixed-gate hitting and two-context moments). *Let F be a one-state full-branch Gibbs–Markov map and let $\mathcal{G}$ be a fixed finite gate cylinder of return depth L . Suppose bounded distortion gives*

$$\inf_C \mu^u(\mathcal{G} \mid C) \geq \eta_G > 0 \quad (75)$$

for every past cylinder C , and suppose

$$\mathbb{E}e^{\epsilon R} < \infty \quad (76)$$

for some $\epsilon > 0$. If N_G is the first gate-hitting time in L -blocks and C^+ is the total collision length of the future entrance context, then there is $\epsilon_G > 0$ such that

$$\mu^u(N_G > k) \leq (1 - \eta_G)^k, \quad \mathbb{E}e^{\epsilon_G C^+} < \infty.$$

The reverse quotient gives the same estimate for C^- . On the natural extension no independence is needed:

$$\mathbb{E}e^{\epsilon'_G (C^- + C^+)} < \infty \quad (77)$$

for every sufficiently small $\epsilon'_G > 0$.

Proof. Condition (75) and the Markov property give $\mu^u(N_G > k + 1 \mid N_G > k) \leq 1 - \eta_G$, proving the geometric tail. Let $K(\theta)$ be the uniform conditional exponential moment of one return; $K(0) = 1$ and $K(\theta) < \infty$ for small $\theta > 0$. Bounded Gibbs distortion gives

$$\mathbb{E}\left[e^{\theta C^+}; N_G = k\right] \leq C\{(1 - \eta_G)K(\theta)^L\}^{k-1}K(\theta)^L.$$

Choose θ so that the quantity in braces is smaller than one and sum in k . Time reversal proves the past estimate. Finally

$$\mathbb{P}(C^- + C^+ > t) \leq \mathbb{P}(C^- > t/2) + \mathbb{P}(C^+ > t/2),$$

so the two marginal exponential tails imply (77) without independence. $\square$

For the marked quotient, choose one fixed finite gate word. The Gibbs ratio of that word after an arbitrary prefix differs from its unconditional weight by at most the uniform distortion factor. Hence (75) holds with $\eta_G = C_D^{-1}\mu^u(\mathcal{G}) > 0$; it is not an additional mixing assumption.

Lemma 6.17 (Joint return–maximal-grazing moment). *Let Y be the base of an exponential Young return, R its return time, and let $B(x)$ be the largest homogeneity rank visited before $R(x)$. In $(r, p) = (r, \sin \varphi)$ coordinates suppose*

$$\mu\{1 - |p| \lesssim 2^{-2b}\} \leq C2^{-2b}, \quad \mu_Y(R > L) \leq Ce^{-cL}.$$

Then B has an exponential tail. More precisely, for every fixed $A > 0$,

$$\mu_Y(B \geq b) \leq Ce^{-cAb} + CA b 2^{-2b}. \quad (78)$$

Consequently, for sufficiently small $\epsilon, \kappa > 0$,

$$\mathbb{E}_Y e^{\epsilon R + \kappa B} < \infty. \quad (79)$$

If the inverse-Jacobian and coarea loss of a return word is bounded by $Ce^{c_0 R + qB}$, then (70) and the marked-amplitude moment hold after decreasing the exponents. No independence of R and B is required.

Proof. On $\{R \leq Ab\}$ take a union bound over the at most Ab collision times. Since $\mu_Y(E) \leq \mu(E)/\mu(Y)$ for every event E and the collision SRB measure is invariant, each visit to a rank- b boundary layer costs at most $C2^{-2b}$. Adding $\mu_Y(R > Ab)$ proves (78). Choose A so that both terms decay exponentially in b ; hence $\mathbb{E}e^{q\kappa B} < \infty$ for some $q > 1$ and small κ . For conjugate $p, q > 1$, Hölder gives

$$\mathbb{E}e^{\epsilon R + \kappa B} \leq (\mathbb{E}e^{p\epsilon R})^{1/p} (\mathbb{E}e^{q\kappa B})^{1/q} < \infty.$$

Absorb $c_0 R + qB$ with smaller exponents and apply Lemma 6.14. The same argument applies to the scale-normalized coarea amplitude. $\square$

Lemma 6.17 verifies the grazing factor from the collision density and strip width, while Proposition 6.11 supplies the moving-face amplitude budget on the same quotient. Lemma 6.16 supplies the entrance/exit moment. Thus the particular full-branch quotient uses one joint return/grazing/context ledger rather than an unverified appeal to the literature theorem.

This route also exposes a flaw in the finite-horseshoe detour. A conservative pasting lemma [33] can locally modify or smooth a volume-preserving map, but it does not automatically extend a finite family of billiard branches to a global area-preserving Axiom–A diffeomorphism whose equilibrium state has the required collision-SRB weights. Even a successful extension can put a fractal Gibbs conditional on a saddle basic set in place of the absolutely continuous unstable conditional of the billiard SRB measure. The direct countable-IFS quotient preserves the correct measure and should therefore be the primary route.

Thus light-tail is not expected to be a new principal theorem once the standard homogeneous inducing scheme is put in the one-state form required above. The exact value $\mathfrak{A} = -1$, the clean homoclinic theorem, and Proposition 6.13 produce the clean polynomial escape seed without naming a connector. Proposition 6.48 records the local physical slope transfer, and the cohomological bracket below identifies its physical collision-SRB object.

6.9 The cohomological bracket and polynomial context blow-ups

The quotient regularity is only $C^{1+\alpha}$. The useful exact object is therefore a four-point difference of Birkhoff sums, not a derivative of a holonomy. Let f denote the billiard map

restricted to the finite clean basic set, let F be its stable factor, and put $\tau_f = \log J^u f$ and $\tau_F = \log |F'|$. As in [28, Lemma 3.15], choose a Hölder function θ such that

$$\tau_f = \tau_F \circ \pi + \theta \circ f - \theta. \quad (80)$$

For p, q in one Markov rectangle define

$$\Delta(p, q) = \sum_{j \in \mathbb{Z}} \{ \tau_f(f^j p) - \tau_f(f^j[p, q]) - \tau_f(f^j[q, p]) + \tau_f(f^j q) \}. \quad (81)$$

Lemma 6.18 (Cohomological bracket identification). *Let $\mathbf{a}, \mathbf{b}$ be the length- N past words of p, q and let $x = \pi p$, $y = \pi q$. With inverse branches $g_{\mathbf{a}}, g_{\mathbf{b}}$ of F , set*

$$\begin{aligned} \mathcal{D}_N(\mathbf{a}, \mathbf{b}; x, y) &:= S_N \tau_F(g_{\mathbf{a}} x) - S_N \tau_F(g_{\mathbf{a}} y) \\ &\quad - S_N \tau_F(g_{\mathbf{b}} x) + S_N \tau_F(g_{\mathbf{b}} y). \end{aligned} \quad (82)$$

Then

$$|\mathcal{D}_N(\mathbf{a}, \mathbf{b}; x, y) - \Delta(p, q)| \leq C \kappa^{\alpha N}. \quad (83)$$

The constants are uniform on the persistent finite basic set and for $|s| < s_0$. Entrance and exit contexts are not included in this statement; their finite jets and holonomy tails are treated explicitly in Proposition 6.19.

Proof. Insert (80) in (81). The four θ -terms telescope. For $j \geq 0$, stable projection gives $\pi(f^j p) = \pi(f^j[p, q])$ and $\pi(f^j q) = \pi(f^j[q, p])$, so the positive-time factor terms cancel exactly. Writing the negative iterates through the past inverse words gives (82). The omitted tail starts at $N + 1$; Hölder regularity of τ_F and inverse contraction give

$$\sum_{j > N} C \kappa^{\alpha j} \leq C' \kappa^{\alpha N}.$$

This is precisely the proof of [28, Lemma 3.15] on the clean basic set. Hyperbolic continuation makes the constants uniform in s . $\square$

The all-scale escape in [28, Propositions 4.31–4.32] supplies an equilibrium/Cantor cutting template only. The actual collision–SRB transfer at the prescribed depth remains the independent interface PPE_N below; it is not a consequence of the quoted propositions.

The preceding bracket must still be matched to the physical endpoint maps. For a stable projection π_s on a clean rectangle, the stable holonomy Jacobian satisfies the convergent identity

$$\log J \pi_s(r) = \sum_{j \geq 0} \{ \tau_f(f^j r) - \tau_f(f^j \pi_s r) \}. \quad (84)$$

This is the Jacobian formula of [28, Lemma C.1]. In particular, entrance and exit holonomies are not discarded as unspecified coboundaries: their four endpoint series are inserted into (81); the common terms cancel and the unpaired tail is bounded by $C \kappa^{\alpha N}$.

Proposition 6.19 (Context–polynomial blow-up). *Assume $\mathfrak{A}(p) = -1$ and let $I_c = t_c + \ell_c[-1, 1]$ be any homogeneous resolving cylinder whose central excursion enters the persistent clean basic set. For the full reduced-slope bracket S_c , including the two endpoint holonomy Jacobians, let N be the total central resolving depth and let $L_c = |c^-| + |c^+|$ be the entrance/exit length. There are a fixed degree d , a rough template $\mathcal{X}_{\omega_c}$ from the clean basic set, a polynomial P_c of degree at most d , and a context cost $H_c \geq 1$ such that*

$$\ell_c^{-1} \{ S_c(t_c + \ell_c u) - S_c(t_c) \} = \mathcal{X}_{\omega_c}(u) + P_c(u) + E_c(u), \quad \|E_c\|_{\infty} \leq C \left(H_c \ell_c^{\alpha} + \frac{\kappa^{\alpha N}}{\ell_c} \right). \quad (85)$$

Moreover $H_c \leq Ce^{aL_c}$, and for some $p > 1$ the context ledger gives

$$\sup_{|s| < s_0} \sum_c w_c H_c^p < \infty. \quad (86)$$

The degree and the escape constants of $\mathcal{X}_{\omega_c}$ modulo $\mathbb{R}_d[X]$ are independent of c and s . In particular, if

$$N \geq q_* |\log \ell_c|, \quad q_* > \frac{1 + \alpha}{\alpha |\log \kappa|}, \quad (87)$$

then the second error in (85) is $O(\ell_c^\alpha)$.

Proof. Use (84) for both endpoint maps and split each series at the entrance and exit times. The four common tails cancel in the bracket. Each remaining finite clean context sum is $C^{1+\alpha}$ on its homogeneous component. Its affine Taylor polynomial is linear (and is regarded as an element of $\mathbb{R}_d[X]$); the normalized remainder is bounded by its $C^{1+\alpha}$ norm times ℓ_c^α . The chain rule and uniform one-step distortion give $H_c \leq Ce^{aL_c}$. Choose $p > 1$ so that pa is smaller than the exponent in (77); this proves (86). Lemma 6.18 identifies the central part with the factor four-point cocycle with error $O(\kappa^{\alpha N})$. Condition (87), rather than an unrecorded fixed acceleration, makes its normalized value $O(\ell_c^\alpha)$. Finally [28, Propositions 4.31–4.32], activated by $\mathfrak{A}(p) \neq 0$, gives uniform all-location, all-scale escape of the rough template from every polynomial of degree at most d . No fixed acceleration is used. $\square$

6.10 Native-coordinate single-source slope energy

The following two conditions isolate the genuinely new clean-gate geometry on the *actual* collision–SRB interval. They are not inferred from a smooth extension of a fractal horseshoe conditional.

Definition 6.20 (Single-source bracket condition SS). *Fix one orientation and condition on the opposite symbolic record, the value t' , and the two finite contexts. For a native interval Q write $z = \Phi_{\beta,Q}(t)$ and $\sigma_Q = \text{diam } \Phi_{\beta,Q}(Q)$. Put $d_Z = \max\{1, d\}$, where d is the fixed degree in Proposition 6.19. Condition $\text{SS}(d_Z)$ means that*

$$S_\beta(t) - S_\beta(t') = \ell_{\beta,Q} \{X_{\beta,Q}^{[N]}(z) - P_{\beta,Q,t'}(z)\} + \mathcal{R}_{\beta,Q,t'}(z), \quad (88)$$

where $P_{\beta,Q,t'} \in \mathbb{R}_{d_Z}[z]$ and $X_{\beta,Q}^{[N]}$ is the finite endpoint series on the actual depth- N collision cylinder. On the Cantor core it is within $C\kappa^N$ of one native Leclerc rough template, and

$$0 < c_\ell \leq |\ell_{\beta,Q}| \leq C_\ell, \quad \|\mathcal{R}_{\beta,Q,t'}\|_\infty \leq C \{H_\beta \sigma_Q^{1+\alpha_B} + \kappa^N\}. \quad (89)$$

The endpoint identity proving that every other t -dependent rough term is absent or belongs to the fixed-degree polynomial is part of the condition; freezing a symbolic name alone is not enough.

Definition 6.21 (Actual endpoint–slope bridge ACTUAL_SLOPE_BRIDGE). *On every labelled carrier to which a balanced phase estimate is applied, let u be the very same carrier coordinate used by BR_{sec} , and let $X_{\lambda,m,s}(u)$ and $Y_{\lambda,n,s}(u)$ be its two actual scalar endpoint maps. The interface requires their derivatives to be nonzero on the labelled homogeneous component and defines, rather than merely names,*

$$\varepsilon_{\text{or},\lambda,m,n,s} := \text{sgn} \left(\frac{\partial_u Y_{\lambda,n,s}}{\partial_u X_{\lambda,m,s}} \right), \quad S_{\lambda,m,n,s}(u) := \log \left| \frac{\partial_u Y_{\lambda,n,s}}{\partial_u X_{\lambda,m,s}} \right|. \quad (90)$$

After the explicitly recorded carrier coordinate changes, this function is required to be simultaneously the S_β in Definition 6.20, the $S_{\lambda,a}$ in every source-clock row, and the S_η^0 in the unselected reference pair law. The branch, orientation, time, and carrier labels in these four occurrences must agree; equality of their distributions is not enough. In particular, for the actual endpoint phase

$$\psi_\xi(u) = \xi_1 X_{\lambda,m,s}(u) + \xi_2 Y_{\lambda,n,s}(u) \quad (91)$$

one has the exact typed identity

$$\partial_u \psi_\xi = \partial_u X_{\lambda,m,s} (\xi_1 + \varepsilon_{\text{or},\lambda,m,n,s} \xi_2 e^{S_{\lambda,m,n,s}}). \quad (92)$$

Condition $\text{SS}(d_Z)$ is imposed on this exact derivative-ratio slope. An auxiliary cocycle or an abstract function satisfying (88) does not satisfy this interface unless the displayed pointwise identification has been proved.

Definition 6.22 (Actual native stopping antichain NST_{phys}). For $0 < r < r_0$ put $\sigma(r) = (r/c_\ell)^{1/(1+\alpha_L)}$. On each actual collision-SRB unstable conditional, follow homogeneous full-branch descendants and stop at the first descendant Q satisfying

$$\text{diam}_u(Q) \leq \sigma(r) < \text{diam}_u(\text{parent } Q). \quad (93)$$

Condition NST_{phys} requires:

- (NST1) the stopped descendants form a measurable antichain with a normalized positive refinement kernel; their union carries all actual conditional mass except a grazing/return set of mass at most $Cr^{\Theta_{\text{st}}}$;
- (NST2) for fixed $0 < c_{\text{st}} < C_{\text{st}} < \infty$,

$$c_{\text{st}} \sigma(r) \leq \sigma_Q \leq C_{\text{st}} \sigma(r) \quad (94)$$

on every stopped interval, with bounded overlap after connected component indexing;

- (NST3) the homogeneous buffer and every retained continuation are mass-preserving, and their total depth is $O(|\log r|)$; along $a(\theta s)$ they belong to UCR_{sh} whenever this depth is at most $L(s)$. There are fixed $c_{\text{buf}}, c_{\text{gate}}, c_{\text{cont}} > 0$ such that the discarded buffer, gate, and exceptional-continuation masses at resolving depth N are bounded respectively by $Ce^{-c_{\text{buf}}N}$, $Ce^{-c_{\text{gate}}N}$, and $Ce^{-c_{\text{cont}}N}$ under the unweighted complete positive tree $\Pi_{s,N}$. Each discarded mass is retained on its named cemetery label. These raw tails are used in the phase proof only after TENERGY_W has placed them under $\Pi_{s,N}^W$;
- (NST4) the context variable H_β , the physical Mark-Z factors, dominant inverse scale, and PPE_N marks have their stated conditional moments under this one complete tree law, including the measurable last-mark extension on cemetery labels. In particular, for some $p > 1$,

$$\sup_{r,\beta} \mathbb{E}_{\text{stop}}[H_\beta^p \mid \mathcal{F}_{\text{parent}}] < \infty. \quad (95)$$

The scale comparison and the complete-tree moments are part of one interface; moments from an unrelated native partition are not substituted into the energy proof. Leclerc's first-stopping/Vitali construction [28, Section 4] gives the comparable-scale model on Markov rectangles. Its transfer to actual collision-SRB conditionals across homogeneity cuts is not automatic.

Definition 6.23 (Prescribed-depth actual physical escape PPE_N). Assume NST_{phys} and let $\mu_{\beta,Q}^u$ be the normalized collision-SRB conditional on one of its stopped native intervals Q . For an energy scale $0 < r < r_0$ set

$$\sigma(r) = (r/c_\ell)^{1/(1+\alpha_L)}, \quad N(r) = \lceil C_{\text{buf}} \log(1/r) \rceil, \quad \kappa^{N(r)} \leq r/100. \quad (96)$$

The family satisfies $\text{PPE}_N(d_Z)$ if there are $\Theta_{\text{PPE}} > 0$ and $C < \infty$ such that the following hold on every stopped actual conditional and every frozen context used in $\text{SS}(d_Z)$, all represented in the retained part of the complete tree $\Pi_{s,N}$ from NST_{phys} .

(PPE1) For every $P \in \mathbb{R}_{d_Z}[z]$ the endpoint series at the prescribed depth in (96) obeys

$$\mu_{\beta,Q}^u \{ |X_{\beta,Q}^{[N(r)]} - P| \leq Cr \} \leq Cr^{\Theta_{\text{PPE}}}. \quad (97)$$

This is an estimate on the full absolutely continuous collision-SRB conditional stopped at the native scale, not on a surviving Cantor conditional.

(PPE2) Multiplication by a retained positive amplitude preserves (97) with its local PPE mark. Endpoint/Jacobian derivatives and the physical Lipschitz cost at depth $N(r)$ are bounded by the marks already charged in (245). This local clause does not control context, stopping, buffer, gate, or continuation failures and does not by itself imply the full- $\mathcal{W}_*$ estimate (154).

(PPE3) The depth- $N(r)$ endpoint is defined on retained open physical descendants, the resolving buffer has length $O(N(r))$, bounded overlap, and retains all continuations. It is the same buffer as in NST_{phys} . No affine jet is absorbed into P and no comparison with an auxiliary depth J is used.

(PPE4) At $s = 0$ the constants are uniform for all r . Along $a(\theta s)$ they are uniform whenever the resolving depth is at most $L(s)$ on UCR_{sh} cells. Endpoint/section buffers at two-time depth N_{time} cost at most $C_{\text{ep}} N_{\text{time}}$. A phase-mode buffer is added only in the bounded intermediate frequency range (235); no such assertion is made for an unbounded Fourier complement.

This is the exact physical escape input used by the energy theorem. In particular no statement about $X^{[J]}$ is transferred to $X^{[N(r)]}$.

Lemma 6.24 (Negative-pressure trial obstruction). Suppose a fixed clean subsystem has relative pressure $P_{\text{cl}} = -\chi < 0$, so a depth- J clean trial has physical survival mass at most $Ce^{-\chi J}$. Then $O(J)$ such trials expose at most $O(Je^{-\chi J})$ of the original conditional mass. In particular they cannot yield an uncovered gap $O(e^{-cJ})$.

Proof. Sum the survival masses of the trials. This is the union bound $CJe^{-\chi J} = o(1)$, so the unexposed mass tends to one. In the Bernoulli model with clean probability $p = e^{-\chi} < 1$ this estimate is exact up to a constant. Conditioning on survival cancels p^J only inside a bad/good ratio; it cannot cancel it in the physical cover. $\square$

Remark 6.25 (Adaptive full-mass alphabet as a nonvacuity route). A falsifiable route to PPE_N is to use the actual one-state quotient rather than a fixed negative-pressure horse-shoe. At depth N take an alphabet

$$\mathcal{A}_N = \{\text{return/grazing/context mark} \leq b_N\} \cup \mathcal{A}_{\text{rough}}, \quad b_N = \lambda N,$$

where the finite set $\mathcal{A}_{\text{rough}}$ contains fixed QNL rough witnesses. If the actual quotient tail is $Ce^{-c_b b}$, the mass lost in the first N steps is at most $CNe^{-c_b \lambda N} = O(e^{-c'N})$; hence the

relative pressure gap tends to zero and $N_{\chi_N} \rightarrow 0$. To obtain (97) one must still prove that rough-template, cutting, distortion, component, and open-descendant constants for $\mathcal{A}_N$ grow slowly enough to leave a strict exponent margin against $c_b \lambda$. This uniform finite-alphabet theorem is not supplied by the existing literature and is not assumed to follow from Leclerc's fixed-alphabet theorem. Failure of that margin leaves PPE_N primitive. Open-system normalization in [22, 2] preserves the survival factor and cannot evade Lemma 6.24.

Proposition 6.26 (Single-source physical bracket). *Assume*

$$\text{SS}(d_Z), \quad \text{ACTUAL_SLOPE_BRIDGE},$$

and the dominant clean-gate chart normalization on the same labelled BR_{sec} carrier. Then (88)–(89) hold on every labelled native interval. In the dominant coordinates there are positive labelled scales a_X, a_Y such that

$$c \leq |X'|/a_X, |Y'|/a_Y \leq C, \quad \Xi = |\xi_1|a_X \asymp |\xi_2|a_Y. \quad (98)$$

Proof. The bracket assertion is $\text{SS}(d_Z)$ applied, by Definition 6.21, to the actual logarithmic derivative ratio (90). The last display is the separately assumed nonvanishing clean-gate chart normalization after contracted coordinates are removed. The proposition does not derive this normalization from SS . No absolute uniform lower bound is asserted: the inverse scale cost is the explicit factor $\mathfrak{d}_\lambda$ in (245). $\square$

Definition 6.27 (Frozen-parent kernel reverse-Hölder interface KRH_p). *Let $\mathfrak{P}_{\text{fr}}$ be the set of every frozen parent actually used below: context and orientation parents, frozen opposite endpoints, actual/split constituents, cemetery continuations, and the normalized branch–source-incidence parents of Definition 6.38. For $\eta \in \mathfrak{P}_{\text{fr}}$ let K_η be the normalized positive descendant kernel on Ξ_η . In a branch kernel the descendant is the complete retained/cemetery label; in a source kernel it is the labelled source occurrence. It never includes a main-cell selector, a reciprocal proposal factor, or the two internal carrier coordinates used by main energy; that complete change of law belongs exclusively to Definition 6.53. In particular, a parent in TENERGY_W may freeze physical context, orientation, or the opposite endpoint, but never a main source cell A . A zero-mass source kernel is omitted. Let $\mathscr{W}_\eta > 0$ be the complete survivor or incidence envelope used on that kernel and define*

$$L_\eta(\xi) := \frac{\mathscr{W}_\eta(\xi)}{\int_{\Xi_\eta} \mathscr{W}_\eta dK_\eta}, \quad \int L_\eta dK_\eta = 1. \quad (99)$$

The interface KRH_p is the uniform normalized bound

$$\sup_{\eta \in \mathfrak{P}_{\text{fr}}} \int_{\Xi_\eta} L_\eta^p dK_\eta \leq C_{\text{KRH}}, \quad p > 1. \quad (100)$$

All constants are independent of m, n, s , scalar mode, polynomial, and final observable. A global complete-tree L^p bound is not a substitute for (100); rare parent cells remain separate kernels.

Lemma 6.28 (Parentwise tilted transfer). *Under KRH_p , every K_η -event B satisfies*

$$K_\eta^W(B) := \int_B L_\eta dK_\eta \leq C_{\text{KRH}}^{1/p} K_\eta(B)^{1/p'}, \quad p' = \frac{p}{p-1}. \quad (101)$$

Consequently a raw polynomial or exponential tail uniform over the frozen parents transfers with exponent divided by p' . The statement applies separately to branch and source-incidence kernels and therefore does not move a global moment through a parent conditioning.

Proof. Apply Hölder on the single normalized kernel K_η and use (100). No global tree disintegration or signed cancellation is involved. $\square$

Definition 6.29 (Tilted endpoint upper-scale grazing ledger $\text{USC}_{\text{end}}^W(q_g^W)$). *On a depth- N homogeneous endpoint branch let k_j^- and k_j^+ be the past and future homogeneity-strip indices and put*

$$G_N^- := \sum_{j < N} \log(1 + k_j^-), \quad G_N^+ := \sum_{j < N} \log(1 + k_j^+). \quad (102)$$

Let $S_\lambda^\pm \geq 1$ dominate the positive derivative scale of the initial physical-face parametrization, same-branch section, scalar endpoint chart, and coarea/holonomy coordinate change in the two orientations, and put

$$S_\lambda := 1 + S_\lambda^- + S_\lambda^+, \quad \tilde{G}_N^\pm := G_N^\pm + \frac{1}{2} \log S_\lambda^\pm. \quad (103)$$

On a retained label define the physical survivor factor

$$\mathcal{W}_{\lambda, \text{surv}} := e^{\epsilon(|c^-| + |c^+|)} (1 + H_\lambda)^{a_H} \mathfrak{d}_\lambda^{A_{\text{dom}}} \text{Mark}(\lambda)^2 \{1 + Z(\mathcal{G}_\lambda^-) + Z(\mathcal{G}_\lambda^+)\} S_\lambda^{a_{\text{scl}}}. \quad (104)$$

Here $a_{\text{scl}} > 0$ is fixed below the available positive-scale moment. For a first-failure label of type $i \in \{\text{ctx}, \text{stop}, \text{buf}, \text{gate}, \text{cont}\}$, let $\mathcal{W}_{\lambda, i}^{\text{cem}}$ be the predictable stopped-source envelope from Definition 6.35. The one weight used everywhere is

$$\mathcal{W}_{\lambda, *} := \mathcal{W}_{\lambda, \text{surv}} \mathbf{1}_G(\lambda) + \sum_i \mathcal{W}_{\lambda, i}^{\text{cem}} \mathbf{1}_{B_i}(\lambda). \quad (105)$$

The events G, B_i are disjoint because the tree records the first failure. We retain the superscript W for tilted laws and rates, but it always means the unique weight $\mathcal{W}_$ in (105); there is no separate survivor or cemetery tilt. For every frozen parent $\eta \in \mathfrak{P}_{\text{fr}}$ whose descendant kernel contains a depth- N retained endpoint, the complete actual/split law extending NST_{phys} satisfies, with the same weight (105), $0 < \int \mathcal{W}_{\eta, *} dK_\eta \leq C_W$ uniformly in the allowed parameters and depths, and*

$$\int_{\Xi_\eta} \mathcal{W}_{\lambda, *} \mathbf{1}_G \mathbf{1}_{\{\tilde{G}_N^\pm > \gamma_g N + u\}} dK_\eta \leq C \left(\int_{\Xi_\eta} \mathcal{W}_{\eta, *} dK_\eta \right) e^{-q_g^W u}, \quad u \geq 0. \quad (106)$$

The indicator $\mathbf{1}_G$ is part of the assertion: the depth- N strip indices, $S_\lambda^\pm$, and $\tilde{G}_N^\pm$ are read only on retained homogeneous endpoint branches. They are not extended to an early cemetery branch. Such branches are charged by CEM_DOM and CLOCK_PROB below; additive cemetery currents are charged on the source-incidence ledger. The same complete label law is used for the native stopping, PPE, endpoint marks, and all cemetery failures; no marginal tail is transferred to a different conditional law, and an unweighted probability tail is not promoted to a marked tail by mere L^1 integrability. Equivalently, under every normalized parent likelihood (99), the survivor subprobability has augmented grazing exponent q_g^W . This is the “normalize the positive source, estimate, then multiply its mass back” organization used in [22, Lemma 5.10]; that lemma is a template and does not prove the present tilted tail. A sufficient, but not necessary, route to (106) is the corresponding raw parentwise tail together with KRH_{p_W} . Lemma 6.28, applied separately on every frozen parent kernel, gives only $q_g^W = q_g/p_W'$, and all ledgers below use this degraded exponent. A complete-tree global L^{p_W} estimate gives only a global aggregate scale tail and is not used here.

Definition 6.30 (Complete positive tilted energy tree $\Pi_{s,N}^W$). *Fix an energy scale r , the prescribed depth $N = N(r)$, one orientation, and the frozen opposite context. Before any mass is discarded, compose the normalized positive refinement kernels for the context gate, actual native stopping, homogeneous buffer, physical gate, continuation, and retained PPE descendant. Whenever a stage fails, send its full positive mass to a distinct cemetery label*

$$\partial_{\text{ctx}}, \quad \partial_{\text{stop}}, \quad \partial_{\text{buf}}, \quad \partial_{\text{gate}}, \quad \partial_{\text{cont}}$$

and keep the successful labels. The resulting probability measure on the complete tree, including the sampled physical point and all actual/split constituent labels, is denoted $\Pi_{s,N}$. A cemetery label carries the last marks measurable before failure. It is not completed by declaring unknown future factors equal to one. Instead, the stopped source and its remaining propagator are enveloped before the branch enters the cemetery, as in Definition 6.35. Thus the piecewise weight $\mathcal{W}_$ in (105) is positive and measurable on the entire tree, not only on the terminal survivor law. Across the depths used in one CM2 quadrant, these laws are required to be the exact tree marginals of the master law in (126). Independently reconstructed depth laws are not admissible, even if their separate tail constants are uniform. Define*

$$d\Pi_{s,N}^W := \frac{\mathcal{W}_*}{\mathbb{E}_{\Pi_{s,N}} \mathcal{W}_*} d\Pi_{s,N}, \quad 1 \leq \mathbb{E}_{\Pi_{s,N}} \mathcal{W}_* \leq C_W. \quad (107)$$

All tree laws below, including those for a split constituent, use this same extension and tilt. A terminal survivor law which has already forgotten discarded branches is not a substitute for $\Pi_{s,N}$.

Definition 6.31 (Concrete oriented source and test spaces). *Fix the common finite physical atlas, the forward/reverse homogeneous cones, and the separation-time exponent α_* . For $\varepsilon \in \{+, -\}$ let $\mathfrak{S}_{\text{sf}}^\varepsilon$ be the completion of finite signed combinations of homogeneous measured standard curves $\Sigma = \sum_j b_j(W_j, \rho_j)$ in the positive-decomposition norm*

$$\|\Sigma\|_{\mathfrak{S}_{\text{sf}}^\varepsilon} := \inf_{\Sigma = \sum_j b_j(W_j, \rho_j)} \sum_j |b_j| M_j \{1 + \text{Reg}_{\alpha_*}(\rho_j) + Z(W_j)\}. \quad (108)$$

Here $M_j = \int_{W_j} \rho_j dm_{W_j}$ and $Z(W_j)$ is the usual inverse-length boundary functional after canonical homogeneous subdivision, and the infimum is over positive measured-curve decompositions of the Jordan parts.

Let $\mathfrak{S}_{\text{flux}}^\varepsilon$ be the completion of finite signed oriented coarea/face currents $\Sigma = \sum_j b_j(\iota_j)_(\rho_j ds)$, with each ι_j a homogeneous C^2 face chart in orientation ε , under*

$$\|\Sigma\|_{\mathfrak{S}_{\text{flux}}^\varepsilon} := \inf \sum_j |b_j| M_j \{1 + \text{Reg}_{\alpha_*}(\rho_j) + Z_\partial(\iota_j) + \text{Atl}_2(\iota_j)\}. \quad (109)$$

Z_∂ counts oriented endpoints and homogeneous cuts with their inverse-length weights, while Atl_2 is the fixed-atlas C^2 chart/coarea mark, including the tangent Jacobian. Coincident artificial faces are assembled with orientation before this norm is taken.

A normalized positive standard family is recorded as $\mathcal{G} = \{p_j, (W_j, \rho_j)\}$ with $p_j \geq 0$, $\sum_j p_j = 1$, and each ρ_j a probability density on W_j . It describes only the carrier shape. Its shape cost is

$$K_{\text{sf}}(\mathcal{G}) := 1 + \text{Reg}_{\alpha_*}(\mathcal{G}) + Z(\mathcal{G}), \quad Z(\mathcal{G}) := \sum_j \frac{p_j}{|W_j|}. \quad (110)$$

For a normalized flux family one analogously defines $K_{\text{flux}} = 1 + \text{Reg}_{\alpha_*} + Z_{\partial} + \text{Atl}_2$. A positive source of exact mass m is $m\mathcal{G}$, and its source-norm cost is $mK(\mathcal{G})$; probability normalization never sets $K(\mathcal{G})$ equal to one. This is the mass-shape organization of standard families used in [22].

The corresponding test spaces are the completions of bounded functions with

$$\|\phi\|_{\mathfrak{T}_{\text{sf}}^\varepsilon} := \|\phi\|_\infty + \sup_{W \in \mathcal{W}^\varepsilon} [\phi|_W]_{\alpha_*, s_\varepsilon}, \quad (111)$$

$$\|\phi\|_{\mathfrak{T}_{\text{flux}}^\varepsilon} := \|\phi\|_{\mathfrak{T}_{\text{sf}}^\varepsilon} + \sup_{\iota \in \mathcal{A}_{\text{face}}^\varepsilon} \|\phi \circ \iota\|_{C^1}. \quad (112)$$

The seminorm in the first line is the dynamic Hölder seminorm determined by separation time. Forward and reverse transfer act by the canonical pushforward of measured curves/currents and by pullback on these tests. Thus $U_s^{k,\varepsilon}$ is an actual operator $\mathfrak{S}_{\text{sf/flux}}^\varepsilon \rightarrow (\mathfrak{T}_{\text{sf/flux}}^\varepsilon)^*$, not a symbol depending on the eventual observable. Any cemetery row not admitting one of these two representations remains an explicitly named SOURCE_PROP_i primitive with its own fully defined Banach source/test pair.

Lemma 6.32 (Source-space separation and operator interface). *The norms (108) and (109) dominate the total variation of every finite representative. Hence they separate currents, and their completions are Banach source spaces. On every common homogeneous atlas used below, SOURCE_PROP_i includes the quantitative assertion*

$$\|U_s^{k,\varepsilon} \Sigma\|_{(\mathfrak{T}_i)^*} \leq C_i \gamma_i^{(k-R_i(\Sigma))_+} \|\Sigma\|_{\mathfrak{S}_i^\varepsilon}, \quad (113)$$

with measurable recovery mark and constants uniform in s, k , the marked occurrence, Fourier mode, polynomial, and final observable. Thus bounded pushforward/pullback is an explicit falsifiable part of the interface; it is not inferred merely from the word “completion.”

Proof. For every positive decomposition, total variation is at most $\sum_j |b_j| M_j$, which is bounded by the displayed source cost. Taking the infimum proves separation and permits completion. The operator assertion is precisely the typed propagation estimate that must be proved for the moving atlas; once it holds on the dense finite representatives, it extends uniquely to the completion. $\square$

Definition 6.33 (Measurable constant-polarity decomposition interface MPD). *Every signed source admitted to a regular source row is required to be represented on a countable standard-Borel component extension $B_{\lambda,a} \subset \mathbb{N}$ by*

$$\Sigma_{\lambda,a} = \int_{B_{\lambda,a}} s(b) m_{\lambda,a}(db) \nu_{\lambda,a,b}, \quad s(b) \in \{+1, -1\}, \quad \nu_{\lambda,a,b} \in \mathcal{P}(X_{\lambda,a,b}). \quad (114)$$

Here $m_{\lambda,a}$ is an atomic countable kernel, the component graph and weights are jointly measurable, each $\nu_{\lambda,a,b}$ is a positive normalized regular section, and $s(b)$ is constant on its entire carrier. The component index b is absorbed into the fixed countable occurrence/component mark below. In particular, no carrier-dependent polar multiplier is applied after the positive section norm has been evaluated.

Joint Borel measurability of the component graph, atomic kernel, masses, normalized sections, source norms, recovery marks, and operator marks is an explicit part of MPD; it is not inferred from fibrewise Jordan decomposition. An essentially uncountable decomposition is not admitted without a separately typed kernel interface. If only an abstract Jordan decomposition is available and either positive part falls outside the declared regular cone, the source is routed to a typed TV or cemetery row.

Lemma 6.34 (Two-component baseline cone split). *There is a directly checkable two-component construction for a $C^{\alpha*}$ signed density f on one homogeneous carrier, with respect to its normalized arclength. Choose*

$$c \geq 2\|f\|_\infty + [f]_{\alpha*}, \quad f = (c + f) - c. \quad (115)$$

Both summands are positive, the first is bounded below by $c/2$, and their normalized densities have a uniform log-Hölder constant. Their two positive masses, source norms, chart/boundary costs, recovery times, and operator costs are charged separately to the envelope kernel. The same construction applies chartwise to a regular flux density whenever the chosen atlas and all normalization data are Borel in the external labels.

Proof. For the displayed construction, $c + f \geq c/2$ and

$$|\log(c + f(x)) - \log(c + f(y))| \leq 2[f]_{\alpha*} c^{-1} d(x, y)^{\alpha*}.$$

The constant component has zero log-Hölder seminorm. Normalize the two positive components and multiply their exact masses back. This proves the claimed cone control without cutting at the zero set of f . It does not prove the Borel selection required by MPD for an arbitrary moving family. $\square$

Definition 6.35 (Cemetery source kernels CEM.DOM). *Let*

$$\mathcal{I}_{\text{cem}} := \{\text{ctx}, \text{stop}, \text{buf}, \text{gate}, \text{cont}\}.$$

For $i \in \mathcal{I}_{\text{cem}}$ let τ_i be the first failure stage. Fix a parameter-independent Banach source/test pair $\mathfrak{S}_i^{\varepsilon_i}, \mathfrak{T}_i$ and a standard-Borel carrier X_i . On B_i require two measurable scalar masses $0 \leq m_i(\lambda) \leq q_i(\lambda) < \infty$ and a weakly measurable probability section $\nu_{i,\lambda} \in \mathcal{P}(X_i)$. Here m_i is the exact positive coefficient/probability mass of that stopped component, while q_i is only its source-norm envelope. Put $K_i(\lambda) := \|\nu_{i,\lambda}\|_{\mathfrak{S}_i^{\varepsilon_i}}$; in general $K_i > 1$. By Definition 6.33, the label space has already been extended to a positive component and carries a measurable constant sign $s_i(\lambda) \in \{+1, -1\}$. Let $C_{U,i}(\lambda) \geq 1$ dominate the remaining typed operator on that component. The exact discarded current is defined by its scalar pairing

$$J_i^{\text{disc}}(\Phi) := \int_{B_i} s_i(\lambda) m_i(\lambda) \langle U_{i,\lambda} \nu_{i,\lambda}, \Phi \rangle d\Pi_{s,N}(\lambda), \quad s_i(\lambda) \in \{+1, -1\}. \quad (116)$$

The carrier is integrated inside the typed source/operator pairing before the label integral is taken. In particular, (116) does not assert absolute integrability of an arbitrary pointwise representative $U_{i,\lambda}^ \Phi$ against $\nu_{i,\lambda}$. Thus m_i is a density on the tree-label space while $\nu_{i,\lambda}$ is the normalized shape on its physical carrier; no Radon-Nikodym identity equates measures living on these different spaces. Labelwise the honest source-norm/recovery envelope is*

$$q_i(\lambda) \geq m_i(\lambda) K_i(\lambda) \gamma_i^{-R_i(\lambda)} \mathcal{A}_i^{\text{opp}}(\lambda) C_{U,i}(\lambda). \quad (117)$$

Let R_i be the recovery time and $\mathcal{A}_i^{\text{opp}} \geq 1$ the already frozen opposite-side amplitude. There must be a predictable cemetery envelope, fixed before m, n , the mode, and the final observable, such that

$$q_i(\lambda) \leq C \mathcal{W}_i^{\text{cem}}(\lambda) \quad \Pi_{s,N}\text{-a.e. on } B_i. \quad (118)$$

Weak measurability means that $\lambda \mapsto \langle U_{i,\lambda} \nu_{i,\lambda}, \Phi \rangle$ is measurable for every $\Phi \in \mathfrak{T}_i$; this is part of the interface. The source-operator bound, followed by the scalar triangle inequality on the label space, gives

$$|J_i^{\text{disc}}(\Phi)| \leq C \|\Phi\|_{\mathfrak{T}_i} \int_{B_i} \mathcal{W}_i^{\text{cem}} d\Pi_{s,N} = C \|\Phi\|_{\mathfrak{T}_i} \mathbb{E}_{\Pi_{s,N}} \mathcal{W}_* \Pi_{s,N}^W(B_i). \quad (119)$$

A future factor may be replaced by one only when the remaining propagator is a positive total-variation contraction. Otherwise its recovery and opposite-amplitude costs are frozen in q_i before the cemetery transition. The fixed owner table is: context and physical-gate failures use oriented flux carriers; stopping, buffer, and continuation failures use measured standard-curve carriers. Each row records its sign, forward/reverse orientation, $m_i, q_i, \nu_{i,\lambda}, K_i, R_i, U_i$, and opposite-side amplitude. Any row not admitting a standard-family or flux representation uses a registered SOURCE_PROP_i Banach pair, but it must still admit the displayed positive CEM factorization. A row admitting no such factorization is a no-go, not a separately floating current. Its contribution is not allowed to live in a parallel ledger: the Borel embedding into the global cemetery owner, including equality of the exact mass, envelope, section, polarity, type, recovery, and operator marks, is required in (144) below.

Lemma 6.36 (Fixed-norm stopped source and summable propagation). *For each $i \in \mathcal{I}_{\text{cem}}$ assume that $U_{i,s}^{k,\varepsilon_i} : \mathfrak{S}_i^{\varepsilon_i} \rightarrow \mathfrak{T}_i^*$ is measurable in the preceding sense and, uniformly in the moving-table parameter, common atlas, time, and orientation,*

$$|\langle U_{i,s}^{k,\varepsilon_i} \Sigma, \phi \rangle| \leq C \|\Sigma\|_{\mathfrak{S}_i^{\varepsilon_i}} \gamma_i^{(k-R_i)_+} \|\phi\|_{\mathfrak{T}_i}, \quad 0 < \gamma_i < 1. \quad (120)$$

Then (117)–(118) imply (119), uniformly in m, n, s , mode, and test. Normalize the positive shape, propagate it, and multiply the exact mass back; domination uses the distinct norm envelope q_i . No signed cancellation or cross-space Radon–Nikodym formula is used, and no pointwise carrier Tonelli estimate is claimed. The results of [21, 22] are templates for this source-first organization, not proofs for moving billiards.

Definition 6.37 (Post-propagation carrier domination POST_CARRIER_DOM). *The operator norm in (113) controls an occurrence after its carrier has been integrated; it does not by itself license restriction to a phase-bad subset of that carrier. Accordingly, every occurrence declared to have passage type J must carry a jointly measurable, predictable mark $C_{\text{car}}(\lambda, a) \geq 1$ and, for every $\Phi_a \in \mathfrak{T}_a$, a canonical measurable representative u_{λ,a,Φ_a} , linear in Φ_a , such that*

$$\Lambda_{\lambda,a,\Phi_a}(E) := \int_E u_{\lambda,a,\Phi_a}(x) d\nu_{\lambda,a}(x), \quad (121)$$

$$\|u_{\lambda,a,\Phi_a}\|_{L^\infty(\nu_{\lambda,a})} + [u_{\lambda,a,\Phi_a}]_{\alpha^*, \text{sc}} \leq C_U(\lambda, a) C_{\text{car}}(\lambda, a) K_{\lambda,a} \|\Phi_a\|_{\mathfrak{T}_a}. \quad (122)$$

Here $E \mapsto \Lambda_{\lambda,a,\Phi_a}(E)$ is the declared finite signed post-propagation carrier measure and

$$\Lambda_{\lambda,a,\Phi_a}(X_{\text{type}(a)}) = \langle U_{\lambda,a} \nu_{\lambda,a}, \Phi_a \rangle. \quad (123)$$

When $\mathbf{1}_E \nu_{\lambda,a}$ belongs to the original source space, this measure agrees with

$$\langle U_{\lambda,a}(\mathbf{1}_E \nu_{\lambda,a}), \Phi_a \rangle.$$

For an arbitrary Borel E the latter is not used as an ill-typed source-space expression. Here the scaled seminorm is the one required by the later fractional nonstationary-phase row; when only measurable restrictions are used its seminorm term may be omitted explicitly. In either case

$$\int_E |u_{\lambda,a,\Phi_a}| d\nu_{\lambda,a} \leq C_U C_{\text{car}} K_{\lambda,a} \nu_{\lambda,a}(E) \|\Phi_a\|_{\mathfrak{T}_a}. \quad (124)$$

The factor C_{car} is fixed before (m, n, ξ, P) and the final test and is absorbed into the propagated q envelope. A row for which (121)–(122) is unavailable is assigned passage type TV and is never cut in the carrier coordinate. Thus post-propagation cancellation cannot be hidden behind a pre-propagation constant-polarity decomposition.

Definition 6.38 (Propagated dual-mass marked source kernel). *Together with MPD, the interface DUAL_MASS_KERNEL first fixes the owner and type registries*

$$\mathcal{O}_{\text{src}} := \{\text{cem}, \text{main}, \text{exc}_J, \text{exc}_{\text{TV}}, \text{word}\},$$

and

$$\mathcal{T}_{\text{src}} := \{\text{sf}^+, \text{sf}^-, \text{flux}^+, \text{flux}^-\} \sqcup \{(i, \varepsilon) : i \in \mathcal{I}_{\text{prop}}^{\text{extra}}, \varepsilon \in \{+, -\}\}.$$

Here $\mathcal{I}_{\text{prop}}^{\text{extra}}$ is a fixed countable registry of every additional SOURCE_PROP_i Banach source/test pair admitted by the theorem. Thus an extra typed row is part of the common kernel rather than an unnamed functional outside it. The standard-Borel mark space is

$$Y_{\text{src}} := \mathcal{O}_{\text{src}} \times \mathbb{N} \times \{r, \text{PPE}, N_{\text{time}}, \text{word}\} \times \mathbb{N}_0 \times \{J, \text{TV}\} \\ \times \{+, -\}_{\text{pol}} \times \{+, -\}_{\text{or}} \times \{-, +, \text{both}, \text{none}\}_{\text{rec}} \times \mathcal{T}_{\text{src}}.$$

The first integer is the occurrence/component index supplied by the common refinement. The second is the numerical word depth ℓ_{word} ; it is set to zero when that coordinate is unused, while the clock field distinguishes an actual depth-zero word row. Let X_{src} be the corresponding countable disjoint union of the standard-family, flux, and registered extra typed carriers.

Before any tree truncation, resolving scale, product time, mode, polynomial, or final test is chosen, the interface fixes one standard-Borel complete tree $(\Lambda_\infty, \Pi_s^\infty)$ and one datum-independent positive incidence law

$$d\overline{M}_s^{q,\infty}(\omega, a) := d\Pi_s^\infty(\omega)q_\omega(da) \quad \text{on a fixed Borel graph } I_\infty \subset \Lambda_\infty \times Y_{\text{src}}. \quad (125)$$

For every displayed N_{tree} there are Borel truncations $p_{N_{\text{tree}}}^{\text{tree}} : \Lambda_\infty \rightarrow \Lambda_{N_{\text{tree}}}$ and record-preserving $p_{N_{\text{tree}}}^{\text{inc}} : I_\infty \rightarrow I_{N_{\text{tree}}}$ such that

$$(p_{N_{\text{tree}}}^{\text{tree}})_* \Pi_s^\infty = \Pi_{s, N_{\text{tree}}}, \quad \overline{M}_{s, N_{\text{tree}}}^{q,0} := (p_{N_{\text{tree}}}^{\text{inc}})_* \overline{M}_s^{q,\infty}, \quad (126) \\ \text{pr}_{\Lambda_N} \circ p_N^{\text{inc}} = p_N^{\text{tree}} \circ \text{pr}_{\Lambda_\infty}, \quad p_{N, N'}^\bullet \circ p_{N'}^\bullet = p_N^\bullet \quad (N \leq N', \bullet \in \{\text{tree}, \text{inc}\}).$$

The occurrence index is retained by these maps, so distinct physical occurrences are not coalesced. All source filters and currents are defined on I_∞ ; a finite-tree display is only the pushforward of a finite-prefix-measurable restriction. Thus changing N_{tree} cannot select a new q . Below we write $I := I_\infty$ and $\lambda := \omega$ on the complete tree unless a finite marginal is explicitly displayed. Its immutable fields include

$$m, q, \nu, U, C_U, C_{\text{car}}, o, j, \kappa, \pi, s_{\text{pol}}, \ell_{\text{word}}, \varepsilon_{\text{or}}, h_{\text{rec}}, \\ \mathcal{P}_{\text{rec}}, \text{type}, R, \gamma, \mathcal{A}^{\text{opp}}.$$

In particular, none is allowed to depend on a resolution query or on a test.

A master query is only an external index

$$\varrho = (r, N_{\text{PPE}}, N_{\text{time}}, m, n, L, \xi, P), \quad \mathfrak{d} = (N_{\text{tree}}, \varrho) \in \mathfrak{D}.$$

The final test is another external index and is never a coordinate of the incidence kernel. The admissible linked queries form a Borel graph $\Gamma \subset \mathfrak{D}$. On every section used in the product-time CM2 argument the time clock is not an independent coordinate: throughout this section and every downstream central-phase call we impose the typed identity

$$N_{\text{time}} = N_{m,n} := \max\{m, n\}. \quad (127)$$

Likewise $N_{\text{PPE}} = N_{\text{PPE}}(r)$, the resolving cutoff L , and every scale r subsequently chosen from (m, n, ξ, P) are Borel maps on Γ , not parameters of the raw kernel. An auxiliary construction may temporarily carry a free N_{time} , but it is not an admissible central query until restricted to (127). Thus no time-tail estimate is silently evaluated at a different depth from the product time. For every $(N_{\text{tree}}, \varrho) \in \Gamma$ put, on the complete tree,

$$\Pi_{s, \mathfrak{d}} := \Pi_s^\infty, \quad \overline{M}_{s, \mathfrak{d}}^q := \overline{M}_s^{q, \infty}.$$

Every *CLOCK*, *main*, *FACE*, *shallow*, and *deep cell* below is a Borel restriction or push-forward of this same base measure. Its indicator may depend on the full linked query ϱ and on the final test, but the measure and its record fields may not. All estimates are uniform in those external indices. When only a scalar tree or resolving depth is displayed, the older notation $\Pi_{s, N}$ and $\overline{M}_{s, N}^q$ abbreviates the consistent marginal (126); it never licenses a separately chosen resolution law.

More explicitly, for any two admissible data $(N_{\text{tree}}, \varrho)$ and $(N'_{\text{tree}}, \varrho')$, including pairs for which changing (m, n, ξ, P) also changes $(r, N_{\text{PPE}}, N_{\text{time}}, L)$, and for any two final tests, the unfiltered complete-tree incidence laws are both exactly (125), and the fields

$$(m, q, \nu, U, C_U, C_{\text{car}}, o, j, \kappa, \pi, s_{\text{pol}}, \ell_{\text{word}}, \varepsilon_{\text{or}}, h_{\text{rec}}, \mathcal{P}_{\text{rec}}, \text{type}, R, \gamma, \mathcal{A}^{\text{opp}}) \quad (128)$$

agree identically on I_∞ . Only Borel clock, phase, and owner-handler filters of this fixed record may depend on the query or test. Equivalently, every displayed $\overline{M}_{s, \mathfrak{d}}^q$ is the same simultaneous positive incidence measure before its displayed filter is applied. A family such as $q_{N_{\text{time}}}$, q_r , or q_ξ chosen separately on admissible sections is not an admissible *DUALMASSKERNEL*, even if all of its sectionwise bounds are uniform.

There are jointly measurable weights $w^m(\lambda, a), w^q(\lambda, a) \geq 0$ on the fixed marked product, supported on the same Borel countable-fibre set $I \subset \Lambda \times Y_{\text{src}}$, with $w^m \leq w^q$, and

$$\begin{aligned} m_\lambda(da) &:= \sum_{a \in I(\lambda)} w^m(\lambda, a) \delta_a(da), \\ q_\lambda(da) &:= \sum_{a \in I(\lambda)} w^q(\lambda, a) \delta_a(da), \end{aligned} \quad \sum_{a \in I(\lambda)} w^q(\lambda, a) < \infty. \quad (129)$$

The support I is restricted to the compatibility diagonal on which $\text{type}(a) = \text{sf}^{\varepsilon_{\text{or}}(a)}$, $\text{flux}^{\varepsilon_{\text{or}}(a)}$, or $(i, \varepsilon_{\text{or}}(a))$ for a registered extra type i . Polarity $s_{\text{pol}}(a)$ is an independent field and imposes no orientation change. The fields $\kappa(a)$ and $\pi(a)$ are the immutable active clock and passage records, not preliminary classifications. If a construction must reroute an occurrence because its J structure is unavailable, it must change the record to $\pi(a) = \text{TV}$ before the base kernel and q are fixed. Conversely, promoting a preliminary *TV* row to a J handler requires recording $\pi(a) = J$ and charging C_{car} in q before any filter is selected. No post-freeze filter may override either field. Thus $m_\lambda \ll q_\lambda$ are finite positive kernels on the fixed standard-Borel space. Write $g_\lambda = dm_\lambda/dq_\lambda$, so $0 \leq g_\lambda \leq 1$. The kernel m_λ records the exact positive coefficient mass in the chosen constant-polarity decomposition; it is not the total variation of the recombined signed current. The kernel q_λ records the larger propagated source-norm envelope. For every Borel A , both kernel evaluations are measurable in λ . This is the measurable weighted-counting construction, rather than an unmeasurable variable-fibre sum; standard kernel composition and Tonelli are used in the sense of [24]. A countable registry may first be s -finite, but every cell used by the central Jensen argument must have finite envelope mass. For q_λ -a.e. mark a , let $\nu_{\lambda, a} \in \mathcal{P}(X_{\text{type}(a)})$ be a normalized weakly measurable source section and put $K_{\lambda, a} := \|\nu_{\lambda, a}\|_{\mathfrak{S}_a^{\varepsilon_{\text{or}}(a)}}$. The section is positive and

its polarity is the occurrence-level mark $s_{\text{pol}}(a) \in \{+1, -1\}$ supplied by Definition 6.33; $s_{\text{pol}}(a)$ is never a function of the carrier coordinate. The record

$$(j(a), o(a), \text{type}(a), \varepsilon_{\text{or}}(a), s_{\text{pol}}(a), h_{\text{rec}}(a), \ell_{\text{word}}(a), \kappa(a), \pi(a), \nu_{\lambda,a}, U_{\lambda,a}, K_{\lambda,a}, R_{\lambda,a}, \gamma_a, C_U(\lambda, a), C_{\text{car}}(\lambda, a), \mathcal{A}_{\lambda,a}^{\text{opp}}, \mathcal{P}_{\text{rec}}(\lambda, a)) \quad (130)$$

is encoded by the mark and the measurable operator kernel. Distinct physical currents have distinct occurrence indices even when their supports coincide.

For a face-time eligible occurrence, the last field is a finite measurable family of alternative factorizations of that same occurrence:

$$\mathcal{P}_{\text{rec}}^{\mathfrak{o}}(\lambda, a) = (\text{type}_{\text{rec}}^{\mathfrak{o}}, \nu_{\text{rec}}^{\mathfrak{o}}, U_{\text{rec}}^{\mathfrak{o}}, K_{\text{rec}}^{\mathfrak{o}}, R_{\text{rec}}^{\mathfrak{o}}, \gamma_{\text{rec}}^{\mathfrak{o}}, C_{U,\text{rec}}^{\mathfrak{o}}, C_{\text{car},\text{rec}}^{\mathfrak{o}}, \mathcal{A}_{\text{rec},\mathfrak{o}}^{\text{opp}}, \mathcal{E}_{\text{rec}}^{\mathfrak{o}}), \quad \mathfrak{o} \in \text{Acc}(a). \quad (131)$$

For every recorded alternative,

$$R_{\text{rec}}^{\mathfrak{o}} \in \mathbb{N}_0, \quad 0 < \gamma_{\text{rec}}^{\mathfrak{o}} < 1. \quad (132)$$

The recovery type $\text{type}_{\text{rec}}^{\mathfrak{o}}(a)$ is an immutable mark-only field, so its source/test Banach pair is independent of λ whenever the same mark occurs in two tree fibres. The typed source in this tuple has orientation $\mathfrak{o}$, is positive and normalized, and carries the same polarity $s_{\text{pol}}(a)$. The auxiliary bilinear lift in the tuple is an immutable pre-query rule $(\mathfrak{d}, f, g) \mapsto \mathcal{E}_{\mathfrak{d},\text{rec}}^{\mathfrak{o}}(f, g)$, rather than a query-evaluated field that could change the source or operator record, and satisfies $\|\mathcal{E}_{\text{rec}}^{\mathfrak{o}}(f, g)\| \leq C\|f\|_{C^r}\|g\|_{C^r}$ uniformly. It represents the exact same physical scalar occurrence:

$$\langle U_{\text{rec}}^{\mathfrak{o}} \nu_{\text{rec}}^{\mathfrak{o}}, \mathcal{E}_{\mathfrak{d},\text{rec}}^{\mathfrak{o}}(f, g) \rangle = \langle U_{\lambda,a} \nu_{\lambda,a}, \mathcal{E}_{\mathfrak{d},a}^{\text{pri}}(f, g) \rangle. \quad (133)$$

It is never added to the primary representation. The query may select one available tuple for a signed recovery estimate, while m and q are each counted exactly once. Predictability is paid by the additional envelope condition

$$w^q(\lambda, a) \geq w^m(\lambda, a) \max_{\mathfrak{o} \in \text{Acc}(a)} \left\{ K_{\text{rec}}^{\mathfrak{o}} (\gamma_{\text{rec}}^{\mathfrak{o}})^{-R_{\text{rec}}^{\mathfrak{o}}} \times \max\{1, \mathcal{A}_{\text{rec},\mathfrak{o}}^{\text{opp}}\} C_{U,\text{rec}}^{\mathfrak{o}} \max\{1, C_{\text{car},\text{rec}}^{\mathfrak{o}}\} \right\}. \quad (134)$$

The maximum over an empty access set is zero. The whole payload is part of the record matched by PROP_Q_MATCH; two separately weighted copies of one physical face are not an admissible substitute.

To make these alternatives test-side typed, let $\mathfrak{T}_a^{\text{pri}}$ denote the previously declared local test space and put

$$\mathfrak{T}_a^{\text{all}} := \mathfrak{T}_a^{\text{pri}} \oplus_{\ell^\infty} \bigoplus_{\mathfrak{o} \in \text{Acc}(a)}^{\ell^\infty} \mathfrak{T}_{a,\text{rec}}^{\mathfrak{o}}. \quad (135)$$

The primary operator acts on the primary projection; an alternative operator acts only on its named recovery projection. The physical-test lift supplies all of these coordinates simultaneously with the same uniform product bound. Thus choosing a recovery representation cannot introduce an unrecorded test norm or a time-dependent lift constant.

To type the global scalar assembly, use the fixed Banach product

$$\mathfrak{T}_{\text{src}} := \prod_{a \in Y_{\text{src}}}^{\ell^\infty} \mathfrak{T}_a^{\text{all}}, \quad \|\Phi\|_{\mathfrak{T}_{\text{src}}} := \sup_a \|\Phi_a\|_{\mathfrak{T}_a^{\text{all}}}. \quad (136)$$

Compatibility with physical observables is quantitative. The interface `PHYS_TEST_LIFT` supplies, before source assembly, one bounded bilinear map for every admissible master datum (equivalently, one bounded linear map on the projective tensor product),

$$\mathcal{E}_\mathfrak{d} : C^r(M) \times C^r(M) \longrightarrow \mathfrak{T}_{\text{src}}, \quad \|\mathcal{E}_\mathfrak{d}(f, g)\|_{\mathfrak{T}_{\text{src}}} \leq C\|f\|_{C^r}\|g\|_{C^r}, \quad (137)$$

uniformly in (s, m, n, ξ, P) . Its coordinates are the actual tests used by the physical bilinear response term. Joint linearity on the Cartesian product is not meant: together with the displayed product bound it would force the map to vanish. Any time- or mode-dependent loss in a concrete realization must be bounded predictably before those choices and absorbed into C_U (hence into q); it cannot be hidden in the lift norm. The lift also closes the physical/source diagram, not merely the norms:

$$B_\mathfrak{d}^{\text{phys}}(f, g) = \langle J_\mathfrak{d}^{\text{src}}, \mathcal{E}_\mathfrak{d}(f, g) \rangle, \quad (138)$$

where the left side is the exact physical occurrence sum before any absolute value and the right side is the grouped source current (143). This identity is required for every admissible master datum and is compatible with the exact current clause of `PROP_Q_MATCH`. Closedness of a putative compatible subspace without (137) is insufficient. The primary $U_{\lambda,a}^*$ acts on $\text{pr}_{\text{pri}} \Phi_a$; an alternative recovery operator acts on its named recovery projection. Below these projections are suppressed only when the active representation is unambiguous. Hence heterogeneous standard-family and flux rows are not silently treated as elements of one undeclared local test space.

The positive envelope measure is

$$dM_{\text{src}}^q(\lambda, a, x) := d\Pi_{s,\mathfrak{d}}(\lambda) q_\lambda(da) d\nu_{\lambda,a}(x). \quad (139)$$

This positive measure is used only for carrier-event energies, measurable coverage, and positive majorants. It does not define the signed current; that current is the grouped occurrencewise pairing (143) below. The two atomic masses obey the pointwise domination with $0 < \gamma_a < 1$, $R_{\lambda,a} \in \mathbb{N}_0$, and

$$\overline{\mathcal{A}}_{\lambda,a}^{\text{opp}} := \max\{1, \mathcal{A}_{\lambda,a}^{\text{opp}}\} : \quad (140)$$

$$w^q(\lambda, a) \geq w^m(\lambda, a) K_{\lambda,a} \gamma_a^{-R_{\lambda,a}} \overline{\mathcal{A}}_{\lambda,a}^{\text{opp}} C_{\text{prop}}(\lambda, a), \quad (141)$$

for every active occurrence, where

$$C_{\text{prop}}(\lambda, a) := C_U(\lambda, a) \max\{1, \mathbf{1}_{\{\pi(a)=J\}} C_{\text{car}}(\lambda, a)\}, \quad (142)$$

$C_U(\lambda, a) \geq 1$ is the measurable operator mark satisfying $\|U_{\lambda,a}\|_{\mathfrak{S}_a \rightarrow (\mathfrak{T}_a^{\text{pri}})^*} \leq C_U(\lambda, a)$, and every J row satisfies `POST_CARRIER_DOM`. Thus operator growth, post-propagation carrier restriction, recovery, and the frozen opposite amplitude are absorbed into the same propagated envelope q used by every source ledger. Because every active J handler is required below to lie in $\{\pi(a) = J\}$, no carrier-restriction cost can be removed by a handler relabelling after this envelope is fixed. For every admissible test define the exact assembled current by first evaluating each occurrence in its typed source/test dual and only then integrating the scalar result:

$$\langle J_{\text{src}}, \Phi \rangle := \int s_{\text{pol}}(a) g_\lambda(a) \langle U_{\lambda,a} \nu_{\lambda,a}, \text{pr}_{\text{pri}} \Phi_a \rangle d\overline{M}_{s,\mathfrak{d}}^q(\lambda, a). \quad (143)$$

The grouped scalar integrand is required to be jointly measurable. Formula (143) is not shorthand for a Tonelli integral of $[U^* \Phi](x)$ over the carrier. Carrier-restricted J pieces

are defined by the signed scalar measure (121); TV pieces are never cut in the carrier coordinate.

The local cemetery kernels are identified with, rather than merely dominated by, the cemetery restriction of this global kernel. Namely there are Borel mark maps $a_i : B_i \rightarrow Y_{\text{src}}$ for which $\lambda \mapsto (\lambda, a_i(\lambda))$ are disjoint injections into I_{cem} and whose graph images exhaust I_{cem} modulo $\overline{M}_{s,0}^q$ -null sets and for which

$$\begin{aligned} w^m(\lambda, a_i(\lambda)) &= m_i(\lambda), & w^q(\lambda, a_i(\lambda)) &= q_i(\lambda), \\ (\nu, s_{\text{pol}}, U, K, R, \gamma, \mathcal{A}^{\text{opp}}, C_U, C_{\text{car}})(\lambda, a_i(\lambda)) &= (\nu_i, s_i, U_i, K_i, R_i, \gamma_i, \mathcal{A}_i^{\text{opp}}, C_{U,i}, 1)(\lambda). \end{aligned} \quad (144)$$

For these cemetery embeddings set $h_{\text{rec}}(\lambda, a_i(\lambda)) = \text{none}$ and $\mathcal{P}_{\text{rec}}(\lambda, a_i(\lambda)) = \emptyset$; they are never routed through a recovered face-time cell. The type and orientation fields match the declared Banach pair as well. Consequently $J_{\text{cem}} = \sum_i J_i^{\text{disc}}$ as an absolutely convergent identity in the fixed test dual; there is no second cemetery current. The global finite-envelope closure is the explicit requirement

$$Q_{s,0}^{\text{prop}} := \int q_\lambda(Y_{\text{src}}) d\Pi_{s,0}(\lambda) < \infty. \quad (145)$$

This is the fixed complete-tree total. Any depth dependence belongs only to its record-compatible restrictions or finite marginals and is charged in the BASE, CLOCK, FACE, and deep ledgers. There is no second fixed-depth-only operator integral whose exponent can disappear from the final minimum. Fibrewise finiteness alone is not enough. Measurability of the grouped scalar pairing and (145) make (143) an absolutely convergent outer scalar integral defining one bounded functional on the fixed test space; no carrier-level Gelfand or Bochner integral is asserted. The outer scalar triangle inequality gives uniformly on its unit ball

$$|\langle J_{\text{src}}, \Phi \rangle| \leq Q_{s,0}^{\text{prop}} \|\Phi\|_{\mathfrak{T}_{\text{src}}}. \quad (146)$$

Finally there is a measurable canonical owner map and, modulo M_{src}^q -null sets on the incidence graph $I \subset \Lambda \times Y_{\text{src}}$, the exhaustive identity

$$I = I_{\text{cem}} \sqcup I_{\text{main}} \sqcup I_{\text{exc},J} \sqcup I_{\text{exc},\text{TV}} \sqcup I_{\text{word}}. \quad (147)$$

The static nonmain word owner is record-compatible:

$$I_{\text{word}} \subset \{a : \kappa(a) = \text{word}\}. \quad (148)$$

For every linked query ϱ with cutoff $L = L(\varrho)$, define the active restrictions of this one owner by the exact Borel identity

$$\begin{aligned} I_{\text{word}} &= I_{\text{face/word}}(\varrho) \sqcup I_{\text{deep}}(\varrho), \\ I_{\text{face/word}}(\varrho) &= I_{\text{word}} \cap \{\ell_{\text{word}} \leq L(\varrho)\}, \\ I_{\text{deep}}(\varrho) &= I_{\text{word}} \cap \{\ell_{\text{word}} > L(\varrho)\}. \end{aligned} \quad (149)$$

The corresponding grouped currents satisfy the absolutely convergent querywise identity

$$J_{\text{word}} = J_{\text{face/word}}(\varrho) + J_{\text{deep}}(\varrho). \quad (150)$$

Thus changing L changes only filters of the datum-independent raw kernel; it never changes a static owner or creates a new q . The clock r , PPE, N_{time} , or word is a measurable field/filter on an owner cell, not a second owner. This static identity does not by itself give resolution-dependent coverage inside I_{main} ; that separate job is performed by (188). This construction replaces every naked sum over continuous labels. Probability first-hit disjointization is used only for survivor unions and never deletes a marked occurrence.

Lemma 6.39 (Globally finite marked-current assembly). *Under (141) and (145), (143) belongs to the fixed test-space dual and obeys (146). Moreover, (147) together with (150) gives, for every linked query, the absolutely convergent dual identity*

$$J_{\text{src}} = J_{\text{cem}} + J_{\text{main}} + J_{\text{exc,J}} + J_{\text{exc,TV}} + J_{\text{face/word}} + J_{\text{deep}}. \quad (151)$$

Thus a local bound on every registered cell cannot leave an unowned current in the central or final assembly.

Proof. Use the exact identity $g_\lambda q_\lambda = m_\lambda$, evaluate the carrier inside the grouped dual pairing, use the operator bound on $m_\lambda \nu_{\lambda,a}$, and apply (141) to dominate it directly by the propagated envelope q_λ . Absolute scalar integration on the outer incidence space bounds the result by $Q_{s,\emptyset}^{\text{prop}}$ uniformly on the test unit ball. Restrict the same positive envelope measure to the five static owner cells, then split the word owner by (149), and apply countable additivity. Absolute convergence in the dual follows from the same common majorant, proving (151) without signed cancellation. $\square$

Remark 6.40 (Operator absorption and rate bookkeeping). *All source rates below are rates of the propagated q , not of the bare physical mass. Thus a factor $C_U \lesssim r^{-a_U}$ (respectively $C_U \lesssim e^{a_U N}$) has already enlarged the declared base/likelihood growth and weakened every direct-TV rate by a_U . In the main selection route it enters the actual a_A^L and a_A^B in (210); in exceptional Jensen rows it enters the displayed BASE-ENERGY gap; and in the face, shallow, and deep routes it is contained respectively in A_{face} , A_{sh} , and A_{loss} . It is never subtracted a second time. Conversely, if an operator mark cannot be predicted before the time, mode, and final test are chosen, it is not a legal q mark and must remain a separately typed primitive. Fixed-depth finiteness alone supplies no rate. Every post-absorption gap is required to be strictly positive.*

Definition 6.41 (Complete tilted energy ledger TENERGY_W). *Let $B_{\text{ctx}}(r)$ be failure of the joint native context gate, $B_{\text{stop}}(r)$ failure of the actual stopping antichain, and let $B_{\text{buf}}(N)$, $B_{\text{gate}}(N)$, and $B_{\text{cont}}(N)$ be their named cemetery events. On a retained good label let $B_{\text{PPE}}(r, P)$ be*

$$|X_{\beta,Q}^{[N(r)]} - P| \leq Cr, \quad P \in \mathbb{R}_Z[z].$$

Condition TENERGY_W requires, uniformly in s , orientation, actual/split constituent, parent context, and admissible polynomial,

$$\Pi_{s,N}^W(B_{\text{ctx}}(r)) \leq Cr^{\Theta_{\text{ctx}}^W}, \quad \Pi_{s,N}^W(B_{\text{stop}}(r)) \leq Cr^{\Theta_{\text{st}}^W}, \quad (152)$$

$$\Pi_{s,N}^W(B_i(N)) \leq Ce^{-c_i^W N}, \quad i \in \{\text{buf, gate, cont}\}, \quad (153)$$

$$\Pi_{s,N}^W(\mathbf{G} \cap B_{\text{PPE}}(r, P)) \leq Cr^{\Theta_{\text{PPE}}^W}. \quad (154)$$

Every additional exceptional event must also satisfy the aggregate clock interfaces below. These estimates are parent-kernel statements: they are either independent primitives or consequences of uniform raw parent tails and KRH_p . A global complete-tree moment alone does not imply them. They concern the unselected physical law $\mathbf{R}_0$: “parent context” may freeze physical context, orientation, or the opposite endpoint, but never the later main selector A . Atomic tail bounds by themselves are not part of TENERGY_W . Nor does this ledger by itself identify the two-endpoint product law: an opposite-endpoint-dependent tilt is only a one-sided conditional estimate. The exact pair-law content is separately imposed in PAIR_TENERGY below.

Definition 6.42 (Probability clock and survivor ledger CLOCK_PROB_W). *The five cemetery failures are the named events in TENERGY_W and are not members of an additional registry. Every other bad event used only to control a union or survivor mass is given a first-failure stage and one clock in $\{r, \text{PPE}, N_{\text{time}}, \text{word}\}$. Inside a stage fix a deterministic order and put*

$$\widehat{B}_{\tau,j} := B_{\tau,j} \setminus \bigcup_{k < j} B_{\tau,k}. \quad (155)$$

This is solely a probability partition; it makes no assertion about how many additive currents coexist on an atom. With word-depth currents assigned to either $\text{FACE_TIME}_{\text{CM2}}$ or $\text{FACE_TIME}_{\text{REC}}$ for product time and $\text{FACE}_{2\text{CUT}}$ for the depth limit, define

$$\begin{aligned} \mathfrak{Z}_r^{\text{prob}}(r) &:= \sum_{j \in \mathcal{J}_r^{\text{prob}}} \Pi^W(\widehat{B}_{r,j}(r)), \\ \mathfrak{Z}_{\text{PPE}}^{\text{prob}}(N) &:= \sum_{j \in \mathcal{J}_{\text{PPE}}^{\text{prob}}} \Pi^W(\widehat{B}_{\text{PPE},j}(N)), \\ \mathfrak{Z}_{\text{time}}^{\text{prob}}(N) &:= \sum_{j \in \mathcal{J}_{\text{time}}^{\text{prob}}} \Pi^W(\widehat{B}_{\text{time},j}(N)), \end{aligned} \quad (156)$$

and require the aggregate tails

$$\mathfrak{Z}_r^{\text{prob}}(r) \leq Cr^{\Theta_r^{\text{prob}}}, \quad \mathfrak{Z}_{\text{PPE}}^{\text{prob}}(N) \leq Ce^{-c_{\text{PPE}}^{\text{prob}}N}, \quad \mathfrak{Z}_{\text{time}}^{\text{prob}}(N) \leq Ce^{-c_{\text{time}}^{\text{prob}}N}, \quad (157)$$

with strictly positive exponents for every nonempty class. These bounds, together with the five named TENERGY_W tails, serve only SURV_NORM . Passage through Jensen is irrelevant to this union ledger.

Definition 6.43 (Exceptional source clock–passage ledger CLOCK_SRC_W). *Put $M_* := \mathbb{E}_{\Pi_{s,N}} \mathcal{W}_*$. For $\kappa \in \{r, \text{PPE}, N_{\text{time}}\}$ and $\pi \in \{J, \text{TV}\}$, the interface first gives jointly Borel, pairwise disjoint non-main filters $\mathcal{A}_{\text{nm}}^{\kappa,\pi}(\mathfrak{d})$. Put $I_{\text{nm}}^0 := I_{\text{cem}} \sqcup I_{\text{exc},J} \sqcup I_{\text{exc},\text{TV}}$. Modulo one common $\overline{M}_{s,\mathfrak{d}}^q$ -null set, they satisfy*

$$\begin{aligned} \mathcal{A}_{\text{nm}}^{\kappa,\pi}(\mathfrak{d}) &\subset I_{\text{nm}}^0 \cap \{a : \kappa(a) = \kappa, \pi(a) = \pi\}, \\ \kappa &\in \{r, \text{PPE}, N_{\text{time}}\}, \quad \pi \in \{J, \text{TV}\}. \end{aligned} \quad (158)$$

In addition $I_{\text{exc},J} \subset \{\pi(a) = J\}$ and $I_{\text{exc},\text{TV}} \subset \{\pi(a) = \text{TV}\}$. These are typed subset identities, not a convention encoded only in the superscripts. The filters satisfy

$$I_{\text{cem}} \sqcup I_{\text{exc},J} \sqcup I_{\text{exc},\text{TV}} = \bigsqcup_{\substack{\kappa \in \{r, \text{PPE}, N_{\text{time}}\} \\ \pi \in \{J, \text{TV}\}}} \mathcal{A}_{\text{nm}}^{\kappa,\pi}(\mathfrak{d}). \quad (159)$$

Restriction of the one propagated envelope then gives the absolutely convergent current identity

$$J_{\text{cem}} + J_{\text{exc},J} + J_{\text{exc},\text{TV}} = \sum_{\kappa, \pi} J_{\text{nm}}^{\kappa,\pi}. \quad (160)$$

The filters and bounds are uniform in the full master datum $\mathfrak{d}$; hence no non-main clock row can be made vacuous by omitting its occurrences.

Put $\mathcal{A}_{\kappa,J} := \mathcal{A}_{\text{nm}}^{\kappa,J}$. After the main coverage below is fixed, let

$$\mathcal{A}_{\kappa,\text{TV}} := \mathcal{A}_{\text{nm}}^{\kappa,\text{TV}} \sqcup \mathcal{A}_{\text{main}}^{\kappa,\text{TV}}.$$

Thus the TV cell contains every clock-owned occurrence, including a main-owner occurrence, charged directly in total variation. Main J cells are excluded from $\mathcal{A}_{\kappa,J}$ and are treated by Theorem 6.55; the main TV filters in (188) are exactly the displayed restrictions.

For a Jensen mark let $S_{\lambda,a} : X_{\text{type}(a)} \rightarrow \mathbb{R}$ be its measurable slope/value coordinate and put $\mu_{\lambda,a} := (S_{\lambda,a})_* \nu_{\lambda,a}$. Whenever this row is used for balanced physical phase, this is not a free measurable coordinate: ACTUAL_SLOPE_BRIDGE identifies it pointwise with the logarithmic derivative ratio (90) on the same labelled carrier. For every complete Jensen filter used by a phase call, its resonant centre $z_{\xi,\kappa,\mathfrak{d}}$ is common to that filter; it does not vary with (λ, a) . Thus the supremum over z below controls the actual lifted carrier event. On the carrier lift of every non-main Jensen filter require the jointly Borel implication

$$\{(\lambda, a, x) \in \hat{\mathcal{A}}_{\text{nm}}^{\kappa,J} : |\psi'_{\xi,\lambda,a}(x)| \leq \Xi r\} \subset \{|S_{\lambda,a}(x) - z_{\xi,\kappa,\mathfrak{d}}| \leq Ch_{\kappa}(t)\}. \quad (161)$$

Here the hat denotes the full carrier lift, as in the main coverage below. A source filter may carry passage J only when every phase-good carrier is uniformly admissible for Lemma 6.63: $\psi_{\xi} \in C^{1+\alpha}$ with the declared derivative Hölder bound, the positive amplitude has the required scaled $\mathcal{A}^{\alpha}$ norm, and the physical endpoint ledger is the one in (241). It must also satisfy POST_CARRIER_DOM, so that both the phase-bad restriction and the resulting positive carrier majorant are defined after propagation and their cost is already in q . A cemetery or extra registered type for which this typed NSP structure is unavailable must be placed in a TV cell. That reassignment is performed at the immutable record level before (125) is fixed; it is not an after-the-fact change of the superscript in $\mathcal{A}_{\text{nm}}^{\kappa,\pi}$. Thus a merely Borel slope coordinate never licenses the phase-good estimate. Define the normalized base, fibre-energy, value, and direct-TV masses

$$\begin{aligned} \mathcal{B}_{\kappa,J}(t) &:= M_*^{-1} \int_{\mathcal{A}_{\kappa,J}(t)} q_{\lambda}(da) d\Pi(\lambda), \\ \mathcal{E}_{\kappa,J}(t) &:= M_*^{-1} \int_{\mathcal{A}_{\kappa,J}(t)} (\mu_{\lambda,a} \otimes \mu_{\lambda,a}) \{|u - v| \leq 2h_{\kappa}(t)\} q_{\lambda}(da) d\Pi(\lambda), \\ \mathcal{V}_{\kappa,J}(t) &:= M_*^{-1} \sup_z \int_{\mathcal{A}_{\kappa,J}(t)} \mu_{\lambda,a}([z - h_{\kappa}(t), z + h_{\kappa}(t)]) q_{\lambda}(da) d\Pi(\lambda), \\ \mathcal{D}_{\kappa,\text{TV}}(t) &:= M_*^{-1} \int_{\mathcal{A}_{\kappa,\text{TV}}(t)} q_{\lambda}(da) d\Pi(\lambda). \end{aligned} \quad (162)$$

Here $h_r(r) = r$. The PPE window is tied to the very scale at which the unselected pair energy is evaluated: with

$$r_N := e^{-N/C_{\text{buf}}}, \quad 0 < h_{\text{PPE}}(N) \leq C_{\text{win}} r_N, \quad N_{\text{PPE}}(r) = \lceil C_{\text{buf}} \log(1/r) \rceil. \quad (163)$$

The time window $h_{\text{time}}(N)$ is the declared window in the separate unselected time-pair estimate, and in a central call its depth is $N = N_{\text{time}} = \max\{m, n\}$ by (127). Writing a cell as a function only of $t = r$ or N means its section at fixed suppressed $\mathfrak{d}$; the sets, source kernels, slope maps, and all bounds are jointly measurable and uniform in the suppressed mode, polynomial, time, and depth coordinates.

For nonempty Jensen classes require BASE-ENERGY bounds

$$\begin{aligned} \mathcal{B}_{r,J}(r) &\leq Cr^{-a_{r,J}^B}, & \mathcal{E}_{r,J}(r) &\leq Cr^{\Theta_{r,J}^E}, \\ \mathcal{B}_{\text{PPE},J}(N) &\leq Ce^{a_{\text{PPE},J}^B N}, & \mathcal{E}_{\text{PPE},J}(N) &\leq Ce^{-c_{\text{PPE},J}^E N}, \\ \mathcal{B}_{\text{time},J}(N) &\leq Ce^{a_{\text{time},J}^B N}, & \mathcal{E}_{\text{time},J}(N) &\leq Ce^{-c_{\text{time},J}^E N}. \end{aligned} \quad (164)$$

For the TV column require directly

$$\mathcal{D}_{r,\text{TV}}(r) \leq Cr^{\Theta_{r,\text{TV}}^{\text{src}}}, \quad \mathcal{D}_{\text{PPE},\text{TV}}(N) \leq Ce^{-c_{\text{PPE},\text{TV}}^{\text{src}}N}, \quad \mathcal{D}_{\text{time},\text{TV}}(N) \leq Ce^{-c_{\text{time},\text{TV}}^{\text{src}}N}. \quad (165)$$

The derived exceptional value rates are

$$\Theta_{r,J}^{\text{exc, val}} := \frac{\Theta_{r,J}^E - a_{r,J}^B}{2}, \quad c_{\text{PPE},J}^{\text{exc, val}} := \frac{c_{\text{PPE},J}^E - a_{\text{PPE},J}^B}{2}, \quad c_{\text{time},J}^{\text{exc, val}} := \frac{c_{\text{time},J}^E - a_{\text{time},J}^B}{2}. \quad (166)$$

Every displayed value gap must be strictly positive. Equations (159)–(160), rather than prose alone, contain cemetery, direct-TV, and retained exceptional remainders. Context, native-stop, buffer, gate, continuation, and adaptive-alphabet sources lie in the appropriate Jensen cells; retained absolute PPE and contracted-coordinate remainders in their TV cells. The main retained Jensen source is not a primitive row. Word-depth sources use one of `FACE_TIME_CM2` and `FACE_TIME_REC` for product-time decay, `FACE2CUT` for growing-depth convergence, and the deep ledger for the omitted tail. An empty class has rate $+\infty$ and disappears from minima.

Source-incidence complexity is the growth of the integrated kernel mass. For example, an atomic raw energy rate c and occurrence growth a first give an aggregate energy rate $c - a$; if the base mass also grows like e^{aN} , the Jensen value rate is $(c - 2a)/2$. The TV column needs only $c > a$. Any nonpositive gap is a genuine no-go. A Hölder transfer is legal only under a normalized incidence-law L^p bound; a complete-tree branch moment is not a substitute.

Lemma 6.44 (One global incidence Cauchy). *For every finite marked Jensen cell, including the exceptional cells of Definition 6.43 and the main cells of Definition 6.52,*

$$\mathcal{V}_{\kappa,J}(t)^2 \leq \mathcal{B}_{\kappa,J}(t)\mathcal{E}_{\kappa,J}(t). \quad (167)$$

Consequently the rates in (166) are valid.

Proof. Let $\rho(d\lambda, da) = M_*^{-1} \mathbf{1}_{\mathcal{A}_{\kappa,J}(t)} q_\lambda(da) d\Pi(\lambda)$. For $I = [z - h_\kappa(t), z + h_\kappa(t)]$, Cauchy on the whole incidence measure gives

$$\left(\int \mu_{\lambda,a}(I) d\rho \right)^2 \leq \rho(1) \int \mu_{\lambda,a}(I)^2 d\rho \leq \mathcal{B}_{\kappa,J}(t)\mathcal{E}_{\kappa,J}(t).$$

Take the supremum in z . No square root is taken before occurrences are integrated, so occurrence multiplicity is paid exactly once through the base and energy masses. $\square$

Remark 6.45 (Global probability transfer only). *If the unweighted complete-tree probability partitions are already aggregated and*

$$\|\mathcal{W}_*\|_{L^{p_W}(\Pi_{s,N})} \leq C\mathbb{E}_{\Pi_{s,N}} \mathcal{W}_*, \quad p_W > 1, \quad (168)$$

then global Hölder transfers only the global `CLOCK_PROB_W` tails, with exponents divided by p'_W . It does not imply parentwise `TENERGY_W` or `USC_end^W`, and it does not control `CLOCK_SRC_W` without the separate incidence-law moment just described.

Lemma 6.46 (Homogeneity marks control physical endpoint scale). *Assume $\text{USC}_{\text{end}}^W(q_g^W)$, including the weighted excess tail (106), on the same complete tree law. On every retained labelled branch,*

$$\log^+ a_X \leq C_{\text{init}} + C_0 m + 2G_m^- + \log S_\lambda^-, \quad \log^+ a_Y \leq C_{\text{init}} + C_0 n + 2G_n^+ + \log S_\lambda^+. \quad (169)$$

Fix $\zeta_g > 0$ and put

$$C_{\text{scl}} := C_0 + 2(\gamma_g + \zeta_g), \quad c_{\text{scl}} := q_g^W \zeta_g. \quad (170)$$

For $N = \max\{m, n\}$ call a label scale-good when

$$\tilde{G}_m^- \leq \gamma_g m + \zeta_g N, \quad \tilde{G}_n^+ \leq \gamma_g n + \zeta_g N. \quad (171)$$

The scale-good survivor labels satisfy $a_X + a_Y \leq C e^{C_{\text{scl}} N}$, while the total marked mass of scale-bad survivor labels, with the full weight (105), is at most $C \mathbb{E}_{\text{stop}}[\mathcal{W}_{\lambda,*}] e^{-c_{\text{scl}} N}$. The latter labels may therefore be estimated by weighted total variation and the absolute Fourier coefficient sum, without invoking PPE_N .

Proof. For a sequential billiard, the homogeneous-branch derivative estimate is [1]

$$\frac{\|DT(x)v\|}{\|v\|} \leq \frac{C}{\cos \varphi(Tx)}, \quad \cos \varphi \asymp k^{-2} \quad \text{on } \mathbb{H}_k.$$

Multiplication along the dynamic part of the branch gives the $C_0 m + 2G_m^-$ and $C_0 n + 2G_n^+$ terms. The initial physical-face parametrization, same-branch section, scalar chart, and coarea/holonomy derivatives give the two explicit $\log S_\lambda^\pm$ terms; they are not consequences of compactness of the phase-space chart. Thus (169) is equivalent, up to C_{init} , to a bound by $2\tilde{G}_N^\pm$. Apply the *weighted* tail (106) at depths m, n with $u = \zeta_g N$ to the two orientations and use the union bound. If the parentwise sufficient route is used, Lemma 6.28 is applied before this split and the exponent is q_g/p'_W . This argument explains the geometric form of the interface; uniformity for the stopped actual/split weighted laws remains the explicit content of $\text{USC}_{\text{end}}^W$. In particular it is not asserted for arbitrary billiard invariant measures. $\square$

Lemma 6.47 (Tilted stopped gate, buffer, and native scale). *Assume $\text{SS}(d_Z)$, NST_{phys} , $\text{PPE}_N(d_Z)$, and TENERGY_W ; if the displayed tilted bounds are obtained from raw parentwise bounds rather than assumed directly, also assume the corresponding KRH_{p_W} interface. Let $\alpha_L < \alpha_B$ and put*

$$\sigma = (r/c_\ell)^{1/(1+\alpha_L)}, \quad \Theta_{\text{ctx}} = \frac{p(\alpha_B - \alpha_L)}{1 + \alpha_L}. \quad (172)$$

On the antichain $\text{NST}_{\text{phys}}(r)$ call a stopped context good when

$$H_\beta \sigma_Q^{1+\alpha_B} \leq r/10. \quad (173)$$

Use the prescribed depth $N = N(r)$ of (96). Then the error in (89) is at most $Cr/10$ on good contexts, while

$$\Pi_{s,N}^W \{H_\beta \sigma_Q^{1+\alpha_B} > r/10\} \leq Cr^{\Theta_{\text{ctx}}^W}. \quad (174)$$

The physical resolving time is $N(r) = O(|\log r|)$ and all buffer continuations are retained, so the refinement preserves conditional mass. There are named constants $c_{\text{buf}}^W, c_{\text{gate}}^W, c_{\text{cont}}^W > 0$ such that the buffer, gate, and exceptional-continuation cemetery tails under the same tilted tree are at most

$$C \left(e^{-c_{\text{buf}}^W N(r)} + e^{-c_{\text{gate}}^W N(r)} + e^{-c_{\text{cont}}^W N(r)} \right). \quad (175)$$

The exceptional native stopping mass under $\Pi_{s,N}^W$ is $O(r^{\Theta_{\text{st}}^W})$.

Proof. The first assertion is (173) together with $\kappa^{N(r)} \leq r/100$. The tilted context, stopping, and auxiliary claims are precisely (152)–(153). If they are derived from the raw NST_{phys} tails, Markov's inequality first gives the unweighted exponent Θ_{ctx} displayed above, and KRH_{p_W} then degrades it, and every named parentwise exponential rate, by $1/p'_W$. The global estimate (168) is insufficient for this conditional claim. No unweighted rate is retained after multiplication by $\mathcal{W}_*$. $\square$

Proposition 6.48 (Tilted PPE). *Assume $\text{PPE}_N(dz)$ and TENERGY_W . Uniformly over admissible context polynomials $P \in \mathbb{R}_{dz}[z]$, the retained part of the complete tree obeys*

$$\Pi_{s,N}^W(\mathbf{G} \cap \{|X_{\beta,Q}^{[N(r)]} - P| \leq Cr\}) \leq Cr^{\Theta_{\text{PPE}}^W}. \quad (176)$$

Equivalently, before normalization its full positive marked mass is at most

$$C \mathbb{E}_{\Pi_{s,N}} \mathcal{W}_* r^{\Theta_{\text{PPE}}^W}.$$

Proof. This is (154). The raw physical geometry is (PPE1)–(PPE3), but the weighted conclusion is supplied by TENERGY_W , either directly or through the parent-kernel transfer Lemma 6.28. It is not a deduction from the fixed clean subsystem. Lemma 6.24 explains why the deleted fixed-pressure trial argument could not provide this estimate. $\square$

Lemma 6.49 (Retained-tree normalization). *Assume*

$$\text{TENERGY}_W, \quad \text{CLOCK_PROB}_W.$$

Let the three active resolution coordinates satisfy $r \downarrow 0$, $N_{\text{PPE}} = N_{\text{PPE}}(r) \rightarrow \infty$, and either the time registry is empty or

$$\mathfrak{Z}_{\text{time}}^{\text{prob}}(N_{\text{time}}) \leq \frac{1}{12}. \quad (177)$$

In a product-time call this is eventually automatic from $N_{\text{time}} = \max\{m, n\} \rightarrow \infty$; it is not a consequence of shrinking r while holding N_{time} fixed. Let $\mathbf{G}$ be the retained-label event and set $M_{\mathbf{G}}^W := \Pi_{s,N}^W(\mathbf{G})$. For sufficiently small r , uniformly in the actual/split parent law,

$$\begin{aligned} M_{\mathbf{G}}^W &\geq 1 - \Pi^W(B_{\text{ctx}}) - \Pi^W(B_{\text{stop}}) - \sum_{i \in \{\text{buf}, \text{gate}, \text{cont}\}} \Pi^W(B_i) \\ &\quad - \mathfrak{Z}_r^{\text{prob}}(r) - \mathfrak{Z}_{\text{PPE}}^{\text{prob}}(N_{\text{PPE}}(r)) - \mathfrak{Z}_{\text{time}}^{\text{prob}}(N_{\text{time}}) \geq \frac{1}{2}. \end{aligned} \quad (178)$$

Consequently the conditional survivor law is $\Pi_{s,N}^W(\cdot \cap \mathbf{G})/M_{\mathbf{G}}^W$ and every transfer from the complete tree to that law costs at most a factor two. Equivalently, all energy estimates may be kept in subprobability form, in which case no division occurs.

Proof. Use (152), (153), and (157). The named failures and the r/PPE probability partition functions tend to zero at their declared scales; the time term is controlled by the separate hypothesis (177). Shrink only the coordinates that actually tend to their limiting regime so the total is at most $1/2$. No source incidence mass is subtracted here, and bad context/native stop are not charged a second time through an additional registry. $\square$

Definition 6.50 (Pair-safe reference). *Assume Definition 6.21 and the retained-tree normalization above. The law and energy clauses below are denoted, respectively,*

$$\text{PAIR_SAFE_REFERENCE}, \quad \text{PAIR_TENERGY}.$$

For each legitimate frozen physical parent, before any main-cell selection, let

$$R_0(d\eta, dx, dx') := K^{\text{prob}}(d\eta)\kappa_\eta(dx)\kappa_\eta(dx') := \sum_{\beta} w_{\beta}^W \delta_{\beta}(d\eta)\mu_{\beta}^W(dx)\mu_{\beta}^W(dx') \quad (179)$$

be the actual branch-mixture product law of that normalized unselected survivor. This identity is PAIR_SAFE_REFERENCE; there is no second reference law. Its slope is fixed by

$$S_{\eta}^0 = S_{\beta} = S_{\lambda, m, n, s} = S_{\lambda, a} \quad R_0\text{-a.e. on the corresponding labelled carrier}, \quad (180)$$

where the equalities include the carrier identifications in Definition 6.21. Put

$$D_h^0 := \{(\eta, x, x') : |S_{\eta}^0(x) - S_{\eta}^0(x')| \leq 2h\}. \quad (181)$$

With

$$\Theta_{\text{par}}^W := \min\{\Theta_{\text{PPE}}^W, \Theta_{\text{ctx}}^W, \Theta_{\text{st}}^W\} > 0, \quad (182)$$

$$\mathcal{T}_{\text{prob}}^W(h) := \sum_{i \in \{\text{buf}, \text{gate}, \text{cont}\}} \Pi^W(B_i(N_{\text{PPE}}(h))) + \mathfrak{Z}_r^{\text{prob}}(h) + \mathfrak{Z}_{\text{PPE}}^{\text{prob}}(N_{\text{PPE}}(h)), \quad (183)$$

PAIR_TENERGY requires the exact, parentwise bound

$$R_0(D_h^0) \leq Ch^{\Theta_{\text{par}}^W} + C\mathcal{T}_{\text{prob}}^W(h), \quad 0 < h < h_0. \quad (184)$$

It may be verified by the following factorized sufficient route: after all retained/cemetery decisions and positive tilts, the conditional carrier law must be the same κ_{η} for both endpoints and, uniformly in x' ,

$$\kappa_{\eta}\{x : |S_{\eta}^0(x) - S_{\eta}^0(x')| \leq 2h\} \leq Ch^{\Theta_{\text{par}}^W} + C\mathcal{T}_{\text{prob}}^W(h). \quad (185)$$

Then Tonelli under the genuine product law proves (184). A family of frozen-endpoint laws $\kappa_{\eta, x'}$ depending on x' does not prove this condition and may not be integrated as if it were $\kappa_{\eta} \otimes \kappa_{\eta}$; in that case (184) remains an independent primitive.

Theorem 6.51 (Complete tilted physical single-source energy tree). Assume $\text{SS}(dZ)$, NST_{phys} , $\text{PPE}_N(dZ)$, TENERGY_W , CLOCK_PROB_W , $\text{ACTUAL_SLOPE_BRIDGE}$, the dominant clean-gate chart normalization of Proposition 6.26, Lemma 6.49, and PAIR_SAFE_REFERENCE together with PAIR_TENERGY. Let w_{β}^W and μ_{β}^W be the branch disintegration in (179), and put $\nu_{\beta}^W = (S_{\beta})_{*}\mu_{\beta}^W$. Then one has

$$\sum_{\beta} w_{\beta}^W \iint \mathbf{1}_{\{|S_{\beta}(t) - S_{\beta}(t')| \leq 2r\}} d\mu_{\beta}^W(t) d\mu_{\beta}^W(t') \leq Cr^{\Theta_{\text{par}}^W} + C\mathcal{T}_{\text{prob}}^W(r), \quad (186)$$

$$\sup_z \sum_{\beta} w_{\beta}^W \nu_{\beta}^W([z - r, z + r]) \leq Cr^{\Theta_{\text{par}}^W/2} + C\mathcal{T}_{\text{prob}}^W(r)^{1/2}. \quad (187)$$

Proof. The first assertion is exactly (184) for the law (179); it is not obtained by recombining opposite-endpoint-dependent conditional laws. If the factorized route is available, integrate (185) first in x' against the same κ_{η} , then in η . Finally use

$$\nu_{\beta}^W(I)^2 \leq (\nu_{\beta}^W \otimes \nu_{\beta}^W)\{|u - v| \leq |I|\}$$

and Jensen in the positive weights w_{β}^W to obtain (187). The conditional normalization and every probability tail have already been paid in the exact law and $\mathcal{T}_{\text{prob}}^W$; no selected source law enters this theorem. $\square$

Definition 6.52 (Main clock coverage and BEV). *This definition states the resolution-exhaustive main-clock interface. Put*

$$I_{\text{main}} := I \cap \{o(a) = \text{main}\}, \quad \widehat{I}_{\text{main}} := \{(\lambda, a, x) : (\lambda, a) \in I_{\text{main}}, x \in X_{\text{type}(a)}\}.$$

For every admissible resolution/mode datum $\vartheta = (r, N_{\text{PPE}}(r), N_{\text{time}}, \xi, P)$, viewed as a section of the master datum $\mathfrak{d}$, there are jointly Borel, pairwise disjoint filters for which, modulo one common $\overline{M}_{s,\mathfrak{d}}^q$ -null set,

$$\begin{aligned} I_{\text{main}} &= G_{\text{main}}(\vartheta) \sqcup \bigsqcup_{\substack{\kappa \in \{r, \text{PPE}, N_{\text{time}}\} \\ \pi \in \{J, \text{TV}\}}} \mathcal{A}_{\text{main}}^{\kappa, \pi}(\vartheta) \\ &\sqcup I_{\text{main}}^{\text{face/word}}(\vartheta) \sqcup I_{\text{main}}^{\text{deep}}(\vartheta). \end{aligned} \quad (188)$$

The displayed handler names are tied to the immutable record by the typed relations

$$\begin{aligned} G_{\text{main}}(\vartheta) &\subset I_{\text{main}} \cap \{a : \pi(a) = J\}, \\ \mathcal{A}_{\text{main}}^{\kappa, \pi}(\vartheta) &\subset I_{\text{main}} \cap \{a : \kappa(a) = \kappa, \pi(a) = \pi\}, \\ I_{\text{main}}^{\text{face/word}}(\vartheta) &\subset I_{\text{main}} \cap \{a : \kappa(a) = \text{word}, \ell_{\text{word}}(a) \leq L\}, \\ I_{\text{main}}^{\text{deep}}(\vartheta) &\subset I_{\text{main}} \cap \{a : \kappa(a) = \text{word}, \ell_{\text{word}}(a) > L\}. \end{aligned} \quad (189)$$

Here L is the Borel cutoff map of the linked query, not a parameter of the base incidence law. An occurrence whose preliminary handler changes must be re-recorded, and its q recomputed, before (125) is frozen. In particular there is no admissible filter which records $\pi(a) = \text{TV}$ but is used as a J row; if such a rerouting is required, the active record is $\pi(a) = J$ and C_{car} enters (142). If several clocks fire, a fixed measurable first-active ordering selects exactly one filter; this ordering partitions additive occurrences and is not the probability first-hit partition. The handler table is part of the interface: G_{main} is treated by nonstationary phase; every J cell by the selection-safe energy lift below; every TV cell by the corresponding row of CLOCK_SRC_W ; and the final two cells respectively by either $\text{FACE_TIME}_{\text{CM2}}$ or $\text{FACE_TIME}_{\text{REC}}$, together with $\text{FACE}_{2\text{CUT}}$, and by $\text{TAIL}_{\text{env}}^+/\text{NREC}_{\text{deep}}$. In particular, membership in G_{main} or in the phase-good complement of a main J cell additionally requires exactly the typed NSP admissibility imposed on non-main J rows: on the same carrier $\psi_\xi \in C^{1+\alpha}$ with the declared Hölder derivative bound, the positive amplitude has the scaled $\mathcal{A}^\alpha$ norm used by Lemma 6.63, and the endpoint contribution is charged in (241). It also requires $\text{ACTUAL_SLOPE_BRIDGE}$ and POST_CARRIER_DOM on that carrier. A main occurrence lacking any one of these properties must be assigned to the corresponding TV , face/word , or deep handler; a lower derivative bound by itself does not license nonstationary phase. In particular, write $\widehat{A} := \{(\lambda, a, x) : (\lambda, a) \in A, x \in X_{\text{type}(a)}\}$ for the carrier lift of a mark filter. The actual phase-bad set is the jointly Borel carrier event

$$\widehat{B}_{\text{phase}}(\vartheta) := \{(\lambda, a, x) \in \widehat{I}_{\text{main}} : |\psi'_{\xi, \lambda, a}(x)| \leq \Xi r\}.$$

On $\widehat{G}_{\text{main}}$ the complementary derivative lower bound holds. For each Jensen filter the interface supplies a measurable centre $z_{\xi, \kappa, \vartheta}$ common to that complete filter and the fibrewise implication

$$\widehat{B}_{\text{phase}}(\vartheta) \cap \widehat{\mathcal{A}}_{\text{main}}^{\kappa, J} \subset \{|S_{\lambda, a}(x) - z_{\xi, \kappa, \vartheta}| \leq Ch_\kappa(t)\}.$$

The remaining phase-bad carrier points obey the typed inclusion

$$\widehat{B}_{\text{phase}}(\vartheta) \subset \bigsqcup_{\kappa} \left(\widehat{\mathcal{A}}_{\text{main}}^{\kappa, J} \cap \{|S - z_{\xi, \kappa, \vartheta}| \leq Ch_\kappa(t)\} \right) \sqcup \bigsqcup_{\kappa} \widehat{\mathcal{A}}_{\text{main}}^{\kappa, \text{TV}} \sqcup \widehat{I}_{\text{main}}^{\text{face/word}} \sqcup \widehat{I}_{\text{main}}^{\text{deep}}. \quad (190)$$

Restriction of the one propagated positive envelope gives the absolutely convergent dual identity

$$J_{\text{main}} = J_{G_{\text{main}}} + \sum_{\kappa, \pi} J_{\text{main}}^{\kappa, \pi} + J_{\text{main}}^{\text{face/word}} + J_{\text{main}}^{\text{deep}}. \quad (191)$$

For a Jensen filter $A = \mathcal{A}_{\text{main}}^{\kappa, J}(\vartheta)$, $\kappa \in \{r, \text{PPE}, N_{\text{time}}\}$, use the same slope laws $\mu_{\lambda, a}$ and $M_* = \mathbb{E}\mathcal{W}_*$ as above and define

$$\begin{aligned} \mathcal{B}_{\text{main}, \kappa}(t) &:= M_*^{-1} \int_{\mathcal{A}_{\text{main}}^{\kappa, J}(t)} q_{\lambda}(da) d\Pi(\lambda), \\ \mathcal{E}_{\text{main}, \kappa}(t) &:= M_*^{-1} \int_{\mathcal{A}_{\text{main}}^{\kappa, J}(t)} (\mu_{\lambda, a} \otimes \mu_{\lambda, a}) \{|u - v| \leq 2h_{\kappa}(t)\} q_{\lambda}(da) d\Pi(\lambda), \\ \mathcal{V}_{\text{main}, \kappa}(t) &:= M_*^{-1} \sup_z \int_{\mathcal{A}_{\text{main}}^{\kappa, J}(t)} \mu_{\lambda, a}([z - h_{\kappa}(t), z + h_{\kappa}(t)]) q_{\lambda}(da) d\Pi(\lambda). \end{aligned} \quad (192)$$

These quantities use the operator- and recovery-charged q , are finite by the declared bounds below, and are defined on the incidence graph rather than by summing currents over labels. Identity (188), not the static owner identity alone, is the coverage statement used by the phase proof. Suppressing ξ, P or unused depth entries in $A(t)$ always means a section at frozen ϑ ; all estimates are uniform in those entries.

Definition 6.53 (Selection-safe joint source likelihood SEL_JSL_p). Fix a legitimate frozen physical parent, but do not select a main cell. Use the exact K^{prob} , κ_{η} , product law R_0 , slope S_{η}^0 , and event D_h^0 from (179)–(181). They are the objects fixed by $\text{PAIR_SAFE_REFERENCE}$ and estimated by PAIR_TENERGY , not newly declared look-alikes. There must exist one probability proposal kernel $\zeta_{\eta}(x, x'; da)$, measurable with respect to the preselection physical parent only, with $\zeta_{\eta}(x, x'; Y_{\text{src}}) = 1$ pointwise. It is fixed before every resolution/mode section ϑ and before every cell A ; symbolically $\exists \zeta \forall (\vartheta, A)$. Put

$$\bar{R}(d\eta, da, dx, dx') := R_0(d\eta, dx, dx') \zeta_{\eta}(x, x'; da). \quad (193)$$

For the a -independent reference resonance event D_h^0 in (181), normalization gives the exact mass-preserving identity

$$\bar{R}(D_h^0 \times Y_{\text{src}}) = R_0(D_h^0). \quad (194)$$

On a same-carrier cell put $\bar{R}^{\text{src}} := \bar{R}$; on a different-carrier cell put $\bar{R}^{\text{src}} := \bar{R}^H$, where the typed pushforward law and event inclusion are those in (202)–(205).

For a nonzero main Jensen filter A of Definition 6.52, define the unnormalized source pair measure

$$d\Sigma_A(\eta, a, x, x') := M_*^{-1} \mathbf{1}_A(\eta, a) d\Pi(\eta) q_{\eta}(da) \nu_{\eta, a}(dx) \nu_{\eta, a}(dx'). \quad (195)$$

Define the actual source pair event by

$$D_h^{\text{src}} := \{(\eta, a, x, x') : |S_{\eta, a}(x) - S_{\eta, a}(x')| \leq 2h\}. \quad (196)$$

Thus $\Sigma_A(1) = \mathcal{B}_A$ and, at the clock scale $h = h_{\kappa}(t)$, $\Sigma_A(D_h^{\text{src}}) = \mathcal{E}_A(h)$. In the PPE row this is exactly the window in (163); it may not be replaced by a larger independently declared window. The interface requires

$$\Sigma_A \ll \bar{R}^{\text{src}}, \quad Z_A := \frac{d\Sigma_A}{d\bar{R}^{\text{src}}}, \quad p \in (1, \infty]. \quad (197)$$

The proposal support must cover the source support, and the source event must satisfy the typed measurable compatibility

$$\begin{cases} D_h^{\text{src}} \subset D_{C_0h}^0 \times Y_{\text{src}}, & \text{on a same-carrier cell,} \\ \mathcal{H}^{-1}D_h^{\text{src}} \subset D_{C_0h}^0 \times Y_{\text{src}}, & \text{on a different-carrier cell.} \end{cases}$$

Both inclusions are understood on the corresponding support of Σ_A . For the PPE row they hold with $h = h_{\text{PPE}}(N) \leq C_{\text{win}}e^{-N/C_{\text{buf}}}$ and hence transfer into $D_{Ce^{-N/C_{\text{buf}}}}^0$, the same scale controlled by PAIR_TENERGY. Uniformly in all frozen data, the scale-dependent likelihood bounds are

$$\|Z_{A,r}\|_{L^p(\bar{\mathbf{R}}^{\text{src}})} \leq Cr^{-a_{\text{main},r}^L}, \quad \|Z_{A,\kappa}\|_{L^p(\bar{\mathbf{R}}^{\text{src}})} \leq Ce^{a_{\text{main},\kappa}^L N}, \quad \kappa \in \{\text{PPE}, N_{\text{time}}\}. \quad (198)$$

The selector $\mathbf{1}_A$, the q reweighting, the reciprocal proposal cost, and all carrier-density costs occur in this single Z_A ; neither $\mathbf{R}_0$ nor ζ may depend on A . Consequently

$$\mathcal{E}_A(h) = \Sigma_A(D_h^{\text{src}}) \leq \|Z_A\|_{L^p(\bar{\mathbf{R}}^{\text{src}})} \mathbf{R}_0(D_{C_0h}^0)^{1/p'}, \quad \frac{1}{p} + \frac{1}{p'} = 1, \quad (p' = 1 \text{ when } p = \infty). \quad (199)$$

Because $\bar{\mathbf{R}}^{\text{src}}$ is a probability, $\mathcal{B}_A = \|Z_A\|_1 \leq \|Z_A\|_p$; the declared likelihood and base growth rates must be compatible with this elementary check. A proposal missing source support or chosen after seeing A is an immediate no-go. This is the unique change of law at the kernel where selection and carrier reweighting occur. Parent KRH_p contains none of $\mathbf{1}_A$, $1/\zeta$, or the two internal carrier coordinates and cannot replace this interface, in agreement with [25].

Lemma 6.54 (Selection factor criterion and typed carrier match). *Suppose first that the proposal is carrier-independent, $\zeta_\eta(x, x'; da) = \zeta_\eta(da)$, and that source and probability sections use the same carrier. Define*

$$w_A^{\text{occ}} := \frac{d\{M_*^{-1}\mathbf{1}_A(\eta, a)\Pi(d\eta)q_\eta(da)\}}{d\{K^{\text{prob}}(d\eta)\zeta_\eta(da)\}}, \quad Z_A = w_A^{\text{occ}}(\eta, a)\ell_{\eta,a}(x)\ell_{\eta,a}(x'), \quad \ell_{\eta,a} := \frac{d\nu_{\eta,a}}{d\kappa_\eta}. \quad (200)$$

Thus w_A^{occ} visibly contains the selector, propagated q weight, and reciprocal proposal cost. If $w_A^{\text{occ}} \in L^{p_0}$ uniformly and $\sup_{\eta,a} \|\ell_{\eta,a}\|_{L^{p_c}(\kappa_\eta)} < \infty$, then direct disintegration gives $Z_A \in L^p(\bar{\mathbf{R}})$ for every $1 < p \leq \min\{p_0, p_c\}$. In particular, when $p_c = \infty$, the endpoint is available without an arbitrary loss:

$$p_{\text{joint}} = p_0, \quad \|Z_A\|_{L^{p_0}(\bar{\mathbf{R}})} \leq \|w_A^{\text{occ}}\|_{L^{p_0}} \sup_{\eta,a} \|\ell_{\eta,a}\|_\infty^2. \quad (201)$$

If $\nu_{\eta,a}$ and κ_η are normalized densities in one uniform log-Hölder cone on the same homogeneous carrier, the cone bounds

$$e^{-\varpi}|W|^{-1} \leq \rho \leq C_{\text{cone}}e^{\varpi}|W|^{-1}$$

from the standard-family construction of [22] give $p_c = \infty$. Hence only the occurrence/selection likelihood is paid. If that factor is also bounded one may take $p_{\text{joint}} = \infty$.

For different carriers, CARRIER_MATCH must instead supply a jointly measurable $H_{\eta,a} : W_{\text{prob}} \rightarrow W_{\text{src}}$ and the actual pushforward reference law

$$\begin{aligned} \mathcal{H}(\eta, a, x, x') &:= (\eta, a, H_{\eta,a}x, H_{\eta,a}x'), \\ \bar{\mathbf{R}}^H &:= \mathcal{H}_*\bar{\mathbf{R}} = K^{\text{prob}}(d\eta)\zeta_\eta(da)(H_{\eta,a*}\kappa_\eta)(dy)(H_{\eta,a*}\kappa_\eta)(dy'). \end{aligned} \quad (202)$$

The entire kernel H is measurable with respect to the same preselection parent data as ζ and is fixed before every resolution/mode section and every cell: symbolically $\exists H \forall(\vartheta, A)$. An A -dependent holonomy is forbidden because it would hide the selected conditioning inside the reference carrier. It requires

$$\nu_{\eta,a} \ll H_{\eta,a*\kappa_\eta}, \quad \ell_{\eta,a}^H := \frac{d\nu_{\eta,a}}{d(H_{\eta,a*\kappa_\eta})}, \quad \sup_{\eta,a} \|\ell_{\eta,a}^H\|_{L^{p_c}(H_{\eta,a*\kappa_\eta})} \leq C, \quad (203)$$

and, for a carrier-independent proposal, the exact likelihood factorization under the push-forward law

$$Z_A^H(\eta, a, y, y') = w_A^{\text{occ}}(\eta, a) \ell_{\eta,a}^H(y) \ell_{\eta,a}^H(y'). \quad (204)$$

On a different-carrier cell the Z_A in (197) is, by definition, this Z_A^H . If $w_A^{\text{occ}} \in L^{p_o}$ and the density ratio has the displayed uniform L^{p_c} bound, then $Z_A^H \in L^p(\bar{\mathbf{R}}^H)$ for every $1 < p \leq \min\{p_o, p_c\}$; when $p_c = \infty$ the endpoint $p = p_o$ holds with the analogue of (201). The interface also requires the target-scale compatibility

$$|S_{\text{src}}(\eta, a, H_{\eta,a}x) - S_{\text{prob}}(\eta, x)| \leq C_0 h_\kappa(t), \quad \mathcal{H}^{-1} D_h^{\text{src}} \subset D_{C_1 h}^0, \quad (205)$$

with $h_\kappa(t)$ equal to the actual r -, PPE-, or time-window scale. A Jacobian bound relative to the two normalized arclength measures is only a geometric sufficient ingredient for (203); it does not replace the Radon–Nikodym condition. If the proposal depends on (x, x') and destroys the product disintegration, one must verify the full $Z_A = d\Sigma_A/d\bar{\mathbf{R}}^H$ bound directly. Without either the same-carrier or this typed pushforward alternative, the lift is a no-go.

Proof. Integrating $|Z_A|^p$ first in x, x' proves the conditional factor claim; (201) is the same computation at $p = p_o$ with bounded carrier factors. The cone claim is the ratio of its uniform upper and lower density bounds. For different carriers, (202) defines the law on which the density ratio in (203) lives, and (204) followed by the same disintegration proves the required selection-safe L^p bound. Finally, (205) transfers the source event to an enlarged unselected reference event. These are separate requirements. $\square$

Theorem 6.55 (Main-source energy lift MAIN_SOURCE_ENERGY_LIFT). Assume ACTUAL_SLOPE_BRIDGE, PAIR_SAFE_REFERENCE, PAIR_TENERGY, the parentwise probability energy estimate of Theorem 6.51, and SEL_JSL $_{p_{\text{joint}}}$ on every nonempty main J filter. Put $p'_{\text{joint}} = p_{\text{joint}}/(p_{\text{joint}} - 1)$, with $p'_{\text{joint}} = 1$ in the bounded-likelihood case. Compress the complete reference energy theorem into the explicit rates

$$\begin{aligned} \Theta_{\text{ref},r} &:= \min\{\Theta_{\text{par}}^W, \Theta_r^{\text{prob}}, C_{\text{buf}} c_{\text{PPE}}^{\text{prob}}, C_{\text{buf}} c_{\text{buf}}^W, C_{\text{buf}} c_{\text{gate}}^W, C_{\text{buf}} c_{\text{cont}}^W\}, \\ c_{\text{ref},\text{PPE}} &:= \min\{\Theta_{\text{par}}^W/C_{\text{buf}}, \Theta_r^{\text{prob}}/C_{\text{buf}}, c_{\text{PPE}}^{\text{prob}}, c_{\text{buf}}^W, c_{\text{gate}}^W, c_{\text{cont}}^W\}. \end{aligned} \quad (206)$$

The second line is the first line at the declared relation $r = r_N = e^{-N/C_{\text{buf}}}$; by (163) the actual PPE source event transfers into an enlarged reference event at this very scale. Every component is already a parentwise tilted probability rate; any likelihood used to obtain those probability rates is paid there. For a nonempty main N_{time} Jensen cell, require the corresponding unselected reference energy rate

$$D_{\text{time},N}^0 := D_{h_{\text{time}}(N)}^0, \quad R_0(D_{\text{time},N}^0) \leq C e^{-c_{\text{ref},\text{time}} N}, \quad c_{\text{ref},\text{time}} > 0. \quad (207)$$

This rate is supplied by the actual unselected probability theorem/clock and is not inferred from the owner or main-clock partition. The unique source selection likelihood is paid

exactly once through p'_{joint} below. Suppose

$$\begin{aligned}\mathcal{B}_{\text{main},r}(r) &\leq Cr^{-a_{\text{main},r}^B}, \\ \mathcal{B}_{\text{main,PPE}}(N) &\leq Ce^{a_{\text{main,PPE}}^B N}, \quad \mathcal{B}_{\text{main,time}}(N) \leq Ce^{a_{\text{main,time}}^B N}.\end{aligned}\quad (208)$$

Together with (198), (199) gives the unnormalized source-energy bounds directly:

$$\begin{aligned}\mathcal{E}_{\text{main},r}(r) &\leq Cr^{\Theta_{\text{ref},r}/p'_{\text{joint}} - a_{\text{main},r}^L}, \\ \mathcal{E}_{\text{main,PPE}}(N) &\leq Ce^{-(c_{\text{ref,PPE}}/p'_{\text{joint}} - a_{\text{main,PPE}}^L)N}, \\ \mathcal{E}_{\text{main,time}}(N) &\leq Ce^{-(c_{\text{ref,time}}/p'_{\text{joint}} - a_{\text{main,time}}^L)N}.\end{aligned}\quad (209)$$

One application of Lemma 6.44 yields

$$\begin{aligned}\mathcal{V}_{\text{main},r}(r) &\leq Cr^{\Theta_{\text{main},r}^{\text{val}}}, \quad \Theta_{\text{main},r}^{\text{val}} := \frac{\Theta_{\text{ref},r}/p'_{\text{joint}} - a_{\text{main},r}^L - a_{\text{main},r}^B}{2}, \\ \mathcal{V}_{\text{main,PPE}}(N) &\leq Ce^{-c_{\text{main,PPE}}^{\text{val}} N}, \quad c_{\text{main,PPE}}^{\text{val}} := \frac{c_{\text{ref,PPE}}/p'_{\text{joint}} - a_{\text{main,PPE}}^L - a_{\text{main,PPE}}^B}{2}, \\ \mathcal{V}_{\text{main,time}}(N) &\leq Ce^{-c_{\text{main,time}}^{\text{val}} N}, \quad c_{\text{main,time}}^{\text{val}} := \frac{c_{\text{ref,time}}/p'_{\text{joint}} - a_{\text{main,time}}^L - a_{\text{main,time}}^B}{2}.\end{aligned}\quad (210)$$

Every nonempty value gap must be positive. If instead one starts from the normalized source law $\mathbf{S}_A = \Sigma_A/\mathcal{B}_A$ with $\|\mathbf{dS}_A/d\bar{\mathbf{R}}^{\text{src}}\|_p \leq Ch^{-d_A}$, then $a_A^L = a_A^B + d_A$ and the polynomial value rate becomes

$$\frac{1}{2} \left(\frac{\Theta_{\text{ref},r}}{p'_{\text{joint}}} - 2a_A^B - d_A \right). \quad (211)$$

Thus the old two-base-cost expression is only the special case $d_A = 0$; it is not inherited after selection. Define the complete Jensen value rates

$$\begin{aligned}\Theta_{r,J}^{\text{val}} &:= \min\{\Theta_{\text{main},r}^{\text{val}}, \Theta_{r,J}^{\text{exc, val}}\}, \\ c_{\text{PPE},J}^{\text{val}} &:= \min\{c_{\text{main,PPE}}^{\text{val}}, c_{\text{PPE},J}^{\text{exc, val}}\}, \\ c_{\text{time},J}^{\text{val}} &:= \min\{c_{\text{main,time}}^{\text{val}}, c_{\text{time},J}^{\text{exc, val}}\}.\end{aligned}\quad (212)$$

An absent class has rate $+\infty$. The main marked value tail is therefore derived from unselected reference probability energy, the single selection/source likelihood, finite base mass, and one global incidence Cauchy step; it is not assumed in `CLOCK_SRCW`.

Proof. Theorem 6.51 on the unselected law $\mathbf{R}_0$ gives the r /PPE rates (206); the separate hypothesis (207) supplies the time rate when that class is nonempty. The pointwise normalization of ζ preserves these reference energies because D_h^0 is a -independent, and the source/reference inclusions use the exact events (181) and (196). Apply Hölder once to the unnormalized Σ_A against $\bar{\mathbf{R}}^{\text{src}}$; the single density Z_A simultaneously pays the selector, propagated occurrence weight, proposal, and both carrier coordinates, giving (209). Apply (167) once to each complete main clock cell. The likelihood and base costs $a^L + a^B$ are paid before the square root. No selected reference law, occurrencewise Jensen, or primitive main source tail is used. $\square$

Corollary 6.56 (Central slope small-ball). *Assume all hypotheses of Theorem 6.51 and of Theorem 6.55. In particular assume PAIR_SAFE_REFERENCE, PAIR_TENERGY, and the ACTUAL_SLOPE_BRIDGE. The complete central collision-SRB quotient, normalized by its positive event weight, satisfies (187) and the main marked-source bounds (210) at $s = 0$. At finite s the constants are uniform precisely for the stopped scales whose total resolving depth is at most $L(s)$. Proposition 6.66 invokes this finite- s statement only in its bounded intermediate frequency range.*

Proof. Apply Theorem 6.51 and Theorem 6.55. The latter already estimates the unnormalized selected source measure. Multiplying the common M_* normalization back gives the complete marked phase mass; no selected reference law, signed cancellation, occurrencewise square root, or unweighted Jensen step is used. $\square$

Lemma 6.57 (Slope small-ball controls the balanced phase). *Assume*

ACTUAL_SLOPE_BRIDGE, CLOCK_SRC $_W$, MAIN_CLOCK_COVERAGE,

Theorem 6.55, and the exceptional source rates (166) on all nonempty displayed rows. For $\psi_\xi = \xi_1 X + \xi_2 Y$ on an affine-normalized gate chart, let

$$\Xi = |\xi_1|a_X \asymp |\xi_2|a_Y, \quad (213)$$

with a_X, a_Y as in (98). On the restricted r /PPE Jensen subcurrent

$$J_{r/\text{PPE},J}^{\text{cen}} := \sum_{\kappa \in \{r, \text{PPE}\}} \left(J_{\text{nm}}^{\kappa,J} + J_{\text{main}}^{\kappa,J} \right),$$

the value mass of $\{|\psi'_\xi| \leq \Xi r\}$ is at most $C\{r^{\Theta_{r,J}^{\text{val}}} + e^{-c_{\text{PPE},J}^{\text{val}} N_{\text{PPE}}(r)}\}$. This is the additive source value mass derived by global incidence Cauchy, not the probability of its support. Smooth value-space cutoffs on its complement have C^α norm $O(r^{-1})$ and satisfy $\inf |\psi'_\xi| \geq c\Xi r$.

Proof. By the exact identity (92) in ACTUAL_SLOPE_BRIDGE,

$$\psi'_\xi = X'(\xi_1 + \varepsilon_{\text{or}} \xi_2 e^S).$$

The lower and upper bounds in (98) convert the bracketed factor to the physical scale (213). Only opposite-oriented coefficients can resonate, and the resonant set is contained in one interval $|S - z_\xi| \leq Cr$. Restrict (160) and (191) to the two displayed Jensen filters, and use the carrier-level implications (161) and (190). Then apply (210), combine it with the exceptional rates in (212), and use (143). Composing a fixed smooth three-term partition with $(\xi_1 + \varepsilon_{\text{or}} \xi_2 e^S)/(\Xi r)$ gives the cutoff and norm bounds. The N_{time} , TV, face/word, and deep restrictions are not claimed by this lemma; they are split off and handled by their typed rows in Proposition 6.66 and the two cutoff/tail theorems. $\square$

Lemma 6.58 (Typed global double projection before disintegration). *Let J be the complete finite signed response measure on $M \times M$ in common physical coordinates, let $m_J^\pm$ be its marginals, and let $\nu^- = \nu^+ = \mu_0$ be the fixed invariant reference supplied by (2). Define the current-side projection*

$$D_{\text{cur}} J =: J^\circ = J - m_J^- \otimes \nu^+ - \nu^- \otimes m_J^+ + J(1,1) \nu^- \otimes \nu^+. \quad (214)$$

For a bounded test F define $D_{\text{test}}F$ by subtracting its two integrated marginals and adding its product mean. Then

$$\int F d(D_{\text{cur}}J) = \int D_{\text{test}}F dJ.$$

Thus J° is double centered and the current projection is the adjoint, not the same operator, as test centering. If the complete response current already has zero marginals, then

$$J^\circ = J. \quad (215)$$

Only after this identity is formed in physical coordinates is the singular part refined by face labels. If the original physical amplitudes obey (351) and the references have fixed C^α densities, then

$$\|D_{\text{cur}}J\|_{\text{TV}} \leq 4\|J\|_{\text{TV}}, \quad \|J\|_{\text{phys,R}} + \|m_{\bar{J}} \otimes \mu_0 + \mu_0 \otimes m_J^\dagger - J(1,1)\mu_0 \otimes \mu_0\|_{\mathfrak{P}_R} < \infty. \quad (216)$$

Proof. Fubini proves the adjoint identity and each marginal has total variation at most $\|J\|_{\text{TV}}$. This gives the factor four. Marginalization is contractive by (9), and tensoring with the fixed probability μ_0 gives exactly the separate product type (10). These operations do not preserve dimension and hence are not assigned the graph-face norm. The positive face ledger dominates signed total variation before recombination; absolute summation of (351) and (9) prove the second assertion. $\square$

Lemma 6.59 (Endpoint unfolding and cutoff commutation). *Let a localized assembled face at parameter s be $J_{\lambda,s} = (H_{\lambda,s})_*(a_{\lambda,s}(u) du)$ with $H_{\lambda,s}(u) = (x_{\lambda,s}(u), y_{\lambda,s}(u))$. Put*

$$X_{\lambda,m,s}^-(u) := T_s^{-m} x_{\lambda,s}(u), \quad Y_{\lambda,n,s}^+(u) := T_s^n y_{\lambda,s}(u). \quad (217)$$

Let $g^\circ = g - \mu_0(g)$ and $f^\circ = f - \mu_0(f)$. After the complete current is assembled and the global current-side double projection is taken,

$$\int (Q_s^m g)(x) (Q_s^{*n} f)(y) dD_{\text{cur}} \left(\sum_{\lambda} J_{\lambda,s} \right) = \sum_{\lambda} \int a_{\lambda,s}(u) g^\circ(X_{\lambda,m,s}^-(u)) f^\circ(Y_{\lambda,n,s}^+(u)) du. \quad (218)$$

The two marginal and constant product blocks are therefore assembled before the face sum is estimated. This is the exact physical endpoint stage. It is distinct from the following scalarization stage.

Let $X_{\lambda,m,s}(u)$ and $Y_{\lambda,n,s}(u)$ be the scalar base coordinates of the same-branch sections in BR_{sec} . Branchwise contraction replaces the two endpoint factors by

$$g^\circ(\iota_\lambda^-(X_{\lambda,m,s}(u))), \quad f^\circ(\iota_\lambda^+(Y_{\lambda,n,s}(u))), \quad (219)$$

with the explicit error (222). Only now do we expand the localized restrictions of the original centered functions $g^\circ \circ \iota_\lambda^-$ and $f^\circ \circ \iota_\lambda^+$. The scaled composition clause in BR_{sec} gives their C^r Fourier decay with the recorded $\mathfrak{d}_\lambda$ and Mark costs, uniformly in m, n, s . A scalar mode has phase

$$\psi_{k,l,\lambda,m,n,s}(u) = kX_{\lambda,m,s}(u) + lY_{\lambda,n,s}(u). \quad (220)$$

Thus the Fourier coefficients belong to the fixed original observables; time dependence lies in the scalar endpoint phase only. Equation (218) and (220) are not asserted to be the same identity: the controlled section-replacement error lies between them. The homogeneous refinement cuts only the amplitude and endpoint phase. Its Jacobian, cutoff derivatives,

and genuine singularity endpoints are included in the physical Mark-Z ledger. Applying the same scalar multiplier to the two endpoint coordinates before or after the global double projection gives the same joint, two axis, and zero blocks. Hence the low multiplier and its high complement are disjoint and exhaustive after assembly.

Proof. Equations (3)–(4) and the adjoint identity of Lemma 6.58 give the displayed formula. The axis and constant terms cancel only after the complete physical current is assembled; no face is separately asserted to have zero marginals. Insert the fixed physical partition of unity, apply (222), and only then Fourier-expand the localized restrictions of the original observables. No derivative is applied to $Q_s^m g$ or $Q_s^{*n} f$. Fubini and the even scalar multiplier show that Fourier cutoff and the four-term double projection commute. The last assertion follows because every boundary created by the homogeneous partition is either paired with its adjacent cell or is a genuine endpoint recorded in the ledger. $\square$

Definition 6.60 (Branch-section interface BR_{sec}). *On every clean graph rectangle, the dominant endpoint coordinates admit branchwise sections*

$$\iota_\lambda^- : X \mapsto (X, \zeta_\lambda^-(X)), \quad \iota_\lambda^+ : Y \mapsto (Y, \zeta_\lambda^+(Y))$$

inside the same homogeneous inverse branches as the physical graph trace. Their images, in the appropriate forward and reversed orientations, are homogeneous measured standard curves of class $C^{r_{\text{obs}}}$. For a fixed $q_{\text{end}} \geq 0$, their scalar composition maps satisfy

$$\begin{aligned} \left| \widehat{\chi_\lambda^-(g \circ \iota_\lambda^-)}(k) \right| &\leq C \mathfrak{d}_\lambda^{q_{\text{end}}} \|g\|_{C^{r_{\text{obs}}}} (1 + |k|)^{-r_{\text{obs}}}, \\ \left| \widehat{\chi_\lambda^+(f \circ \iota_\lambda^+)}(l) \right| &\leq C \mathfrak{d}_\lambda^{q_{\text{end}}} \|f\|_{C^{r_{\text{obs}}}} (1 + |l|)^{-r_{\text{obs}}}. \end{aligned} \quad (221)$$

The powers of $\mathfrak{d}_\lambda$ are charged to A_{dom} ; cutoff, Jacobian, and face-amplitude costs already form the single squared mark in (245) and are not inserted a second time through the Fourier coefficients. Their arclength/coordinate densities, endpoint Z values, and scaled Hölder marks are bounded by $\text{Mark}(\lambda)\{1 + Z(\mathcal{G}_\lambda^-) + Z(\mathcal{G}_\lambda^+)\}$. The label includes the endpoint depths and the parameter. With the unfolded endpoint maps of (217), branchwise contraction gives

$$\begin{aligned} \int |g(X_{\lambda,M,s}^-(\bar{s})) - g(\iota_\lambda^-(X_{\lambda,M,s}(\bar{s})))| d|\omega_\lambda| &\leq C e^{-c-M} \|g\|_{C^1} |\omega_\lambda|, \\ \int |f(Y_{\lambda,N,s}^+(\bar{s})) - f(\iota_\lambda^+(Y_{\lambda,N,s}(\bar{s})))| d|\omega_\lambda| &\leq C e^{-c+N} \|f\|_{C^1} |\omega_\lambda|, \end{aligned} \quad (222)$$

where $d\omega_\lambda = M_\lambda \omega_\lambda d\bar{s}$. This is a same-branch condition: no conditional measure is averaged across different symbolic pasts or singularity cuts.

Lemma 6.61 (Dominant-coordinate branch-section reduction). *Assume BR_{sec} . After assembly and the recentering of Lemma 6.58, retain the rank-one and regular-product blocks as ordinary product kernels and evolve both factors by their centered operators. On a clean graph rectangle in the diagonal band $D^{-1}M < N < AM$, the graph-current pairing decomposes as*

$$B_{M,N} = B_{M,N}^{\text{dom,dom}} + B_{M,N}^{\text{dom,rem}} + B_{M,N}^{\text{rem,dom}} + B_{M,N}^{\text{rem,rem}}. \quad (223)$$

The first term is the scalar quotient integral obtained by evaluating the two tests on $\iota_\lambda^-(X)$ and $\iota_\lambda^+(Y)$. Every other term is bounded by

$$C e^{-c \max\{M,N\}} \|g\|_{C^1} \|f\|_{C^1}. \quad (224)$$

This bound remains valid after multiplication by the complete positive event weight and integration over $\Pi_{s,N}^W$. The displayed remainder is a genuine $N_{\text{time}} = \max\{M, N\}$ source occurrence in the time-TV cell of CLOCK_SRC_W . Any label on which uniform contraction is unavailable is separately owned according to the stage where failure is first observed and entered in exactly one clock-passage cell. It is never assigned an untyped exponential rate. Thus no unweighted exceptional tail or clock conversion is hidden in the contracted-coordinate remainder. The global joint constant and axis blocks vanish or remain in the explicitly separated rank-one summands before the graph-rectangle sum is estimated. Every graph-axis mode created by a boundary cutoff is uniformly nonstationary and is charged to its cutoff norm.

Proof. In stable-unstable endpoint coordinates, use the same-branch sections of BR_{sec} and telescope the two replacements. Equation (222) gives remainders bounded by $C\lambda_-^M \|g\|_{C^1}$ and $C\lambda_+^N \|f\|_{C^1}$ after the physical amplitude is integrated. Disintegrating first under the positive complete tilt preserves the same uniform factors; the named exceptional complement is charged by its corresponding typed clock in (164) or (165). In the diagonal band these imply (224). The two frozen functions depend only on X, Y , proving (223). Global projection separates the constant and two axis blocks before any label-dependent chart is chosen. They are estimated by their explicit product formulas, not redescribed as graph currents; Lemma 6.64 supplies product decay. A graph boundary cutoff can reintroduce an axis term, but it has phase $\xi_1 X$ or $\xi_2 Y$; (98) gives the physical derivative lower bound, and its cutoff ledger is absolutely summable by (216). $\square$

Proposition 6.62 (Scaled physical face density). *On a homogeneous physical face cell $Q = t_Q + \ell_Q[-1, 1]$, the collision-SRB conditional is $d\mu_Q^u = \rho_Q(t) dt$, where, before normalization,*

$$\rho_Q^{\text{raw}}(t) = c_\mu \cos \varphi(t) J_{\text{coa}}(t) J_{\text{hol}}(t) J_{\text{chart}}(t), \quad J_{\text{coa}} = |\nabla G|^{-1}. \quad (225)$$

Put $\hat{\rho}_Q(u) = \ell_Q \rho_Q(t_Q + \ell_Q u)$. There is a mark

$$\text{Mark}(Q) = 1 + \mathfrak{h}_Q + \mathfrak{b}_Q + [\log J_{\text{hol}}]_{\alpha, Q} + [\log J_{\text{coa}}]_{\alpha, Q} + [\log J_{\text{chart}}]_{\alpha, Q} + \|w_Q\|_{\mathcal{A}^\alpha} \quad (226)$$

such that

$$C^{-1} \text{Mark}(Q)^{-1} \leq \hat{\rho}_Q \leq C \text{Mark}(Q), \quad [\log \hat{\rho}_Q]_\alpha \leq C \text{Mark}(Q). \quad (227)$$

For some $p_d > 2$,

$$\sum_Q M_Q \text{Mark}(Q)^{p_d} < \infty \quad (228)$$

uniformly in the small table family. The same statement holds after the positive/negative decomposition of a signed face amplitude.

Proof. In collision coordinates the invariant density is $c_\mu \cos \varphi dr d\varphi$. Coarea along the physical graph, followed by stable projection to the gate coordinate, gives (225). On a homogeneous strip, $\cos \varphi$, the chart Jacobian, and the holonomy Jacobian have the standard scaled log-Hölder distortion. Dividing by $\int_Q \rho_Q^{\text{raw}}$ gives (227); the factor ℓ_Q removes the cell scale.

The homogeneity contribution has a fixed polynomial cost, while the strip mass is exponentially summable. The holonomy and chart contributions are summed by the context moment, and the amplitude contribution by the marked face ledger. Generalized Hölder with slightly smaller exponents proves (228). For a signed amplitude q , write $q = (2\|q\|_\infty + q) - 2\|q\|_\infty$; multiplication by the positive masses restores the scaled norm and introduces no reciprocal small-mass factor. $\square$

Lemma 6.63 (Fractional nonstationary phase). *Let I be an interval, $a \in C^\alpha(I)$, and $\phi \in C^{1+\alpha}(I)$. On each component of the support of a , suppose ϕ' has one sign and $|\phi'| \geq \lambda > 0$. Put $\Phi = \sum_C \int_C |\phi'(t)| dt$, the total phase length of the support. For $C = (l_C, r_C)$ define the endpoint functional $\mathfrak{E}(a) = \sum_C (|a(l_C+)| + |a(r_C-)|)$.*

$$\left| \int_I e^{i\phi(t)} a(t) dt \right| \leq C_\alpha \left\{ \frac{\mathfrak{E}(a)}{\lambda} + \frac{\Phi[a]_\alpha}{\lambda^{1+\alpha}} + \frac{\Phi \|a\|_\infty [\phi']_\alpha}{\lambda^{2+\alpha}} \right\}. \quad (229)$$

The estimate adds over a countable union of sign components without a component-count loss.

Proof. On one sign component use $y = \phi(t)$ and write $b(y) = a(\phi^{-1}y)/\phi'(\phi^{-1}y)$. Then

$$\|b\|_\infty \leq \|a\|_\infty / \lambda, \quad [b]_\alpha \leq [a]_\alpha \lambda^{-1-\alpha} + \|a\|_\infty [\phi']_\alpha \lambda^{-2-\alpha}.$$

For a shift $h = \pi$, the identity

$$2 \int e^{iy} b(y) dy = \int e^{iy} \{b(y) - b(y+h)\} dy + \text{endpoint terms}$$

gives $C\Phi[b]_\alpha$ plus the two endpoint values of b on that component. Since $|b| \leq |a|/\lambda$, summing the endpoint values gives $\mathfrak{E}(a)/\lambda$, while the sum of the component y -lengths is Φ . This proves (229); no number-of-components estimate is hidden in the supremum term. $\square$

Lemma 6.64 (Rank-one product under double centering). *If $\eta_\pm$ are admissible dual collision currents and $R = \eta_- \otimes \eta_+$, then*

$$R(Q_-^m g \otimes Q_+^n f) = \eta_-(Q_-^m g) \eta_+(Q_+^n f).$$

Thus the two one-sided gaps give $C\tau^m \vartheta^n$.

Proof. This is the universal property of the projective tensor product followed by the two admissible dual bounds. In the principal proof the rank-one corrections are instead included in the low-frequency functional of Proposition 6.73; this lemma is only an algebraic consistency check. $\square$

Lemma 6.65 (Balanced–unbalanced exhaustion in physical frequency). *Assume the clean-gate chart bounds*

$$c_0 a_X \leq |X'| \leq C_0 a_X, \quad c_0 a_Y \leq |Y'| \leq C_0 a_Y \quad (230)$$

with the fixed orientation signs on one homogeneous face cell. Put $C_{\text{bal}} = 4C_0/c_0$. Every nonzero scalar mode belongs to exactly one of the following regions (with the boundary assigned once):

- (i) $C_{\text{bal}}^{-1} \leq |\xi_1| a_X / (|\xi_2| a_Y) \leq C_{\text{bal}}$ (balanced);
- (ii) $|\xi_1| a_X > C_{\text{bal}} |\xi_2| a_Y$ (past-dominant);
- (iii) $|\xi_2| a_Y > C_{\text{bal}} |\xi_1| a_X$ (future-dominant).

On either unbalanced region the phase $\psi_\xi = \xi_1 X + \xi_2 Y$ has one-sign derivative after at most the fixed orientation cut and

$$|\psi'_\xi| \geq c \max\{|\xi_1|a_X, |\xi_2|a_Y\}. \quad (231)$$

Consequently the complete unbalanced contribution is bounded by

$$C \text{Mark}(Q)^2 \{1 + Z(\mathcal{G}_Q)\} \max\{|\xi_1|a_X, |\xi_2|a_Y\}^{-\alpha}, \quad (232)$$

and its two-dimensional lattice sum converges under the same observable regularity used for the balanced sum.

Proof. Consider the past-dominant case. The orientation table and (230) give

$$|\psi'_\xi| \geq c_0 |\xi_1|a_X - C_0 |\xi_2|a_Y \geq \frac{1}{2} c_0 |\xi_1|a_X.$$

The other case is symmetric. Cut only at the fixed orientation endpoints; no frequency-dependent sign partition is introduced. Apply Lemma 6.63 with this lower bound and with the actual amplitude and endpoint ledger. Its three terms are bounded by the right side of (232); the exponent α may be decreased once to absorb fixed chart losses. By Lemma 6.59, multiplication is by the Fourier coefficients of the two localized original $C^{r_{\text{obs}}}$ observables, with the scaled restriction cost in (221), not by coefficients of evolved tests. Summation over the two unbalanced cones is finite as soon as $r_{\text{obs}} > 2$; the stronger balanced threshold imposed below therefore suffices. The three regions are disjoint and exhaustive. $\square$

Proposition 6.66 (Central balanced estimate with explicit budget). *Assume*

$$\begin{aligned} & \text{ACTUAL_SLOPE_BRIDGE}, \quad \text{PAIR_SAFE_REFERENCE}, \\ & \text{PAIR_TENEGY}, \quad \text{CLOCK_SRC}_W, \quad \text{MAIN_CLOCK_COVERAGE}, \\ & \text{SEL_JSL}_{p_{\text{joint}}}, \quad \text{BR}_{\text{sec}}, \quad \text{USC}_{\text{end}}^W(q_g^W), \\ & \text{POST_CARRIER_DOM}, \quad \text{PHYS_TEST_LIFT}, \quad \text{PROP_Q_MATCH}, \end{aligned}$$

the hypotheses of Theorems 6.51 and 6.55, and the typed NSP admissibility in both the non-main and main clock definitions, as well as all hypotheses of Proposition 6.73 and Lemmas 6.72 and 6.71 for the low block, with the main-source likelihood and base bounds (198) and (208). For every nonempty main time-Jensen cell also assume (207) on the same frozen parent. Use the full weighted upper-scale tail (106), the joint Mark-Z estimate (283), and the global physical ledger (351) on the same projectively fixed incidence law. Use the complete additive source rates (164), (165), and (212). Set

$$\begin{aligned} A_{\text{frac}} &= 3 + \alpha, \quad 0 < \delta < \frac{\alpha}{2A_{\text{frac}}}, \\ \beta_{\text{PPE},J}^{\text{val}} &:= \delta C_{\text{buf}} c_{\text{PPE},J}^{\text{val}}, & \beta_{\text{PPE},\text{TV}}^{\text{src}} &:= \delta C_{\text{buf}} c_{\text{PPE},\text{TV}}^{\text{src}}, \\ \beta_{\text{phase}}^W &:= \min \left\{ \begin{array}{c} \delta \Theta_{r,J}^{\text{val}}, \delta \Theta_{r,\text{TV}}^{\text{src}}, \alpha - A_{\text{frac}} \delta, \\ \beta_{\text{PPE},J}^{\text{val}}, \beta_{\text{PPE},\text{TV}}^{\text{src}} \end{array} \right\} > 0. \end{aligned} \quad (233)$$

Put $N = N_{\text{time}} = \max\{m, n\}$ as required by (127), and choose

$$K_N = e^{\eta_{\text{lo}} N}, \quad H_N = e^{\eta_{\text{hi}} N}, \quad 0 < \eta_{\text{lo}} < \eta_{\text{hi}}. \quad (234)$$

For a scalar mode put $\Lambda = \max\{1, |k|, |l|\}$ and retain $\Xi = \max\{|k|a_X, |l|a_Y\}$. Every balanced mode in the intermediate range

$$\Xi > K_N, \quad \Lambda \leq H_N, \quad (235)$$

is first split as follows. Split every Jensen carrier into its phase-bad fibre restriction and its complement. Let $J_{\text{ph}}^{\text{clock}}$ be the sum of $J_{G_{\text{main}}}$, all phase-good Jensen restrictions, and the r/PPE phase-bad Jensen and TV restrictions, with main and non-main parts both included. Let $J_{\text{time}}^{\text{bad/TV}}$ be the N_{time} phase-bad Jensen restriction plus the complete time-TV restriction. These two currents, together with the separately owned face/word and deep currents, give the exact restriction obtained from the global owner identity and the two clock identities. Every balanced mode of $J_{\text{ph}}^{\text{clock}}$ in (235) has saving $C\Xi^{-\beta_{\text{phase}}^W}$ times a fixed power of $\text{Mark}(Q)$ on scale-good labels. The scale-bad label mass is $O(e^{-c_{\text{scl}}N})$ and is estimated without phase or PPE. Modes with $\Lambda > H_N$ are likewise estimated without PPE $_N$ and have total bound

$$CH_N^{-\rho_\infty} \|g\|_{C^{r_{\text{obs}}}} \|f\|_{C^{r_{\text{obs}}}}, \quad \rho_\infty := r_{\text{obs}} - 2 > 0. \quad (236)$$

If $r_{\text{obs}} > 2 + \beta_{\text{phase}}^W$, the complete high lattice sum is exponentially small in the two induced depths. Here all modes are those of the original localized observables furnished by Lemma 6.59; the induced depths occur only in the endpoint phase. Identity (253) assigns every localized quotient mode exactly once. Low graph modes, rank-one corrections, and the regular product block are treated together by Proposition 6.73. If $a_{\text{ultra}} \geq 0$ denotes the explicitly recorded time-complexity exponent outside the original-observable tail (in the present organization $a_{\text{ultra}} = 0$), the required ultra-high inequality is

$$\eta_{\text{hi}} \rho_\infty > a_{\text{ultra}}. \quad (237)$$

The original-observable lattice sum of the remaining time subcurrent obeys

$$\sum_{k,l} |\widehat{g}_k \widehat{f}_l| \left| \langle J_{\text{time}}^{\text{bad/TV}}, \Phi_{k,l} \rangle \right| \leq C e^{-\min\{c_{\text{time},J}^{\text{val}}, c_{\text{time},\text{TV}}^{\text{src}}\}N} \|g\|_{C^{r_{\text{obs}}}} \|f\|_{C^{r_{\text{obs}}}}. \quad (238)$$

This proposition makes no estimate of $J_{\text{main}}^{\text{face/word}} + J_{\text{face/word}}$ or $J_{\text{main}}^{\text{deep}} + J_{\text{deep}}$. The former restrictions are routed to the selected face-time interface and $\text{FACE}_{2\text{CUT}}$; the latter are routed to $\text{TAIL}_{\text{env}}^+/\text{NREC}_{\text{deep}}$.

Proof. For a mode in (235), take $r = \Xi^{-\delta}$. The energy theorem bounds the resonant mass by

$$C\Xi^{-\delta\Theta_{r,J}^{\text{val}}} + C\Xi^{-\delta C_{\text{buf}}^{\text{val}} c_{\text{PPE},J}^{\text{val}}}. \quad (239)$$

The owner identity (151), followed by identities (160) and (191), together with the lifted inclusion (190), first give exactly the split stated in the proposition. The face/word and deep restrictions are removed before this phase estimate. Then $N_{\text{PPE}}(r) = C_{\text{buf}} \log(1/r) + O(1)$; no time-depth symbol is substituted for this resolving depth. The r - and PPE-depth source occurrences in the TV column are outside Jensen and contribute, by weighted absolute summation,

$$C\Xi^{-\delta\Theta_{r,\text{TV}}^{\text{src}}} + C\Xi^{-\delta C_{\text{buf}}^{\text{src}} c_{\text{PPE},\text{TV}}^{\text{src}}}; \quad (240)$$

this is the origin of $\beta_{\text{PPE},\text{TV}}^{\text{src}}$ and has no square-root loss. The same support may contribute several occurrences, all of which are present in (162). The N_{time} phase-bad Jensen restriction is bounded by $c_{\text{time},J}^{\text{val}}$ and the time-TV restriction by $c_{\text{time},\text{TV}}^{\text{src}}$. Since the coefficients are those of the original localized observables, their absolute lattice sum is finite; weighted absolute summation proves (238). On every phase-good Jensen carrier use two smooth cutoffs in $\psi'_\xi/(\Xi r)$, one for each sign. They vanish on the resonant boundaries. Sum adjacent physical cells before taking absolute values, so their artificial endpoint terms cancel

with opposite orientation. The remaining face, singularity, and homogeneity endpoints obey

$$\sum_Q \mathfrak{E}(a_Q) \leq Cr^{-1} \sum_Q M_Q \text{Mark}(Q) \{1 + Z(\mathcal{G}_Q)\}, \quad (241)$$

by Proposition 6.80. Proposition 6.62 gives

$$\|a_{\pm}\|_{\infty} + [a_{\pm}]_{\alpha} \leq C \text{Mark}(Q) r^{-1}.$$

Furthermore

$$\lambda = c\Xi r, \quad \Phi \leq C\Xi, \quad [\psi'_{\xi}]_{\alpha} \leq C\Xi \text{Mark}(Q).$$

Lemma 6.63 therefore gives

$$\begin{aligned} |\mathcal{I}_Q(\xi)| &\leq C \text{Mark}(Q)^2 \{(1 + Z(\mathcal{G}_Q))\Xi^{-1}r^{-2} + \Xi^{-\alpha}r^{-(3+\alpha)}\} \\ &\leq C \text{Mark}(Q)^2 (1 + Z(\mathcal{G}_Q))\Xi^{-\beta_{\text{phase}}^W}, \end{aligned}$$

where (239) and (240) are included through the two typed PPE rates in (233). The joint estimate (283) sums the cells without separating the density and boundary moments. The two restriction estimates (221) add at most $\mathfrak{d}_{\lambda}^{2q_{\text{end}}}$, which is absorbed by $A_{\text{dom}} \geq 2 + 2q_{\text{end}}$ in the global ledger.

First restrict to the scale-good labels of Lemma 6.46. Then $a_X + a_Y \leq Ce^{C_{\text{scl}}N}$ and $\Lambda \leq H_N$, hence $\Xi \leq Ce^{(C_{\text{scl}} + \eta_{\text{hi}})N}$. A mode in (235) therefore calls the prescribed escape estimate only to depth

$$N_{\text{PPE}} \leq C_{\text{buf}} \delta \log \left(Ce^{(C_{\text{scl}} + \eta_{\text{hi}})N} \right) \leq \{C_{\text{buf}} \delta (C_{\text{scl}} + \eta_{\text{hi}}) + o(1)\} N. \quad (242)$$

Scale-bad labels are never sent through PPE: total variation, the absolute Fourier coefficient sum, and the tilted estimate (106) applied to the complete weight $\mathscr{W}_{\lambda,*}$ give $Ce^{-c_{\text{scl}}N}$. Neither the unweighted scale-bad probability nor the mere finiteness of $\sum_{\lambda} M_{\lambda} \mathscr{W}_{\lambda,*}$ is used for this step. For $\Lambda > H_N$ use no phase cutoff and no physical escape. Sum the two original-observable coefficient estimates (221) directly. After decreasing the exponent once, a dyadic shell of radius R in the two scalar modes has $O(R^2)$ terms of total size at most $CR^{2-r_{\text{obs}}}$. The shells $R > H_N$ therefore give $CH_N^{-(r_{\text{obs}}-2)}$, proving (236). The fixed inverse-scale and cutoff costs remain in A_{dom} ; there is no frequency-dependent physical mark in this tail.

For completeness, put $K = K_N = e^{\eta_0 N}$ with η_0 in the window (270). After global centering and the dominant-coordinate reduction, the face-adapted projector $\mathbf{S}_K^{\text{dom}}$ retains precisely the scalar modes satisfying $|\xi_1|a_X \leq K$ and $|\xi_2|a_Y \leq K$. Proposition 6.73 treats this finite sum by delayed recovery and separate label tails. Every balanced mode of the complementary dominant sum has $\Xi > K$ by definition, so the preceding saving is $e^{-\eta_0 \beta_{\text{phase}}^W N}$ in the intermediate range. The ultra-high range contributes $e^{-(\eta_{\text{hi}} \rho_{\infty} - a_{\text{ultra}})N}$ by (236). Contracted-coordinate terms were removed before the cutoff and are exponentially small by Lemma 6.61. The difference between the graph trace and its same-branch section is already part of that contracted error; no fixed-fiber average is used. The two unbalanced scalar cones are estimated, with their full lattice sum, by Lemma 6.65. The two-dimensional dominant lattice sum of the original g, f coefficients converges because $r_{\text{obs}} > 2 + \beta_{\text{phase}}^W$. Together with scale-bad labels the complete central exponent is

$$c_{\text{cen}}^W := \min\{\eta_0 \beta_{\text{phase}}^W, c_{\text{time}, \text{J}}^{\text{val}}, c_{\text{time}, \text{TV}}^{\text{src}}, \eta_{\text{hi}} \rho_{\infty} - a_{\text{ultra}}, c_{\text{scl}}, c_{\text{low}}\} > 0, \quad (243)$$

where c_{low} is supplied by the low recovery theorem and $c_{\text{time,J}}^{\text{val}}, c_{\text{time,TV}}^{\text{src}} > 0$ are the two genuine N_{time} value/TV source rates. The first already contains the global base-energy Cauchy loss; it is not halved a second time. The second, which includes the contracted-coordinate main remainder in the present organization, is summed by TV. All N_{PPE} and r -polynomial exceptional events have already entered β_{phase}^W and do not appear a second time. Lemma 6.61 supplies the actual typed remainder, while Lemma 6.47 supplies the resolving depth only in the intermediate range, where (242) is finite. Global marginal corrections are inserted through the typed current/test adjoint identity before any absolute value is taken. The regular $M \times M$ density is not placed in the face lattice: Corollary 6.74 uses its independent $W^{R,1}$ smoothing tail. $\square$

6.11 Low frequencies after dominant-coordinate reduction

The cutoff is imposed only after the graph term has been reduced to its two scalar dominant coordinates. This avoids the false implication that a large ambient Fourier vector must have a large restriction to a one-dimensional face. Contracted coordinates are estimated before the cutoff.

Definition 6.67 (Dominant event norm and separate dynamic-Hölder norm). *Fix a finite physical product atlas and assemble coincident artificial faces before inserting its partition of unity. Use the already fixed references $\nu^- = \nu^+ = \mu_0$ and set $dm_{M \times M} = d\nu^- \otimes d\nu^+$. For a face label λ let $a_{X,\lambda}, a_{Y,\lambda} > 0$ be the two scales in (98) and put*

$$\mathfrak{d}_\lambda = 1 + a_{X,\lambda}^{-1} + a_{Y,\lambda}^{-1}.$$

Write

$$\mathcal{K}_{\text{ev}} = k_{\text{reg}} dm_{M \times M} + \sum_{\lambda} M_{\lambda}(H_{\lambda})_*(w_{\lambda} d\bar{s})$$

. Fix a dominant-scale ledger exponent $A_{\text{dom}} \geq 2 + 2q_{\text{end}}$. It covers the $O(K^2 \mathfrak{d}_{\lambda}^2)$ scalar mode count and the fixed scaled chart costs; it does not encode either boundary complexity. Define the two parameter-free recovery marks

$$\mathcal{R}_{\lambda}^{\pm} := C_R [1 + \log \mathfrak{d}_{\lambda} + \log \text{Mark}(\lambda) + \log\{1 + Z(\mathcal{G}_{\lambda}^{\pm})\}]. \quad (244)$$

For an integer $1 \leq R \leq 3$ allowed by the C^3 table regularity, define

$$\|\mathcal{K}_{\text{ev}}\|_{\text{phys,R}} := \sum_{\chi} \|\zeta_{\chi} k_{\text{reg}}\|_{W^{R,1}} + \sum_{\lambda} M_{\lambda} \|w_{\lambda}\|_{\mathcal{A}^{\alpha}} \mathfrak{d}_{\lambda}^{A_{\text{dom}}} \text{Mark}(\lambda)^2 \{1 + Z(\mathcal{G}_{\lambda}^{-}) + Z(\mathcal{G}_{\lambda}^{+})\}. \quad (245)$$

For bounded $F : M^q \rightarrow \mathbb{C}$ let

$$\|F\|_{\text{SH}} := \|F\|_{\infty} + \sum_{i=1}^q \sup_{x_j: j \neq i} \sup_{x_i \neq y_i} \frac{|F(\dots, x_i, \dots) - F(\dots, y_i, \dots)|}{\theta_0^{s_i(x_i, y_i)}}. \quad (246)$$

The stable or unstable separation time appropriate to the i th slot is used, as in [10, Definition 2.2].

We now define the low-frequency object on the scalar quotient on which it actually lives. Fix an even cutoff $\chi \in C_c^{\infty}((-2, 2))$ with $\chi = 1$ on $[-1, 1]$. After the branch-section reduction, the assembled graph piece induces the finite signed scalar measure

$$\nu_{\lambda} = (X_{\lambda}, Y_{\lambda})_*(M_{\lambda} w_{\lambda} d\bar{s}), \quad \widehat{\nu}_{\lambda}(k, l) = \int e^{-i(kX_{\lambda} + lY_{\lambda})} M_{\lambda} w_{\lambda} d\bar{s}. \quad (247)$$

Define the anisotropic multiplier and its polynomial density by

$$m_{\lambda,K}(k,l) = \chi(a_{X,\lambda}k/K)\chi(a_{Y,\lambda}l/K), \quad (248)$$

$$A_{\lambda,K}^{\text{dom}}(x,y) = (2\pi)^{-2} \sum_{(k,l) \in \mathbb{Z}^2} m_{\lambda,K}(k,l) \widehat{\nu}_\lambda(k,l) e^{i(kx+ly)}. \quad (249)$$

This is a density on the scalar quotient and nowhere else. Define the two mode currents on the branch sections by

$$\eta_{\lambda,k}^-(h) := \int h(\iota_\lambda^-(X)) \chi_\lambda^-(X) e^{ikX} dX, \quad (250)$$

$$\eta_{\lambda,l}^+(f) := \int f(\iota_\lambda^+(Y)) \chi_\lambda^+(Y) e^{ilY} dY, \quad (251)$$

where the fixed scalar cutoffs are part of the localized quotient chart. They equal one on a neighborhood of $\text{supp } \nu_\lambda$ and their supports stay inside the homogeneous section domains. The branch-section lift $\mathfrak{L}_{\lambda,K}^{\text{sec}}$ is

$$\int h(x)f(y) d\mathfrak{L}_{\lambda,K}^{\text{sec}}(x,y) := \int h(\iota_\lambda^-(X))f(\iota_\lambda^+(Y))\chi_\lambda^-(X)\chi_\lambda^+(Y)A_{\lambda,K}^{\text{dom}}(X,Y) dX dY. \quad (252)$$

If $P_{\lambda,K}$ denotes the even quotient multiplier in (248) and $H(X,Y) = h(\iota_\lambda^-(X))f(\iota_\lambda^+(Y))$, then self-adjointness of the multiplier and the support clause give the exact localized identity

$$\int H d\nu_\lambda - \int \chi_\lambda^- \chi_\lambda^+ H A_{\lambda,K}^{\text{dom}} dX dY = \int (I - P_{\lambda,K})(\chi_\lambda^- \chi_\lambda^+ H) d\nu_\lambda. \quad (253)$$

Thus the only localization error is the same quotient high-pass term treated by the balanced/unbalanced phase estimates; it is not an unrecorded leakage of the Fourier polynomial outside the branch chart. Fourier inversion on the quotient gives the exact finite-rank identity

$$\mathfrak{L}_{\lambda,K}^{\text{sec}} = (2\pi)^{-2} \sum_{k,l} m_{\lambda,K}(k,l) \widehat{\nu}_\lambda(k,l) \eta_{\lambda,k}^- \otimes \eta_{\lambda,l}^+. \quad (254)$$

Thus finite rank is obtained in the tensor product of two concrete measured standard-curve currents. The original graph current is not claimed to be nuclear or absolutely continuous.

For a finite signed kernel σ on $M \times M$ put

$$\langle \text{Op}(\sigma)h, f \rangle := \int_{M \times M} h(x)f(y) d\sigma(x,y). \quad (255)$$

Fix once and for all a finite product atlas, a subordinate smooth product partition $\{\zeta_\chi := \zeta_\chi^-(x)\zeta_\chi^+(y)\}$, product super-cutoffs $\zeta_\chi = 1$ on $\text{supp } \zeta_\chi$, and extension maps E_χ supported where the corresponding super-cutoff is one and bounded simultaneously on L^1 and on $W^{r,1}$ for every $1 \leq r \leq R \leq 3$. We use the fixed periodic box $\mathbb{T}^4 = (\mathbb{R}/2\pi\mathbb{Z})^4$ with normalized Haar density $(2\pi)^{-4}dz$ and Fourier convention $\widehat{h}(q) = (2\pi)^{-4} \int_{\mathbb{T}^4} h(z) e^{-iq \cdot z} dz$. Choose a real even $\vartheta \in C_c^\infty(\mathbb{R}^4)$ with $\vartheta = 1$ on $[-1,1]^4$ and $\vartheta = 0$ outside $[-2,2]^4$, and set

$$\mathcal{P}_K^{(4)}k := \sum_\chi \widetilde{\zeta}_\chi \kappa_\chi^* \left[\sum_{q \in \mathbb{Z}^4} \vartheta(q/K) \widehat{E_\chi(\zeta_\chi k)}(q) e^{iq \cdot z} \right]. \quad (256)$$

This is a signed flat spectral smoother, not a positive ordinary mollifier. On each chart it contains at most CK^4 characters and satisfies

$$\|(I - \mathcal{P}_K^{(4)})k\|_{L^1} \leq C_R K^{-R} \|k\|_{W^{R,1}}, \quad 1 \leq R \leq 3. \quad (257)$$

Let $\check{\vartheta}$ be the Euclidean inverse Fourier transform. Flatness at the origin gives $\int_{\mathbb{R}^4} z^\alpha \check{\vartheta}(z) dz = 0$ for $1 \leq |\alpha| \leq R-1$, while its Schwartz decay gives a finite absolute R th moment. The Fourier series in (256) is convolution by the periodization of $K^4 \check{\vartheta}(Kz)$, equivalently Euclidean convolution of the periodic extension. Taylor's integral remainder on $\mathbb{R}^4$ therefore gives the $K^{-R} W^{R,1}$ bound without asserting polynomial moment cancellation on a chosen torus fundamental domain. The simultaneous $L^1/W^{R,1}$ extension bounds and the fixed partition give (257). The super-cutoffs preserve the localized input and only multiply the finitely many characters by fixed product chart functions. Hence every four-dimensional character remains a product of two endpoint characters, and the super-cutoffs create neither a K -dependent mode count nor an extra derivative loss.

Lemma 6.68 (Oscillatory section currents are recoverable). *For each mode current in (250)–(251), its real and imaginary parts are finite signed differences of positive measured standard families. Their total masses are bounded by the section mark, their scaled density marks are at most $CK\mathfrak{d}_\lambda \text{Mark}(\lambda)$, and they are proper after*

$$R_{\lambda,K}^\pm \leq C_0 + C_1 \log K + \mathcal{R}_\lambda^\pm. \quad (258)$$

The same assertion holds for the reversed collision map. If $\tau_\pm \in (0,1)$ are the two proper-family coupling rates, then the centered mode currents obey the delayed one-sided bounds

$$|(\eta_{\lambda,k}^\pm)^\circ(Q_\pm^j h)| \leq C \text{Mark}(\lambda) \tau_\pm^{(j-R_{\lambda,K}^\pm)_+} \|h\|_{C^1}. \quad (259)$$

Proof. Relative to the positive coordinate/arclength density on the section, the multiplier is a fixed cutoff times e^{ikX} . Split real and imaginary parts and add twice their supremum to obtain positive densities, exactly as in Lemma 6.83. One scalar derivative costs $C|k| \leq CK/a_{X,\lambda}$ (and similarly in Y), which gives the displayed scaled mark. The density recursion contracts this mark in $O(\log(K\mathfrak{d}_\lambda \text{Mark}(\lambda)))$ iterates. The boundary growth recursion has the affine form $Z(F^{n*}\mathcal{G}) \leq \gamma Z(\mathcal{G}) + Z_0$ and therefore costs $O(\log(1+Z))$ iterates. Forward/reverse proper-family growth and coupling are uniform on the compact table class, proving (258). Apply exponential coupling only after that time and total variation before it. This proves (259) without converting the two recovery windows into a product of initial boundary complexities. $\square$

There are two different centering operations, and we keep their types separate. If σ is a finite signed measure on $M \times M$, write $\sigma_\pm = (\text{pr}_\pm)_* \sigma$ and $|\sigma|_0 = \sigma(M \times M)$, and define

$$\mathbf{D}_{\text{cur}} \sigma := \sigma - \sigma_- \otimes \nu^+ - \nu^- \otimes \sigma_+ + |\sigma|_0 \nu^- \otimes \nu^+. \quad (260)$$

Its test-side adjoint is

$$\begin{aligned} \mathbf{D}_{\text{test}} F(x, y) &:= F(x, y) - \int F(x, y') d\nu^+(y') - \int F(x', y) d\nu^-(x') \\ &\quad + \int F(x', y') d\nu^-(x') d\nu^+(y'), \end{aligned} \quad (261)$$

so that

$$\int F d(\mathbf{D}_{\text{cur}} \sigma) = \int \mathbf{D}_{\text{test}} F d\sigma. \quad (262)$$

In particular a current is never silently fed to an operator defined on functions. On the regular density use the flat band-limited smoother (256) and define

$$\begin{aligned} \mathcal{S}_K^{\text{dom}} \mathcal{K}_{\text{ev}} &:= \mathcal{D}_{\text{cur}} \left([\mathcal{P}_K^{(4)} k_{\text{reg}}] dm_{M \times M} + \sum_{\lambda} \mathfrak{L}_{\lambda, K}^{\text{sec}} \right), \\ \mathcal{H}_K^{\text{dom}}(\mathcal{D}_{\text{cur}} \mathcal{K}_{\text{ev}}) &:= \mathcal{D}_{\text{cur}} \mathcal{K}_{\text{ev}} - \mathcal{S}_K^{\text{dom}} \mathcal{K}_{\text{ev}}. \end{aligned} \quad (263)$$

All artificial endpoints are assembled before this definition. Expanding $\mathcal{D}_{\text{cur}}$ centers each pair of section-mode currents and includes the two marginal and constant product blocks explicitly. The definition of $\mathcal{H}_K^{\text{dom}}$ gives the exact operator decomposition

$$\text{Op}(\mathcal{D}_{\text{cur}} \mathcal{K}_{\text{ev}}) = \text{Op}(\mathcal{S}_K^{\text{dom}} \mathcal{K}_{\text{ev}}) + \text{Op}(\mathcal{H}_K^{\text{dom}}(\mathcal{D}_{\text{cur}} \mathcal{K}_{\text{ev}})). \quad (264)$$

Theorem 6.69 (Branch-section low projection and exact finite rank). *For $K \geq 2$ the centered current in (263) has the exact finite-rank expansion*

$$\mathcal{S}_K^{\text{dom}} \mathcal{K}_{\text{ev}} = \sum_{j \in \mathcal{J}_K} c_j (\eta_j^-)^{\circ} \otimes (\eta_j^+)^{\circ}, \quad (265)$$

where $\eta^{\circ} = \eta - \eta(1)\nu^{\pm}$. Face factors are the concrete section currents (250)–(251); regular-density factors are ordinary absolutely continuous currents. Before either recovery estimate is used, the face modes have the total-variation mass budget

$$\sum_{\lambda} \sum_{k, l} |c_{\lambda, k, l}| \|(\eta_{\lambda, k}^-)^{\circ}\|_{\text{TV}} \|(\eta_{\lambda, l}^+)^{\circ}\|_{\text{TV}} \leq CK^2 \sum_{\lambda} M_{\lambda} \|w_{\lambda}\|_{L^1} \mathfrak{d}_{\lambda}^2 \text{Mark}(\lambda)^2. \quad (266)$$

The constant is independent of return and grazing truncations, and (264) holds before any absolute value is taken. This is a finite-rank statement only after scalar cutoff; no nuclear or absolutely continuous representation of the original graph current is asserted. In particular, no product $(1 + Z(\mathcal{G}_{\lambda}^-))(1 + Z(\mathcal{G}_{\lambda}^+))$ is hidden in (266).

Proof. The coefficient bound $|\widehat{\nu}_{\lambda}(k, l)| \leq M_{\lambda} \|w_{\lambda}\|_{L^1}$ follows directly from (247). The multiplier has at most $CK^2 \mathfrak{d}_{\lambda}^2$ nonzero pairs. Formula (254) gives their exact product-current expansion. Each centered section current has total variation at most a fixed multiple of $\text{Mark}(\lambda)$, independently of its recovery time. Absolute summation of the coefficients and modes proves (266). Formula (260) replaces each factor by its centered version, and (262) proves the four marginal identities. The definition of the remainder proves the exact low/high operator identity; Fourier inversion is used only on the scalar quotient. The preliminary replacement of the physical graph trace by the branch sections is charged separately to (224) before the cutoff. $\square$

Definition 6.70 (Separate recovery-tail input $\text{REC}(q_{\text{rec}})$). *For $q_{\text{rec}} > 0$, the terminal face-label law satisfies $\text{REC}(q_{\text{rec}})$ if, separately for the two orientations,*

$$\sum_{\lambda} M_{\lambda} \|w_{\lambda}\|_{\mathcal{A}^{\circ}} \mathfrak{d}_{\lambda}^{A_{\text{dom}}} \text{Mark}(\lambda)^2 e^{q_{\text{rec}} \mathcal{R}_{\lambda}^{\pm}} < \infty. \quad (267)$$

The two displayed sums are taken under the same terminal label law, but their integrands are not multiplied. Thus this input is a pair of marginal exponential recovery tails, not a pointwise product- Z moment.

Lemma 6.71 (Regular high-frequency tail). *For the regular $M \times M$ density,*

$$\|(I - \mathcal{P}_K^{(4)})k_{\text{reg}}\|_{L^1(M \times M)} \leq CK^{-R}\|k_{\text{reg}}\|_{W^{R,1}(M \times M)}. \quad (268)$$

Consequently its pairing with $Q^m g$ and $(Q^)^n f$ is bounded by the right-hand side times $\|g\|_\infty \|f\|_\infty$, uniformly in m, n .*

Proof. This is (257); Markov normalization gives $\|Q^m h\|_\infty \leq 2\|h\|_\infty$. $\square$

Lemma 6.72 (Regular low projector). *Let $c_-, c_+ > 0$ be the two one-state mixing exponents. The centered regular low block satisfies*

$$\left| \left\langle Q^n \text{Op}\left(\mathcal{D}_{\text{cur}}[(\mathcal{P}_K^{(4)}k_{\text{reg}})dm_{M \times M}]\right) Q^m g, f \right\rangle \right| \leq CK^6 e^{-c_- m - c_+ n} \|k_{\text{reg}}\|_{L^1} \|g\|_{C^1} \|f\|_{C^1}. \quad (269)$$

Proof. In a fixed product atlas the four-dimensional cutoff has $O(K^4)$ Fourier modes and every coefficient is bounded by $C\|k_{\text{reg}}\|_{L^1}$. Each two-dimensional endpoint character has C^1 cost at most CK . After current centering, Fubini makes every mode a product of two centered one-state correlations. Their two coupling bounds contribute $e^{-c_- m - c_+ n}$, while the mode count and endpoint costs contribute $K^4 K^2$. Summing the fixed atlas proves (269). $\square$

Proposition 6.73 (Central low frequency by delayed bi-recovery). *Let $c_-, c_+ > 0$ be the uniform past and future two-state mixing rates supplied by proper-family coupling, let $D \geq 1$ define the diagonal band $D^{-1}m \leq n \leq Dm$, and assume $\text{REC}(q_{\text{rec}})$. Fix $0 < \epsilon < 1/2$, put $N = \max\{m, n\}$ and $K = e^{\eta_0 N}$, and choose*

$$6\eta_0 < \frac{1}{2} \min\{c_-, c_+\}, \quad 2\eta_0 < (1-\epsilon) \min\{c_-, c_+\}, \quad (q_{\text{rec}}C_1 + 2)\eta_0 < q_{\text{rec}}\epsilon/D. \quad (270)$$

Then, for some $c_{\text{low}} > 0$ independent of m, n ,

$$\left| \left\langle Q^n \text{Op}(\mathcal{S}_K^{\text{dom}} \mathcal{K}_{\text{ev}}) Q^m g, f \right\rangle \right| \leq C e^{-c_{\text{low}}(m+n)} \|g\|_{C^r} \|f\|_{C^r}. \quad (271)$$

Proof. Insert (265) into the product-current pairing. After the face modes of one label are summed, Fubini, (259), and (266) give

$$CM_\lambda \|w_\lambda\|_{L^1} K^2 \mathfrak{d}_\lambda^2 \text{Mark}(\lambda)^2 e^{-c_-(m-R_{\lambda,K}^-)_+} e^{-c_+(n-R_{\lambda,K}^+)_+} \|g\|_{C^1} \|f\|_{C^1}.$$

There is no condition $y = Tx$: the two factors are independent one-time correlations. Split the labels into the good set $R_{\lambda,K}^- \leq \epsilon m$, $R_{\lambda,K}^+ \leq \epsilon n$ and the two bad sets where one of these inequalities fails. On the good set the first two inequalities in (270) absorb the K^2 mode count. On the past-bad set, (258) implies

$$\mathcal{R}_\lambda^- > \epsilon m - C_0 - C_1 \eta N.$$

Use total variation for both factors and the minus instance of (267). Its contribution is at most a constant times

$$\exp\{-[q_{\text{rec}}\epsilon/D - (q_{\text{rec}}C_1 + 2)\eta]N\}.$$

The plus-bad set is identical and uses only the plus recovery moment. A union bound, rather than a product of the two initial complexities, completes the face estimate. The flat band-limited four-dimensional low projector of the regular density is bounded by Lemma 6.72 and is absorbed by the first inequality in (270). Since $N \asymp_D m+n$, decreasing the exponent proves (271). $\square$

Corollary 6.74 (Regular defect via functional correlation). *Assume all hypotheses of Proposition 6.73 and Lemmas 6.72 and 6.71, including the proper-family coupling, recovery moments, common event/test norms, and a uniform $W^{R,1}$ regular-density ledger. Then The regular part of K_V has central-band exponential decay: its finite-rank low part is covered by Proposition 6.73, and Lemma 6.71 gives $K^{-R} = e^{-\eta_0 R N}$ for its high part. No tensor, wavelet, or face-phase estimate is used.*

6.12 Radial faces: direct flux boundary control

We retain every connected component. Let $\mathcal{W}_\lambda(j, k) = \{W_{\lambda,j,k,r}\}_r$ be the maximal components of one coarea face with incoming strip j and outgoing strip k . Put

$$p_{\lambda,j,k,r} = \int_{W_{\lambda,j,k,r}} \rho_\lambda^{\text{raw}} ds, \quad M_\lambda = \sum_{j,k,r} p_{\lambda,j,k,r}, \quad \bar{p}_{\lambda,j,k,r} = p_{\lambda,j,k,r} / M_\lambda.$$

Thus $\bar{p}$, not p , is the weight of the normalized standard family.

Lemma 6.75 (Oriented refinement bookkeeping). *Assume a face has been cut into $N_{\text{or}}(\lambda)$ signed charts on each of which either the forward or the reversed input and output angle coordinates are monotone. Then for fixed incoming/outgoing signed homogeneity ranks (j, k) there are at most $4N_{\text{or}}(\lambda)$ connected components and*

$$\sum_{j,k,r} \frac{1}{(1+j)^2(1+k)^2} \leq C N_{\text{or}}(\lambda). \quad (272)$$

The existence of such a cover and every estimate for N_{or} are not deduced from cone transversality or a qualitative jet condition; their weighted form is an explicit part of FZ_{geom} below.

Proof. On an unstable graph the cone bounds give dr/dt and $d\varphi/dt$ one fixed sign after the orientation of t is chosen. Invariance of the unstable cone gives the same statement after one collision; the reversed cone gives the backward version. A signed homogeneity strip is an interval in the angle coordinate. The preimages of an incoming and an outgoing signed strip are therefore intervals on the oriented chart, and their intersection is an interval. The two grazing signs in each time orientation account for the factor four. Summing over charts and then over j, k proves (272). This proves only the bookkeeping implication from an already constructed oriented cover; it makes no analytic zero-count assertion. $\square$

Lemma 6.76 (Full tangent coarea Jacobian). *Let $\gamma(t) = (r_-(t), \varphi_-(t))$ be a unit-speed parametrization of an input coarea face and let $W = T\gamma$. On every oriented component the unnormalized density with respect to output arclength is*

$$\rho_{W,\lambda}^{\text{raw}}(T\gamma(t)) = \frac{|a_\lambda| J_{\text{chart},\lambda} J_{\text{hol},\lambda} \cos \varphi_-(t)}{|\nabla G_\lambda(\gamma(t))| |DT(\gamma(t))\gamma'(t)|}. \quad (273)$$

If output impact coordinate r_+ is used instead of arclength, the last factor in the denominator is $|d(r_+ \circ \gamma)/dt|$. For the vertical family $\gamma(t) = (r_0, t)$ this becomes

$$\rho_{r_+}^{\text{raw}} = \frac{|a_\lambda| J_{\text{chart},\lambda} J_{\text{hol},\lambda} \cos \varphi_- \cos \varphi_+}{|\nabla G_\lambda| \tau}, \quad (274)$$

which is precisely the calculation of Lemma 5.1 of [22]. Formula (273), not the vertical special case, is used for a general physical face. The reversed orientation has the same change-of-variables form.

Proof. Input arclength coarea and the collision-SRB density give the measure

$$b_\lambda(t) dt, \quad b_\lambda(t) = \frac{|a_\lambda| J_{\text{chart}, \lambda} J_{\text{hol}, \lambda} \cos \varphi_-(t)}{|\nabla G_\lambda(\gamma(t))|}.$$

The one-dimensional area formula for the injective oriented branch $t \mapsto T\gamma(t)$ divides this density by the curve Jacobian $J_\gamma T(t) = |DT(\gamma(t))\gamma'(t)|$, proving (273). Parametrizing the image by r_+ instead divides by $|d(r_+ \circ \gamma)/dt|$. When r_- is fixed, the billiard differential identity $dr_+/d\varphi_- = -\tau/\cos \varphi_+$ yields (274). Applying the same area formula to T^{-1} proves the reversed statement without asserting that a general tangent is vertical. $\square$

Definition 6.77 (Geometric flux and component condition FZ_{geom}). *A face family satisfies $\text{FZ}_{\text{geom}}(\theta_Z)$ if the density of Lemma 6.76 has marks $\mathfrak{M}_\lambda, \mathfrak{C}_\lambda \geq 1$ such that*

$$p_{\lambda, j, k, r} \leq C \mathfrak{M}_\lambda \frac{|W_{\lambda, j, k, r}|}{(1+j)^2(1+k)^2}, \quad (275)$$

$$\sum_{j, k, r} \frac{1}{(1+j)^2(1+k)^2} \leq \mathfrak{C}_\lambda := C N_{\text{or}}(\lambda), \quad (276)$$

and, for some $q_F > 1$,

$$\sum_\lambda M_\lambda \left[\text{Mark}(\lambda)^2 \left\{ 1 + \frac{\mathfrak{M}_\lambda \mathfrak{C}_\lambda}{M_\lambda} \right\} \right]^{q_F} < \infty. \quad (277)$$

Pieces whose tangent makes angle at most ϵ with both usable cones have total marked mass at most $C\epsilon^{\theta_Z}$. This is the required near-stable angle-sublevel input. On the complementary good-angle pieces, the product of all inverse-angle costs in the graph chart, density, nonstationary phase, and recovery maps is at most $C\epsilon^{-A_{\text{ang}}}$ for a fixed $A_{\text{ang}} > 0$. Both constants are part of FZ_{geom} . This condition contains no multi-time mixing or split-law assertion.

Definition 6.78 (Split-law standard-family and positive-envelope interface). *Let $F_{s,L}^{\text{col}}$ be the sharp four-state functional of the assembled finite-parameter collar, after coincident artificial faces have cancelled. For every (m, n) , expansion of the global test-side double projection gives the exact signed law*

$$\Sigma_{m,n}^{\text{law}} = \rho_{m,n}^0 - \sum_{j=1}^J a_j \rho_{m,n}^j, \quad \rho_{m,n}^0 = \rho_{m,n}^{\text{orig}}, \quad (278)$$

where the $\rho_{m,n}^j$ are the block-product laws in the functional-correlation split; all $\rho_{m,n}^j$, including $\rho_{m,n}^0$, are probability laws, $J \leq J_0$, and $Z_{\text{law}} := 1 + \sum_j |a_j| \leq Z_0$ uniformly. Define the positive envelope

$$\bar{\rho}_{m,n} := Z_{\text{law}}^{-1} \left(\rho_{m,n}^0 + \sum_{j=1}^J |a_j| \rho_{m,n}^j \right). \quad (279)$$

The interface SF_{split} requires that the restriction of every constituent law in (278), in either usable time orientation, disintegrates into bounded-mark proper standard families with the same conditional mark- Z ledger as the original collar. This is a primitive joint-law statement, not a consequence of the one-coordinate marginals.

Let $\mathcal{P}_r^{m,n}$ be the common rank- r product-cylinder partition, refined into connected event-chart components. For a bounded F put

$$\Omega_r^{\text{env}}(F) := \sum_{C \in \mathcal{P}_r^{m,n}} \bar{\rho}_{m,n}(C) \underset{C, \bar{\rho}_{m,n}}{\text{ess osc}} F. \quad (280)$$

The connected-cylinder pullback condition CP_{env} says that, outside pieces of total envelope mass $Ce^{-c_{\text{bad}}r}$, every such component pulls back to a good-angle homogeneous event cell of physical diameter at most $Ce^{-\chi r}$, with the conditional density and component marks in FZ_{geom} and SF_{split} . Grazing, near-stable, and homogeneity-boundary cells are included in the displayed bad mass, with every component counted. There are fixed $A_{\text{amp}} > 0$ and $\alpha_{\text{amp}} \in (0, 1]$ such that, on a good cell, the smooth coefficient multiplying each sharp collar indicator has oscillation at most $C|s|^{-1}e^{A_{\text{amp}}L}e^{-\chi\alpha_{\text{amp}}r}$. The connected refinement is required to be separation-time admissible: two distinct retained atoms either receive the same cylinder value or differ in some coordinate before rank r . Hence a function constant on its atoms has separate dynamic-Hölder cost at most $Ce^{q_{\text{cyl}}r}$ times its supremum. In particular no artificial chart boundary may create an uncharged jump between symbolically indistinguishable points.

Proposition 6.79 (Geometric envelope dynamical tube). *Assume $\text{FZ}_{\text{geom}}(\theta_Z)$, SF_{split} , and CP_{env} . Then there are $A_{\text{col}} > 0$ and $\alpha_{\text{col}} \in (0, 1]$ such that*

$$\sup_{m,n} \Omega_r^{\text{env}}(F_{s,L}^{\text{col}}) \leq C|s|^{-1}e^{A_{\text{col}}L}e^{-\alpha_{\text{col}}r}, \quad r \geq 1. \quad (281)$$

Proof. On a good connected component, split the oscillation into variation of the smooth coefficient and variation of a sharp indicator. The first part is bounded by the scaled Hölder clause in CP_{env} . The second produces two points on opposite sides of a genuine face. Connectedness and the diameter bound imply that the whole component lies in the $Ce^{-\chi r}$ tube of that face. By (279), its envelope mass is the fixed positive combination of the actual and split-law masses. Apply the full tangent area formula and the standard-family conditional density bound to every constituent. The good-angle part costs $C|s|^{-1}e^{A_{\text{col}}L}e^{-\chi\alpha_0r}$ for a geometric tube exponent $\alpha_0 > 0$, after taking $A_{\text{col}} \geq A_{\text{amp}}$ and decreasing α_0 to at most α_{amp} . The complement costs $C|s|^{-1}e^{A_{\text{col}}L}e^{-c_{\text{bad}}r}$ by CP_{env} . Balancing the angle cutoff with the ϵ^{θ_Z} sublevel and decreasing the exponent once gives (281). Artificial boundaries were cancelled before (278), and connected components were indexed before the diameter argument, so neither can create an uncharged crossing. The standard-pair strip coverage of [21] is a template for verifying CP_{env} ; it is not used as a black box. $\square$

Proposition 6.80 (Flux-weighted radial boundary lemma). *Assume $\text{FZ}_{\text{geom}}(\theta_Z)$ and decompose the signed amplitude into positive pieces. Then the normalized standard family obeys*

$$M_\lambda Z(\mathcal{G}_\lambda) = \sum_{j,k,r} \frac{p_{\lambda,j,k,r}}{|W_{\lambda,j,k,r}|} \leq C\mathfrak{M}_\lambda \mathfrak{C}_\lambda, \quad (282)$$

and, for the actual physical labels,

$$\sum_\lambda M_\lambda \text{Mark}(\lambda)^2 \{1 + Z(\mathcal{G}_\lambda^-) + Z(\mathcal{G}_\lambda^+)\} < \infty. \quad (283)$$

and the near-stable pieces have the marked ϵ^{θ_Z} tail in Definition 6.77.

Proof. Divide (275) by the component length and sum over the full triple (j, k, r) . Equation (276) gives (282); no component is suppressed. Multiply by the squared density/amplitude mark and use (277) (first with power one) to obtain (283). The full tangent Jacobian, weighted component complexity, and angle sublevel are the three falsifiable geometric inputs; none is deduced merely from the homogeneity labels. $\square$

Lemma 6.81 (Explicit angle balance). *Suppose the good-angle core of a central or off-diagonal estimate is $Ce^{-c_{\text{core}}N}$ before inverse-angle loss. Set*

$$\epsilon_N = e^{-\zeta N}, \quad \zeta = \frac{c_{\text{core}}}{A_{\text{ang}} + \theta_Z}.$$

Then the sum of the bad-angle and good-angle contributions is at most $Ce^{-c_Z N}$, where

$$c_Z = \frac{c_{\text{core}}\theta_Z}{A_{\text{ang}} + \theta_Z} > 0. \quad (284)$$

Proof. The bad-angle mass is $e^{-\zeta\theta_Z N}$, while the good-angle estimate is $e^{-(c_{\text{core}} - A_{\text{ang}}\zeta)N}$. The displayed choice makes the two exponents equal. $\square$

Corollary 6.82 (Uniform radial recovery). *For each time orientation, the radial family is regular after at most*

$$R^\pm \leq \alpha_1 \log\{1 + Z(\mathcal{G}_{\text{rad}}^\pm)\} + \alpha_2 \quad (285)$$

iterates. After multiplication by physical mass, the recovery-window sum in all four time quadrants is finite and is controlled by (283).

Proof. Apply [22, Corollary 6.15(c)] to the positive standard families in each orientation. Use $\log(1 + Z) \leq C_\varepsilon(1 + Z)^\varepsilon$ and multiply by M_λ . The joint mark- Z ledger then sums both orientations and all pre-recovery iterates. This is the weighted hypothesis of Proposition 4.1. $\square$

Lemma 6.83 (Observable multiplier to positive proper families). *Let (W, ρ) be a positive standard pair and let $q_m = (g \circ T^{-m})\rho$ on the time orientation for which W is unstable. There is $C_m \leq C\|g\|_{C^\alpha}$ such that*

$$q_m = (q_m + C_m\rho) - C_m\rho$$

is a difference of positive densities. Their total masses are bounded by $C\|g\|_{C^\alpha}$ and their scaled log-Hölder marks are at most $Ce^{\chi m} \text{Mark}(W, \rho)$. For every admissible slowly moving sequence in the compact table class, after

$$R_m \leq C_0 + C_1 m + C_2 \log\{1 + Z(\mathcal{G})\} \quad (286)$$

iterates, both positive families are proper. The reversed collision map has the analogous statement.

Proof. Take $C_m = 2\|g\|_\infty$. Positivity and the mass bound are immediate. Hyperbolicity and bounded distortion give $[g \circ T^{-m}]_{\alpha, \text{sc}} \leq Ce^{\chi m}\|g\|_{C^\alpha}$. Since the added density is at least $C_m\rho/2$, division by it gives the same scaled log-Hölder bound. More explicitly, the density recursion in Lemma 13 of [6] has the form $H_{j+1} \leq \theta H_j + C$ on every homogeneous descendant. Starting from $H_0 \leq Ce^{\chi m} \text{Mark}(W, \rho)$, it reaches the fixed regular cone after $C_0 + C_1 m$ iterates. The normalized boundary recursion in Lemma 16 of [6] is $Z_j/M \leq C\{1 + \theta_p^j Z_0/M\}$, so a further $C_2 \log(1 + Z_0/M)$ iterates make the family proper. Theorems 1'-2' of [6] make all constants uniform for the compact slowly moving table class. The fixed-map statement follows by taking a constant sequence, and time reversal gives the last assertion. $\square$

Proposition 6.84 (Signed proper-family mapping and off-diagonal bounds). *Assume the forward and reversed billiard growth/coupling theorem on positive proper standard families, $\text{FZ}_{\text{geom}}(\theta_Z)$, and the scaled density bound of Proposition 6.62. Decompose every signed face amplitude as the difference of two positive amplitudes before normalization. Then there are positive exponents $\alpha_{\pm}, \beta_{\pm}$ such that the complete face current satisfies*

$$|B_{m,n}| \leq Ce^{\beta-m-\alpha_+n}, \quad (287)$$

$$|B_{m,n}| \leq Ce^{-\alpha_-m+\beta_+n}. \quad (288)$$

The constants include all recovery windows and are uniform under the physical and flux ledgers.

Proof. Apply Lemma 6.83 on each positive face piece. Its recovery time is bounded by (286). Equations (282) and (285) control the boundary part of that recovery. Coupling for the remaining $n - R_{\lambda}^+$ iterates gives

$$CM_{\lambda} \text{Mark}(\lambda)^2 (1 + Z_{\lambda}^+) e^{\beta-m-\alpha_+(n-R_{\lambda}^+)_+}.$$

Before recovery use total variation. The inequality $e^{\alpha_+R_{\lambda}^+} \leq Ce^{\alpha_+C_1m}(1 + Z_{\lambda}^+)^{\alpha_+C_2}$ and (283), with a weakened coupling exponent chosen below the available mark- Z moment, sum all labels and recovery windows. Absorb $e^{\alpha_+C_1m}$ into $e^{\beta-m}$. This is (287). Apply the same argument to the reversed collision map for (288); positive/negative pieces are recombined only after the two bounds are established. $\square$

7 Slope-energy and fractional-phase proof architecture

The proof now has six noncircular links.

- (F1) Quotient the exponential Young return on the regular magnet to the one-state countable $C^{1+\alpha}$ IFS of Proposition 6.10. Physical side labels remain in the signed amplitude fiber.
- (F2) Lemma 6.17 combines the exponential return tail with the maximal grazing rank and proves the joint moment needed for the countable quotient and the marked coarea amplitudes.
- (F3) The normal-cycle value $\mathfrak{A} = -1$ motivates the clean template; SS and the primitive physical escape PPE_N , CEM_DOM , and $\text{TENERGY}_W + \text{CLOCK_PROB}_W + \text{CLOCK_SRC}_W$, together with the mass-preserving complete stopped-source tree, give the weighted energy theorem and Jensen.
- (F4) Proposition 6.26 and smooth value-space cutoffs and Lemma 6.63 give the balanced Fourier estimate without curvature or componentwise phase partitions.
- (F5) Global centering is performed before the physical chart sum. Lemma 6.61 removes contracted modes; Proposition 6.73 treats the remaining low modes, while Proposition 6.66 treats the high modes.
- (F6) At each cutoff coarea currents converge on a fixed core in the explicit bounded-Lipschitz current topology (24), with the boundary tightness (25). The no- $|s|^{-1}$ middle tail in $\text{FACE}_{2\text{CUT}}$ then upgrades this to growing depth. The component-indexed FZ_{geom} estimate, the SF_{split} and CP_{env} interfaces, QT_{path} , and the strict scalar-tail margin remove L, H before recovery summation.

This order makes clear which external results are used. Leclerc’s Lemma 3.15 supplies the cohomological bracket and his Propositions 4.31–4.33 supply an equilibrium/Cantor cutting template with all-scale polynomial escape on that factor. They do not supply the actual-collision-SRB prescribed-depth estimate PPE_N used here; that estimate remains a primitive input, with the adaptive full-mass alphabet recorded only as a nonvacuity strategy. The final oscillatory estimate is the elementary fractional phase lemma proved above; LPS is retained only as background for the quotient route. Bertozzi’s triple inducing supplies a useful model for the one-state gate architecture, not the slope conclusion. The positive entrance/exit comparison is the symbolic local product argument of [36]. The one-step expansion, linear singularity complexity, and uniform billiard constants used by the master atlas are summarized in [2].

The route deliberately does not use the recent curvilinear two-ends theorem or L^2 flattening: their nonlinear applications require C^2 or higher geometry, whereas the billiard quotient available here is only $C^{1+\alpha}$. It also does not infer a product tower tail from two marginal Chernoff bounds: the diagonal product exponent comes from time comparability and functional correlation, while the off-diagonal regions are handled before the three-region assembly.

7.1 Execution audit

The central analytic chain is now represented by explicit propositions. The remaining primitive geometric inputs are

$$\begin{aligned} & \text{SS, ACTUAL_SLOPE_BRIDGE,} \\ & \text{PAIR_SAFE_REFERENCE, PAIR_TENERGY, PPE}_N, \\ & \text{BR}_{\text{sec}}, \text{CEM_DOM, TENERGY}_W, \text{DUAL_MASS_KERNEL,} \\ & \text{MPD, POST_CARRIER_DOM,} \\ & \text{PHYS_TEST_LIFT, PHYS_TREE_MATCH,} \\ & \text{PROP_Q_MATCH,} \\ & \text{MAIN_CLOCK_COVERAGE, SEL_JSL}_{p_{\text{joint}}}, \\ & \text{CLOCK_PROB}_W \\ & \text{CLOCK_SRC}_W, \text{KRH}_{p_W}, \text{FZ}_{\text{geom}}, \text{SF}_{\text{split}}, \\ & \text{CP}_{\text{env}}, \text{FACE}_{2\text{CUT}}, \text{FACE_TIME}_{\text{CM2}} \text{ or } \text{FACE_TIME}_{\text{REC}}, \text{QT}_{\text{JET}}, \text{QT}_{\text{LEDGER}}. \end{aligned}$$

The exact gate interface is Lemma 6.18, Proposition 6.19, and Lemma 6.47; the slope-energy theorem is Theorem 6.51; the balanced phase is Proposition 6.66; and the low-frequency collision-time estimate is Proposition 6.73. Radial recovery is applied only after these estimates through Corollary 6.82 and Proposition 4.1.

The finite-parameter collar is treated below by the actual/split-law cylinder approximation theorem; no fixed-slice smoothing estimate is used.

Lemma 7.1 (Parameter derivatives of contracting return words). *Let $h_{j,s}$ be the inverse one-step branches in a fixed compact regular atlas and assume, uniformly in j and $|s| < s_0$,*

$$\|Dh_{j,s}\|_\infty \leq \lambda < 1, \quad \|\partial_s h_{j,s}\|_{C^1} + \|\partial_s \log |h'_{j,s}|\|_\infty \leq A.$$

For a clean word $a = (a_1, \dots, a_R)$ let $h_{a,s} = h_{a_1,s} \circ \dots \circ h_{a_R,s}$. Then

$$\|\partial_s h_{a,s}\|_\infty \leq \frac{A}{1-\lambda}, \quad \|\partial_s \log |h'_{a,s}|\|_\infty \leq C(1+R). \quad (289)$$

The same linear-in- R bound holds for products of uniformly C^1 Jacobian, stable-holonomy, and coarea factors. Hence an exponential return moment dominates the difference quotients of all clean marked amplitudes and their return tail.

Proof. Differentiate the composition. The contribution created at position k is multiplied on the left by at least $k - 1$ contracting derivatives, so its C^0 norm is bounded by $A\lambda^{k-1}$; summation gives the first bound. The logarithmic derivative of a composition is a sum of R one-step logarithmic derivatives. Differentiating it gives a uniformly bounded direct term at every position and orbit-displacement terms controlled by the first bound and uniform C^2 distortion. This gives $C(1+R)$. The product statement follows after taking logarithms. Finally $R^p e^{-\epsilon R}$ is summable for every fixed p , so the exponential return moment absorbs all these factors. $\square$

This lemma shows that long return words do not create an exponential parameter-derivative loss on the inverse-branch side. The remaining issue is not the clean word calculus but the common refinement of words that approach or cross a moving singularity; those pieces must be charged to the already constructed face/recovery tree rather than differentiated as clean branches.

That singular refinement is most naturally made before fixing the parameter.

7.2 Parameter-state geometry and cutoff removal

Definition 7.2 (Pathwise quantitative jet condition QT_{JET}). *Let $a(s)$ be the chosen analytic or circular table deformation with $a(0) = a_0$. For every face pair of depth at most L , let $\Delta_{\lambda\lambda'}(s)$ be a finite collection of jet minors covering the compact common-zero set after coincident germs are grouped. Add the individual face-cone boundary determinants and their critical-point jets; these are the functions used to form the oriented tangent cuts in Lemma 6.75. The path satisfies QT_{JET} if there are $A, B, c_{\text{lab}} > 0$ and a base-dependent $c_{\text{base}} > 0$ such that*

$$|\Delta_{\lambda\lambda'}(0)| \geq c_{\text{base}} e^{-AL}, \quad \sup_{|s| < \bar{s}} |\partial_s \Delta_{\lambda\lambda'}(s)| \leq C e^{BL}, \quad (290)$$

$$\#\{(\lambda, \lambda') : |\lambda| \leq L\} \leq C e^{c_{\text{lab}} L}. \quad (291)$$

The constant c_{base} absorbs the finitely many depths not covered by an asymptotic Borel–Cantelli statement. No return-tail or geometric-loss claim is part of QT_{JET} .

Definition 7.3 (Pathwise geometric-loss and return-tail ledger $\text{QT}_{\text{LEDGER}}$). *Condition $\text{QT}_{\text{LEDGER}}$ requires constants $A_{\text{loss}}, c_{\text{tail}} > 0$ such that all inverse-angle, coarea, chart, density, parameter-derivative, component, and Mark–Z losses on the depth- L good restriction are at most $C e^{A_{\text{loss}} L}$. Separately, the actual unsmoothed positive-envelope mass of the complementary return/grazing failure owner whose first required depth exceeds L is at most $C e^{-c_{\text{tail}} L}$. No decay of the good mass itself is asserted. Relative to the A, B furnished by QT_{JET} , the strict path margin is*

$$c_{\text{tail}} - A_{\text{loss}} > A + B. \quad (292)$$

Parameter randomness for the jet minors does not prove either ledger bound.

Definition 7.4 (Complete quantitative path condition). *Write*

$$\text{QT}_{\text{path}} := \text{QT}_{\text{JET}} \wedge \text{QT}_{\text{LEDGER}}.$$

For a finite-dimensional ambient family, the sublevel estimate

$$\text{Leb}\{a : |\Delta_{\lambda\lambda'}(a)| < e^{-AL}\} \leq Ce^{-A\tau L}, \quad A\tau > c_{\text{lab}},$$

is a sufficient Borel–Cantelli mechanism for the first inequality at almost every base parameter, but it is not a substitute for the derivative bound or for QT_{LEDGER}.

Theorem 7.5 (Product-a.e. compact analytic realization of the pathwise jet bound). *Let each scatterer boundary vary in a fixed complex analytic strip of width $\sigma_a > 0$. Fix a strip slack $\delta_a > 0$, put $\sigma_p = \sigma_a + \delta_a$, and parametrize a neighborhood by*

$$a_q = e^{-\sigma_p|q|}b_q, \quad b_q \in [-1, 1]. \quad (293)$$

Equip the b_q with an independent compactly supported product law whose one-dimensional densities are uniformly bounded. The support of (293) is compact in the width- σ_a analytic topology and hence in the chosen compact C^3 table chart. When this theorem is used with the fixed-gauge CM2 criterion, this product chart is additionally required to be an analytic coordinate chart on the constraint manifold (1); independent unconstrained modes on different components do not qualify. Suppose the depth- L face atlas has at most $Ce^{c_{\text{lab}}L}$ noncoincident jet charts and that

$$\begin{aligned} &\text{each compact jet chart has a cover by at most } Ce^{c_{\text{zero}}L} \text{ fiber boxes;} \\ &\text{on every box one index } q \leq C_0L \text{ is fixed for the } b_q \text{ restriction;} \\ &\text{that restriction has at most } Ce^{c_{\text{zero}}L} \text{ monotonicity intervals, and} \end{aligned} \quad (294)$$

$$|\partial_{b_q}\Delta| \geq e^{-(B_0+\sigma_p C_0)L} \quad \text{whenever } |\Delta| \leq 2e^{-AL}.$$

The fiber condition may equivalently be supplied by a uniform Remez–Cartan/valency estimate with the same exponent c_{zero} . Then, for every

$$B_0 + \sigma_p C_0 + c_{\text{lab}} + 2c_{\text{zero}} < A, \quad (295)$$

product-almost every analytic base table satisfies $|\Delta_{\lambda\lambda'}(0)| \geq e^{-AL}$ for all sufficiently large L . After decreasing a base-dependent constant $c_{\text{base}} > 0$, the bound $|\Delta_{\lambda\lambda'}(0)| \geq c_{\text{base}}e^{-AL}$ holds at every finite depth as well. (The finitely many noncoincident minors have zero sets of product measure zero under the fiber hypothesis.) For every analytic direction whose depth- L derivative cost is at most e^{BL} , the resulting path satisfies QT_{JET}. The theorem does not supply QT_{LEDGER}, the return/grazing tail, or the strict combined margin. Small analytic perturbations remain in the finite-horizon dispersing class.

For a fixed finite branch family, the adapted boundary-perturbation construction of [12] supplies the relevant finite-jet surjectivity template: perturb a boundary arc at the earliest differing collision and then realize the required finite jet by an analytic trigonometric polynomial. Condition (294) is the quantitative, simultaneous version needed here. Finite-branch jet surjectivity alone does not control fiber valency. For analytic minors one may verify the stated condition by a quantitative propagation-of-smallness/Remez estimate such as [18]; if the branch equations form a Nash family, the Bernstein–Remez theorem [19] is an alternative. The theorem applies only when the corresponding strip, valency, and normalized-coordinate costs are verified for every face and cone-angle jet in the master atlas. It makes no assertion for the finite-dimensional circular subfamily.

Proof. Condition on all normalized Fourier coefficients except b_q furnished by (294). The one-dimensional change of variables on each monotonicity interval and the bounded conditional density give

$$\mathbb{P}\{|\Delta| < e^{-AL}\} \leq Ce^{2c_{\text{zero}}L}e^{-\{A-B_0-\sigma_p C_0\}L}.$$

The first $e^{c_{\text{zero}}L}$ counts fiber boxes and the second counts monotonicity intervals; the exponential Fourier weight is already present in ∂_{b_q} . Union over at most $Ce^{c_{\text{lab}}L}$ labels. The first inequality in (295) makes the resulting series in L summable, so Borel–Cantelli gives the base-minor bound. Analytic strip norms control the derivative along a fixed analytic direction by Ce^{BL} . The finitely many remaining noncoincident minors give c_{base} after excluding their null zero sets. Finally, strict convexity, disjointness, and finite horizon are open in the chosen compact C^3 neighborhood. $\square$

Lemma 7.6 (Persistence up to a depth depending on the parameter). *Assume QT_{path} and choose*

$$\frac{1}{c_{\text{tail}} - A_{\text{loss}}} < \kappa_L < \frac{1}{A + B}, \quad L(s) = \lfloor \kappa_L \log(1/|s|) \rfloor. \quad (296)$$

For all sufficiently small s and all $L \leq L(s)$,

$$|\Delta_{\lambda\lambda'}(\theta s)| \geq \frac{1}{2}c_{\text{base}}e^{-AL}, \quad 0 \leq \theta \leq 1.$$

Thus every noncoincident depth- $L(s)$ face family remains transverse along the whole coarea segment from 0 to s .

Proof. The mean-value theorem and (290) give

$$|\Delta(\theta s) - \Delta(0)| \leq C|s|e^{BL}.$$

For $L \leq L(s)$ the ratio of this error to $c_{\text{base}}e^{-AL}$ is at most $C(c_{\text{base}})|s|^{1-\kappa_L(A+B)} = o(1)$. The interval for κ_L is nonempty precisely because of (292). $\square$

Definition 7.7 (Shallow spacetime common refinement UCR_{sh}). *For every sufficiently small s and every $L \leq L(s)$, cut $[0, 1]_\theta \times M$ by all transported singularity, homogeneity, gate, and coarea hypersurfaces used by words of depth at most L . Condition UCR_{sh} requires a measurable master refinement with the following properties.*

- (UCR1) *On each shallow spacetime cell the word, orientation, connected-component index, and actual/split-law labels are independent of θ ; the branch and endpoint maps extend to the cell and obey the same cone, distortion, scaled-density, and parameter-derivative bounds uniformly for $0 \leq \theta \leq 1$.*
- (UCR2) *The actual native stopping antichains, branch sections, PPE_N labels, full-tangent coarea charts, positive envelopes, and connected cylinder partitions used at depth at most L are restrictions of this one refinement. Adjacent artificial boundaries cancel before absolute values, and every genuine moving boundary keeps its component index and Mark–Z charge. No assertion about all deeper words is included in this clause.*
- (UCR3) *The union of shallow cells on which such a continuation fails has, for every actual or split constituent law, total marked mass at most $Ce^{-c_{\text{ucr}}L}$ after the loss already recorded in A_{loss} . We replace c_{tail} everywhere by $\min\{c_{\text{tail}}, c_{\text{ucr}}\}$ and require the same strict margin (292). The shallow refinements are compatible when L increases.*

The constants in

$$\begin{aligned} & \text{NST}_{\text{phys}}, \text{BR}_{\text{sec}}, \text{PPE}_N, \\ & \text{ACTUAL_SLOPE_BRIDGE}, \text{PAIR_SAFE_REFERENCE}, \\ & \text{PAIR_TENERGY}, \text{FZ}_{\text{geom}}, \text{SF}_{\text{split}}, \text{CP}_{\text{env}}, \text{REC}(q_{\text{rec}}) \end{aligned}$$

CEM_DOM, TENERGY_W, DUAL_MASS_KERNEL, MPD,
 POST_CARRIER_DOM, PHYS_TEST_LIFT,
 PHYS_TREE_MATCH, PROP_Q_MATCH,
 MAIN_CLOCK_COVERAGE, SEL_JSL_{p_{joint}},
 CLOCK_PROB_W,
 CLOCK_SRC_W, KRH_{p_W}, USC_{end}^W(q_g^W)

on the complete actual/split tree laws are required to be uniform on these cells, including (188), the selection-safe likelihood bounds (198), and the main-source base bounds (208), together with (207) whenever the main time-Jensen class is nonempty. This is a primitive moving-singularity interface; compactness of the table chart alone does not imply it.

Definition 7.8 (Normalized shallow long-time source SHCOL_{env}). *Let $\mathcal{K}_s^{\leq L}$ be the assembled difference quotient carried by face words of depth at most L . Condition SHCOL_{env} is the conjunction of the following three typed interfaces.*

- (SH1) SHSRC⁺: *before signed recombination, the exact actual/split expansion is dominated by propagated positive-envelope pieces. If $C_{\lambda,s,L}^{\text{sh}}$ are their shallow owner cells, put $M_{\lambda,s,L}^{\text{sh}} := \overline{M}_{s,0}^q(C_{\lambda,s,L}^{\text{sh}})$, uniformly in every suppressed master coordinate, and*

$$\Delta_{\text{sh}}(s, L) := \sum_{\lambda} M_{\lambda,s,L}^{\text{sh}} \leq C|s|^{-1}e^{A_{\text{sh}}L}, \quad \|\mathcal{K}_s^{\leq L}\|_{\text{TV}} \leq \Delta_{\text{sh}}(s, L). \quad (297)$$

Artificial-face cancellation is used only after this positive size is recorded. Thus A_{sh} already includes the operator, source norm, recovery, and opposite-amplitude growth absorbed in q ; no unweighted physical mass is substituted for this envelope.

- (SH2) SHCP_{env}: *the corresponding actual/split laws have one positive envelope and one connected, separation-time-admissible cylinder approximant. For the bounded shallow collar functional $F_{s,L}^{\text{sh}}$,*

$$\|F_{s,L}^{\text{sh}}\|_{\infty} \leq C|s|^{-1}e^{A_{\text{sh}}L}, \quad \Omega_r^{\text{env}}(F_{s,L}^{\text{sh}}) \leq C|s|^{-1}e^{A_{\text{sh}}L}e^{-\alpha_{\text{sh}}r}. \quad (298)$$

The constants $c_{\text{FC}}, q_{\text{cyl}}$ are those in Theorem 7.15.

- (SH3) SHREC(q_{sh}): *when $\Delta_{\text{sh}}(s, L) > 0$, put $\pi_{s,L}^{\text{sh}}(\lambda) = M_{\lambda,s,L}^{\text{sh}}/\Delta_{\text{sh}}(s, L)$ and attach the two recovery times $R_{\lambda,s,L}^{\pm, \text{sh}}$ and a coupling mark $\mathcal{W}_{\lambda,s,L}^{\text{sh}} \geq 1$. Separately in the two orientations,*

$$\sup_{s,L} \sum_{\lambda} \pi_{s,L}^{\text{sh}}(\lambda) \mathcal{W}_{\lambda,s,L}^{\text{sh}} e^{q_{\text{sh}} R_{\lambda,s,L}^{\pm, \text{sh}}} \leq C, \quad q_{\text{sh}} > c_{\text{cpl}}^{\text{sh}}. \quad (299)$$

After recovery the forward and reverse standard families couple at rate $c_{\text{cpl}}^{\text{sh}}$.

This is an all-time positive-source condition; it is not inferred from a depth- L common partition. The small-hole estimate $Ct\gamma^n$ and the affine boundary recursion in [22] are its structural model: normalize the small source, recover, and only then multiply back its mass.

Lemma 7.9 (Shallow recovery keeps the positive source prefactor). *Fix the central-band constant $\rho_{\text{sh}} > 0$ and choose $0 < \varepsilon < \rho_{\text{sh}}/2$. Under SHREC(q_{sh}) put*

$$\begin{aligned} c_{\text{cen}}^{\text{sh}} &:= \min\{c_{\text{cpl}}^{\text{sh}}(1 - 2\varepsilon), q_{\text{sh}}\varepsilon\}, \\ c_{\text{one}}^{\text{sh}} &:= \min\{c_{\text{cpl}}^{\text{sh}}(1 - \varepsilon), q_{\text{sh}}\varepsilon\}, \\ c_{\text{off}}^{\text{sh}} &:= \min\{c_{\text{cen}}^{\text{sh}}, c_{\text{one}}^{\text{sh}}\}. \end{aligned} \quad (300)$$

In the central band $\min\{m, n\} \geq \rho_{\text{sh}}(m+n)$,

$$\sum_{\lambda} M_{\lambda,s,L}^{\text{sh}} \mathcal{W}_{\lambda,s,L}^{\text{sh}} \exp\{-c_{\text{cpl}}^{\text{sh}}[(m - R_{\lambda}^{-})_{+} + (n - R_{\lambda}^{+})_{+}]\} \leq C \Delta_{\text{sh}}(s, L) e^{-c_{\text{off}}^{\text{sh}}(m+n)}. \quad (301)$$

The one-sided versions with rate $c_{\text{one}}^{\text{sh}}$ hold in the two off-diagonal cones, measured in their respective long time coordinate.

Proof. Split each orientation into $R^{\pm} \leq \varepsilon(m+n)$ and its complement. On the central good set the recovered time is at least $(1 - 2\varepsilon)(m+n)$; the two bad sets each cost $e^{-q_{\text{sh}}\varepsilon(m+n)}$. Markov's inequality under $\pi_{s,L}^{\text{sh}}$ and (299) charges the two bad sets separately. This proves the first minimum in (300). In an off-diagonal cone only the long orientation is coupled; splitting at $R^{\pm} \leq \varepsilon$ times that long coordinate gives the second minimum. Multiply the normalized estimate by $\Delta_{\text{sh}}(s, L)$. No product of the two initial recovery marks is formed. $\square$

Definition 7.10 (Positive deep envelope tail $\text{TAIL}_{\text{env}}^{+}$). Let $\mathcal{R}_s^{>L}$ be the complete double-centered remainder formed by words deeper than L . Before signed recombination, dominate every actual/split constituent by the restriction of the propagated positive envelope to its deep owner cell $C_{\lambda,s,L}^{\text{deep}}$. This cell is required to lie in

$$(I_{\text{deep}}(\varrho) \sqcup I_{\text{main}}^{\text{deep}}|_{\mathfrak{d}}) \cap \{a : \kappa(a) = \text{word}, \ell_{\text{word}}(a) > L\}, \quad (302)$$

with the main term interpreted at the active linked query as in (189). Put $M_{\lambda,s,L} := \overline{M}_{s,\mathfrak{d}}^q(C_{\lambda,s,L}^{\text{deep}})$, uniformly in every suppressed master coordinate. Define

$$\Delta_{s,L} := \sum_{\lambda} M_{\lambda,s,L} \leq C|s|^{-1} e^{-(c_{\text{tail}} - A_{\text{loss}})L}, \quad \|\mathcal{R}_s^{>L}\|_{\text{TV}} \leq \Delta_{s,L}. \quad (303)$$

The finite split coefficients and all operator/source/recovery costs are included in this same q -mass, so A_{loss} is the post-absorption loss and the gap $c_{\text{tail}} - A_{\text{loss}}$ must be recomputed after any operator growth. Connected-pullback and boundary-oscillation constants are uniform in θ, s, L . Artificial faces cancel only in the final signed recombination; no reverse comparison of positive envelope mass by the assembled signed TV norm is assumed.

Definition 7.11 (Normalized positive-envelope deep recovery $\text{NREC}_{\text{deep}}(q_{\text{deep}})$). For $\Delta_{s,L} > 0$ put $\pi_{s,L}(\lambda) = M_{\lambda,s,L}/\Delta_{s,L}$ and attach recovery times $R_{\lambda,s,L}^{\pm}$ and coupling mark $\mathcal{W}_{\lambda,s,L} \geq 1$. Condition $\text{NREC}_{\text{deep}}(q_{\text{deep}})$ requires, separately in the two orientations,

$$\sup_{s,L} \sum_{\lambda} \pi_{s,L}(\lambda) \mathcal{W}_{\lambda,s,L} e^{q_{\text{deep}} R_{\lambda,s,L}^{\pm}} \leq C, \quad q_{\text{deep}} > \max\{\alpha_{+}, \alpha_{-}\}. \quad (304)$$

Equivalently,

$$\sum_{\lambda} M_{\lambda,s,L} \mathcal{W}_{\lambda,s,L} e^{q_{\text{deep}} R_{\lambda,s,L}^{\pm}} \leq C \Delta_{s,L}. \quad (305)$$

The small positive source is normalized first, recovered second, and multiplied back last.

Lemma 7.12 (Normalized standard-family criterion for deep recovery). Assume that, after division by $\Delta_{s,L}$, every actual/split deep constituent is a probability standard family and

$$\sup_{s,L} \sum_{\lambda} \pi_{s,L}(\lambda) \mathcal{W}_{\lambda,s,L} (1 + Z_{\lambda,s,L})^{q_{\text{deep}} C_1} < \infty. \quad (306)$$

If the growth lemma gives $R_\lambda^\pm \leq C_0 + C_1 \log(1 + Z_\lambda^\pm)$ after a refinement with the same conditional tail, then $\text{NREC}_{\text{deep}}(q_{\text{deep}})$ holds.

Proof. Exponentiate the recovery inequality and integrate against $\pi_{s,L}$. Equation (306) gives (304); multiplying by $\Delta_{s,L}$ gives (305). The affine boundary recursion and leaky standard-family construction of [22] are the model for the normalized estimate. No cancellation-stability inequality is used. $\square$

Proposition 7.13 (Depth-dependent coarea and pathwise scalar tail). *Assume $|\nabla_x G_\lambda| \geq c_G > 0$ on every active face and QT_{path} , UCR_{sh} , $\text{TAIL}_{\text{env}}^+$, the joint Mark–Z ledger (283), and the physical/source identification (348). In particular the return/grazing tail and A_{loss} in the conclusion are the actual positive-envelope quantities furnished by $\text{QT}_{\text{LEDGER}}$, not consequences of UCR_{sh} . For all depths $L \leq L(s)$,*

$$\frac{\mathbf{1}_{\{G_\lambda(s, \cdot) > 0\}} - \mathbf{1}_{\{G_\lambda(0, \cdot) > 0\}}}{s} = \int_0^1 \delta(G_\lambda(\theta s, \cdot)) \partial_s G_\lambda(\theta s, \cdot) d\theta \quad (307)$$

is a uniformly controlled coarea current. The omitted deeper words satisfy

$$|s|^{-1} C e^{-(c_{\text{tail}} - A_{\text{loss}})L(s)} \leq C |s|^{\kappa_L(c_{\text{tail}} - A_{\text{loss}}) - 1} = o(1). \quad (308)$$

This is a scalar total-variation tail. Its conversion to an $o(1)$ tail in the infinite CM2 double sum is a separate time–depth argument below; no such conversion is inferred from (308) alone. No fixed neighborhood is asserted to be uniformly transverse at every depth.

Proof. Lemma 7.6 preserves noncoincident faces, while UCR_{sh} supplies the common labels, components, branches, and uniform geometric bounds along θs for shallow words. The return/grazing tail, the joint mark–Z ledger, and the allowed geometric loss give $C e^{-(c_{\text{tail}} - A_{\text{loss}})L}$ before the difference quotient. Apply the total-variation bound to both endpoints for words deeper than $L(s)$ and divide by $|s|$. The first inequality in (296) gives (308). $\square$

Lemma 7.14 (Finite-parameter collar as a law-adapted functional). *Assume $\text{FZ}_{\text{geom}}(\theta_Z)$, SF_{split} , CP_{env} , UCR_{sh} , $\text{TAIL}_{\text{env}}^+$, and $\text{NREC}_{\text{deep}}(q_{\text{deep}})$. Let $\mathcal{R}_s^{>L}$ be the complete double-centered part of $s^{-1}(P_s - P_0)$ carried by return/grazing words deeper than L . Before the coarea limit it has a full-dimensional density in every local event chart, but its CM2 pairing is*

$$B_{m,n}^{\text{deep}}(s) = \int F_{s,L}^{\text{col}} d\Sigma_{m,n}^{\text{law}}, \quad (309)$$

where $\Sigma_{m,n}^{\text{law}}$ is the exact finite expansion (278). It is therefore not an integral against μ^4 . There are fixed $A_{\text{col}} > 0$ and $\alpha_{\text{col}} \in (0, 1]$ such that

$$\|\mathcal{R}_s^{>L}\|_{\text{TV}} \leq \Delta_{s,L} \leq C |s|^{-1} e^{-(c_{\text{tail}} - A_{\text{loss}})L}, \quad (310)$$

$$\|F_{s,L}^{\text{col}}\|_\infty \leq C |s|^{-1} e^{A_{\text{col}}L}, \quad \Omega_r^{\text{env}}(F_{s,L}^{\text{col}}) \leq C |s|^{-1} e^{A_{\text{col}}L} e^{-\alpha_{\text{col}}r}. \quad (311)$$

The same pieces, disintegrated in either time orientation, are signed differences of regular measured unstable families after their recovery times, and their recovery-weighted mass satisfies (305) with $\Delta_{s,L}$ bounded by the right side of (310).

Proof. On the spacetime cells supplied by UCR_{sh} , the difference of the two moving domains is a collar of width $O(|s|)$ and division by s costs exactly one power of $|s|^{-1}$. Expanding the test-side projection before taking an absolute value gives (309) and (278). The sharp

indicator is not dynamically Hölder. Apply instead Proposition 6.79; this gives (311). Absolute summation with the return/grazing mass gives (310). Disintegrating the collar along the usable unstable orientation preserves positive envelope mass. Normalize this disintegration by $\Delta_{s,L}$ before applying the affine Z recursion, the short-curve tail, and Lemma 6.83. The primitive normalized estimate $\text{NREC}_{\text{deep}}(q_{\text{deep}})$ then gives (305); only afterwards is the factor $\Delta_{s,L}$ restored. Eventual recovery of each piece alone would not imply this weighted estimate. $\square$

Theorem 7.15 (Positive-envelope cylinder approximation). *Let F be a bounded four-state functional and let the signed law and positive envelope be (278)–(279). Suppose $\Omega_r^{\text{env}}(F) \leq V e^{-\alpha r}$ and the common connected partition is separation-time admissible as in CP_{env} . On every rank- r connected product cylinder choose a value in the $\bar{\rho}_{m,n}$ -essential range of F and call the resulting common cylinder function F_r . There is a fixed $q_{\text{cyl}} > 0$ such that*

$$\|F_r\|_{\text{SH}} \leq C \|F\|_{\infty} e^{q_{\text{cyl}} r}, \quad (312)$$

$$\left| \int (F - F_r) d\Sigma_{m,n}^{\text{law}} \right| \leq Z_0 V e^{-\alpha r}. \quad (313)$$

Consequently a functional-correlation split across a gap N obeys

$$|\text{FC}_N(F)| \leq C \left\{ V e^{-\alpha r} + \|F\|_{\infty} e^{-c_{\text{FC}} N + q_{\text{cyl}} r} \right\}. \quad (314)$$

If two mutually singular constituent laws see the values zero and one on the same cylinder, their positive envelope sees both values and the envelope oscillation is one. For the diagonal example, a value constant on every constituent support is represented exactly; if a split constituent sees the opposite side, the envelope charges that discrepancy. Thus the construction never replaces a law-adapted zero error by coordinatewise mollification across the diagonal.

Proof. On each cylinder, the envelope-essential difference from the chosen value is bounded by the envelope-essential oscillation. Using (279),

$$\int |F - F_r| d\rho^0 + \sum_j |a_j| \int |F - F_r| d\rho^j = Z_{\text{law}} \int |F - F_r| d\bar{\rho}_{m,n},$$

which proves (313). Separation-time admissibility in CP_{env} says that a first possible jump occurs only after some separation time drops below r ; it costs at most $C\theta_0^{-r} = C e^{q_{\text{cyl}} r}$ in that slot. This proves (312). Apply [10, Theorems 2.3–2.4] to F_r for the exact signed split and use (313). The same F_r serves all constituent laws because it was constructed from their common positive envelope. No assertion about conditional coordinate marginals is used. $\square$

Theorem 7.16 (Two-threshold removal of shallow long times). *Let*

$$L(s) = \lfloor \kappa_L \log(1/|s|) \rfloor, \quad T_{\text{sh}}(s) = \lfloor \kappa_T^{\text{sh}} \log(1/|s|) \rfloor,$$

and put

$$c_{\text{sh}} := \min \left\{ c_{\text{off}}^{\text{sh}}, \frac{\alpha_{\text{sh}} c_{\text{FC}}}{q_{\text{cyl}} + \alpha_{\text{sh}}} \right\}. \quad (315)$$

Let

$$C_{\text{mid}}(\eta_{\text{hi}}) := C_{\text{ep}} + C_{\text{hom}} + C_{\text{buf}} \delta(C_{\text{scl}} + \eta_{\text{hi}}), \quad (316)$$

where C_{hom} contains the fixed homogeneous and component buffers. Assume the complete labelled ledger (351) and

$$\begin{aligned} & \text{UCR}_{\text{sh}}, \text{NST}_{\text{phys}}, \text{SS}, \text{ACTUAL_SLOPE_BRIDGE}, \\ & \text{PAIR_SAFE_REFERENCE}, \text{PAIR_TENERGY}, \\ & \text{PPE}_N, \text{CEM_DOM}, \text{TENERGY}_W, \\ & \text{DUAL_MASS_KERNEL}, \text{MPD}, \\ & \text{POST_CARRIER_DOM}, \text{PHYS_TEST_LIFT}, \\ & \text{PHYS_TREE_MATCH}, \text{PROP_Q_MATCH}, \\ & \text{MAIN_CLOCK_COVERAGE}, \text{SEL_JSL}_{p_{\text{joint}}} \\ & \text{CLOCK_PROB}_W, \text{CLOCK_SRC}_W, \text{KRH}_{p_W}, \text{USC}_{\text{end}}^W, \text{BR}_{\text{sec}}, \text{REC}, \text{MT}_{\text{DQ}}, \\ & \text{FACE_TIME}_{\text{CM2}} \text{ or } \text{FACE_TIME}_{\text{REC}}, \text{SHCOL}_{\text{env}}. \end{aligned}$$

Assume also $\text{MAIN_CLOCK_COVERAGE}$ and the main-source likelihood and base bounds (198) and (208), uniformly on the same shallow refinement. For every nonempty main time-Jensen cell also assume (207) on the same frozen parents. Require also the strict window

$$C_{\text{mid}}(\eta_{\text{hi}})\kappa_T^{\text{sh}} < \kappa_L, \quad \kappa_T^{\text{sh}}c_{\text{sh}} > 1 + \kappa_L A_{\text{sh}}, \quad \eta_{\text{hi}}\rho_{\infty} > a_{\text{ultra}}. \quad (317)$$

For $N = \max\{m, n\} \leq T_{\text{sh}}(s)$ the $r/\text{PPE}/\text{time-Jensen}$ and TV owners have the product-CM2 majorant from the three-frequency proof, while every main and nonmain face/word owner has the independent majorant (41). Thus the complete shallow difference quotient has the same product-CM2 majorant as the limiting shallow current. For $N > T_{\text{sh}}(s)$,

$$|B_{m,n}^{\text{sh}}(s)| \leq C|s|^{-1-A_{\text{sh}}\kappa_L}e^{-c_{\text{sh}}N}, \quad (318)$$

and consequently

$$\sum_{\max\{m,n\} > T_{\text{sh}}(s)} |B_{m,n}^{\text{sh}}(s)| = o(1). \quad (319)$$

For fixed parameters satisfying the third inequality, the two-sided κ_T^{sh} interval is nonempty precisely when $C_{\text{mid}}(\eta_{\text{hi}})(1 + \kappa_L A_{\text{sh}}) < \kappa_L c_{\text{sh}}$.

Proof. If $N \leq T_{\text{sh}}(s)$, the low block uses only the endpoint/section buffers. In the intermediate block, (242) and the fixed homogeneous buffers give total depth at most $C_{\text{mid}}(\eta_{\text{hi}})N < L(s)$. The ultra-high block uses (236) and no escape or resolving refinement. Scale-bad labels are removed by Lemma 6.46; on scale-good labels the C_{scl} term in C_{mid} is exactly the upper endpoint-scale cost. Thus the common shallow refinement supports the complete three-frequency proof for its Jensen/TV handlers without an infinite-depth call. The face/word handlers removed from that proof are estimated, including their fixed core, by Lemma 1.5; $\text{FACE}_{2\text{CUT}}$ is used only for their word-depth convergence. For $N > T_{\text{sh}}(s)$ do not continue that Fourier proof. In the central band apply Theorem 7.15 to the finite- s collar in Definition 7.8 with

$$r_N = \left\lfloor \frac{c_{\text{FC}}N}{q_{\text{cyl}} + \alpha_{\text{sh}}} \right\rfloor.$$

The two off-diagonal cones use Lemma 7.9; its retained factor $\Delta_{\text{sh}}(s, L)$ is bounded by (297). The definition of c_{sh} makes (318) valid there as well. Since there are $O(N)$ pairs with maximum N , its sum is bounded by

$$C|s|^{-1-A_{\text{sh}}\kappa_L} \sum_{N > T_{\text{sh}}(s)} N e^{-c_{\text{sh}}N} = o(1)$$

by the second inequality in (317). The first inequality is an upper bound for κ_T^{sh} and the second a lower bound, which gives the last assertion. $\square$

Remark 7.17 (Noncircular order of exponent choices). *First prove CEM_DOM, fix the complete event weight $\mathcal{W}_{\lambda,*}$, and fix the five refinement-moment exponents and the genuine complete-tree cemetery exponent q_C . Compute χ_W in (329); if $\chi_W < 1$, choose the actual p_W from (330). For a registry, first subtract aggregate complexity and then degrade the resulting partition rate by $1/p'_W$ only under the appropriate global probability or normalized incidence-law moment. Context, stopping, PPE, and grazing rates are degraded on their corresponding parent kernels only after Lemma 7.20, or an independent KRH_{p_W} primitive, has been verified. Otherwise TENERGY_W , CLOCK_PROB_W , CLOCK_SRC_W , KRH_{p_W} , and $\text{USC}_{\text{end}}^W$ remain independent primitive inputs. Complete both the probability partition and the source-incidence clock matrix before choosing any phase parameter; no bare atomic event rate survives this step. Next fix the coefficient of $\log S_\lambda^\pm$ in $\tilde{G}_N^\pm$ and choose ζ_g ; this simultaneously sets $c_{\text{scl}} = q_g^W \zeta_g$ and enlarges C_{scl} . Then choose δ so that $\alpha - A_{\text{frac}}\delta > 0$. Next choose the smallest admissible C_{buf} that gives $\kappa^{\text{NPE}(r)} \leq r/100$ and the named tail bounds in (233). Then choose η_{hi} so that (237) holds, and verify that*

$$\frac{1 + \kappa_L A_{\text{sh}}}{c_{\text{sh}}} < \kappa_T^{\text{sh}} < \frac{\kappa_L}{C_{\text{mid}}(\eta_{\text{hi}})}$$

is genuinely nonempty. Only afterwards are η_{lo} and the recovery splitting parameters chosen. Increasing C_{buf} is not monotone for the proof: it improves the derived Jensen value rate $\beta_{\text{PPE},J}^{\text{val}}$ and the direct TV rate $\beta_{\text{PPE},\text{TV}}^{\text{src}}$ but enlarges C_{mid} . Likewise increasing ζ_g improves the weighted scale-bad exponent while enlarging C_{scl} and hence C_{mid} ; neither choice may be made after the shallow window has been declared nonempty. If the new cemetery moment makes $\chi_W \geq 1$, or if any nonempty clock class has zero degraded rate, this route is a genuine no-go unless the corresponding weighted interface is assumed directly. In particular one may not conceal the loss by choosing δ , C_{buf} , or η_{lo} after the fact.

Theorem 7.18 ($((L, T, r)$ removal of the deep tail). Assume $\text{FZ}_{\text{geom}}(\theta_Z)$, SF_{split} , CP_{env} , UCR_{sh} , $\text{TAIL}_{\text{env}}^+$, and $\text{NREC}_{\text{deep}}(q_{\text{deep}})$, together with the finite-horizon functional-correlation estimate and the forward/reverse moving-family coupling bounds used in Lemma 6.2. Assume also QT_{path} , including the constants A, B and the path window (296). Thus Lemma 7.14 and Theorem 7.15 apply uniformly to the same master cells. No conclusion of this theorem is available from a scalar TV tail alone. Put $c_d = c_{\text{tail}} - A_{\text{loss}}$ and choose $L(s) = \lfloor \kappa_L \log(1/|s|) \rfloor$ with $\kappa_L c_d > 1$ and $\kappa_L(A + B) < 1$, as in (296). Set

$$c_{\text{col}} := \frac{\alpha_{\text{col}} c_{\text{FC}}}{q_{\text{cyl}} + \alpha_{\text{col}}}, \quad T(s) = \lfloor \kappa_T \log(1/|s|) \rfloor, \quad \kappa_T c_{\text{col}} > 1 + \kappa_L A_{\text{col}}. \quad (320)$$

Let

$$B_{m,n}^{\text{deep}}(s) = \langle Q_s^n \mathcal{R}_s^{>L(s)} Q_s^m g, f \rangle$$

with the fixed-gauge centered operators; replacing one side by Q_0 gives the same conclusion. Then

$$\sum_{m,n \geq 0} |B_{m,n}^{\text{deep}}(s)| = o(1), \quad s \rightarrow 0. \quad (321)$$

Proof. Write $\Delta_s = C|s|^{-1}e^{-c_d L(s)} = O(|s|^{\kappa_L c_d - 1}) = o(1)$. For $N = \max\{m, n\} \leq T(s)$, total variation and Markov normalization give

$$\sum_{m,n \leq T(s)} |B_{m,n}^{\text{deep}}(s)| \leq C \Delta_s T(s)^2 = o(1).$$

In the two off-diagonal regions, disintegrate the collar as in Lemma 7.14. Forward and reverse moving-table coupling for one piece gives, in the two orientations,

$$CM_\lambda \mathcal{W}_\lambda e^{\beta-m-\alpha_+n+\alpha_+R_\lambda^+}, \quad CM_\lambda \mathcal{W}_\lambda e^{-\alpha_-m+\beta_+n+\alpha_-R_\lambda^-}.$$

Because $q_{\text{deep}} > \max\{\alpha_+, \alpha_-\}$, (305) bounds each sum over λ by $C\Delta_s$ times the corresponding cone exponential. Choosing the cone slopes as in Lemma 6.2 therefore makes the complete off-diagonal sums $O(\Delta_s)$. This is the only step where the normalized deep-recovery hypothesis is used; total mass plus eventual recovery would not suffice.

It remains to consider the central band with $N > T(s)$. The collar itself, not its singular coarea limit, is the sharp four-state function $F_{s,L(s)}^{\text{col}}$. It is not dynamically Hölder. Apply Theorem 7.15 with the N -dependent rank

$$r_N = \left\lfloor \frac{c_{\text{FC}}N}{q_{\text{cyl}} + \alpha_{\text{col}}} \right\rfloor.$$

Global double centering kills both product terms, and (311)–(314) give

$$|B_{m,n}^{\text{deep}}(s)| \leq C|s|^{-1-A_{\text{col}}\kappa_L} e^{-c_{\text{col}}N}.$$

There are $O(N)$ pairs with maximum N , so

$$\sum_{\substack{m \lesssim n \\ \max(m,n) > T(s)}} |B_{m,n}^{\text{deep}}(s)| \leq C|s|^{-1-A_{\text{col}}\kappa_L} \sum_{N > T(s)} N e^{-c_{\text{col}}N} = o(1)$$

by (320). The three regions exhaust the quadrant, proving (321). $\square$

This construction is why a differentiable parameter family of Young towers is unnecessary. The dynamical return alphabet is fixed on clean master cells, while every atom which changes its itinerary is represented by the physical coarea current that the response formula already contains.

The clean cohomological bracket and the master refinement can now be put on one ledger. The point of the next proposition is not to create a second symbolic dynamics. It records, on the amplitude side, which already existing full branches and recovery descendants contain a physical face piece.

Lemma 7.19 (Global complete-tree refinement and full-product L^{pw}). *Let $\mathcal{F}_0 \subset \dots \subset \mathcal{F}_5$ be the finite or countable label filtration. Its successive increments record the physical chart/return word, the two endpoint contexts, the native stopping antichain, the homogeneity cuts with component index, and the time orientation/path-depth label. At every stage let $R_i(\lambda_{i-1}, d\lambda_i)$ be the normalized mass kernel of the refinement. Assume*

$$R_i(\lambda_{i-1}, \Lambda_i) = 1 \quad \text{for every parent, including cemetery continuations,} \quad (322)$$

and assume the complete-tree tilt normalization already required in (107):

$$1 \leq \mathbb{E}_{\Pi_{s,N}} \mathcal{W}_* \leq C_W. \quad (323)$$

and, with $\mathfrak{F}$ denoting a positive mark- Z envelope and ρ_{PT} the predictable factor from (334), require the pointwise domination

$$(1 + \rho_{\text{PT}}) \text{Mark}^2\{1 + Z(\mathcal{G}^-) + Z(\mathcal{G}^+)\} \leq C\mathfrak{F} \quad (324)$$

on every actual, split, survivor, and cemetery continuation. Put $\mathcal{C} = 1$ on survivors and $\mathcal{C} = \mathcal{C}_i$ on B_i . With unobserved pre-failure factors replaced by the harmless upper-bound value one only after $\mathcal{C}_i$ has been inserted, one has on the entire tree

$$(1 + \rho_{\text{PT}})\mathscr{W}_* \leq e^{\epsilon(|c^-|+|c^+|)}(1 + H)^{a_H} \mathfrak{d}^{A_{\text{dom}}} \mathfrak{F} S^{a_{\text{scl}}} \mathcal{C}. \quad (325)$$

Assume

$$\begin{aligned} \mathbb{E}[(1 + H)^{q_H} \mid \mathcal{F}_1] &\leq C, & \mathbb{E}[\mathfrak{F}^{q_F} \mid \mathcal{F}_2] &\leq C, \\ \mathbb{E}[\mathfrak{d}^{A_{\text{dom}} q_D} \mid \mathcal{F}_1] &\leq C, & \mathbb{E}[S_\lambda^{a_{\text{scl}} q_S} \mid \mathcal{F}_3] &\leq C, \\ \mathbb{E}[e^{q_R \epsilon(|c^-|+|c^+|)} \mid \mathcal{F}_0] &\leq C. \end{aligned} \quad (326)$$

For the piecewise cemetery factor require instead the genuine complete-tree joint moment

$$\sup_{s, N, \text{actual/split}} \mathbb{E}_{\Pi_{s, N}}[\mathcal{C}^{q_C}] \leq C. \quad (327)$$

Equivalently one may specify a strictly earlier parent sigma-field $\mathcal{G}_i$ and a normalized child kernel $K_i(\omega, d\xi)$ which has not yet revealed $\mathcal{C}_i$, and require

$$\int \mathcal{C}_i(\omega, \xi)^{q_C} K_i(\omega, d\xi) \leq C. \quad (328)$$

Conditioning on $\mathcal{F}_{\tau_i}$ itself is not an alternative: because $\mathcal{C}_i$ is already $\mathcal{F}_{\tau_i}$ -measurable, such a condition would say $\mathcal{C}_i^{q_C} \leq C$ pointwise and would not express an unbounded moment. These estimates are imposed on each complete actual/split tree law with its cemetery continuations; they are not asserted after conditioning on every rare frozen parent. The conditional q_F estimate is a primitive stronger than the unconditional flux sum (277), unless a separate transfer theorem to all these laws has been proved. If $\mathcal{C}$ is pointwise absorbed by an already listed factor, take $q_C = \infty$ and omit the last reciprocal cost; otherwise q_C is a genuinely new primitive moment. Define the full reciprocal cost

$$\chi_W := \frac{a_H}{q_H} + \frac{1}{q_F} + \frac{1}{q_D} + \frac{1}{q_S} + \frac{1}{q_R} + \frac{1}{q_C}. \quad (329)$$

If $\chi_W < 1$, then every

$$1 < p_W < \chi_W^{-1} \quad (330)$$

is available uniformly on the complete actual/split tree as a global law, and

$$\|(1 + \rho_{\text{PT}})\mathscr{W}_*\|_{L^{p_W}(\Pi_{s, N})} \leq C, \quad 1 \leq \mathbb{E}_{\Pi_{s, N}} \mathscr{W}_* \leq C_W. \quad (331)$$

Thus both the bare and the physical-Radon–Nikodym-weighted versions have the same L^{p_W} margin. In particular (168) follows from the displayed moment assumptions with this same p_W . This is a statement on the complete global law: it does not imply KRH_{p_W} on rare frozen-parent cells and it does not control source-incidence mass unless an incidence-space product moment is separately assumed. All coarser global marginals, including the positive cemetery masses, are unchanged by the refinement.

Proof. Equation (322) and Ionescu–Tulcea construct one complete probability law and show by iterated conditional expectation that each earlier marginal is preserved. The displayed conditional estimates imply their global moments on this same complete law. Raise the piecewise weight $(1 + \rho_{\text{PT}})\mathscr{W}_*$ to p_W . On a survivor use (104); on B_i use (118). Generalized Hölder on the complete tree uses reciprocal costs p_W/q_F , p_W/q_D , p_W/q_S ,

p_W/q_R , and $p_W a_H/q_H$, while the cemetery step uses p_W/q_C . Their sum is $p_W \chi_W < 1$, leaving one strict conjugate slot. The genuine joint moment (327), or its earlier-parent kernel realization (328), therefore proves (331). The normalization (323), not a false pointwise floor on every cemetery envelope, gives the second part of that display. This proves an actual L^{p_W} margin rather than mere integrability and does not multiply moments taken under different label laws. Coincident artificial faces are cancelled before R_1 is defined, so the construction also preserves the signed physical marginal. $\square$

Lemma 7.20 (Kernelwise product criterion for KRH_p). *For every frozen parent η of Definition 6.27, suppose the normalized descendant kernel K_η carries the same pointwise product envelope as (325), with all component moments in (326)–(328) valid under K_η , uniformly in η . The assertion includes actual, split, survivor, cemetery, and source-incidence parents. If the reciprocal cost is bounded uniformly by $\chi_W < 1$, also require the genuinely relative normalization*

$$0 < c_K \leq \int W_\eta dK_\eta \leq C_K < \infty \quad \text{uniformly in } \eta. \quad (332)$$

Then every $1 < p_W < \chi_W^{-1}$ satisfies

$$\sup_\eta \int \left(\frac{W_\eta}{\int W_\eta dK_\eta} \right)^{p_W} dK_\eta \leq C. \quad (333)$$

Hence KRH_{p_W} holds.

Proof. Generalized Hölder is applied separately inside each normalized kernel K_η , using the same strict reciprocal margin. Its numerator has a uniform L^{p_W} bound, while (332) controls the denominator. On a source-incidence parent one first normalizes the positive incidence mass; the same displayed lower normalization is required for the resulting likelihood. The constants are independent of m, n, s , Fourier mode, polynomial, and final observable. No global-to-conditional passage is used. $\square$

Corollary 7.21 (Global full-product Hölder transfer). *Assume the complete hypothesis package of Lemma 7.19: normalized refinement kernels, pointwise Mark–Z domination, the full product envelope, and $\chi_W < 1$. Assume also global unweighted aggregate probability-clock bounds on that same complete tree. Choose p_W by (330). Then the corresponding global CLOCK_PROB_W bounds hold with every already aggregated polynomial and exponential rate degraded by*

$$\Theta_{\text{agg}}^W = \Theta_{\text{agg}}/p'_W, \quad c_{\text{agg}}^W = c_{\text{agg}}/p'_W,$$

provided registry complexity has already been subtracted before this single Hölder step. A separate global incidence-space L^{p_W} bound gives the analogous global CLOCK_SRC_W transfer. Neither conclusion implies parentwise TENERGY_W , $\text{USC}_{\text{end}}^W$, or PPE tails.

Proof. For any tree event B , Hölder and (331) give

$$\Pi_{s,N}^W(B) = \frac{\mathbb{E}_{\Pi_{s,N}}[\mathcal{W}_* \mathbf{1}_B]}{\mathbb{E}_{\Pi_{s,N}} \mathcal{W}_*} \leq C \Pi_{s,N}(B)^{1/p'_W}.$$

Apply this once to each already aggregated global probability event. For source occurrences the same calculation is valid only after integrating the positive envelope measure M_{src}^q of Definition 6.38. No branchwise Jensen, conditional transfer, or signed cancellation occurs here. $\square$

Corollary 7.22 (Parent-kernel Hölder transfer). *Assume the raw context, stopping, buffer, gate, continuation, retained PPE, and upper-scale tails uniformly under every frozen parent kernel K_η . If KRH_{p_W} holds, then the corresponding parentwise tilted tails in TENERGY_W , retained PPE, and $\text{USC}_{\text{end}}^W$ hold, with each raw exponent divided by p'_W . If Lemma 7.20 is used as the sufficient route, all of its component moments and the uniform lower normalization (332) must be verified under every such kernel; the global moment Lemma 7.19 is not a substitute.*

Proof. Apply Lemma 6.28 inside each normalized kernel, retaining the survivor indicator on PPE and upper-scale events. Uniformity in η gives exactly the quantifiers required by the three parentwise interfaces. $\square$

Definition 7.23 (Physical/tree-law identification `PHYS_TREE_MATCH`). *Let $\mathfrak{M}_s^{\text{phys}} := \sum_\lambda M_\lambda \delta_\lambda$ be the positive physical label measure used by the face, shallow, and deep ledgers, and let p_{phys} be the Borel projection from the complete refinement tree to its physical label. The interface requires a predictable Radon–Nikodym factor $\rho_{\text{PT}} \geq 0$ and a normalization $0 < Z_- \leq Z_{s,\mathfrak{d}} \leq Z_+ < \infty$ such that, for every nonnegative Borel F ,*

$$\sum_\lambda M_\lambda F(\lambda) = Z_{s,\mathfrak{d}} \int F(p_{\text{phys}}\omega) \rho_{\text{PT}}(\omega) d\Pi_{s,\mathfrak{d}}(\omega). \quad (334)$$

Because p_{phys} may be many-to-one, the tree weight is not inserted into this formula as though it were a label function. Disintegrate the finite measure $\rho_{\text{PT}}\Pi_{s,\mathfrak{d}}$ over p_{phys} :

$$\rho_{\text{PT}}(\omega) d\Pi_{s,\mathfrak{d}}(\omega) = Z_{s,\mathfrak{d}}^{-1} d\mathfrak{M}_s^{\text{phys}}(\lambda) K_{s,\mathfrak{d},\lambda}^{\text{PT}}(d\omega), \quad p_{\text{phys}}\omega = \lambda, \quad (335)$$

where $K_{s,\mathfrak{d},\lambda}^{\text{PT}}$ is a probability kernel. Define the physical label weight, rather than assuming descent, by

$$\mathscr{W}_{\lambda,*}^{\text{phys}} := \int \mathscr{W}_*(\omega) K_{s,\mathfrak{d},\lambda}^{\text{PT}}(d\omega). \quad (336)$$

Then the exact commuting identity is

$$\sum_\lambda M_\lambda \mathscr{W}_{\lambda,*}^{\text{phys}} = Z_{s,\mathfrak{d}} \int \rho_{\text{PT}}(\omega) \mathscr{W}_*(\omega) d\Pi_{s,\mathfrak{d}}(\omega). \quad (337)$$

Below, $\mathscr{W}_{\lambda,}$ in a physical-label sum means precisely $\mathscr{W}_{\lambda,*}^{\text{phys}}$ from this display. If the complete weight is already p_{phys} -measurable, this reduces to its ordinary descent. The factor ρ_{PT} is fixed before (m, n, ξ, P) and the final test. It does not alter the exact definition of $\mathscr{W}_*$. Instead the single domination (324) controls both $\mathscr{W}_*$ and $\rho_{\text{PT}}\mathscr{W}_*$ in the declared $\mathfrak{F}$ moment. No undeclared separate reciprocal Hölder slot is allowed. Thus no proof may pass from a Π -moment to a physical M_λ -sum without using (334). In the common special case $\rho_{\text{PT}} = 1$, the physical positive measure is exactly the bounded normalization of the physical marginal of the complete tree.*

Definition 7.24 (Exact propagated physical/source bridge `PROP_Q_MATCH`). *Let Ω_{tree} be the complete refinement tree and let $\Omega_{\text{occ}}^{\text{phys}}$ be a standard-Borel exact occurrence/component refinement with maps $p_{\text{tree}} : \Omega_{\text{occ}}^{\text{phys}} \rightarrow \Omega_{\text{tree}}$ and $p_{\text{lab}} = p_{\text{phys}} \circ p_{\text{tree}}$. The interface supplies the actual finite positive MPD coefficient kernel $D_\omega^m(dv)$, supported on $p_{\text{tree}}v = \omega$, and defines, rather than freely chooses,*

$$\int F(v) d\mathfrak{M}_{s,\mathfrak{d}}^{m,\text{phys}}(v) := Z_{s,\mathfrak{d}} \int \rho_{\text{PT}}(\omega) \int F(v) D_\omega^m(dv) d\Pi_{s,\mathfrak{d}}(\omega). \quad (338)$$

Writing $c_{\text{MPD}}(\omega) := D_{\omega}^m(\Omega_{\text{occ}}^{\text{phys}})$, the forgetful marginal is the exact identity

$$(p_{\text{lab}})_* \mathfrak{M}^{m, \text{phys}}(A) = Z_{s, \mathfrak{d}} \int \mathbf{1}_{\{p_{\text{phys}} \omega \in A\}} \rho_{\text{PT}}(\omega) c_{\text{MPD}}(\omega) d\Pi_{s, \mathfrak{d}}(\omega). \quad (339)$$

The coefficient kernel is normalized by the physical signed-recombination identity

$$J_{\omega}^{\text{phys}} = \int s_{\text{pol}}(v) U_v \nu_v D_{\omega}^m(dv) \quad (340)$$

in the fixed physical test dual. Hence multiplying D_{ω}^m by an arbitrary scalar is not allowed.

There is also one predictable Borel component envelope density $\mathcal{C}_{\text{occ}}^{\text{phys}}(v)$, fixed before the forbidden choices and containing the exact component-specific multiplicity, source/operator, recovery, and response costs. It is not an extra factor appended after the tree ledger: the defining disintegration bound is

$$\int \mathcal{C}_{\text{occ}}^{\text{phys}}(v) D_{\omega}^m(dv) \leq \mathscr{W}_*(\omega) \quad \text{for } \Pi_{s, \mathfrak{d}}\text{-a.e. } \omega. \quad (341)$$

The same inequality is required after inserting every record-measurable owner/handler restriction used to transfer a physical rate; otherwise that rate is assumed directly on the propagated q -restriction. Define

$$d\mathfrak{M}_{s, \mathfrak{d}}^{q, \text{phys}} := \mathcal{C}_{\text{occ}}^{\text{phys}} d\mathfrak{M}_{s, \mathfrak{d}}^{m, \text{phys}}. \quad (342)$$

Consequently

$$\mathfrak{M}_{s, \mathfrak{d}}^{q, \text{phys}}(\Omega_{\text{occ}}^{\text{phys}}) \leq Z_{s, \mathfrak{d}} \int \rho_{\text{PT}} \mathscr{W}_* d\Pi_{s, \mathfrak{d}} = \sum_{\lambda} M_{\lambda} \mathscr{W}_{\lambda, *}^{\text{phys}}. \quad (343)$$

Thus no component cost can be introduced after the proved label ledger. The two measures have Borel record map Rec_{phys} whose fields are

$$(j, o, \text{type}, \varepsilon_{\text{or}}, s_{\text{pol}}, h_{\text{rec}}, \ell_{\text{word}}, \kappa, \pi, \nu, U, K, R, \gamma, C_U, C_{\text{car}}, \mathcal{A}^{\text{opp}}, \mathcal{P}_{\text{rec}}). \quad (344)$$

Let Rec_{src} be the identical record extracted from (130). There is one jointly Borel Markov occurrence kernel $\mathcal{K}^{\text{occ}}$, projectively fixed before (m, n, ξ, P) and the final test, whose support lies in the exact compatibility graph

$$\text{supp } \mathcal{K}^{\text{occ}} \subset \{(v; \lambda, a) : \text{Rec}_{\text{phys}}(v) = \text{Rec}_{\text{src}}(\lambda, a)\}. \quad (345)$$

and it is mass preserving:

$$\mathcal{K}^{\text{occ}}(v; I) = 1 \quad \text{for } \mathfrak{M}^{m, \text{phys}}\text{-a.e. } v. \quad (346)$$

In particular the kernel cannot swap two labels having the same mass but different owner, type, orientation, polarity, certified recovery side or alternative recovery payload, word depth, clock, passage, or occurrence index. It satisfies the exact positive coefficient identity and propagated envelope domination

$$\int \mathcal{K}^{\text{occ}}(v; B) d\mathfrak{M}_{s, \mathfrak{d}}^{m, \text{phys}}(v) = \int_B d\Pi_{s, \mathfrak{d}}(\lambda) m_{\lambda}(da), \quad (347)$$

$$\overline{M}_{s, \mathfrak{d}}^q(B) \leq C \int \mathcal{K}^{\text{occ}}(v; B) d\mathfrak{M}_{s, \mathfrak{d}}^{q, \text{phys}}(v) \quad (348)$$

for every Borel $B \subset I$. Finally, for every $\Phi \in \mathfrak{T}_{\text{src}}$ the physical occurrence current equals the grouped source current obtained from (143); equivalently, after inserting $\mathbf{1}_B$, both sides

of (347) integrate the same recorded scalar $s_{\text{pol}}\langle U\nu, \text{pr}_{\text{pri}} \Phi_a \rangle$. This exact current identity is part of the interface and is stronger than arbitrary measure domination. The physical q -measure in (348) is exactly (342). Equations (338)–(340), the Markov identity, and the two pushforward relations form one commuting physical–tree–source diagram. No parallel physical mass or independently rescaled occurrence decomposition may be substituted.

Proposition 7.25 (Global physical ledger). *Assume all hypotheses of Lemma 7.19, including its normalized refinement kernels, pointwise domination, full product envelope, return/context moments, and complete-tree cemetery moment, and assume*

$$\begin{aligned} & \text{REC}(q_{\text{rec}}), \text{NST}_{\text{phys}}, \text{SS}(d_Z), \\ & \text{ACTUAL_SLOPE_BRIDGE}, \text{PAIR_SAFE_REFERENCE}, \text{PAIR_TENERGY}, \\ & \text{PPE}_N(d_Z), \text{CEM_DOM}, \text{TENERGY}_W \\ & \text{DUAL_MASS_KERNEL}, \text{MPD}, \text{POST_CARRIER_DOM}, \text{PHYS_TEST_LIFT}, \\ & \text{PHYS_TREE_MATCH}, \text{PROP_Q_MATCH}, \\ & \text{MAIN_CLOCK_COVERAGE}, \text{SEL_JSL}_{p_{\text{joint}}}, \\ & \text{CLOCK_PROB}_W, \text{CLOCK_SRC}_W, \\ & \text{KRH}_{p_W}, \text{USC}_{\text{end}}^W(q_g^W), \text{BR}_{\text{sec}} \\ & \text{FZ}_{\text{geom}}(\theta_Z), \text{SF}_{\text{split}}, \text{CP}_{\text{env}}, \text{UCR}_{\text{sh}}. \end{aligned}$$

Assume in addition the exact physical/tree and record-preserving physical/source bridges of Definitions 7.23 and 7.24. In particular, (334) is used before any tree moment is converted to an M_λ -sum, while (347) identifies the exact occurrence current and (348) controls its propagated envelope. Neither statement follows merely from $w^q \geq w^m K \gamma^{-R} \overline{\mathcal{A}}^{\text{opp}} C_{\text{prop}}$. Assume MAIN_CLOCK_COVERAGE and the selection-safe likelihood and base bounds (198) and (208) with the same constants. For every nonempty main time–Jensen cell also assume the unselected parentwise reference energy (207); it is not supplied by CLOCK_PROB_W. Assume also the complete path condition QT_{path} of Definition 7.4, including both the jet condition and the independent loss/return-tail ledger. Let $q_H, q_F, q_D, q_S, q_R, q_C > 1$ be the available context, flux, inverse-scale, positive-scale, return/context, and complete-tree cemetery moment exponents, and assume

$$\chi_W = \frac{a_H}{q_H} + q_F^{-1} + q_D^{-1} + q_S^{-1} + q_R^{-1} + q_C^{-1} < 1. \quad (349)$$

After artificial-face cancellation, the complete first defect has a disjoint signed refinement

$$\lambda = (\chi, a, k, c^-, g, c^+; Q; j, k', r, \sigma), \quad (350)$$

where χ is a physical chart, a a return word, k the insertion time, c^-, g, c^+ the single-source gate data, Q a native gate ball, (j, k', r) the incoming/outgoing strips and connected-component index, and σ the chosen time orientation. The refinement does not promote amplitude cuts to new inverse branches. For $a_H > 0$ and $a_{\text{scl}} > 0$ below their respective moment thresholds,

$$\sum_{\lambda} M_\lambda \mathcal{W}_{\lambda,*} < \infty, \quad (351)$$

where $\mathcal{W}_{\lambda,*}$ means the physical conditional weight (336) of the exact tree weight (105); no descent through a many-to-one physical projection is assumed, and no smaller unweighted ledger is substituted when scale-bad labels are removed. The occurrence-level bound (341)

and total identity (343) first convert this label ledger to the physical occurrence q -ledger. Then PROP_Q_MATCH makes its record-preserving occurrence-kernel pushforward dominate the corresponding restriction of $d\bar{M}_{s,\mathfrak{d}}^q = d\Pi_{s,\mathfrak{d}}q_\lambda$. Conversely, every source-rate assertion below is made directly for that propagated q -restriction, not inferred from a parallel bare mass. This identification is the one used by the *FACE*, shallow, and deep clauses above. The exact m -pushforward and grouped current identity are simultaneously those of (347); domination alone is not used to identify a physical occurrence. All constants in the functional-correlation, native-energy, fractional-phase, return, grazing, and direct- Z estimates are uniform on the spacetime master cells of the quantitative table family.

Proof. Use the normalized refinement kernels of Lemma 7.19. The context and H moments are transported through the normalized stopped kernel in NST_{phys} ; the underlying bounds come from the full-branch quotient and exponential gate hitting; Proposition 6.80 gives the component-indexed mark-boundary conditional moment before the boundary-oscillation theorem is applied; and the dominant chart hypothesis gives the conditional $\mathfrak{d}^{A_{\text{dom}}qD}$ moment. The two separate recovery moments (267) are retained on the same complete law but are not multiplied with one another. Return/grazing tails and Proposition 7.13 provide the last normalized kernel; the strict margin (292) absorbs its finite-depth loss. Apply Lemma 7.19 and the strict full reciprocal margin (349). The resulting p_W in (330) proves the required complete-tree moment. Insert the charged factor ρ_{PT} and use the exact identity (337) to obtain (351); without this step the tree moment does not control the physical M_λ -measure. This transfers only global probability aggregates. Parentwise energy and upper-scale tails use KRH_{p_W} , while additive remainders use the independent incidence-space ledger. No unrecorded $p > 2$ moment and no intersection of unrelated label laws is used. The ledger is not assumed again in the main theorem. The positive initial endpoint scale is included through $S_\lambda^{a_{\text{scl}}}$. Its excess is then split off by the *tilted* estimate (106), on that same weighted complete law, before the phase estimate is applied. Context, stopping, buffer, gate, continuation, and PPE losses are separately charged by TENERGY_W on this identical tilt, while each additional bad-support class is charged through CLOCK_PROB_W and every additive occurrence through CLOCK_SRC_W . For cemetery pieces the passage from weighted probability to actual discarded current is exactly (119); it is not inferred from the flux sum. The occurrence-level disintegration (341) prevents a component cost from being appended after the tree moment. The exact physical-to-source identification is (347) together with its compatibility graph, and its envelope domination is precisely (348); none of the moment arguments in this proof is claimed to derive either statement. $\square$

Theorem 7.26 (Conditional pathwise quantitative CM2 criterion). *Let an analytic-boundary or circular deformation $a(s)$ satisfy QT_{path} and the gauge-compatibility restriction (1). Assume a compact finite-horizon chart and, along every segment $a(\theta s)$ and every $L \leq L(s)$, one spacetime refinement satisfying UCR_{sh} , with:*

- (H1) *all hypotheses of Lemma 7.19 hold on the same complete actual/split tree. This explicitly includes the normalized refinement kernels (322), the weight normalization (323), the pointwise Mark-Z domination (324), the full cemetery product envelope (325), all conditional moments, and the genuine complete-tree cemetery moment; no collection of separate marginal moments is substituted for this package. In addition assume the finite-horizon functional-correlation bound of [10], the standard one-state marked quotient, and exponential return/context moments; a dominant inverse-scale moment $\sum_\lambda M_\lambda \mathfrak{d}_\lambda^{A_{\text{dom}}qD} < \infty$ and a positive initial-scale moment $\sum_\lambda M_\lambda S_\lambda^{a_{\text{scl}}qS} < \infty$; the separate terminal-law recovery input $\text{REC}(q_{\text{rec}})$, with $q_{\text{rec}}, \epsilon, \eta_{\text{lo}}$ chosen to satisfy (270); and exponents $q_H, q_F, q_D, q_S, q_R, q_C$ satisfying (349), with p_W chosen*

in (330); the refinement moments hold conditionally along the common refinement as in (326), whereas the possibly unbounded cemetery factor satisfies the genuine joint law (327) (or the earlier-parent kernel version (328));

- (H2) forward/reverse proper-family growth and coupling uniformly for the compact slowly moving table class, in the form of Theorems 1'–2' of [6]; Lemma 6.83 then supplies the signed multiplier bridge;
- (H3) On the persistent clean gate assume

NST_{phys}, SS(d_Z), ACTUAL_SLOPE_BRIDGE,
PAIR_SAFE_REFERENCE, PAIR_TENERGY, PPE_N(d_Z),

together with

CEM_DOM, TENERGY_W,
DUAL_MASS_KERNEL, MPD, POST_CARRIER_DOM, PHYS_TEST_LIFT,
PHYS_TREE_MATCH, PROP_Q_MATCH,
MAIN_CLOCK_COVERAGE, SEL_JSL_{p_{joint}},
CLOCK_PROB_W, CLOCK_SRC_W, KRH_{p_W}
USC_{end}^W(q_g^W), BR_{sec}.

The marked kernel satisfies the global finite-envelope closure (145), the exhaustive owner identity (147), the resolution coverage (159)–(188), the exact product reference (179)–(184), the bounded bilinear physical-test lift and its exact commutation (137)–(138), the physical/tree identity (334), and the record-preserving physical/source bridge (338)–(348). It also satisfies the projective consistency (128) and the main-source likelihood/base bounds (198) and (208) uniformly on the same parent cells. For every nonempty main time–Jensen cell the unselected reference energy (207) holds uniformly on those same frozen parents. The selection-safe joint source likelihood is verified either by the same-carrier clause of Lemma 6.54 or by its explicit CARRIER_MATCH alternative. Here KRH_{p_W} is uniform on every frozen parent kernel. These interfaces and the dominant chart bounds (230) have the parameter-uniform constants required by UCR_{sh} for $C_{\text{mid}}(\eta_{\text{hi}})N \leq L(s)$. The cutoffs (234) satisfy $0 < \eta_{\text{lo}} < \eta_{\text{hi}}$ and $r_{\text{obs}} > 2 + \beta_{\text{phase}}^W$, (237), and the complete phase exponent (233); longer shallow times are not estimated by this interface;

- (H4) the two-chart nondegeneracy $|\nabla G| \geq c_G$ and FZ_{geom}(θ_Z), including the full tangent coarea identity, weighted component complexity, angle sublevel, and good-angle exponent A_{ang} , together with SF_{split} and CP_{env} on the same spacetime cells; Proposition 6.79 then supplies (281); the shallow finite-s source satisfies SHCOL_{env} and the strict nonempty window (317); the terminal tail satisfies TAIL_{env}⁺ and NREC_{deep}(q_{deep}) with $q_{\text{deep}} > \max\{\alpha_+, \alpha_-\}$;
- (H5) MT_{DQ} holds for the fixed-gauge operator quotient, including its exact typed decomposition and (15)–(27). The regular defect satisfies, for some $1 \leq R \leq 3$,

$$\sum_{\chi} \|\zeta_{\chi} k_{\text{reg}}\|_{W^{R,1}} < \infty.$$

The local face amplitudes have the scaled $\mathcal{A}^{\alpha}$ regularity used in Proposition 6.62; observable regularity is above the dominant two-dimensional Fourier threshold. The

joint face part of (245) is concluded from Proposition 7.25, not assumed here. The growing-depth face assembly satisfies $\text{FACE}_{2\text{CUT}}$, in particular the no- $|s|^{-1}$ middle estimate (30). Independently, all main and nonmain face/word owners satisfy either $\text{FACE_TIME}_{\text{CM2}}$ or $\text{FACE_TIME}_{\text{REC}}$ on the same propagated q -restriction.

Then the complete double-centered fixed-gauge first operator defect satisfies

$$|\langle Q^n K_V Q^m g, f \rangle| \leq C \tau^m \vartheta^n \|g\|_{C^r} \|f\|_{C^r}.$$

The assembled difference quotients converge after the independent word-depth and time-depth splits. At every fixed (m, n) the response, product-current, regular, and face blocks converge respectively by (21), (22), (23), and Theorem 1.6. The shallow long-time part converges to zero by Theorem 7.16, and the deep remainder does so by Theorem 7.18; hence the CM2 series converges term by term as $s \rightarrow 0$. More precisely, if c_{off} is the off-diagonal coupling exponent and c_Z is the exponent obtained by balancing the FZ_{geom} angle sublevel, one may choose

$$c_* < \min \left\{ \begin{array}{l} c_{\text{off}}, c_{\text{low}}, -2 \log \theta_-, -2 \log \theta_+, \\ \eta_{\text{lo}} \beta_{\text{phase}}^W, \eta_{\text{hi}} \rho_{\infty} - a_{\text{ultra}}, c_{\text{scl}}, \\ c_{\text{time}, J}^{\text{val}}, c_{\text{time}, \text{TV}}^{\text{src}}, \eta_{\text{lo}} R, c_{\text{face}, \text{time}}, c_{\text{tail}} - A_{\text{loss}}, c_Z \end{array} \right\} \quad (352)$$

and then take $\tau = \vartheta = e^{-c_*/2}$ after changing the constant.

Proof. Lemma 1.7 puts the limiting operator and every finite- s operator quotient under the same Π_0 . Coordinate commutators remain in the response block, and their uniform product decay is derived from (19)–(20), while (21)–(27) gives each fixed two-time limit; no transported observable changes the target. The mixed marginal/rank-one terms remain in the fourth product-current type and form the exact centered cluster (14) with their graph face. They obey (10) and vanish only against the two individually centered endpoint tests; they are not absorbed into the regular or face norm. Every graph-supported commutator boundary remains in the physical face atlas under (H5). The regular fixed-term limit is (23). The face fixed-core limit follows from (24) and the boundary-neighbourhood estimate (25); Theorem 1.6 then removes the intermediate triangular band before the terminal deep remainder is invoked. Its uniform CM2 majorant is instead supplied, for every fixed core and every other face/word owner, by Lemma 1.5; the event/section ledgers alone are not used to infer time decay. Thus no uniform norm bound is used as a substitute for DQ convergence in either typed block. Proposition 7.25 first supplies the complete face part of (245) from the primitive moment hypotheses. Conditions NST_{phys} , SS , $\text{ACTUAL_SLOPE_BRIDGE}$, $\text{PAIR_SAFE_REFERENCE}$, PAIR_TENERGY , PPE_N , CEM_DOM , and $\text{TENERGY}_W + \text{CLOCK_PROB}_W + \text{CLOCK_SRC}_W$, together with the complete positive stopped-source tree and its cemetery labels, give Theorem 6.51. The high-frequency balanced estimate is Proposition 6.66. The regular four-dimensional $M \times M$ defect is included in the low-frequency Proposition 6.73. Choosing $K = e^{\eta_{\text{lo}} \max\{m, n\}}$ and $H = e^{\eta_{\text{hi}} \max\{m, n\}}$ produces the low, intermediate-phase, ultra-high observable, scale-bad, and regular-tail exponents in (352).

Outside the diagonal band use Proposition 6.84; Lemma 6.2 converts the three estimates to product decay. For radial pieces Proposition 6.80, Lemma 6.81, and Corollary 6.82 give the common physical majorant without a component-length estimate and with every component explicitly summed. Proposition 7.25 gives the global physical ledger and its L^{PW} margin from the primitive hypotheses. Lemma 7.6 and Proposition 7.13 give the common coarea atlas at each fixed depth. Theorem 1.6 alone upgrades those fixed cores

through the middle band to $L(s)$. Theorem 7.16 removes shallow times beyond $T_{\text{sh}}(s)$. Lemma 7.14 and Theorem 7.18 give the required $o(1)$ CM2-norm tail beyond $L(s)$. No fixed neighborhood is claimed to be transverse at every depth, and no qualitative residual assertion is used in this step. The four fixed-term topologies, FACE_{2CUT}, and the fixed-term uniform ledgers meet in (44). The shallow exponent window converts its triangular bound into a common summable exponential array. The deep block is different: Theorem 7.18 gives direct $\ell^1(\mathbb{N}^2)$ -norm convergence to zero. Thus the final limit–double-sum exchange is dominated convergence for the fixed/shallow blocks plus this separate uniform-integrability estimate for the deep remainder; it is not claimed to be one literal application of (44) to every triangular block. $\square$

Corollary 7.27 (Product-a.e. analytic paths and the circular boundary). *Assume all hypotheses of Theorem 7.26 other than QT_{JET}, uniformly on the product-a.e. bases and directions to which the analytic realization below applies. In particular this includes QT_{LEDGER} and (1), as well as*

$$\begin{aligned} & \text{MT}_{\text{DQ}}, \text{NST}_{\text{phys}}, \\ & \text{ACTUAL_SLOPE_BRIDGE}, \text{PAIR_SAFE_REFERENCE}, \\ & \text{PAIR_TENERGY}, \\ & \text{PPE}_N, \text{CEM_DOM}, \text{TENERGY}_W, \\ & \text{DUAL_MASS_KERNEL}, \text{MPD}, \text{POST_CARRIER_DOM}, \text{PHYS_TEST_LIFT}, \\ & \text{PHYS_TREE_MATCH}, \text{PROP_Q_MATCH}, \\ & \text{MAIN_CLOCK_COVERAGE}, \text{SEL_JSL}_{p_{\text{joint}}}, \\ & \text{CLOCK_PROB}_W, \text{CLOCK_SRC}_W, \text{KRH}_{p_W}, \text{USC}_{\text{end}}^W, \text{BR}_{\text{sec}}, \\ & \text{REC}, \text{FZ}_{\text{geom}}, \text{SF}_{\text{split}}, \text{CP}_{\text{env}}, \text{UCR}_{\text{sh}}, \\ & \text{FACE}_{2\text{CUT}}, \text{FACE_TIME}_{\text{CM2}} \text{ or } \text{FACE_TIME}_{\text{REC}}, \text{SHCOL}_{\text{env}}, \text{TAIL}_{\text{env}}^+, \text{NREC}_{\text{deep}}. \end{aligned}$$

If the analytic boundary family satisfies the localized jet-direction condition (294) and the exponent window (295), then for almost every base table in the analytic product model, restricted to the gauge-compatible analytic chart, and every admissible tangent direction with the recorded derivative exponent, CM2 holds along the path. The product randomness in this statement realizes only QT_{JET}; it supplies neither QT_{LEDGER} nor any dynamical/source interface in the preceding display. For the finite-dimensional circular family, assume instead all hypotheses of the theorem including the explicitly stated QT_{path}; then the same conclusion holds only for center motions with fixed radii or for constrained radius/shape paths satisfying (1). No automatic circular transversality or independent-radius gauge is asserted.

Proof. Apply Theorem 7.5 to obtain QT_{JET}. Combine it with the separately assumed QT_{LEDGER}, and then apply Theorem 7.26. The last sentence is the constrained circular branch of the latter theorem. $\square$

Proposition 7.28 (Pathwise coarea-current convergence). *In a compact table chart sat-*

isfying (H1)–(H5), assume QT_{path} and

$UCR_{\text{sh}}, CEM_DOM, TENERGY_W,$
 $ACTUAL_SLOPE_BRIDGE, PAIR_SAFE_REFERENCE, PAIR_TENERGY,$
 $DUAL_MASS_KERNEL, MPD, POST_CARRIER_DOM, PHYS_TEST_LIFT,$
 $PHYS_TREE_MATCH, PROP_Q_MATCH,$
 $MAIN_CLOCK_COVERAGE, SEL_JSL_{p_{\text{joint}}},$
 $CLOCK_PROB_W, CLOCK_SRC_W, KRH_{p_W}, FACE_{2\text{CUT}},$
 $one\ of\ FACE_TIME_{CM2}, FACE_TIME_{\text{REC}},$
 $SHCOL_{\text{env}}, TAIL_{\text{env}}^+, NREC_{\text{deep}}.$

Then all constants in the functional-correlation, native-energy, fractional-phase, direct-Z, and parameter-tail estimates are uniform up to $L(s)$. The shallow coarea current converges at every fixed depth in (24) and satisfies (25); the middle band is uniformly integrable by (30), hence the growing-depth current converges by Theorem 1.6. Independently, it has the CM2 majorant (41), while the deep remainder is $o(1)$ in the CM2 sum.

Proof. The spacetime cells in UCR_{sh} , rather than compactness alone, make hyperbolicity, distortion, the clean gate, moving branch sections, split laws, connected cylinders, and single-face nondegeneracy uniform along the whole coarea segment. The upper-scale excess tail and every energy-tree failure are uniform on the same complete tilted labels by USC_{end}^W and $TENERGY_W + CLOCK_PROB_W + CLOCK_SRC_W$; no raw lattice cutoff or unweighted tail is used to bound the complete marked phase mass. The fixed-depth coarea formula gives chartwise convergence of orientations and amplitudes, hence bounded–Lipschitz convergence of the assembled current. The uniform component-indexed boundary- Z ledger bounds the mass of a ρ -neighbourhood of the finite singularity set by $C_{L,m,n}\rho^{\theta_{L,m,n}}$, which is exactly (25). Thus the current convergence may be tested against the fixed-time branchwise observables; this is the same weak-* plus boundary-neighbourhood mechanism as [22, Lemma 5.4], now required on the common moving-face atlas. The per-depth positive coarea/return estimate (29) is then summed before any signed recombination to obtain the middle tail with no $|s|^{-1}$ loss. The functional-correlation constants are uniform for a compact finite-horizon family, and the return/context moments and both one-sided coupling estimates are uniform by (H1)–(H2). Proposition 6.80 fixes the component-indexed density/boundary majorant. Use the strict path margin (292). Proposition 7.13 supplies the depth-dependent scalar quotient tail on the same labels. Apply the low/high central estimates and off-diagonal couplings only through $T_{\text{sh}}(s)$, sum (351), use Theorem 7.16 for the remaining shallow times, and use Theorem 7.18 for the deep part. $\square$

The construction does not require a differentiable parameter family of countable towers. The coarea depth grows like $L(s)$, while the remaining collision/grazing words are removed in the CM2 norm by the time–depth split of Theorem 7.18, not by scalar total variation alone.

8 A concrete relative-translation programme

The abstract criterion becomes mathematically useful only after its hardest interfaces are verified on one nonempty moving-table class. This section records the smallest class found in the present audit for which the known billiard technology and the remaining new work can be separated cleanly.

Definition 8.1 (Clean relative-translation class $\mathcal{Q}_{\text{tr}}$). *Let $O_1, \dots, O_d$ be strictly convex analytic scatterers in the flat two-torus, with $d \geq 2$. Fix i_* and a nonzero vector v and put*

$$O_{i_*}(s) = O_{i_*}(0) + sv, \quad O_i(s) = O_i(0) \quad (i \neq i_*). \quad (353)$$

Assume that, for $|s| \leq s_0$, the tables have no overlap, uniform finite horizon, uniform strictly positive free flight, curvature in a fixed compact interval, a common C^3 bound, and no triple singular intersection on the finite clean gate used below. All shapes and hence every component perimeter are fixed. Translation of only one scatterer is not a global conjugacy of the billiard unless the remaining configuration has a special symmetry, so this is a genuinely moving class.

Use arclength and outgoing angle on every component. Then all collision spaces are the same disjoint union M and, exactly rather than up to a Radon–Nikodym factor,

$$d\hat{\mu}_s = d\mu_0 = \frac{\cos \varphi \, dr \, d\varphi}{2 \sum_i |\partial O_i|}. \quad (354)$$

Thus the component-mass obstruction (1) is absent and the measure-trivializing gauge may be taken to be the common arclength gauge.

Proposition 8.2 (Stationary and sequential literature package). *On a sufficiently small compact subchart of $\mathcal{Q}_{\text{tr}}$, the uniform hyperbolicity, growth, distortion, and proper-family recovery constants can be chosen independently of the table. For each fixed s , stationary repetition of F_s preserves the common collision measure (354); once a coarea component has independently been shown to be a proper regular standard family, stationary coupling gives the forward centered estimate in (39), and the reverse estimate follows after applying the billiard time-reversal involution to the separately typed reverse representation. For an admissible nonautonomous small-step sequence, the literature instead gives uniform two-law loss of memory. It does not assert that the cross-configuration maps preserve μ_0 or directly give a fixed- μ_0 centered sequential correlation.*

Proof. The common phase space and the invariant probability for each stationary F_s are (354). Uniform perturbation and spectral stability for movements and deformations of Lorentz scatterers are proved in [7]; uniform exponential two-law loss of memory for arbitrary small-step moving-scatterer sequences is proved in [6]. The latter comparison is not recast as invariance of the usual collision measure across different configurations. The positive standard-family mechanism needed after restriction is consistent with the long-curve/small-bad-part hypothesis of [21] and with the conditional invariance and exponential mixing of standard families in [22]. Those papers do *not* prove that every moving coarea face in the present problem is such a family, nor do they prove the actual-SRB polynomial escape or the occurrencewise likelihood below. These remain explicit gates rather than being hidden inside this proposition. $\square$

The first new gate replaces the full prescribed-depth escape statement by a concrete derivative test. Write $D = d_Z + 1$.

Definition 8.3 (Relative-translation prescribed-depth jet gate $\text{TR_PPE}_{\text{JET}}$). *On each stopped conditional Q , use a normalized native coordinate $z \in [-1, 1]$. At depth N there is an exceptional subset of conditional mass at most $Ce^{-b_{\text{tr}}N}$. Its complement is the union of at most $Ce^{a_{\text{cmp}}N}$ intervals. On each interval require a clean finite-holonomy truncation at $L = \lceil \kappa_h N \rceil$,*

$$X_{\beta, Q}^{[N]} = X_{\beta, Q}^{[N, L]} + \mathcal{R}_{N, L}, \quad \|\mathcal{R}_{N, L}\|_{\infty} \leq Ce^{-c_h L} \leq r/100. \quad (355)$$

The finite-prefix function $X_{\beta,Q}^{[N,L]}$ is analytic on the clean analytic-flight chart and satisfies

$$\inf_z \left| \partial_z^D X_{\beta,Q}^{[N,L]}(z) \right| \geq e^{-a_{\text{jet}}N}. \quad (356)$$

The normalized collision-SRB density in this coordinate is bounded by a uniform constant. The same constants work on every frozen context and retained positive amplitude used by $\text{SS}(d_Z)$, with its amplitude cost charged in the existing PPE mark. Require

$$C_{\text{buf}}(a_{\text{jet}} + Da_{\text{cmp}}) < 1. \quad (357)$$

The truncation window also requires $c_h k_h > C_{\text{buf}}^{-1}$.

Lemma 8.4 (Jet gate supplies the PPE small-ball clauses). *Assume Definition 8.3. Then clause (PPE1) holds with every exponent*

$$0 < \Theta_{\text{PPE}} < \min \left\{ b_{\text{tr}} C_{\text{buf}}, \frac{1 - C_{\text{buf}}(a_{\text{jet}} + Da_{\text{cmp}})}{D} \right\}. \quad (358)$$

The retained positive-amplitude sentence in the gate transfers this same sublevel bound, but the full endpoint/Jacobian derivative and physical Lipschitz mark requirements of (PPE2) remain additional. If those marks, the retained-continuation and resolving-buffer clause (PPE3) from NST_{phys} , and the finite- s UCR_{sh} uniformity clause (PPE4) are verified on the same descendants, then the full $\text{PPE}_N(d_Z)$ interface holds. None of these three additions follows from the derivative gate.

Proof. For $P \in \mathbb{R}_{d_Z}[z]$ the D th derivative of $X_{\beta,Q}^{[N,L]} - P$ is the left side of (356). The one-dimensional finite-type sublevel estimate on each good interval gives

$$|\{X_{\beta,Q}^{[N,L]} - P \leq Cr\}| \leq C(re^{a_{\text{jet}}N})^{1/D}.$$

Sum over at most $Ce^{a_{\text{cmp}}N}$ intervals and add the exceptional mass. With $N = \lceil C_{\text{buf}} \log(1/r) \rceil$, the two terms have the powers displayed in (358). This proof uses the actual absolutely continuous conditional and never renormalizes a thin surviving Cantor set. The last inequality in (355) transfers the same bound, after changing the fixed constant, to $X_{\beta,Q}^{[N]}$. $\square$

Definition 8.5 (Terminal rough-gate alternative $\text{TR_PPE}_{\text{TERM}}$). *Put $D = d_Z + 1$ and $N = \lceil C_{\text{buf}} \log(1/r) \rceil$. Suppose the native conditional Q , of diameter comparable to $\sigma(r) = r^{1/(1+\alpha_L)}$, has a native recovery coefficient $C_g > 0$ and enters a uniform regular-family class after at most*

$$R_Q \leq \rho_{\text{rec}}N + C, \quad \rho_{\text{rec}} := \frac{C_g}{(1 + \alpha_L)C_{\text{buf}}}, \quad (359)$$

outside mass $Ce^{-b_{\text{rec}}N}$. Fix $0 < \theta_0 < \theta_1 < 1 - \rho_{\text{rec}}$ and one of the following two gate mechanisms, before the linked query.

(G1) *There are a finite rough gate $\mathcal{G}$, a block length n_0 , and $p_0 > 0$ such that every normalized family in the regular class hits $\mathcal{G}$ in the next n_0 -block with conditional probability at least p_0 . Restriction to a miss and renormalization stays in the same class. Put*

$$b_{\text{gate}} := \frac{|\log(1 - p_0)|}{n_0}. \quad (360)$$

(G2) *There is a uniform, possibly time-dependent, rough-gate family $\mathcal{G}_j$ and a record-preserving stopped antichain of its first entrance pieces such that, from every recovered proper family, the mass not yet assigned to an entrance piece after t collisions is at most $Ce^{-b_{\text{gate}}t}$. No assertion that each normalized remainder is immediately regular is made in this renewal version.*

Ordinary mixing by itself supplies neither mechanism. At

$$j_- := \lceil (1 - \theta_1)N \rceil \quad (361)$$

restart the chosen gate/renewal mechanism on the then-proper image family; properness at this restart is part of the terminal input. Ignore earlier gate visits and take the first accepted entrance after j_- , requiring $j \leq \lfloor (1 - \theta_0)N \rfloor$. Thus its remaining length satisfies $k = N - j \in [\theta_0 N + O(1), \theta_1 N + O(1)]$.

For each admissible first-hit time j , the union of all first-hit entrance cells, after removal of original conditional mass $Ce^{-b_{\text{rough}}k}$, continues to a disjoint union of at most $Ce^{a_{\text{cmp}}k}$ clean intervals in the original native z coordinate. Uniformly on those intervals, in the frozen context, retained positive amplitude, and $P \in \mathbb{R}_{d_Z}[z]$, require

$$\inf_z |\partial_z^D X_{\beta, Q}^{[N, L]}(z)| \geq e^{-a_{\text{jet}}k}, \quad L = \lceil \kappa_h N \rceil, \quad \|\mathcal{R}_{N, L}\|_\infty \leq Ce^{-c_h L} \leq r/100. \quad (362)$$

The collision-SRB conditional density in this same coordinate is uniformly bounded. All entrance cells, misses, and continuation records are Borel restrictions of the same stopped descendant law used by $\text{PPE}_N(d_Z)$.

Lemma 8.6 (Terminal gate supplies the PPE small-ball clause). *Under $\text{TR_PPE}_{\text{TERM}}$, clause (PPE1) holds with every*

$$0 < \Theta_{\text{PPE}} < \min \left\{ C_{\text{buf}} b_{\text{rec}}, C_{\text{buf}} b_{\text{gate}}(\theta_1 - \theta_0), C_{\text{buf}} b_{\text{rough}} \theta_0, \frac{1 - C_{\text{buf}} \theta_1 (a_{\text{jet}} + D a_{\text{cmp}})}{D} \right\}, \quad (363)$$

provided

$$\rho_{\text{rec}} + \theta_1 < 1, \quad C_{\text{buf}} \theta_1 (a_{\text{jet}} + D a_{\text{cmp}}) < 1, \quad c_h \kappa_h C_{\text{buf}} > 1. \quad (364)$$

The endpoint/Jacobian mark, continuation/buffer, and finite-s clauses (PPE2)–(PPE4) remain additional on exactly the same entrance cells. If continuation has another bad exponent, it is inserted as one more term $C_{\text{buf}} b_{\text{cont}} \theta_0$ in the minimum.

Proof. Under (G1) there are at least $\lfloor (\theta_1 - \theta_0)N/n_0 \rfloor$ admissible attempts, and iteration of the block hazard gives missed-gate mass $Ce^{-b_{\text{gate}}(\theta_1 - \theta_0)N}$. Under (G2) this is the stated stopped-antichain tail over the same window. Sum the finite-type sublevel bound over the $O(N)$ possible first-hit blocks and at most $Ce^{a_{\text{cmp}}\theta_1 N}$ clean cells. The good mass is at most

$$CNe^{a_{\text{cmp}}\theta_1 N} (re^{a_{\text{jet}}\theta_1 N})^{1/D}.$$

The strict last exponent in (363) absorbs the factor N . The recovery and missed-gate terms give the first two exponents. Summing the rough-continuation mass over the $O(N)$ hit times gives the third; its strict inequality absorbs this polynomial factor as well. The last inequality in (364) transfers the truncated phase estimate to the full phase. $\square$

Remark 8.7 (What the terminal gate still requires). *Propositions 4.31–4.33 of [28] supply an all-polynomial, all-scale roughness mechanism on the unstable Cantor conditional of an Axiom–A basic set. They do not supply (362) for the absolutely continuous collision–SRB conditional. A valid billiard application must separately prove that the gate has uniform positive hazard for the actual regular families together with miss-conditioned stability or an equivalent stopped renewal tail, and that the gate-to-native coordinate transfer preserves degree- d_Z polynomials modulo the charged remainder. Thus the terminal construction reduces the length over which losses accumulate; it does not replace the missing rough-transfer theorem. The time-dependent coupling magnet of [6, Sections 5–7 and Corollary 34] supplies a model for (G2) after proper recovery: its uncoupled remainder has a uniform geometric renewal tail even though individual leftovers need not remain proper. In its block notation one may take $b_{\text{gate}} = \Delta^{-1} |\log(1 - \zeta/2)|$. It does not put the rough jet (362) on the extracted magnet reference pieces; that identification remains new.*

The second gate makes the change of law and the physical/source identification one calculation rather than two unrelated assumptions.

Definition 8.8 (Relative-translation matching gate TR_MATCH). *For every preselection parent η , let the source occurrence envelope q_η be the restriction of the datum-independent master kernel and set*

$$\zeta_\eta(da) = \frac{q_\eta(da)}{q_\eta(Y_{\text{src}})} \quad (365)$$

when the denominator is nonzero. On $\{q_\eta(Y_{\text{src}}) = 0\}$ set $\zeta_\eta := \zeta_0$ for one fixed $\zeta_0 \in \mathcal{P}(Y_{\text{src}})$. Thus $\zeta_\eta(Y_{\text{src}}) = 1$ for every parent, including a zero-source parent; this harmless convention is Borel and is fixed before every main selector. On each clean coarea component require a record-preserving Borel holonomy/translation map $H_{\eta,a}$ from the probability carrier to the physical source carrier such that

$$\frac{d\Pi^\eta}{dK^{\text{prob}}} \leq Ce^{a_{\text{tree}}N}, \quad q_\eta(Y_{\text{src}}) \leq Ce^{a_qN}, \quad \left\| \frac{d\nu_{\eta,a}}{d(H_{\eta,a*}\kappa_\eta)} \right\|_\infty \leq Ce^{a_cN}. \quad (366)$$

Here Π^η is the exact η -marginal of the complete-tree law used in the numerator of the source occurrence measure, so the displayed Radon–Nikodym derivative is between measures on the same space. The map preserves the full immutable occurrence record and the exact coarea m -current, and its actual slope obeys the event preimage inclusion of CARRIER_MATCH at the exact active window for every $\kappa \in \{r, \text{PPE}, N_{\text{time}}\}$; an r -clock row is not inferred from the PPE inclusion. Moreover the physical coefficient and propagated occurrence measures admit one record-preserving Markov kernel K^{occ} , fixed before the linked query, for which (347), (348), and the exact scalar-current identity following those displays hold on every record-measurable restriction. Thus the source q is the dominated pushforward of the physical occurrence envelope, not an independently enlarged finite measure.

Lemma 8.9 (Concrete selection-safe likelihood). *Under TR_MATCH, the physical/source bridge is exact and SEL_JSL_∞ holds with*

$$a_L \leq a_{\text{tree}} + a_q + 2a_c. \quad (367)$$

The same statement with L^{p_o} and L^{p_c} bounds gives the joint exponent of Lemma 6.54.

Proof. Use (365) in the selection-safe lifted reference law. The occurrence factor contains the selector, which is at most one, the tree density, and $q_\eta(Y_{\text{src}})$; the two carrier coordinates

each contribute the last density in (366). Up to the uniformly bounded M_*^{-1} normalization, their product is the density Z_A , giving (367). The support, event-inclusion, exact-current, and record statements are part of TR_MATCH, so no selected conditional law or parallel physical ledger is introduced. $\square$

Proposition 8.10 (Diagonal common-law realization of TR_MATCH). *Suppose the complete physical tree and the source construction are made on one unselected parent law, and, parent by parent,*

$$\Pi^\eta = K^{\text{prob}}, \quad \nu_{\eta,a} = H_{\eta,a*} \kappa_\eta, \quad (m, q)_{\text{src}} = \mathcal{K}_*^{\text{occ}}(m, q)_{\text{phys}} \quad (368)$$

as signed current and dominating positive envelope on every record-measurable restriction. Assume that the identity occurrence kernel is record preserving, that the exact active-slope event pulls back under $H_{\eta,a}$, and that $q_\eta(Y_{\text{src}}) \leq Ce^{a_q N}$. Then TR_MATCH holds with $a_{\text{tree}} = a_c = 0$, and its unique lifted likelihood is, up to the uniformly bounded normalization,

$$Z_A = M_*^{-1} \mathbf{1}_A q_\eta(Y_{\text{src}}), \quad a_L \leq a_q. \quad (369)$$

On a zero-source parent the selected source restriction is zero and the fixed convention ζ_0 above changes neither current nor likelihood.

Proof. All three nontrivial Radon–Nikodym ratios in (366) are one. The proposal cancels the conditional q_η factor and leaves only its total mass. The exact signed-current and positive-envelope identities in (368) are precisely PROP_Q_MATCH, rather than a comparison of independently chosen source and physical ledgers. The event inclusion and immutable record statements complete CARRIER_MATCH and give (369). $\square$

The third gate joins face time to the shallow/deep mechanism instead of postulating decay of the full positive face mass.

Definition 8.11 (Relative-translation coarea recovery gate TR_FACE_REC). *The moving coarea current, after the constant-polarity decomposition, has the immutable side-specific recovery payload $\mathcal{P}_{\text{rec}}$ and a fixed number $0 < \delta < 1$. Its constants obey*

$$c_R, c_{\text{or}}, c_{\text{mix}}, c_{\text{ret}}, A_{\text{face}}, c_{\text{tail}}, A_{\text{loss}} > 0, \quad \kappa_{\text{rec}}, a_{\text{fw}} \geq 0.$$

At the linked query define the complete deep restriction

$$I_{\text{deep}, > L}(s, m, n) := (I_{\text{deep}}(\varrho) \sqcup I_{\text{main}}^{\text{deep}}(\vartheta)) \cap \{a : \ell_{\text{word}}(a) > L\}. \quad (370)$$

It contains both main and nonmain word owners. Require

$$\sup_{\mathfrak{o} \in \{-, +\}} \overline{M}^q \{a : \mathfrak{o} \in \text{Acc}(a), R_{\text{rec}}^{\mathfrak{o}}(a) > u\} \leq Ce^{-c_R u}, \quad (371)$$

$$\overline{M}^q(I_{\text{fw}}(s, m, n)) \leq Ce^{a_{\text{fw}} \max\{m, n\}}, \quad (372)$$

$$\overline{M}^q\{\ell_{\text{word}} = \ell\} \leq C_{m,n} e^{-(c_{\text{ret}} - A_{\text{face}})\ell}, \quad (373)$$

$$\overline{M}^q(I_{\text{deep}, > L}) \leq Ce^{-(c_{\text{tail}} - A_{\text{loss}})L}. \quad (374)$$

For $N = \max\{m, n\}$ put

$$\text{Long}(m, n) := \{\mathfrak{o} \in \{-, +\} : d_{\mathfrak{o}}(m, n) = N\}$$

and define the active-side mismatch filter

$$I_{\text{mis}}(s, m, n) := \{a \in I_{\text{fw}}(s, m, n) : \text{Acc}(a) \cap \text{Long}(m, n) = \emptyset\}. \quad (375)$$

It is required that

$$\overline{M}^q(I_{\text{mis}}(s, m, n)) \leq Ce^{-c_{\text{or}}N}. \quad (376)$$

On $I_{\text{fw}}(s, m, n) \setminus I_{\text{mis}}$ choose measurably $\mathfrak{o}_*(a; m, n)$ to minimize $R_{\text{rec}}^{\mathfrak{o}}(a)$ over $\text{Acc}(a) \cap \text{Long}(m, n)$, with a fixed tie order, and put

$$I_{R, \text{tail}}(s, m, n) := \{a \in I_{\text{fw}}(s, m, n) \setminus I_{\text{mis}} : R_{\text{rec}}^{\mathfrak{o}_*(a; m, n)}(a) > \delta N\}. \quad (377)$$

The two-sided bound (371) gives $\overline{M}^q(I_{R, \text{tail}}) \leq 2Ce^{-\delta c_R N}$. Every recovered cell below uses this selected tuple and its actual $R_{\text{rec}}^{\mathfrak{o}_*}$. The first image of every recovered component is a regular standard family in the uniform class of Proposition 8.2; the forward or reverse long-side test is centered and obeys its coupling estimate with exponent c_{mix} . More precisely, outside I_{mis} every recovered face component uses a pre-recorded tuple on a side in $\text{Acc}(a) \cap \text{Long}(m, n)$ and, at the recorded integer time $R_{\text{rec}}^{\mathfrak{o}}$, its tangent lies in the corresponding unstable cone and its normalized density is regular, with

$$R_{\text{rec}}^{\mathfrak{o}} \leq \kappa_{\text{rec}} \ell_{\text{word}} + C. \quad (378)$$

The two alternative tuples, when present, both obey (133) and share one coefficient; no new side, source decomposition, or positive mass is created by the query. A component for which neither orientation is available must belong to $I_{\text{deep}, > L}$ and is controlled by (374); it may not remain in the shallow face/word core. This `ORIENTED_FACE_ACCESS` clause is not inferred from analyticity: a preimage-of-grazing face can run directly into the phase boundary. The word estimate is the positive coarea estimate before recombination and contains no $|s|^{-1}$. More precisely, for both the finite- s current and its $s = 0$ limit, the exact derivative-weighted coarea density $\mathcal{M}_{\lambda, m, n}$ is the record-preserving pushforward of the same physical/source occurrence restriction supplied by `TR_MATCH`, and it satisfies the full per-depth estimate (29) with the same c_{ret} , A_{face} and $C_{m, n}$. Thus (373) is not an unweighted count of labels and does not replace either positive-current clause of `FACE2CUT`. The shallow collar is required, on every linked shallow cell and with the same restricted q , to satisfy all three clauses SHSRC^+ , SHCP_{env} , and $\text{SHREC}(q_{\text{sh}})$ of Definition 7.8, including the normalized exponential moment (299). The deep remainder similarly satisfies the positive-envelope and normalized recovery clauses of $\text{TAIL}_{\text{env}}^+$ and $\text{NREC}_{\text{deep}}$, rather than only their unnormalized global tails. Finally the strict windows

$$c_{\text{ret}} > A_{\text{face}}, \quad \kappa_L(c_{\text{tail}} - A_{\text{loss}}) > 1, \quad (1 - \delta)c_{\text{mix}} > a_{\text{fw}}, \quad (317) \quad (379)$$

hold.

Definition 8.12 (Normalized boundary-complexity recovery subgate `TR_Z_REC`). For every accessible orientation $\mathfrak{o}$, push the corresponding positive coarea component for a uniformly bounded geometric entry time to a measured unstable family $\mathcal{G}_a^{\mathfrak{o}}$. This entry map, orientation, and family are Borel, record preserving, and fixed before the linked query. Let $H_a^{\mathfrak{o}}$ be its scaled log-Hölder density mark and put

$$\mathfrak{Z}_a^{\mathfrak{o}} := 1 + H_a^{\mathfrak{o}} + Z(\mathcal{G}_a^{\mathfrak{o}}) \geq 1. \quad (380)$$

There are $C_{\text{rec}} \geq 1$, $A_Z \geq 0$, and $\chi_Z > 0$ such that the recovery clock obtained from the density and growth recursions satisfies

$$R_{\text{rec}}^{\mathfrak{o}}(a) \leq C + C_{\text{rec}} \log \mathfrak{Z}_a^{\mathfrak{o}}, \quad \log \mathfrak{Z}_a^{\mathfrak{o}} \leq C + A_Z \ell_{\text{word}}(a). \quad (381)$$

With $\mathcal{W}_a \geq 1$ the common immutable coupling mark for the two orientations, require the global recovery moment

$$\sup_{s,m,n,\mathbf{o}} \int_{\{a:\mathbf{o} \in \text{Acc}(a)\}} \mathcal{W}_a(\mathfrak{Z}_a^{\mathbf{o}})^{\chi_Z} d\overline{M}^q(a) < \infty. \quad (382)$$

The same moment is required after normalizing each shallow collar and each complete main/nonmain deep owner by its positive q -mass: in the notation of (299) and (304),

$$\begin{aligned} \sup_{s,L,\mathbf{o}} \sum_{\lambda} \pi_{s,L}^{\text{sh}}(\lambda) \mathcal{W}_{\lambda,s,L}^{\text{sh}}(\mathfrak{Z}_{\lambda,s,L}^{\mathbf{o},\text{sh}})^{\chi_Z} &< \infty, \\ \sup_{s,L,\mathbf{o}} \sum_{\lambda} \pi_{s,L}(\lambda) \mathcal{W}_{\lambda,s,L}(\mathfrak{Z}_{\lambda,s,L}^{\mathbf{o}})^{\chi_Z} &< \infty, \end{aligned} \quad (383)$$

for the shallow and deep normalized laws respectively. Zero-mass cells are omitted. Here the two clocks $R_{\text{rec}}^{\mathbf{o}}$ restrict exactly to the existing $R_{\lambda,s,L}^{\pm,\text{sh}}$ and $R_{\lambda,s,L}^{\pm}$, and the two displayed common marks are exactly the $\mathcal{W}$ fields in (299) and (304); no parallel recovery payload is introduced. Likewise $\mathfrak{Z}_{\lambda,s,L}^{\mathbf{o},\text{sh}}$ and $\mathfrak{Z}_{\lambda,s,L}^{\mathbf{o}}$ are the exact restrictions of $\mathfrak{Z}_a^{\mathbf{o}}$ under those same two q -disintegrations, so (381) holds constituent by constituent after normalization. All integrals use the propagated q law already present in TR_MATCH; no family is renormalized into a new occurrence law. Finally there are exponents q_{sh} and q_{deep} with

$$c_{\text{cpl}}^{\text{sh}} < q_{\text{sh}} < \frac{\chi_Z}{C_{\text{rec}}}, \quad \max\{\alpha_+, \alpha_-\} < q_{\text{deep}} < \frac{\chi_Z}{C_{\text{rec}}}. \quad (384)$$

Proposition 8.13 (One common Z -moment criterion closes all three recovery ledgers). *Assume TR_Z_REC and every geometric-access, exact-current, and positive-envelope clause of TR_FACE_REC other than the four recovery conclusions displayed next. Then (371), (378), SHREC(q_{sh}), and NREC_{deep}(q_{deep}) hold. More precisely, every $c_R < \chi_Z/C_{\text{rec}}$ is admissible after changing the constant, and $\kappa_{\text{rec}} = C_{\text{rec}}A_Z$ is admissible. This proposition does not derive oriented access, the coarea-to-unstable entry map, either positive mass tail, or any PPE clause.*

Proof. From (381), $\{R_{\text{rec}}^{\mathbf{o}} > u\}$ implies $\mathfrak{Z}_a^{\mathbf{o}} > \exp((u - C)/C_{\text{rec}})$. Markov's inequality in the global moment (382) gives (371) with every stated c_R . The second inequality in (381) gives the recorded word-clock bound. For $qC_{\text{rec}} < \chi_Z$,

$$\mathcal{W}_a e^{qR_{\text{rec}}^{\mathbf{o}}} \leq C \mathcal{W}_a(\mathfrak{Z}_a^{\mathbf{o}})^{qC_{\text{rec}}},$$

so the two normalized instances of (383) give respectively (299) and (304). The lower inequalities in (384) are exactly the existing coupling windows. The density recursion of [6, Lemma 13] and the normalized boundary recursion of [6, Lemma 16] furnish (381) once the displayed entry family has been constructed; they do not construct that family from an arbitrary moving coarea face. $\square$

Proposition 8.14 (Coarea recovery supplies face-time, shallow, and deep). *Under TR_FACE_REC and TR_MATCH, with the fixed δ of the former, FACE-TIME_REC holds with*

$$c_{\text{tail},\text{fw}} = \min\{\delta c_R, c_{\text{or}}\}, \quad c_{\text{fw}}^{\text{rec}} \geq \min\{\delta c_R, c_{\text{or}}, (1 - \delta)c_{\text{mix}} - a_{\text{fw}}\}. \quad (385)$$

If, in addition, the fixed-core and boundary-tightness clauses of MT_DQ hold on the same coarea atlas, then (373) gives FACE_2CUT. The shallow and deep clauses give SHCOL_{env}, TAIL_{env}⁺, and NREC_{deep} with the strict rates used by Theorems 7.16 and 7.18.

Proof. Put $I_{R,\text{tail}} \cup I_{\text{mis}}$ in the positive tail and use (377) and (376). On its complement, use the certified long-side coupling after $R_{\text{rec}}^{\circ*}$ iterates; this is (39). This proves (385). Sum (373) in word depth. The remaining conclusions are precisely the normalized shallow/deep estimates and the strict gaps in (379); the small positive mass is multiplied back only after recovery. $\square$

Theorem 8.15 (Conditional three-gate interface reduction for relative translations). *For the class $\mathcal{Q}_{\text{tr}}$, the literature-backed package in Proposition 8.2, the explicit fixed-gauge/coarea identities of MT_{DQ} , and the three concrete gates*

$$\text{TR_PPE}_{\text{JET}} \text{ or } \text{TR_PPE}_{\text{TERM}}, \quad \text{TR_MATCH}, \quad \text{TR_FACE_REC}$$

verify (PPE1), selection-safe change of law, physical/source matching, face-time, and the explicitly assumed shallow/deep cluster. The full mark clause (PPE2), continuation/buffer clause (PPE3), and finite-s clause (PPE4) remain additional hypotheses for full PPE_N ; the fixed-core and boundary-tightness clauses of MT_{DQ} remain additional hypotheses for FACE_2CUT . Let $c_{\text{ref,PPE}}^{\text{tr}}$ denote the full minimum in (206) after inserting the PPE exponent from (358) or (363), according to the active gate, and every remaining context, stopping, buffer, gate, continuation, and probability-clock rate. Set $a_L^N := a_{\text{main,PPE}}^L$ and $a_B^N := a_{\text{main,PPE}}^B$, the actual PPE-row likelihood and base losses per collision time in (198) and (208), and use the same p'_{joint} as in Theorem 6.55. Independently, on the parameter clock $t_s := \log(1/|s|)$, put $L(s) = \lfloor \kappa_L t_s \rfloor$ and $A_D := A_{\text{loss}}$ from (374). The exact new gaps are

$$\begin{aligned} c_{\text{main}}^{\text{tr}} &= \frac{1}{2} \left\{ \frac{c_{\text{ref,PPE}}^{\text{tr}}}{p'_{\text{joint}}} - a_L^N - a_B^N \right\} > 0, \\ c_{\text{face}}^{\text{tr}} &= \frac{1}{2} \min\{\delta c_R, c_{\text{or}}, (1 - \delta)c_{\text{mix}} - a_{\text{fw}}\} > 0, \\ c_{\text{deep,cut}}^{\text{tr}} &= \kappa_L(c_{\text{tail}} - A_D) - 1 > 0. \end{aligned} \tag{386}$$

Here every symbol is measured on the same physical/source occurrence law; none is a parallel unweighted word count. In particular an exceptional PPE rate smaller than the finite-type good-set rate remains inside $c_{\text{ref,PPE}}^{\text{tr}}$, and no per-time likelihood loss is rescaled by the independent parameter cutoff. The first two lines use collision time $N = \max\{m, n\}$; the last uses the independent parameter clock t_s . The last line is the finite-s difference-quotient cutoff exponent after the explicit $|s|^{-1}$ loss; it is not a product-time mixing rate. If their exponents satisfy (357) in the direct-jet case or (364) in the terminal-gate case, together with (379) and all remaining strict gaps in (233) and (352), and if all remaining hypotheses of Theorem 7.26 are verified on the same base incidence kernel, then that genuinely moving relative-translation path satisfies CM2. Those remaining hypotheses include QT_{path} , UCR_{sh} , FZ_{geom} , SF_{split} , CP_{env} , the full convergence and regular/face clauses of MT_{DQ} , and every other probability, response, and finite marked-tree input listed there. The three gates do not prove those conditions and this theorem is an interface reduction, not a nonvacuity theorem.

Remark 8.16 (Exact current frontier). *No cited theorem currently proves the three displayed gates simultaneously for the actual collision-SRB descendants of a moving dispersing billiard. The Fourier-decay results [29, 28] concern smooth Axiom-A/conformal or parabolic iterated systems and do not supply (356) through billiard homogeneity cuts. The terminal alternative lowers the accumulated jet and component losses, but still needs the actual-SRB block hazard, stability after miss-conditioning, and the native-coordinate rough*

transfer in Remark 8.7. The random-scatterer model of [8] and the random Lorentz-gas local limit theorem [9] suggest an annealed route: condition on all but one refreshed center coordinate and prove a one-block coarea derivative bound. Concretely, suppose an active refreshed coordinate c_j , with conditional density bounded above and below, is independent of the frozen opposite context, native chart, and coefficients of every admissible P , and combine this with the one-block derivative bound on $\partial_{c_j} X^{[N]}$. More generally, without that independence, require directly and uniformly in the admissible polynomials that there are at most $e^{a_{\text{env}, \text{cmp}} N}$ monotonicity pieces and

$$|\partial_{c_j}(X^{[N]} - P)| \geq e^{-a_{\text{env}} N}$$

outside joint environment-SRB mass $Ce^{-b_{\text{env}} N}$. Conditional coarea then gives

$$(\mathbb{P} \otimes \mu_Q^u)\{|X^{[N]} - P| \leq r\} \leq Cre^{(a_{\text{env}} + a_{\text{env}, \text{cmp}})N} + Ce^{-b_{\text{env}} N}.$$

Thus $N = C_{\text{buf}} \log(1/r)$ gives a positive annealed exponent if $C_{\text{buf}}(a_{\text{env}} + a_{\text{env}, \text{cmp}}) < 1$. A quenched conclusion still requires weighted Borel–Cantelli for the bad descendant mass, not a union bound over all symbolic words. This would be a useful testbed, but it does not by itself prove the autonomous pathwise theorem above. Likewise, [22] proves conditional mixing only after the shrinking boundary source is shown to be a regular standard family. The obstruction is not cosmetic: the 2026 review [2, Problem 8.7] records that small-set indicators can concentrate densities on atypical trajectories, and general aperiodic Lorentz-gas memory loss remains open. The main new geometric work is therefore sharply localized: prove the uniform depth- N derivative gate or its terminal-gate replacement, the occurrencewise coarea holonomy density, and the recovery/tail estimates on one common relative-translation refinement. In particular, one should not try to verify $\text{FACE_TIME}_{\text{CM2}}$ by showing that the entire positive face mass decays; the recovered-core alternative is the nonvacuous route.

9 Global ledger and quantitative falsification checks

For reference, the logical dependencies of the active proof are

- (D1) (H1)–(H5), UCR_{sh} , the global physical ledger (including the joint Mark– Z consequence of FZ_{geom}), and the clock/source interfaces give the prescribed-depth Jensen/TV low–high rows.
- (D2) Either face-time interface and the bounded test lift give every main/nonmain face/word product-time bound.
- (D3) (H1),(H2),(H4),(H5), the scaled face-density ledger, $\text{SHCOL}_{\text{env}}$, and $\text{NREC}_{\text{deep}}$ give shallow long-time and the two off-diagonal removals.
- (D4) The functional-correlation and forward/reverse coupling inputs in (H1),(H2),(H4), together with FZ_{geom} , SF_{split} , CP_{env} , UCR_{sh} , QT_{JET} , $\text{QT}_{\text{LEDGER}}$, $\text{TAIL}_{\text{env}}^+$, $\text{NREC}_{\text{deep}}$, and the Mark– Z ledger give envelope crossing and the (L, T, r) deep-tail removal.
- (D5) MT_{DQ} and $\text{FACE}_{2\text{CUT}}$ give the four typed fixed limits and the independent word-depth face limit.

These outputs are combined only after their types and time ranges have been fixed.

The proof has seven independent exponent ledgers.

(A1) For $N \leq T_{\text{sh}}(s)$, the low-frequency ledger is

$$K^2 e^{-c_-(1-\epsilon)m-c_+(1-\epsilon)n} + K^2 e^{-q_{\text{rec}}\epsilon m + q_{\text{rec}}C_1 \log K} + K^2 e^{-q_{\text{rec}}\epsilon n + q_{\text{rec}}C_1 \log K}, \quad K = e^{\eta_{\text{lo}}N},$$

plus the regular-density term $K^6 e^{-c_-m-c_+n}$.

(A2) The prescribed-depth intermediate face ledger is $K^{-\beta_{\text{phase}}^W}$; the ultra-high original-observable tail is $e^{a_{\text{ultra}}N} H^{-\rho_\infty}$ with $H = e^{\eta_{\text{hi}}N}$, the scale-bad tail is $e^{-c_{\text{scl}}N}$, and the regular high tail is K^{-R} . The definition (233) charges every r - or PPE-depth energy tail with its registered Jensen status; genuine-time events enter (243) separately and no event is counted twice.

(A3) The first inequality in (317) guarantees that every resolution used in items 1–2 lies inside UCR_{sh} . The face/word owner is not contained in those two ledgers; it has the independent face-time rate $e^{-c_{\text{face,time}}(m+n)}$ from (41).

(A4) The shallow long-time collar has ledger

$$|s|^{-1-A_{\text{sh}}\kappa_L} e^{-c_{\text{sh}}T_{\text{sh}}(s)},$$

which vanishes by the second inequality in (317).

(A5) Lemma 6.2 supplies the two off-diagonal coupling ledgers, while the pathwise return/grazing quotient tail is multiplied by the joint mark- Z sum (283).

(A6) The deep positive source is removed by

$$|s|^{\kappa_L(c_{\text{tail}}-A_{\text{loss}})-1} \log^2(1/|s|) \quad \text{and} \quad |s|^{-1-A_{\text{col}}\kappa_L} e^{-c_{\text{col}}T(s)}.$$

The off-diagonal recovery prefactor retains $\Delta_{s,L}$ by (305).

(A7) The fixed-gauge operator quotient is typed before absolute values. Its regular and face terms use their strong corresponding ledgers for the majorant and the weaker topologies (23) and (24) for the regular limit and fixed face core. The middle face band is charged by (30), while the product-current terms use (10)–(22) and vanish only against the two centered endpoint tests. The response term is charged to the derived estimate (43); its difference quotient converges in the weaker topology (21), rather than through an assumed CM2 norm.

All exponents are required strictly positive; in particular the κ_T^{sh} interval must be nonempty.

The primitive inventory is falsifiable term by term. MT_{DQ} must retain the fixed-gauge commutator, double-project regular and response blocks, keep every graph face with its exact marginal product correction, and separate response-space mixing from difference-quotient convergence. Its fourth type uses the concrete strong one-coordinate TV space $\mathfrak{R}_R = \mathcal{M}(M)$ and the contractive marginal maps (9), while fixed-term factor convergence is taken in the weaker $\mathfrak{R}_{\text{BL}}$ topology. No unnamed stronger marginal norm is imported from the face ledger. NST_{phys} must provide a mass-preserving actual stopping law with $\sigma_Q \asymp \sigma(r)$. SS must contain exactly one rough source, and $\text{ACTUAL_SLOPE_BRIDGE}$ must identify that source pointwise with the logarithmic derivative ratio of the two actual BR_{sec} endpoints. PPE_N must give actual-SRB escape at the prescribed depth, while $\text{PAIR_SAFE_REFERENCE}$ fixes the exact unselected branch-mixture product law and

slope and PAIR_TENERGY controls their energy at the same PPE window. CEM_DOM must dominate every discarded current by its predictable stopped-source envelope, while TENERGY_W must put that escape and every failure cemetery label under the one full event tilt. DUAL_MASS_KERNEL must measurably retain every additive occurrence even on overlapping supports, separate exact positive-component mass from norm envelope, absorb operator growth into that same propagated q , and exhaust all owners before scalar weak assembly. MPD must resolve every regular signed source into measurable positive components with occurrence-constant polarity. PROP_Q_MATCH must identify every occurrence with a record-preserving exact- m pushforward and a uniformly dominated q pushforward of the physical ledger; finiteness of the latter alone does not bound an independently enlarged q . POST_CARRIER_DOM separately controls the variation created by propagation before a J row is restricted in its carrier coordinate. PHYS_TEST_LIFT supplies one uniformly bounded lift of the two physical observables into the heterogeneous product test space, and PHYS_TREE_MATCH gives the exact Radon–Nikodym identity between the physical M_λ ledger and the complete-tree law. The master source kernel and all these records are projectively fixed before time, mode, polynomial, and final test. MAIN_CLOCK_COVERAGE supplies the separate resolution-dependent partition and handler identity inside the main owner. SEL_JSL _{p_{joint}} must compare each unnormalized selected source measure with one unselected probability proposal law; the selector, reciprocal proposal, and carrier changes all belong to the unique density Z_A . Theorem 6.55 then derives the main marked value mass from unselected reference energy, selection likelihood, declared base-mass bounds, and one global incidence Cauchy step; it is not a primitive source-clock row. CLOCK_PROB_W controls only bad-support unions and survivor mass, while CLOCK_SRC_W controls the full clock–passage matrix of positive current masses after source complexity. KRH _{p_W} supplies the uniform frozen-parent event-weight bound; a global L^{p_W} moment is not substituted for it. USC_{end}^W must charge positive endpoint-scale growth to the homogeneity indices and the initial scale mark S_λ under the complete tilted event weight on that same complete actual/split law. BR_{sec} and REC control the same-branch sections and their two separate recovery tails. FZ_{geom} counts every component; SF_{split} represents every constituent law; CP_{env} gives connected cells; UCR_{sh} continues only finite shallow labels; FACE_{2CUT} supplies uniform integrability between a fixed face core and $L(s)$, while the selected face-time interface separately supplies either propagated positive-mass decay or a recovered centered signed core plus a positively controlled unrecovered tail for every active main/nonmain face/word restriction in product time; SHCOL_{env} gives all-time shallow positive-envelope mixing; TAIL_{env}⁺ gives the deep positive source mass; NREC_{deep} preserves that source under recovery weighting; QT_{JET} supplies only minor separation and path derivatives, and QT_{LEDGER} separately supplies the actual loss/tail margin. Their conjunction is QT_{path}. They are primitive quantitative conditions, not consequences hidden in compactness or a Banach norm.

Proposition 9.1 (One hundred and three obstruction tests). *The active interfaces reject the following countermodels.*

- (T1) *If $\nu \neq \mu_0$ and $J = \delta_{(x,y)}$, fixed and moving centerings disagree. The measure-trivializing gauge excludes it.*
- (T2) *If the physical dynamics is fixed but the auxiliary gauge is changed, the transported-observable physical derivative can vanish while the fixed-gauge commutator is nonzero. MT_{DQ} retains the commutator and therefore does not identify the two targets.*

- (T3) *A clean Bernoulli gate with $p < 1$ has depth- J mass p^J ; Lemma 6.24 forbids an $O(J)$ -trial exponential physical cover.*
- (T4) *If $X^{[J]}$ has good sublevel bounds but $X^{[N]}$ collapses to P off a zero-volume clean core, an existential-depth PPE would pass while the phase energy fails. PPE_N tests $X^{[N(r)]}$ directly.*
- (T5) *If a purported energy cell has $\sigma_Q = 1$ while $\sigma(r) \rightarrow 0$, the remainder $H_\beta \sigma_Q^{1+\alpha_B}$ need not be small even when the marginal H_β gate passes. NST_{phys} enforces (94) and the joint gate (173).*
- (T6) *A fixed shallow word with $N > L(s)$ lies outside the finite-depth refinement. Theorem 7.16 sends it to the all-time collar estimate instead of the Fourier proof.*
- (T7) *For a fixed shallow word and $N \leq T_{\text{sh}}(s)$, a Fourier mode with $\Lambda = e^{L(s)^2}$ would require a PPE depth larger than $L(s)$. The cutoff H_N sends it to (236), which uses no PPE.*
- (T8) *The family $k_s = sh_s$ may have a limiting weak derivative while $\|h_s\|_{\text{strong}} \rightarrow \infty$. MT_{DQ} requires the finite-DQ uniform bound rather than only a limiting derivative.*
- (T9) *Two equal positive pieces with opposite signs have assembled TV zero but positive envelope mass two. $\text{TAIL}_{\text{env}}^+$ uses the latter and never infers it from signed TV.*
- (T10) *A deep piece of mass Δ and recovery time $q^{-1} \log(1/\Delta)$ has bounded unnormalized moment but divergent normalized moment. $\text{NREC}_{\text{deep}}$ rejects it.*
- (T11) *The same example for a shallow collar would make its off-diagonal prefactor diverge despite a uniform post-recovery rate. SHREC rejects it through (299).*
- (T12) *If one symbolic cylinder has two disconnected event components, CP_{env} assigns distinct component indices before oscillations are taken.*
- (T13) *A graph current may have nonzero marginals even though its complete current-side projection is centered. Applying the response-side zero-mass estimate to the graph and marginal corrections separately is illegal. The exact joint identity (14), together with the individual projection rule (12) only for the regular and response rows, rejects this model.*
- (T14) *Two zero-mass operators $K^A, K^B : \mathcal{B}_2 \rightarrow \mathcal{B}_1$ may have the same uniform response bound while $\widehat{K}_{s,\text{sw}}^{\text{DQ}}$ alternates between them as $s \rightarrow 0$. Equation (21) rejects the resulting nonconvergent difference quotient.*
- (T15) *A raw mode with $\Lambda = 1$ may have $a_X = e^{X^N}$ and hence physical frequency $\Xi = e^{X^N}$. The raw cutoff alone misses it; $\text{USC}_{\text{end}}^W$ either puts it in the scale-good cost C_{scl} or charges it to $e^{-C_{\text{scl}}N}$.*
- (T16) *If the derived main or exceptional Jensen value rate is smaller than the direct TV or fractional-integration rate, a phase minimum formed from probability energy alone overstates the saving. The complete value rates (212) and the typed minimum (233) return the correct smaller exponent without applying a second square root.*
- (T17) *Let $\mathbb{P}(j) \asymp e^{-qj}$, $\widetilde{G} = j$, and $\mathcal{W}_j = e^{qj}/j^2$. The unweighted exponential tail and L^1 integrability both hold, but $\mathbb{E}[\mathcal{W} \mathbf{1}_{\{\widetilde{G} > u\}}] \asymp u^{-1}$. The tilted estimate (106), or its explicit parent-kernel KRH_{pw} sufficient condition, rejects this correlated-mark model.*

- (T18) *Let an auxiliary failure have raw probability e^{-qN} and let the full event weight grow like e^{aN} on precisely those branches, with an available L^{pw} moment. Its marked tail is degraded and cannot retain exponent q . Equations (153) and Corollary 7.22 use only the degraded parentwise rate.*
- (T19) *A weight may concentrate on context- or PPE-bad branches while every unweighted energy estimate and the unweighted Jensen step remain valid. The complete law (107), including cemetery labels, and TENERGY_W force Jensen to be taken in the actual positive phase law.*
- (T20) *A regular DQ kernel may alternate between two $W^{R,1}$ -bounded densities, and a face DQ current may alternate between two amplitudes on the same common atlas. Their uniform majorant ledgers hold but neither difference quotient converges. Equations (23) and (24) reject both versions.*
- (T21) *At $m = 0$ one can have $G_0^- = 0$ but $S_\lambda^- = e^J$ from an initial coordinate change, and hence unbounded a_X . Equation (169) records $\log S_\lambda^-$, and USC_{end}^W charges its excess under the same event weight.*
- (T22) *A triangular family may converge on every fixed face depth and have no mass beyond $L(s)$ while carrying unit oscillating mass exactly at depth $L(s)$. Fixed-depth BL* convergence does not reject it; (30) does.*
- (T23) *A proposed middle-tail estimate of the form $|s|^{-1}e^{-cL_0}$ cannot be made small by first fixing L_0 and then taking $s \rightarrow 0$. The positive coarea estimate (29) and FACE_{2CUT} explicitly forbid this factor.*
- (T24) *Let a cemetery label set every unknown future factor to one, while the actual discarded current carries density e^{aN} on a failure set of mass e^{-qN} . Its probability tail has rate q , but its current has only rate $q - a$. Equation (118) rejects the missing envelope.*
- (T25) *If a remaining propagator U_i expands total variation, replacing it by a unit future factor underestimates the discarded source. The predictable C_i and Lemma 6.36 charge that expansion before the cemetery transition.*
- (T26) *A tail $e^{-cN_{PPE}(r)}$ with $r = \Xi^{-\delta}$ and $\Xi = e^{\eta_0 N_{time}}$ contributes only $e^{-\eta_0 \delta C_{buf} c N_{time}}$ before Jensen; it cannot be entered as $e^{-cN_{time}}$. The clock registry rejects this substitution.*
- (T27) *If the same PPE-depth event passes through the slope-energy Jensen step and its base occurrence mass is uniformly bounded, its saving is $\Xi^{-\delta C_{buf} c/2}$, not $\Xi^{-\delta C_{buf} c}$. The class $\mathcal{J}_{PPE}^J$ records the factor $1/2$.*
- (T28) *Counting native stopping both as a named failure and again in an additional probability registry falsely improves a minimum. Its named row and CLOCK_PROB_W forbid this double charge.*
- (T29) *At depth N , let there be e^{aN} disjoint events, each of mass e^{-cN} . Every atomic rate is positive, but their union has only rate $c - a$ and is order one when $a = c$. The aggregate probability partition bounds (157) reject the latter case.*
- (T30) *In the preceding model, summing the raw partition and applying Hölder once gives $(c - a)/p'_W$, whereas tilting every atom first gives only $c/p'_W - a$. Corollary 7.21 fixes the global order and prevents the stronger rate from being borrowed by the weaker route.*

- (T31) *If an unbounded C_i is already $\mathcal{F}_{\tau_i}$ -measurable, then $\mathbb{E}[C_i^{qC} \mid \mathcal{F}_{\tau_i}] = C_i^{qC}$ and no uniform conditional bound exists. Equation (327), or the genuinely earlier kernel (328), rejects this filtration tautology.*
- (T32) *A branch entering cemetery before depth N has no future strip indices, endpoint scale, or $\tilde{G}_N^\pm$. The survivor indicator in (106) prevents the upper-scale ledger from evaluating a nonexistent random variable; cemetery mass is charged by CEM.DOM and CLOCK.PROB_W.*
- (T33) *A hierarchy may have exponentially decaying atomic activations but a partition function with nonpositive pressure gap. Then neither survivor normalization nor a response majorant follows. The positive gaps in (157), (164), (165), and (166) declare this an explicit no-go rather than a tunable parameter loss.*
- (T34) *Let one rare support have weighted probability e^{-cN} but carry e^{aN} distinct positive remainder currents of unit envelope. A first-hit probability partition sees only e^{-cN} , whereas (146) and (162) record the true $e^{-(c-a)N}$ source mass.*
- (T35) *It is possible that every bad-support union in CLOCK.PROB_W is exponentially small while the sum of additive currents is order one. The separate incidence partition (162) makes CLOCK.SRC_W, rather than survivor probability, the required phase input.*
- (T36) *A genuine N_{time} source with bounded base occurrence mass and energy e^{-cN} contributes only $e^{-cN/2}$ after the energy/Jensen passage. The two time cells in (164) and (165), together with the final minimum (352) reject use of the full rate c .*
- (T37) *Let parent cells A_N have mass 2^{-2N} , put a raw bad subset of conditional mass 2^{-N} inside A_N , and set $W = 2^N$ only there. A global L^p bound may hold for $p < 3$, while the tilted conditional bad probability is order one. Definition 6.27 rejects this model; Lemma 7.19 alone does not.*
- (T38) *If two overlapping named events activate two different currents, assigning the overlap to the first event deletes the second current from a probability first-hit partition. The canonical owner map in Definition 6.38 identifies physical occurrences, not event supports, and therefore retains both.*
- (T39) *Let $\lambda \in [0, 1]$, one occurrence be active for every label, and let its current be δ_λ . The naked expression $\sum_{\lambda \in [0, 1]} \langle \delta_\lambda, 1 \rangle$ is undefined, while (143) is the Lebesgue integral and equals one.*
- (T40) *A countable set $I(\lambda)$ chosen by a nonmeasurable subset of the label space has finite fibres but does not define a kernel. The Borel graph and jointly measurable weighted counting kernel (129) reject it before Tonelli is invoked.*
- (T41) *Let $M = \epsilon^{-1}$ occurrences have energies $E_a = \epsilon/M$. Then $\sum_a E_a = \epsilon$ but $\sum_a \sqrt{E_a} = 1$. The global inequality (167) takes one square root after the base and energy masses are integrated and therefore does not accept the occurrencewise argument.*
- (T42) *If e^{aN} occurrences each have raw energy e^{-cN} , then direct TV summability may hold for $c > a$, whereas the Jensen value gap is $(c - 2a)/2$ and fails whenever $c \leq 2a$. The BASE-ENERGY ledger (164) declares exactly this no-go.*

- (T43) *Parentwise PPE may hold uniformly while the main occurrence base mass grows faster than its energy saving. Then the probability theorem is true but the marked value tail is false. The selection-likelihood and base bounds (198) and (208), together with positivity of (210) reject this model; the main source is not silently reinserted into CLOCK_SRC_W .*
- (T44) *There may be one source occurrence, so the occurrence-level KRH_p is trivial, while the collision-SRB slope law is uniform and the source-section slope law is δ_0 . Probability PPE then has energy $O(r)$ but source energy is one. $\text{SEL_JSL}_{p_{\text{joint}}}$ in (197) rejects this internal-law mismatch.*
- (T45) *Let $W_{\text{prob}} = W_{\text{src}} = [0, 1]$, $H = \text{id}$, $\kappa = \text{Leb}$, $\nu = \delta_0$, and $S(x) = x$. The Jacobian is one and the slopes match exactly, but $\nu \not\ll H_*\kappa$; reference energy is $O(h)$ while source energy is one. The pushforward law and RN condition (202)–(203), not a bounded Jacobian alone, reject it.*
- (T46) *If the joint likelihood is only L^p and the reference energy is r^Θ , transferring it as r^Θ misses the Hölder loss. Equations (199) and (210) use Θ/p'_{joint} before the global square root.*
- (T47) *Let the exact source mass be one while its boundary/atlas complexity tends to infinity. Setting the envelope mass equal to one loses the source norm although the probability section remains normalized. Equations (110) and (141) force this complexity into q .*
- (T48) *Every fibre can satisfy $q_\lambda(Y_{\text{src}}) < \infty$ while $\int q_\lambda(Y_{\text{src}})d\Pi(\lambda) = \infty$. Then pointwise scalar integrals do not define a bounded current on the test unit ball. The single propagated global requirement in (145) reject this model.*
- (T49) *A unit occurrence may have owner main, so (147) holds, while at one resolution every main clock filter is empty. All local rate hypotheses are then vacuous although an order-one resonant current remains. Identity (188), not the owner partition, rejects this model.*
- (T50) *Let the unselected reference law satisfy $R_0(D_h^0) = \delta_h \rightarrow 0$, and let a main selector isolate the D_h^0 branch on which the source energy is one. Conditioning a reference law on that cell would inherit a false order- δ_h bound. The fixed-before-selection proposal (193) and the unnormalized density Z_A in (197) force the selector rarity into a_A^L .*
- (T51) *If $R_0(A) = \delta$, then $\|dR_0(\cdot \mid A)/dR_0\|_p = \delta^{-1/p'}$. A scale-free normalized selected likelihood would hide exactly this blow-up. Equation (198) records it, and (210) rejects a nonpositive resulting gap.*
- (T52) *Suppose $\Sigma_A(B) > 0$ but the proposal gives $\bar{R}(B) = 0$. Every separate occurrence and carrier moment may look finite, yet no source likelihood exists. The absolute continuity and support clause in (197) makes the main lift an explicit no-go.*
- (T53) *On one fixed carrier let ν be normalized Lebesgue measure and let $\epsilon_N(x) = \text{sgn} \sin(2\pi 2^N x)$. The positive section has bounded source cost, while $\epsilon_N \nu$ has exponentially many cuts (or a smooth approximation has $C^{\alpha*}$ cost of order $2^{\alpha*N}$). The constant-polarity decomposition (114) forbids applying this uncharged carrier-dependent sign.*

- (T54) *A naive Jordan split of a regular signed density may produce positive parts vanishing at their zero sets and hence outside a uniform log-Hölder cone. It cannot be sent through SOURCE_PROP. The baseline split (115) gives two strictly positive regular pieces, while the final clause of MPD sends an unavailable regular split to a TV/cemetery row.*
- (T55) *At depth N , let bare physical mass be e^{-cN} and let the operator norm be $C_U = e^{aN}$. Every fixed-depth operator integral is finite, but the propagated current has only rate $c-a$ and does not decay when $a \geq c$. The envelope inequality (141) and Remark 6.40 put this growth into q and declare the zero gap a no-go.*
- (T56) *A normalized proposal may attach a mark whose source slope is constant on the resonant set even though the a -free reference slope has small energy. Pointwise normalization of ζ alone then preserves the wrong event. The event compatibility in (199), or the transported inclusion (205), rejects it.*
- (T57) *Positive decompositions may exist separately on every fibre while the component choice is non-Borel. Then neither the marked kernel nor Tonelli assembly is defined. The standard-Borel measurable component extension in Definition 6.33 rejects this model before any signed recombination.*
- (T58) *Let the complete positive physical ledger have total mass one, but attach to its single source occurrence an independent propagated envelope of mass e^{aN} . Fibrewise q is finite and the physical moment ledger is finite, yet neither controls the other and every claimed source rate can lose a . The occurrence-kernel domination (348) rejects this model; finiteness on the two spaces is not a substitute for their typed identification.*
- (T59) *Let the only face/word current have word depth zero and $B_{m,n}^{\text{fw}} = \mathbf{1}_{\{m=n\}}$. Every fixed-time FACE_{2CUT} tail is then trivial, but the CM2 double sum diverges. Either the propagated-envelope decay (31) or the recovered signed estimate (39) with its recovery tail, not word-depth convergence, rejects it.*
- (T60) *For two collision components, vary one perimeter while fixing the other. A near-identity component-preserving diffeomorphism cannot change either component's total mass, so no A_s can satisfy (2). Condition (1) rejects the path before any dynamical estimate.*
- (T61) *A random analytic base may satisfy every jet-minor lower bound while its unsmoothed return mass is order one or its coarea loss is $e^{2c_{\text{tail}}L}$. Theorem 7.5 supplies only QT_{JET}; the separate QT_{LEDGER} rejects this overclaim.*
- (T62) *Double projection of a graph current produces $-m_j^- \otimes \mu_0$ and $-\mu_0 \otimes m_j^+$, which are neither a four-dimensional density nor a one-dimensional graph face. The fourth type (10), its exact joint cluster (14), and its convergence (22) reject either misclassification.*
- (T63) *Let $J_s = \delta_{2s} \rightarrow \delta_0$ in BL^* , and choose a continuous bump Φ_s supported in $(s, 3s)$ with $\Phi_s(2s) = 1$ but pointwise limit zero. Uniform sup norm and branchwise pointwise convergence do not imply convergence of the pairing. The common off-boundary BL modulus and convergence in (28) reject this model.*
- (T64) *Take the actual endpoints $X(u) = Y(u) = u$ and the balanced mode $(1, -1)$, so the physical phase derivative vanishes, but attach an unrelated abstract slope satisfying*

an excellent polynomial sublevel estimate. Conditions SS and PPE_N alone do not see the mismatch. The pointwise identities (90)–(92) reject it.

- (T65) *On the circle let a frozen-opposite-endpoint carrier law have density proportional to $d(x, x')^k$ relative to arclength. Every conditional diagonal window can have size $O(h^{k+1})$, while the two endpoint marginals may both be uniform and their genuine iid diagonal energy is only $O(h)$. The common-kernel requirement (185), or the direct pair bound (184), forbids integrating the x' -dependent laws as $\kappa \otimes \kappa$.*
- (T66) *Let R_0 be uniform with slope $S(x) = x$, all likelihood and base costs equal one, but declare $h_{\text{PPE}}(N) = 1$. The small- r reference energy holds while the PPE source energy is one for every N . The scale identity (163) and the event inclusion in SEL_JSL reject this model.*
- (T67) *For occurrence j take $X_j = \{+, -\}$, $\nu_j = (1/2, 1/2)$, $\|(x, y)\|_{\mathfrak{S}_j} = |x + y| + 2^j|x - y|$, $U_j(x, y) = 2^j(x - y)$, and $m_j = q_j = 2^{-j}$. The grouped operator pairing is bounded (indeed it cancels for the unit test), but $\sum_j q_j \int |U_j^* 1| d\nu_j = \infty$. The occurrencewise definition (143) makes the grouped scalar integral legitimate, while POST_CARRIER_DOM forbids any subsequent carrier restriction not controlled by positive variation.*
- (T68) *Let the j th declared local test space be $\mathbb{R}$ with norm $2^j|t|$, and let the physical unit observable generate coordinate one in every row. All coordinates exist separately but the tuple is not a bounded product test. The uniform lift (137) rejects this model.*
- (T69) *For each mode datum $\mathfrak{d}$ let $q^{\mathfrak{d}} = \delta_{a_{\mathfrak{d}}}$ on a new mark. Every fibrewise kernel is finite, yet no finite simultaneous kernel governs all modes. The projective identity (128) rejects this post-selection construction.*
- (T70) *Two physical occurrences may have equal masses but different owner, orientation, or word depth. An arbitrary domination kernel can swap them and still preserve total mass. The support condition (345) and the exact current identity following (347) reject the swap.*
- (T71) *Let $\Pi(j) = 2^{-j}$, let the physical label mass be $M_j \asymp j^{-2}$, and attach a physical factor $C_j = j$. Every Π -moment of $\mathcal{C}$ is finite, whereas $\sum_j M_j C_j = \infty$. The exact physical/tree Radon–Nikodym identity (334), with its density charged in the full product envelope, rejects the untyped passage between these measures.*
- (T72) *A jointly linear map on $C^r(M) \times C^r(M)$ satisfying a product norm bound vanishes: its values on $(f, 0)$ and $(0, g)$ are zero, hence so is its value on (f, g) . The bounded bilinear lift (137) and the commuting identity (138) reject this formally empty interface.*
- (T73) *Let the physical Radon–Nikodym factor tend to zero while the bare Mark–Z factor grows like its reciprocal. Controlling only their product can leave every physical moment finite while the bare tree weight has no $L^{1+\epsilon}$ margin. The factor $(1 + \rho_{\text{PT}})$ in (324)–(331) controls both laws and rejects the model.*
- (T74) *An independently declared physical occurrence measure can be multiplied by any $\varepsilon > 0$ and still satisfy a pair of homogeneous domination displays. The defining formula (338), its forgetful marginal, and the signed recombination identity (340) fix its scale before the Markov occurrence kernel is applied.*

- (T75) *A conventional even mollifier has $1 - \widehat{\rho}(1/K) \asymp K^{-2}$ and therefore cannot furnish a $K^{-3}W^{3,1}$ remainder, while a generic mollifier is not band-limited at all. The explicitly flat finite-frequency multiplier (256) and its Taylor estimate (257) reject both substitutions.*
- (T76) *Take one TV occurrence with $K = m = q = 1$, $U = 100 \text{id}$, $C_U = 100$, $R = 0$, $\gamma = 1/2$, and $\mathcal{A}^{\text{opp}} = 10^{-2}$. An envelope using the unfloored opposite amplitude would falsely bound a current of size 100 by a unit q -mass. The global restrictions $0 < \gamma < 1$, $R \in \mathbb{N}_0$ and the floor (140) force the full operator cost into (141).*
- (T77) *Let two complete-tree points have the same physical label but different charged weights. Choosing either value as a putative descended label weight can undercount the physical sum while the Radon–Nikodym identity for label functions still holds. The fibre kernel (335) and conditional definition (336) replace this invalid descent by the exact conditional average.*
- (T78) *On one tree fibre let component coefficients be $D^m(j) = 2^{-2j}$ and append costs $\mathcal{C}_{\text{occ}}(j) = 2^{2j}/j$. The exact signed decomposition can have finite coefficient mass while its propagated occurrence envelope diverges. The disintegration bound (341), required again after every handler restriction, rejects this post-ledger component cost.*
- (T79) *A map may be bounded on $W^{1,1}$ but not from L^1 to L^1 : add to a valid extension a fixed exterior bump multiplied by an interior trace functional. Narrow input bumps then have vanishing L^1 mass but a nonvanishing zero Fourier coefficient after extension. The simultaneous $L^1/W^{r,1}$ bounds imposed before (256) rule out this false low-mode estimate.*
- (T80) *A positive graph ledger may have uniformly bounded total mass while an independently declared stronger marginal norm diverges on its projected marginals. The concrete choice $\mathfrak{R}_R = \mathcal{M}(M)$, the contractive identities (9), and the separate weak $\mathfrak{R}_{\text{BL}}$ convergence close the product-current type without importing such an uncharged strong norm or demanding TV convergence of a moving collar.*
- (T81) *For a checkerboard or fat-Cantor carrier event E , the restricted measure $\mathbf{1}_E \nu$ need not belong to the regular source cone even when ν does. The scalar post-propagation measure (121)–(123), rather than an application of U to that ill-typed source, defines the J restriction.*
- (T82) *Let a complete law be concentrated on a cemetery row and let a positive weight concentrate on events of probability $\delta_N \rightarrow 0$. Without a uniform normalization floor, its L^p norm divided by its mean grows like $\delta_N^{-1/p'}$. The explicit normalization (323) together with the full product envelope rejects this hidden rare conditioning; no pointwise cemetery floor is asserted in its proof.*
- (T83) *Separate moments of ρ_{PT} and of a cemetery weight can be finite while their correlated product has infinite mean. The main theorem now imports the complete package of Lemma 7.19, in particular (325); listing only the component moments is not an admissible substitute.*
- (T84) *Inside normalized parent kernels let W_η equal one on a set of mass $\delta_\eta \downarrow 0$ and δ_η^2 elsewhere. Uniform upper moments do not prevent the normalized likelihood $W_\eta / \int W_\eta dK_\eta$ from having divergent L^p norm. The parentwise lower bound (332),*

which is independent of the global tree normalization, rejects this rare-parent countermodel.

- (T85) *For every admissible time section let $N_{\text{time}} = \max\{m, n\} = N$ and choose a different unit envelope $q_N = \delta_{a_N}$; similarly one may choose q_r after the phase mode fixes r and hence $N_{\text{PPE}}(r)$. All sectionwise masses and rate bounds can be uniform although no occurrence law was fixed before the linked tuple was known. The datum-independent base law (125) and the full linked consistency (128) reject both constructions; derived resolving coordinates select filters of one law and never select a new q .*
- (T86) *Record one occurrence with $\pi(a) = \text{TV}$ and unit m, q, K, C_U , but let its post-propagation carrier density have height N on a set of ν -mass N^{-1} , so $C_{\text{car}} \asymp N$ and the restricted current is order one. Routing this recorded TV mark through $\mathcal{A}_{\text{main}}^{r, J}$ would omit C_{car} from (142) and falsely use a carrier-event mass of order N^{-1} . The record identities (158) and (189) forbid the routing; a genuine promotion to J must be recorded before freezing the base law and must charge C_{car} in q .*
- (T87) *Let a moving face have order-one positive q -mass for every N , but suppose every component has an accessible active-long-side payload recovering by time δN and its centered signed pairings decay like $e^{-c_{\text{mix}}(1-\delta)N}$. The positive interface (31) would reject this natural mixing model, whereas FACE_TIME_REC accepts it and charges cancellation only after recovery. This test prevents the sufficient positive interface from being advertised as necessary.*
- (T88) *In the preceding model let the recovered propagated envelope grow as $e^{a_{\text{fw}}N}$ with $a_{\text{fw}} \geq (1-\delta)c_{\text{mix}}$. Every component separately mixes, but their total signed bound need not decay. The strict rate (40) rejects the missing aggregate complexity gap.*
- (T89) *Let an analytic finite-prefix slope satisfy the derivative gate (356), but let the omitted holonomy tail have size $r^{1/2}$ at an energy window r . The truncated function can have a good r -sublevel estimate while the actual slope is shifted across the entire window. The same-scale remainder bound (355) rejects this substitution of analytic finite words for the actual slope.*
- (T90) *An analytic moving singularity face can point into the stable cone in forward time and hit the phase boundary in backward time. Analyticity and a bounded coarea density alone do not make it a regular unstable standard family. The orientation-specific access clause in TR_FACE_REC sends every component without a clean orientation to the positive deep tail.*
- (T91) *An annealed fresh-center estimate can have summable mean bad mass while a sparse set of environments carries all bad descendants at infinitely many prescribed depths. Averaged coarea therefore does not imply the quenched PPE_N required by the pathwise theorem. The weighted Borel–Cantelli step in Remark 8.16 is an additional gate, not a consequence of the random local limit theorem.*
- (T92) *Let one order-one physical face be usable in both recovery directions. Replacing it by two additive typed occurrences leaves the unused occurrence as an order-one positive tail in each one-sided time cone, so no exponential face-time rate follows. The single coefficient occurrence with two exact alternative factorizations (131)–(133), charged once by (134), rejects this duplication.*

- (T93) Give marks a_k masses proportional to e^{-2k} and recovery time k , but let a query cell select only a_k . The unnormalized global recovery tail is exponential while the normalized cell exponential moment diverges with k . The cellwise normalized clause (299), explicitly retained in TR_FACE_REC, rejects this false implication.
- (T94) Let a physical coefficient occurrence have mass $m_N = e^{-N}$ but source envelope $q_N = 1$, while a proposed matching kernel preserves only the exact m -current. The coefficient identity is true but it cannot transfer any physical envelope rate to the source. The q -pushforward domination (348), now part of TR_MATCH, rejects the model.
- (T95) Let the finite-type good-set exponent be one, an exceptional/context reference exponent be 0.01, and the total per-time likelihood/base loss be 0.1. A schematic formula using only the good exponent can be positive although the true main value rate is negative. The full minimum $c_{\text{ref,PPE}}^{\text{tr}}$ and the per-time formula in (386) reject both that omission and a second spurious multiplication of the likelihood loss by κ .
- (T96) A bare mark count may satisfy $q\{\ell_{\text{word}} = \ell\} \leq e^{-c\ell}$ while the finite- s derivative-weighted coarea density, or its limiting positive current, grows like $e^{2c\ell}$. Such a count does not imply FACE_{2CUT}. The two exact positive-current clauses of (29), tied to TR_MATCH, reject it.
- (T97) A depth- N analytic jet can satisfy the uniform polynomial sublevel estimate while its selected descendant omits a continuation or its finite- s resolving buffer exits the UCR_{sh} range. The jet lemma supplies only (PPE1); the separately required mark, continuation, and uniformity clauses (PPE2)–(PPE4) reject the overclaim that a state-variable derivative alone proves full PPE_N.
- (T98) In a refreshed environment take $X^{[N]}(c_j) = c_j$ and allow the opposite-context polynomial to be $P(c_j) = c_j$. A lower bound on $|\partial_{c_j} X^{[N]}|$ holds, but the sublevel event has probability one. The frozen-context independence, or the direct uniform bound on $|\partial_{c_j}(X^{[N]} - P)|$, in Remark 8.16 rejects this invalid annealed coarea step.
- (T99) Let $q_\eta(Y_{\text{src}}) = 0$. Defining $\zeta_\eta = q_\eta/q_\eta(Y_{\text{src}})$ only on positive-mass parents leaves no probability kernel on this parent, even though its selected current is zero. The fixed ζ_0 convention following (365) makes the preselection kernel total and Borel without changing any current or rate.
- (T100) A regular family can hit a fixed gate with positive unconditional frequency while restriction to successive misses concentrates on shorter and shorter curves and leaves the regular cone. Ordinary mixing then does not justify multiplying the same block hazard. The miss-conditioned stability in (G1), or the stopped renewal tail in (G2), of TR_PPE_{TERM} rejects this survivor-selection model.
- (T101) At each terminal hit time let there be $e^{a_{\text{ent}}N}$ entrance cells, each admitting $e^{a_{\text{cmp}}k}$ clean continuations and the stated derivative bound. A proof that counts only the $O(N)$ hit times misses the entrance multiplicity. Definition 8.5 bounds the disjoint union over all entrance cells at each time, before the sublevel estimate is summed.
- (T102) A power- Z moment only on I_{fw} cannot control the recovery tail on every accessible shallow/deep owner; likewise newly attached recovery clocks can satisfy that moment without controlling the clocks in SHREC or NREC_{deep}. The global domain in (382) and the exact clock/mark identifications following (383) reject both substitutions.

(T103) *Suppose every orbit visits the rough gate once before the terminal search window and never afterwards. Its global first entrance lies outside the window, so a renewal estimate started at time zero does not supply a terminal entrance. The explicit restart at (361), with earlier visits ignored, makes the gate tail apply to exactly the active window.*

Proof. Items 1–3, 8–11, and 13–103 are direct finite-measure, frequency, or norm calculations. Item 4 is the missing endpoint-depth countermodel, item 5 is the native-scale countermodel, item 6 is the word/time-depth countermodel, item 7 is the ultra-high-frequency countermodel, and item 12 is exactly the connected-refinement clause. $\square$

References

- [1] M. Demers and C. Liverani, *Projective cones for sequential dispersing billiards*, Commun. Math. Phys. 401 (2023), arXiv:2104.06947.
- [2] M. Demers and C. Liverani, *Recent progress in the application of transfer operators to dispersing billiards*, arXiv:2606.10155 (2026).
- [3] G. Froyland and M. Phalempin, *Optimal linear response for Anosov diffeomorphisms*, arXiv:2504.16532 (2025).
- [4] W. Bahsoun and S. Galatolo, *Linear response due to singularities*, arXiv:2301.02301 (2023).
- [5] M. Demers and C. Liverani, *Central limit theorem for sequential dynamical systems*, arXiv:2502.07765v2 (2025).
- [6] M. Stenlund, L.-S. Young and H.-K. Zhang, *Dispersing billiards with moving scatterers*, arXiv:1210.0011.
- [7] M. Demers and H.-K. Zhang, *A functional analytic approach to perturbations of the Lorentz gas*, Commun. Math. Phys. 324 (2013), 767–830, arXiv:1210.1261.
- [8] M. Stenlund, *A vector-valued almost sure invariance principle for Sinai billiards with random scatterers*, Commun. Math. Phys. 325 (2014), 879–916, arXiv:1210.0902.
- [9] M. Demers, F. Pène and H.-K. Zhang, *Local limit theorem for randomly deforming billiards*, Commun. Math. Phys. 375 (2020), 2281–2334, arXiv:1902.06850.
- [10] J. Leppänen and M. Stenlund, *A note on the finite-dimensional distributions of dispersing billiard processes*, arXiv:1702.01461.
- [11] N. Fleming-Vázquez, *Functional correlation bounds and optimal iterated moment bounds for slowly-mixing nonuniformly hyperbolic maps*, arXiv:2106.06486v2.
- [12] I. Baldomá, A. Florio, M. Leguil and T. M.-Seara, *Chaoticity of generic analytic convex billiards*, arXiv:2605.19897 (2026).
- [13] C. Wormell, *Conditional mixing in deterministic chaos*, arXiv:2206.09291; and *On convergence of linear response formulae in piecewise hyperbolic maps*, arXiv:2206.09292.
- [14] S. Baker and A. Banaji, *Polynomial Fourier decay for fractal measures and their pushforwards*, Math. Ann. 392 (2025), 209–261, arXiv:2401.01241.

- [15] S. Baker, O. Khalil and T. Sahlsten, *Fourier decay from L^2 -flattening*, arXiv:2407.16699.
- [16] V. Climenhaga and J. Day, *Every finite horizon Sinai billiard map has a unique measure of maximal entropy*, arXiv:2604.25881.
- [17] P. Bálint, P. Nándori, D. Szász and I. P. Tóth, *Equidistribution for standard pairs in planar dispersing billiard flows*, Ann. Henri Poincaré 19 (2018), arXiv:1707.03068.
- [18] S. Huang, G. Wang and M. Wang, *Observability inequality, log-type Hausdorff content and heat equations*, arXiv:2411.11573v2 (2024).
- [19] S. Barbieri and L. Niederman, *Bernstein–Remez inequality for Nash functions: a complex analytic approach*, arXiv:2207.13922 (2022).
- [20] O. Butterley, G. Canestrari and R. Castorrini, *Discontinuities cause essential spectrum on surfaces*, Ann. Henri Poincaré (2024), arXiv:2306.00484.
- [21] G. Canestrari, *On linear response for discontinuous perturbations of smooth endomorphisms*, Advances in Mathematics 497 (2026), 111008, arXiv:2411.16628v3.
- [22] G. Canestrari, *Linear response for Sinai billiards with small holes*, arXiv:2604.19671v2 (2026).
- [23] J. F. Alves and W. Bahsoun, *Linear response for skew-product maps with contracting fibres*, arXiv:2602.02317v2 (2026).
- [24] M. Vákár and L. M. Ong, *On s -finite measures and kernels*, Mathematical Foundations of Programming Semantics (2018), arXiv:1810.01837.
- [25] A. Ni, *Ergodic and foliated kernel-differentiation method for linear responses of random systems*, arXiv:2410.10138 (2024).
- [26] S. Galatolo and V. Lucarini, *A mathematical framework for linear response theory for nonautonomous systems*, arXiv:2603.19509v3 (2026).
- [27] G. Leclerc, *Fourier decay of equilibrium states for bunched attractors*, arXiv:2301.10623.
- [28] G. Leclerc, *Fourier decay of equilibrium states and the Fibonacci Hamiltonian*, arXiv:2507.23731v2 (2025; revised 2026).
- [29] G. Leclerc, S. Paukkonen and T. Sahlsten, *Fourier decay in parabolic $C^{1+\alpha}$ systems with overlaps*, arXiv:2505.15468v3 (2025).
- [30] A. Algom, F. Rodriguez Hertz and Z. Wang, *Polynomial Fourier decay and a cocycle version of Dolgopyat’s method for self conformal measures*, arXiv:2306.01275v2 (2024).
- [31] Y. Lima, D. Obata and M. Poletti, *Measures of maximal entropy for non-uniformly hyperbolic maps*, arXiv:2405.04676v2 (2025).
- [32] N. Chernov, *Decay of correlations and dispersing billiards*, J. Stat. Phys. 94 (1999), 513–556.

- [33] P. Teixeira, *On the conservative pasting lemma*, Ergodic Theory Dynam. Systems 40 (2020), 1402–1440, arXiv:1611.01694.
- [34] R. Rühr, *Pressure inequalities for Gibbs measures of countable Markov shifts*, arXiv:2012.13226v3.
- [35] N. Bertozzi, *Exponential mixing for hyperbolic flows on non-compact spaces*, arXiv:2603.11852v2 (2026).
- [36] V. Climenhaga, *Gibbs measures have local product structure*, arXiv:2310.17495v2 (2025).
- [37] S. Gouëzel, *Berry–Esseen theorem and local limit theorem for non uniformly expanding maps*, Ann. Inst. H. Poincaré Probab. Statist. 41 (2005), 997–1024, arXiv:math/0310391.
- [38] D. Galli, *A cohomological approach to Ruelle–Pollicott resonances and speed of mixing of Anosov diffeomorphisms*, arXiv:2405.17045.